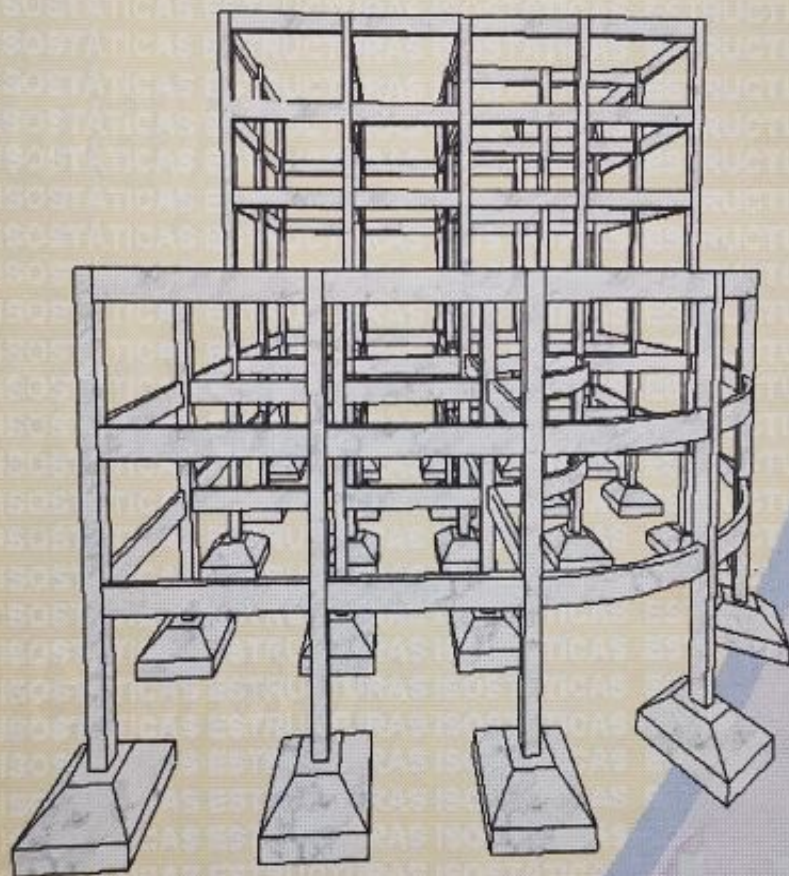




estructuras Isostáticas

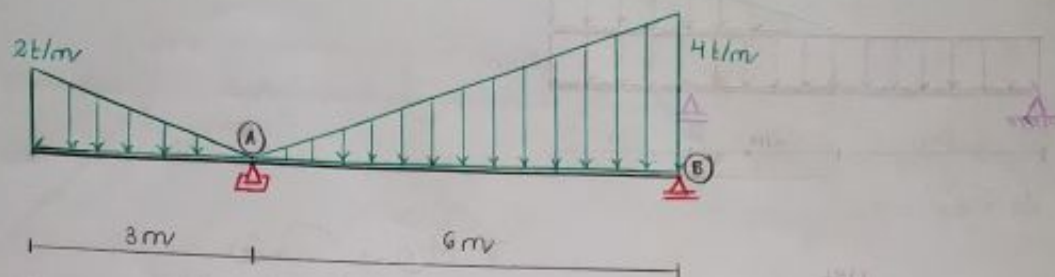
CIENTO POR CIENTO PRÁCTICO



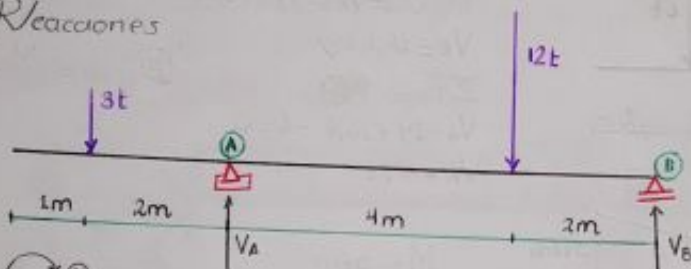
**La mayor colección de
ejercicios prácticos**

Tomás Wilson Alemán Ramírez

3 Graficar los diagramas de esfuerzos característicos utilizando el método gráfico-numérico.



1.- Cálculo de Reacciones



$$\sum M_A = 0 \quad \curvearrowright \oplus$$

$$-3(2) + 12(4) - V_B(6) = 0$$

$$V_B = 7t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$-3 + V_A - 12 + 7 = 0$$

$$V_A = 8t //$$

2.- Momentos

$$M_A = -3(2)$$

$$M_A = -6 \text{ tmv}$$

$$M_B = 0 \text{ tmv}$$

3.- Cortantes

$$Q_A = -3t \text{ (Izg)}$$

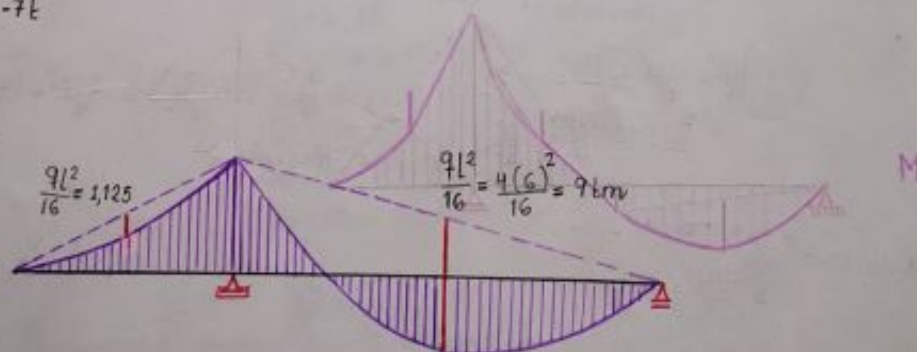
$$Q_A = -3 + 8 = 5t \text{ (Der)}$$

$$Q_B = -7t$$

4.- Diagramas

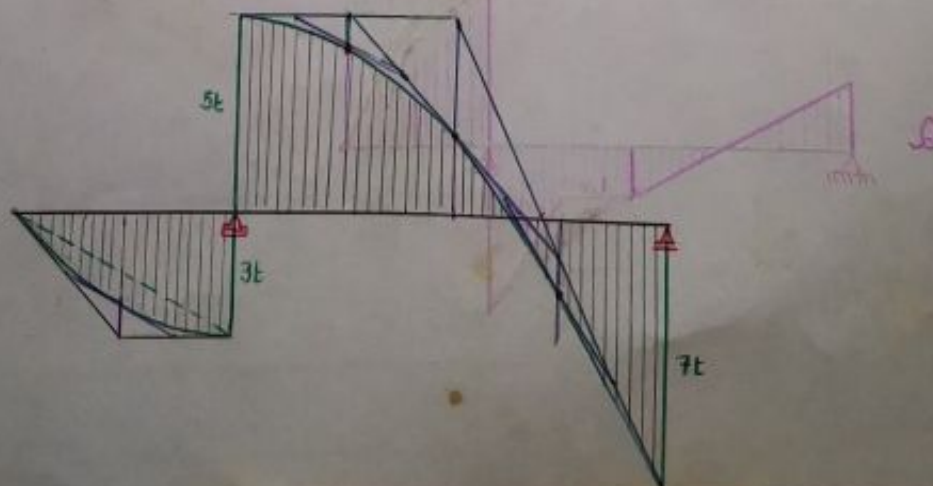
- Momento

Escala = 4 tm/cm

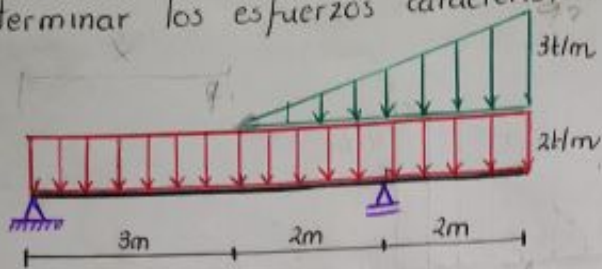


Cortante

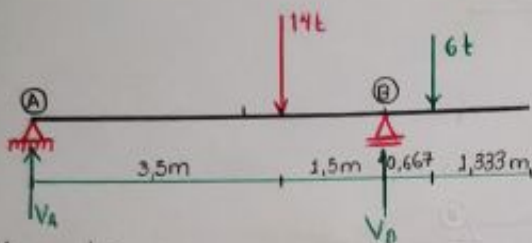
Escala = 2 tm/cm



14. Determinar los esfuerzos característicos

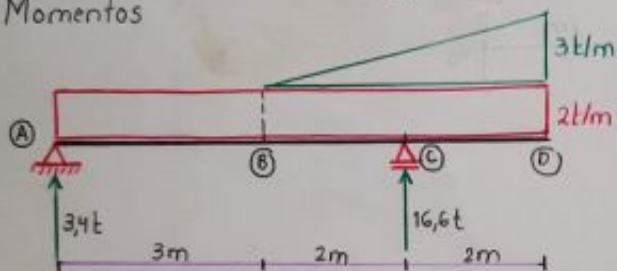


1.- Cálculo de Reacciones



$$\begin{aligned}\sum M_A &= 0 \quad \curvearrowright + \\ 14(3,5) - V_B(5) + 6(5,667) &= 0 \\ V_B &= 16,6t \\ \sum F_V &= 0 \quad \uparrow + \\ V_A - 14 + 16,6 - 6 &= 0 \\ V_A &= 3,4t\end{aligned}$$

2.- Momentos

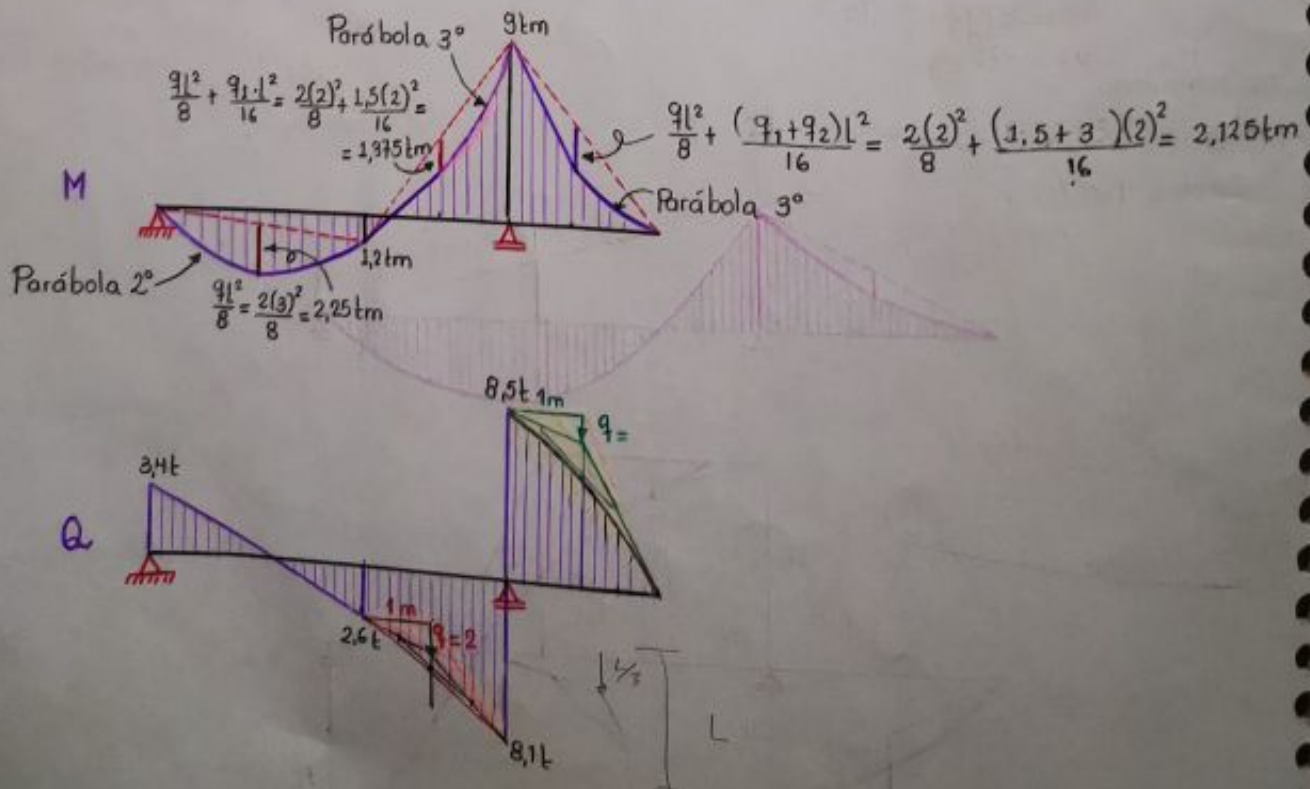


$$\begin{aligned}M_A &= 0tm \\ M_B &= 3,4(3) - 2(3)(1,5) = 1,2tm \\ M_C &= 3,4(5) - 2(5)(2,5) - \frac{1,5(2)}{2} \left(\frac{1}{3} \cdot 2 \right) = -9tm \\ M_D &= 0tm\end{aligned}$$

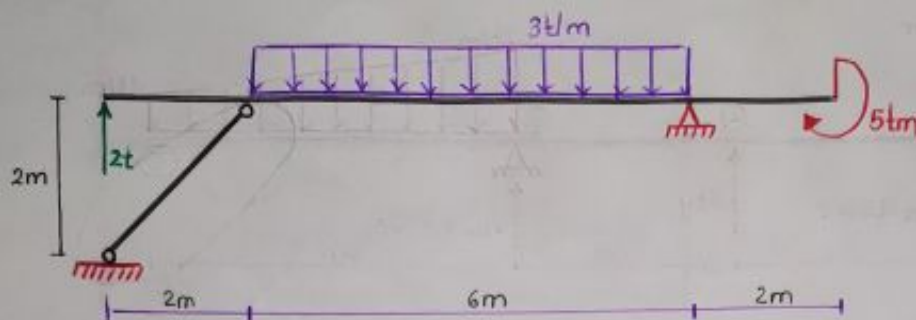
3.- Cortantes

$$\begin{aligned}Q_A &= 3,4t \\ Q_B &= 3,4 - 2(3) = -2,6t \\ Q_C (Izq) &= 3,4 - 2(5) - \frac{1,5(2)}{2} = -8,1t \\ Q_C (Der) &= Q_C (Izq) + 16,6 = -8,1 + 16,6 = 8,5t \\ Q_D &= 0t\end{aligned}$$

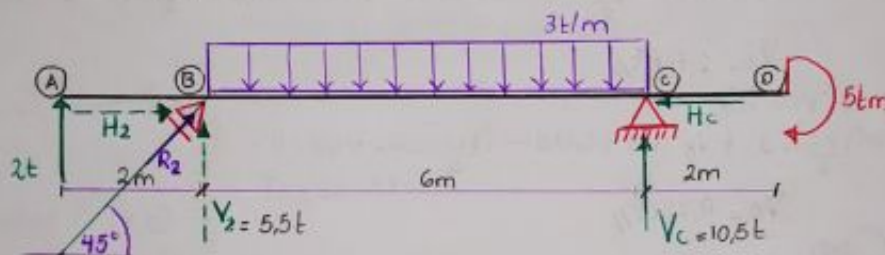
4.- Diagramas



15.- Graficar los diagrama de Momento, cortante y normal



1.- Idealización de apoyo



2.- Reacciones

$$\sum M_B = 0 \quad \curvearrowright \oplus$$

$$2(2) + 3(6)(3) + 5 - V_c(6) = 0$$

$$V_c = 10.5t$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$2 + V_2 - 3(6) + 10.5$$

$$V_2 = 5.5t //$$

$$\text{Sen } 45^\circ = \frac{V_2}{R_2} \Rightarrow R_2 = \frac{V_2}{\text{Sen } 45}$$

$$R_2 = 5.5 / \text{Sen } 45 = 7.778t //$$

$$H_2 = R_2 \cos 45 = 5.5t$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$+5.5 - H_c = 0$$

$$H_c = 5.5t //$$

3.- Momentos

$$M_A = 0$$

$$M_B = 2(2) = 4tm$$

$$M_C = -5tm$$

$$M_D = -5tm$$

4.- Cortantes

$$Q_A = 2t$$

$$Q_B (\text{Izq}) = 2t$$

$$Q_B (\text{Der}) = 2 + 5.5 = 7.5t$$

$$Q_C (\text{Izq}) = 2 + 5.5 - 3(6) = -10.5t$$

$$Q_C (\text{Der}) = -10.5 + 10.5 = 0t$$

$$Q_D = 0t$$

5.- Normales

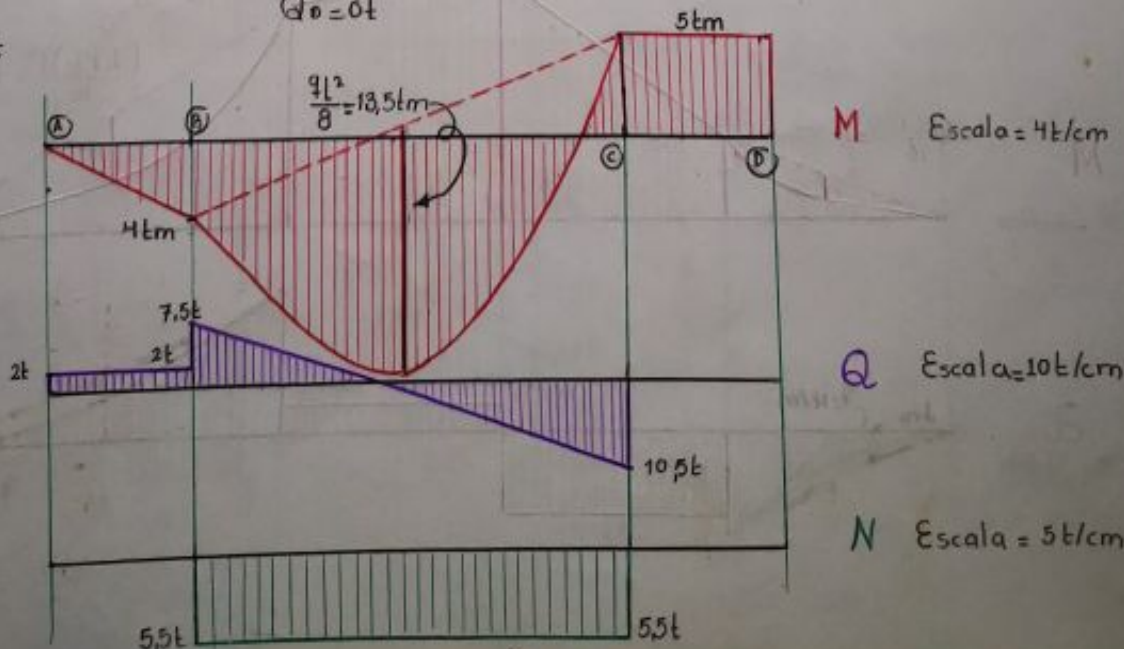
$$N_A = 0$$

$$N_B = -5.5t$$

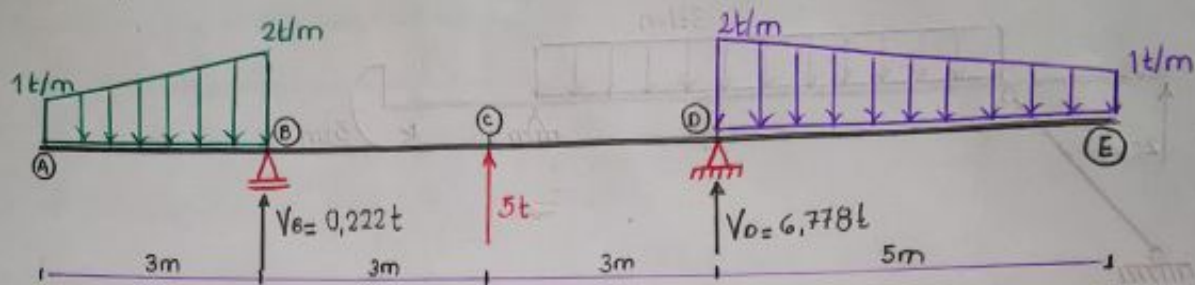
$$N_C = -5.5t$$

$$N_D = 0t$$

6.- Diagramas



16.- Para la siguiente viga obtener los diagrama de Momento y cortante.



1.- Reacciones

$$\sum M_B = 0 \quad \curvearrowright \oplus$$

$$-1(3)(1,5) - \frac{1(3)(1)}{2} - 5(3) - V_D(6) + 1(5)(8,5) + \frac{1(5)}{2} \left(6 + \frac{1}{3} \cdot 5\right) = 0$$

$$V_D = 6,778t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$-\left(\frac{1+2}{2}\right) \cdot 3 + V_B + 5 + 6,778 - \left(\frac{1+2}{2}\right) \cdot 5 = 0$$

$$V_B = 0,222t //$$

2.- Momento $\curvearrowright \oplus$

$$M_A = 0 \text{ tm}$$

$$M_B = -1(3)(1,5) - \frac{1(3)(1)}{2} = -6 \text{ tm}$$

$$M_C = -1(3)(4,5) - \frac{1(3)(1)}{2} + 0,222(3) = -18,834 \text{ tm}$$

$$M_D = -1(5)(2,5) - \frac{1(5)(5)}{2} = -16,667 \text{ tm}$$

3.- Cortantes $\uparrow \oplus$

$$Q_A = 0t$$

$$Q_B(\text{IZA}) = -\left(\frac{1+2}{2}\right) \cdot 3 = -4,5t$$

$$Q_B(\text{DER}) = -4,5 + 0,222 = -4,278t$$

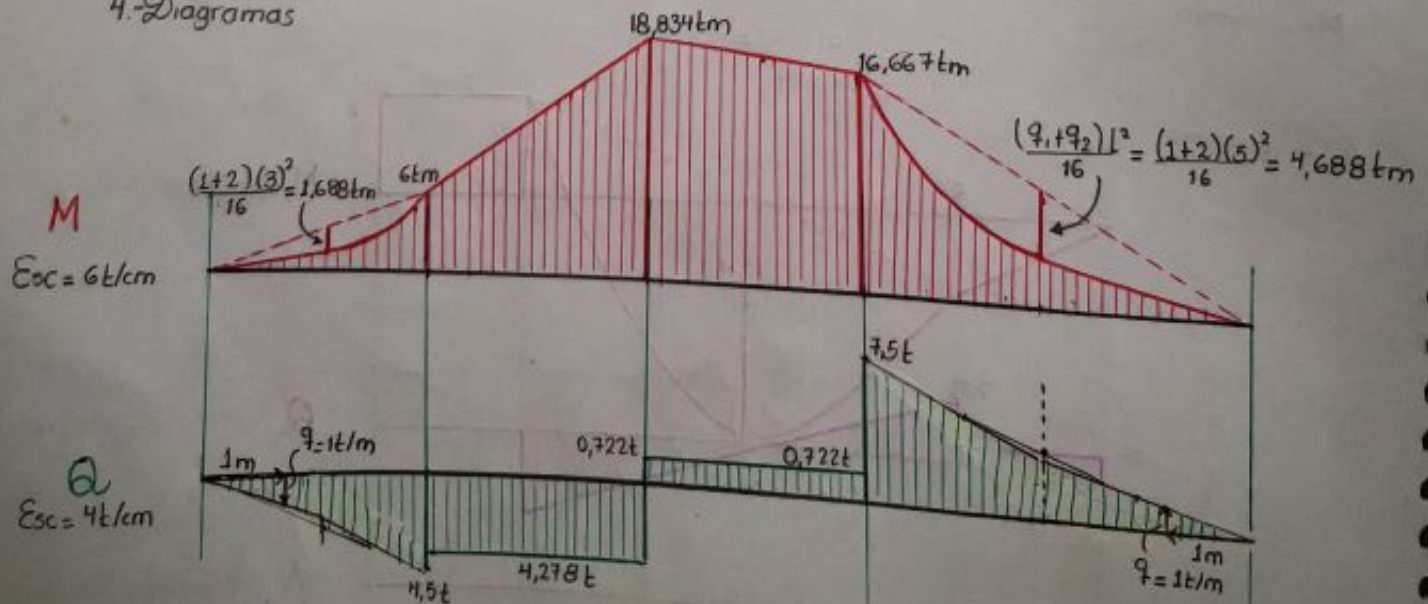
$$Q_C(\text{IZA}) = -4,278t$$

$$Q_C(\text{DER}) = -4,278 + 5 = 0,722t$$

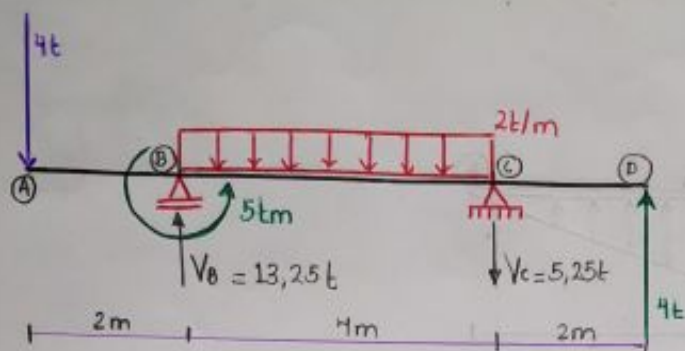
$$Q_D(\text{IZA}) = 0,722t$$

$$Q_D(\text{DER}) = \left(\frac{1+2}{2}\right) \cdot 5 = 7,5t$$

4.- Diagramas



17. Diagramar los esfuerzos característicos



1.- Reacciones

$$\sum M_B = 0 \quad (\curvearrowright \oplus)$$

$$-4(2) - 5 + 2(4)(2) + V_C(4) - 4(6) = 0$$

$$V_C = 5,25t$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$-4 + V_B - 2(4) - 5,25 + 4 = 0$$

$$V_B = 13,25t$$

2.- Momentos

$$M_A = 0tm$$

$$(IZQ) M_B = -4(2) = -8tm$$

$$(DER) M_B = -8 - 5 = -13tm$$

$$M_C = 4(2) = 8tm$$

3.- Cortantes

$$Q_A = -4t$$

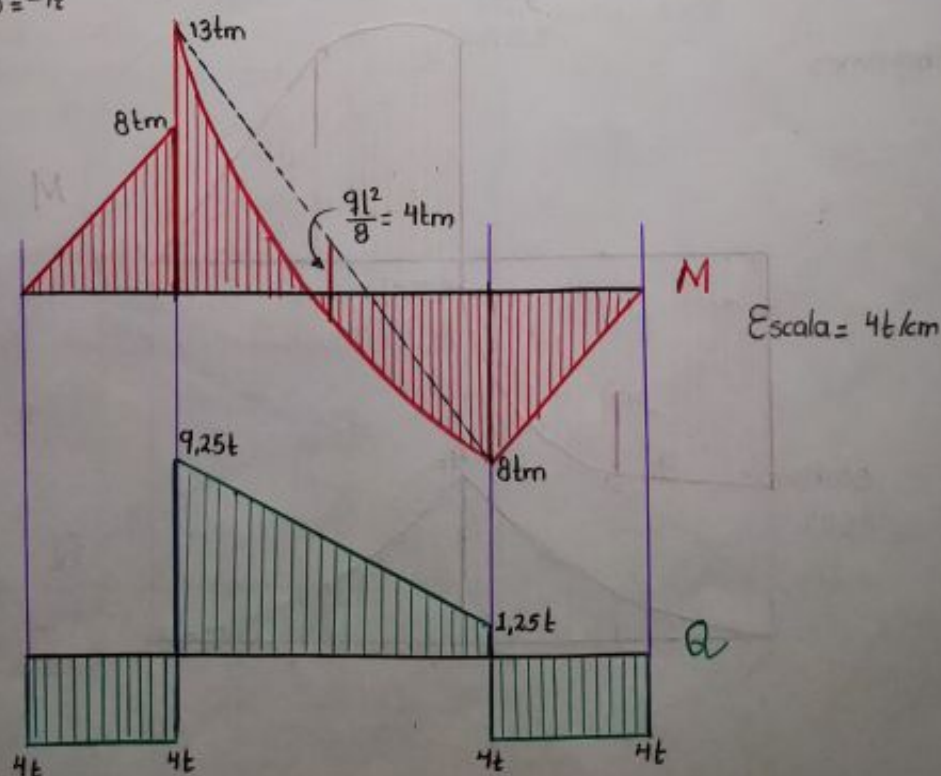
$$Q_B(IZQ) = -4t$$

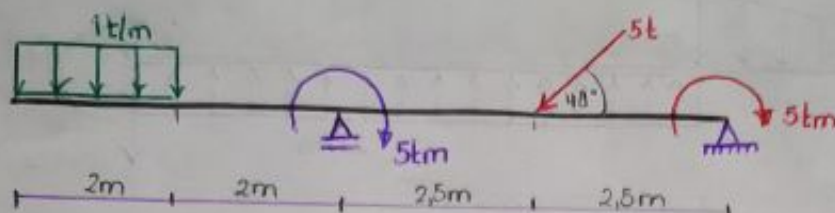
$$Q_B(DER) = -4 + 13,25 = 9,25t$$

$$Q_C(IZQ) = -4 + 5,25 = 1,25t$$

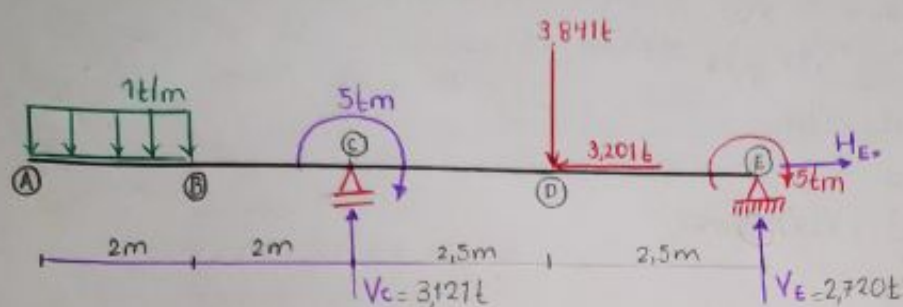
$$Q_C(DER) = -4t$$

$$Q_D = -4t$$





1- Cálculo de Reacciones



$$\begin{aligned}\sum M_{\odot} &= 0 \quad \curvearrowright \oplus \\ -1(2)(3) + 5 + 3,841(2,5) + 5 - V_E(5) &= 0 \\ V_E &= 2,720t \\ \sum F_v &= 0 \quad \uparrow \oplus \\ -1(2) + V_C - 3,841 + 2,720 &= 0 \\ V_C &= 3,121t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ -3,201 + H_E &= 0 \\ H_E &= 3,201t\end{aligned}$$

2- Momentos

$$\begin{aligned}M_A &= 0 \\ M_B &= -1(2)(1) = -2tm \\ M_C(IZQ) &= -1(2)(3) = -6tm \\ M_C(DER) &= -6 + 5 = -1tm \\ M_D &= -1(2)(5,5) + 5 + 3,121(2,5) = 1,803tm \\ M_E &= -5tm \quad (\text{de derecha a izquierda})\end{aligned}$$

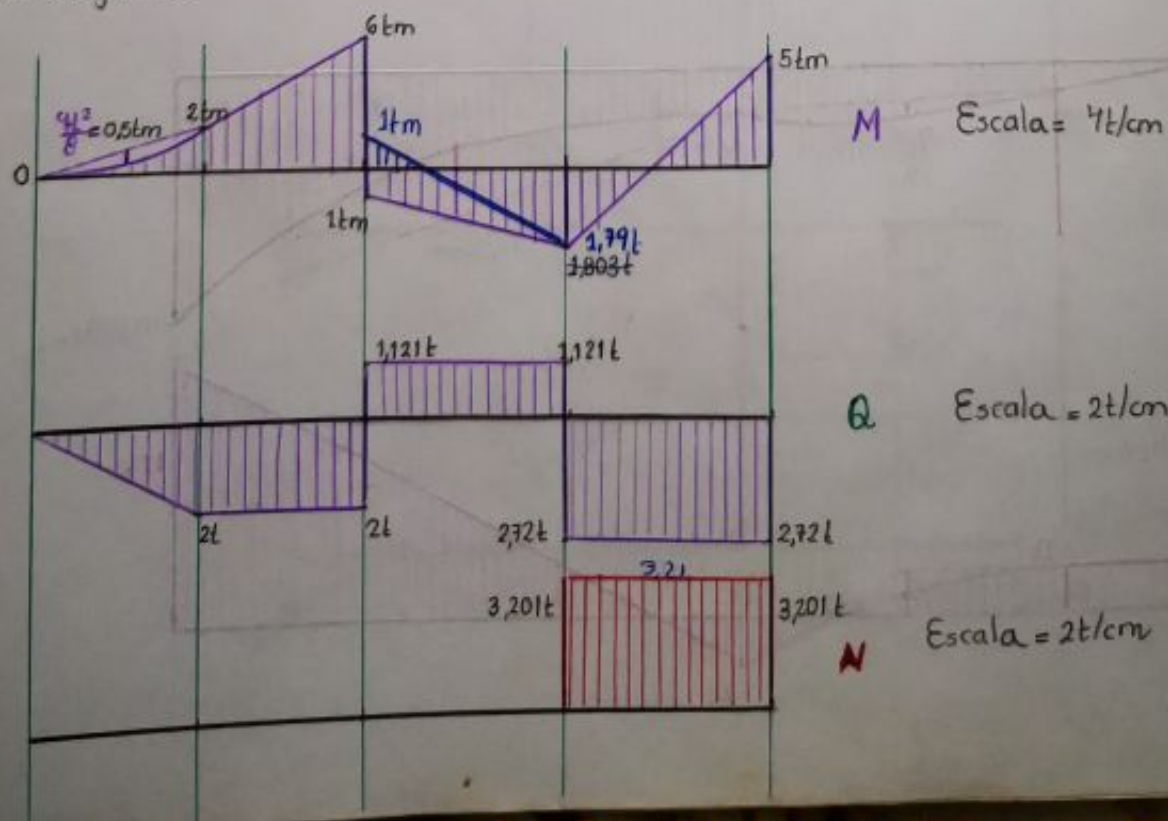
3- Cortante

$$\begin{aligned}Q_A &= 0 \\ Q_B &= -2t \\ Q_C(IZQ) &= -2t \\ Q_C(DER) &= -2 + 3,121 = 1,121t \\ Q_D(IZQ) &= 1,121t \\ Q_D(DER) &= 1,121 - 3,841 = -2,720t \\ Q_E &= -2,720t\end{aligned}$$

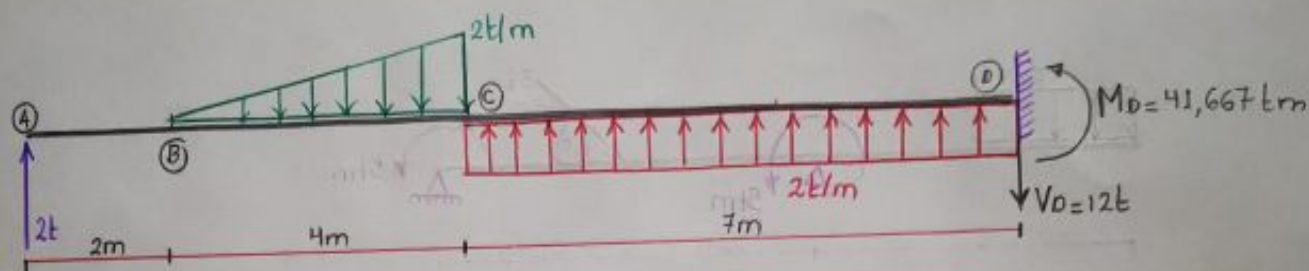
4- Normales

$$\begin{aligned}N_A &= 0 \\ N_B &= 0 \\ N_C &= 0 \\ N_D &= 3,201t \\ N_E &= 3,201t\end{aligned}$$

5- Diagramas



20 Determinar los esfuerzos característicos



1- Cálculo de Reacciones

$$\sum M_D = 0 \quad (\curvearrowright \oplus)$$

$$2(13) - \frac{2(4)}{2} \left(7 + \frac{4}{3}\right) + 2(7)(3,5) - M_D = 0$$

$$M_D = 41,667 \text{ tm}$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$2 - \frac{2(4)}{2} + 2(7) - V_D = 0$$

$$V_D = 12t$$

2- Momentos

$$M_A = 0$$

$$M_B = 2(2) = 4 \text{ tm}$$

$$M_C = 2(6) - \frac{2(4)}{2} \left(\frac{4}{3}\right) = 6,667 \text{ tm}$$

$$M_D = 41,667 \text{ tm}$$

3- Cortantes

$$Q_A = 2t$$

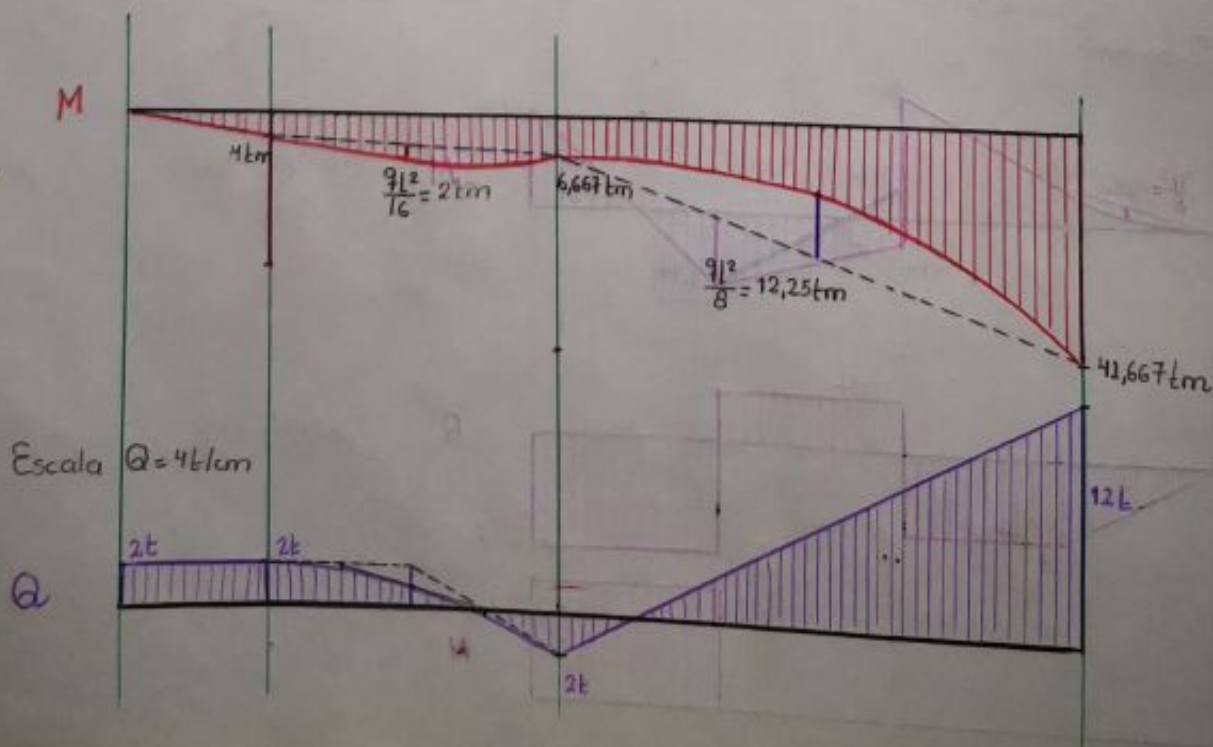
$$Q_B = 2t$$

$$Q_C = 2 - \frac{2(4)}{2} = -2t$$

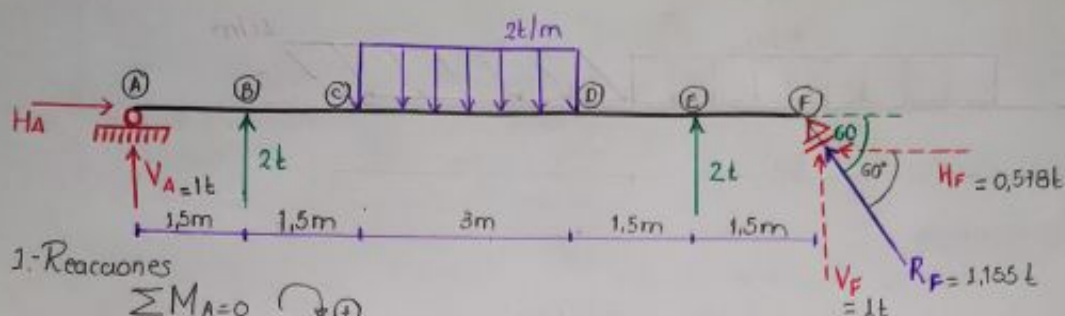
$$Q_D = 2 - \frac{2(4)}{2} + 2(7) = 12t //$$

4- Diagramas

Escala $M = 15t/cm$



21. Diagramar los esfuerzos característicos



1.- Reacciones

$$\sum M_A = 0 \quad \curvearrowright \oplus$$

$$-2(1,5) + 2(3)(4,5) - 2(7,5) - V_F(9) = 0$$

$$V_F = 1t$$

$$\text{Sen } 60^\circ = \frac{V_F}{R_F} \Rightarrow R_F = \frac{1}{\text{Sen } 60} = 1,155t$$

$$H_F = R_F \text{ Cos } 60 = 1,155 \text{ Cos } 60 = 0,578t$$

$$\sum F_v = 0 \rightarrow \oplus$$

$$V_A + 2 - 2(3) + 2 + 1 = 0$$

$$V_A = 1t$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$H_A = 0,578t$$

2.- Momentos

$$M_A = 0 \text{ tm}$$

$$M_B = 1(1,5) = 1,5 \text{ tm}$$

$$M_C = 1(3) + 2(1,5) = 6 \text{ tm}$$

$$M_D = 1(3) + 2(1,5) = 6 \text{ tm}$$

$$M_E = 1(1,5) = 1,5 \text{ tm}$$

$$M_F = 0 \text{ tm}$$

3.- Cortantes

$$Q_A = 1t$$

$$Q_B(\text{Izq}) = 1t$$

$$Q_B(\text{Der}) = 3t$$

$$Q_C = 3t$$

$$Q_D = 3 - 2(3) = -3t$$

$$Q_E(\text{Izq}) = -3t$$

$$Q_E(\text{Der}) = -3 + 2 = -1t$$

$$Q_F = -1t$$

4.- Normales

$$N_A = -0,578t$$

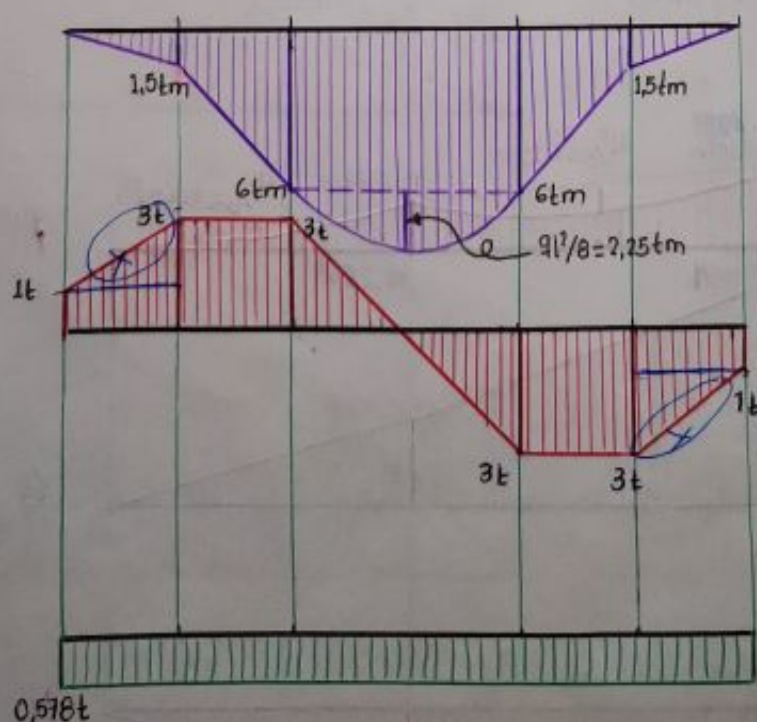
$$N_B = -0,578t$$

$$N_C = -0,578t$$

$$N_D = -0,578t$$

$$N_E = -0,578t$$

$$N_F = -0,578t$$

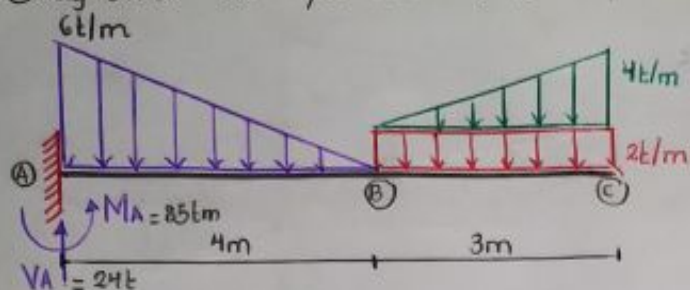


M Escala = 3t/cm

Q Escala = 2t/cm

N Escala = 1t/cm

23.- Diagramar los esfuerzos característicos



1.- Reacciones

$$\sum F_v = 0 \uparrow \oplus$$

$$V_A - \frac{6(4)}{2} - \frac{4(3)}{2} - 2(3) = 0$$

$$V_A = 24t //$$

$$\sum M_A = 0 \curvearrowright \oplus$$

$$-M_A + \frac{6(4)}{2} \left[\frac{1}{3} \cdot 4 \right] + 2(3)(5.5) + \frac{4(3)}{2} \cdot (6) = 0$$

$$M_A = 85tm$$

2.- Momentos

$$M_A = -85tm$$

$$M_B = -85 + 24(4) - \frac{6(4)}{2} \left(\frac{2}{3} \cdot 4 \right) = -21tm$$

$$M_C = 0tm$$

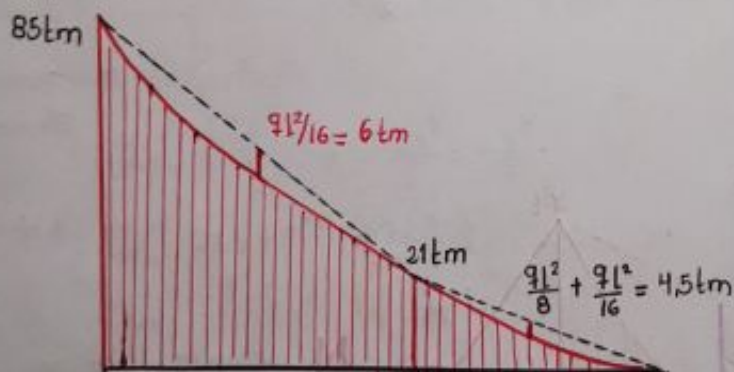
3.- Cortantes

$$Q_A = 24t$$

$$Q_B = 24 - \frac{6 \cdot 4}{2} = 12t$$

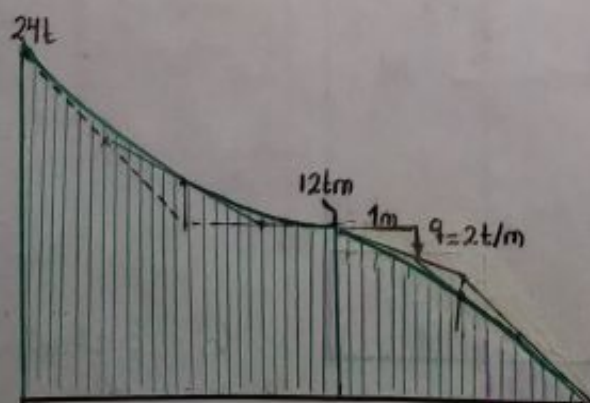
$$Q_C = 0tm$$

4.- Diagramas



M

Escala = 20 tm/cm

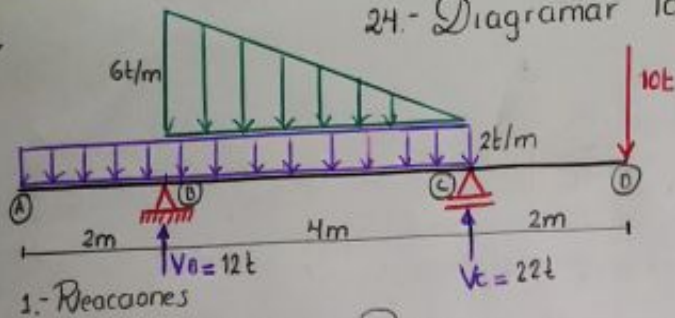


Q

Escala = 6 t/cm

24.

24.- Diagramar los esfuerzos característicos



1.- Reacciones

$$\sum M_B = 0 \quad \curvearrowright \oplus$$

$$-2(2)(1) + 2(4)(2) + \frac{6(4)}{2} \left(\frac{1}{3} \cdot 4 \right) - V_C(4) + 10(6) = 0$$

$$V_C = 22t //$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$-2(2) + V_B - 2(4) - \frac{6(4)}{2} + 22 - 10 = 0$$

$$V_B = 12t$$

2.- Momentos

$$M_A = 0tm$$

$$M_B = -2(2)(1) = -4tm$$

$$M_C = -10(2) = -20tm$$

$$M_D = 0tm$$

3.- Cortantes

$$Q_A = 0t$$

$$Q_{B(IZQ)} = -2(2) = -4t$$

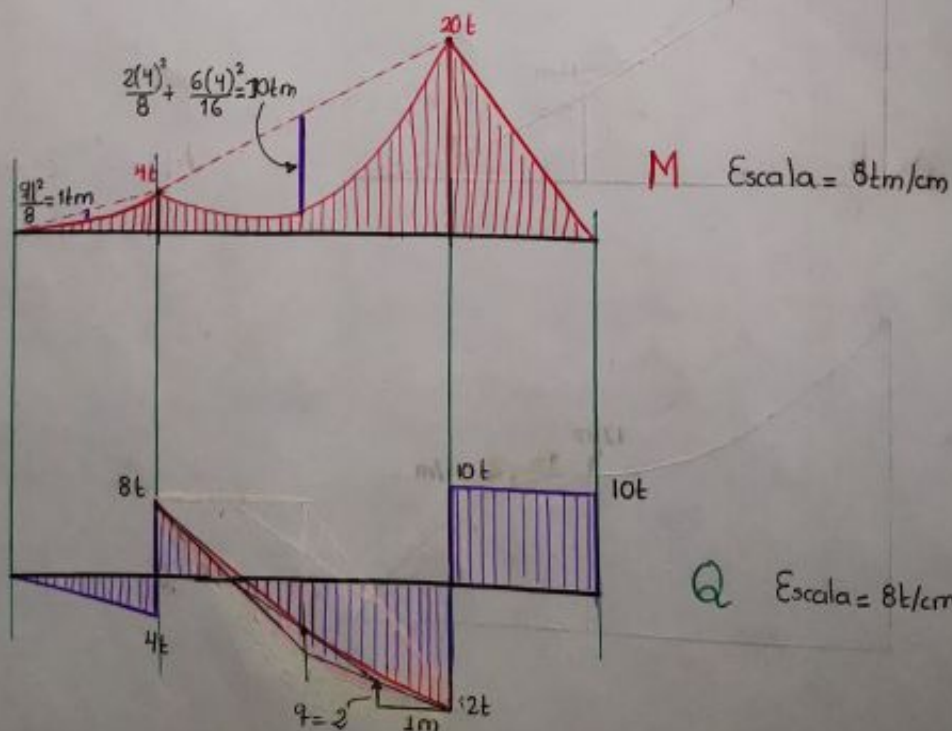
$$Q_{B(DER)} = -4 + 12 = 8t$$

$$Q_C(IZQ) = 10 - 22 = -12t$$

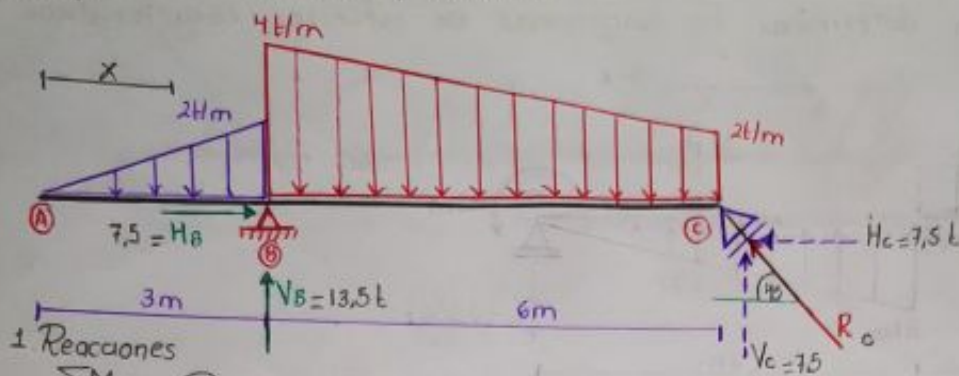
$$Q_{C(DER)} = 10t$$

$$Q_D = 10t$$

4.- Diagramas



25.- Diagramar Momento y cortante



1.- Reacciones

$$\sum M_B = 0 \quad (\curvearrowright)$$

$$-\frac{2(3)}{2} \cdot (1) + 2(6)(3) + \frac{2(6)}{2} \cdot 2 - V_C \cdot 6 = 0 \quad \therefore V_C = 7.5t$$

$$\sum F_V = 0 \quad \uparrow$$

$$-\frac{2(3)}{2} + V_B - \frac{(2+4)}{2} \cdot 6 + 7.5 = 0 \quad \therefore V_B = 13.5t$$

$$\sum F_H = 0 \quad \rightarrow \quad \therefore H_B = 7.5t$$

$$H_C = V_C = 7.5t$$

$$R_C = 10.607t$$

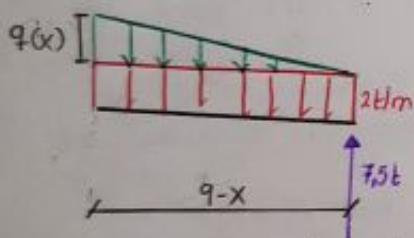
2.- Ecuaciones de Momento

* Tramo A-B

$$\frac{q(x)}{x} = \frac{2}{3} \Rightarrow q(x) = \frac{2x}{3} \Rightarrow M_{AB} = \frac{-\frac{2x}{3} \cdot x}{2} \cdot \frac{x}{3} = -\frac{2x^3}{18}$$

$$M_{AB} = -\frac{x^3}{9} //$$

* Tramo BC



$$\frac{q(x)}{9-x} = \frac{2}{6} \Rightarrow q(x) = 3 - \frac{x}{3} //$$

$$M_{BC} = 7.5(9-x) - \frac{2(9-x)^2}{2} - \frac{(3-\frac{x}{3})(9-x)}{2} \cdot \frac{(9-x)}{3}$$

$$M_{BC} = 0.0556x^3 - 2.5x^2 + 24x - 54$$

3.- Ecuaciones de Cortante

* Tramo AB

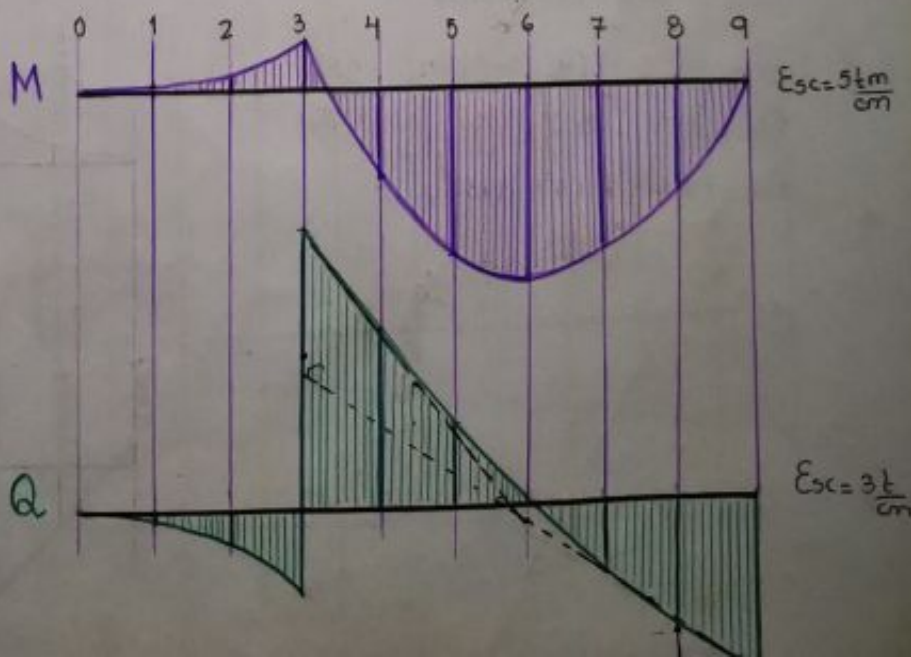
$$Q_{AB} = \frac{(-\frac{2x}{3}) \cdot x}{2} = -\frac{2x^2}{6} = -\frac{x^2}{3}$$

$$* \text{Tramo BC} \quad Q_{BC} = -7.5 + 2(9-x) + \frac{(3-\frac{x}{3})(9-x)}{2}$$

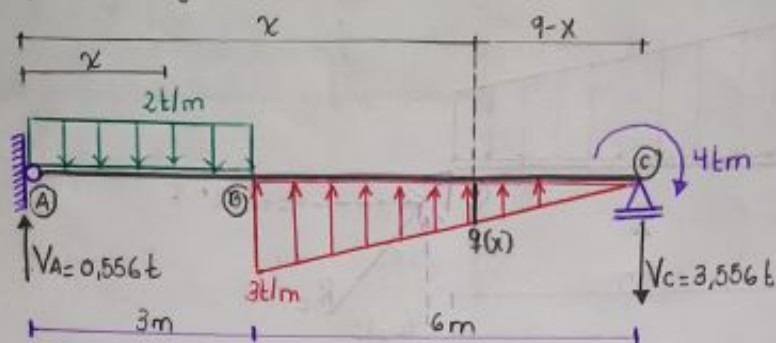
$$Q_{BC} = 0.1667x^2 - 5x + 24$$

4.- Diagramas

x	Momento	Cortante
0	0,000	0,000
1	-0,111	-0,333
2	-0,889	-1,333
3	-3	-3 / 10,50
4	5,558	6,667
5	10,45	3,168
6	12,01	0,00
7	10,571	-2,832
8	6,467	-5,3312
9	0,000	-7,5



26.- Para la siguiente viga determinar los diagramas de esfuerzos característicos



1.- Reacciones

$$\begin{aligned}\sum M_A &= 0 \quad \curvearrowright \oplus \\ 2(3)(1.5) - \frac{3(6)}{2}(5) + 4 + V_C(9) &= 0 \\ V_C &= 3.556 \text{ t} // \\ \sum F_V &= 0 \quad \uparrow \oplus \\ V_A - 2(3) + \frac{3(6)}{2} - 3.556 &= 0 \\ V_A &= 0.556 \text{ t} //\end{aligned}$$

2.- Ecuaciones de Momento

Tramo A-B

$$M_{AB} = 0.556x - 2(x)(\frac{x}{2})$$

$$M_{AB} = 0.556x - x^2 //$$

Tramo BC

$$\frac{q(x)}{9-x} = \frac{3}{6} \Rightarrow q(x) = 4.5 - 0.5x$$

$$M_{BC} = -3.556(9-x) - 4 + \frac{(4.5-0.5x)(9-x)}{2} \cdot \frac{1}{3}(9-x)$$

$$M_{BC} = -32.004 + 3.556x - 4 + \frac{1}{6}(1.5-0.5x)(81-18x+x^2)$$

$$M_{BC} = -36.004 + 3.556x + \frac{1}{6}[364.5 - 81x + 4.5x^2 - 40.5x + 9x^2 - 0.5x^3]$$

$$M_{BC} = -36.004 + 3.556x + \frac{1}{6}[364.5 - 121.5x + 13.5x^2 - 0.5x^3]$$

$$M_{BC} = -36.004 + 3.556x + 60.75 - 20.25x + 2.25x^2 - 0.083x^3$$

$$M_{BC} = 24.746 - 16.694x + 2.25x^2 - 0.083x^3 //$$

3.- Ecuaciones de Cortante

$$Q_{AB} = 0.556 - 2(x) //$$

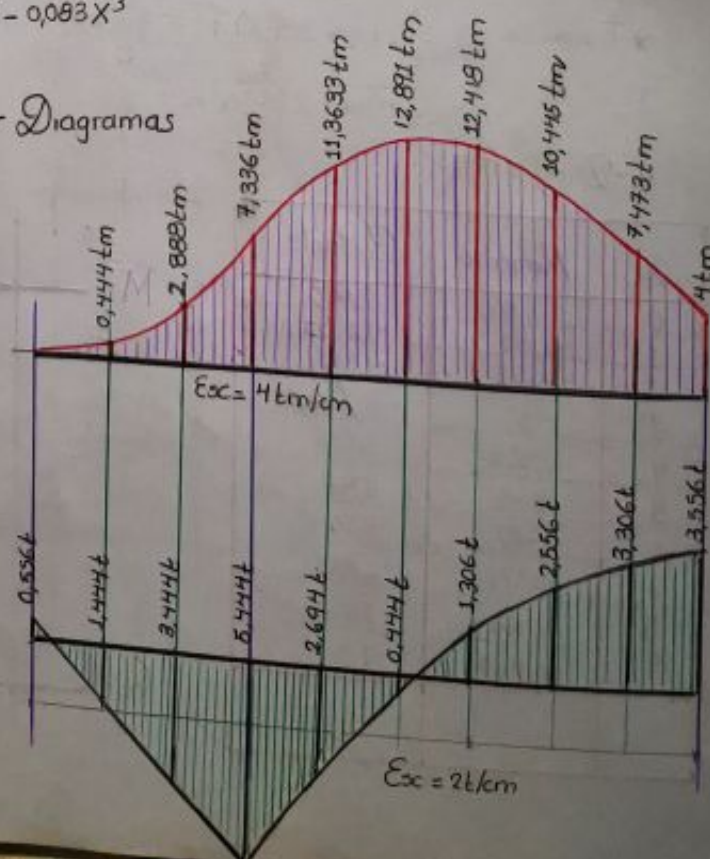
$$Q_{BC} = 3.556 - \frac{(4.5-0.5x)(9-x)}{2}$$

$$Q_{BC} = 3.556 - \frac{1}{2}(40.5 - 4.5x - 4.5x + 0.5x^2)$$

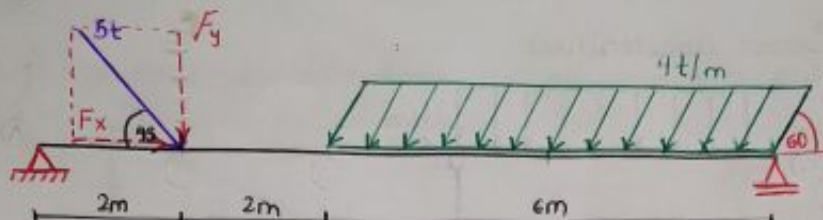
$$Q_{BC} = 3.556 - 20.25 + 2.25x + 0.25x^2$$

$$Q_{BC} = -16.694 + 2.25x + 0.25x^2 //$$

4.- Diagramas



27.- Diagramar los esfuerzos característicos

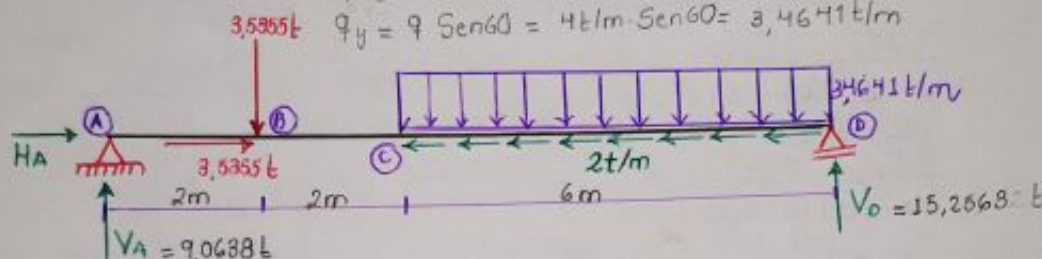


1.- Idealizar Cargas

$$F_x = F_y = 5t \cdot \cos 45 = 3,5355t //$$

$$q_x = q \cdot \cos 60 = 4t/m \cdot \cos 60 = 2t/m$$

$$q_y = q \cdot \sin 60 = 4t/m \cdot \sin 60 = 3,4641t/m$$



2.- Reacciones

$$\sum M_A = 0 \quad \curvearrowright \oplus$$

$$3,5355(2) + 3,4641(6)(7) - V_D(10) = 0 \quad \therefore V_D = 15,2563t$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_A - 3,5355 - 3,4641(6) + 15,2563 = 0 \quad \therefore V_A = 9,0638t$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$H_A + 3,5355 - 2(6) = 0 \quad \therefore H_A = 8,4645t$$

3.- Ecuaciones de Momento

* Tramo A-B

$$M_x = V_A \cdot x = 9,0638 \cdot x$$

* Tramo BC

$$M_x = 9,0638 \cdot x - 3,5355(x-2)$$

$$M_x = 5,5283x + 7,0710 //$$

* Tramo CD

$$M_x = 15,2563(10-x) - 3,4641 \frac{(10-x)^2}{2}$$

$$M_x = -1,7321x^2 + 19,3847x - 20,642 //$$

4 Ecuaciones de Corte

* Tramo AB

$$Q_x = 9,0638t$$

* Tramo BC

$$Q_x = 9,0638 - 3,5355$$

$$Q_x = 5,5283$$

* Tramo CD

$$Q_x = -15,2563 + 3,4641(10-x)$$

$$Q_x = -3,4641x + 19,3842$$

5.- Ecuaciones de Normal

* Tramo AB

$$N_x = -8,4645t$$

* Tramo BC

$$N_x = -8,4645 - 3,5355$$

$$N_x = -12t$$

* Tramo CD

$$N_x = -2(10-x)$$

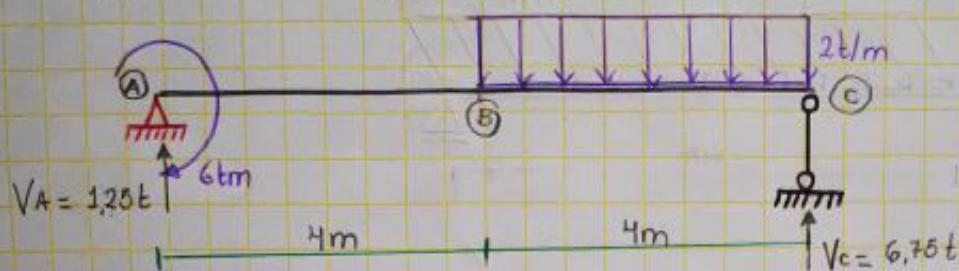
$$N_x = 2x - 20$$

6.- Diagramas

x	Mto	Corte	Normal
0	0	9,0638	-8,4645
1	9,0638	9,0638	-8,4645
2	18,1276	5,5283	-12
3	27,1914	5,5283	-12
4	36,2552	5,5283	-12
5	45,3190	2,0637	-10
6	54,3828	-1,4004	-8
7	63,4466	-4,8645	-6
8	72,5104	-8,3286	-4
9	81,5742	-11,7927	-2
10	90,6380	-15,2563	0



28. Diagramar los esfuerzos característicos



1.- Reacciones

$$\sum M_A = 0 \quad \curvearrowright \oplus$$

$$6 + 2(4)(6) - V_C(8) = 0$$

$$V_C = 6,75t //$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$V_A - 2(4) + 6,75 = 0$$

$$V_A = 1,25t$$

2.- Ecuaciones de Momento

* Tramo AB $M_x = 1,25 \cdot x + 6 //$

* Tramo BC $M_x = 6,75 \cdot (8-x) - \frac{2(8-x)^2}{2}$

$$M_x = -x^2 + 9,25x - 10 //$$

3.- Ecuaciones de Corte

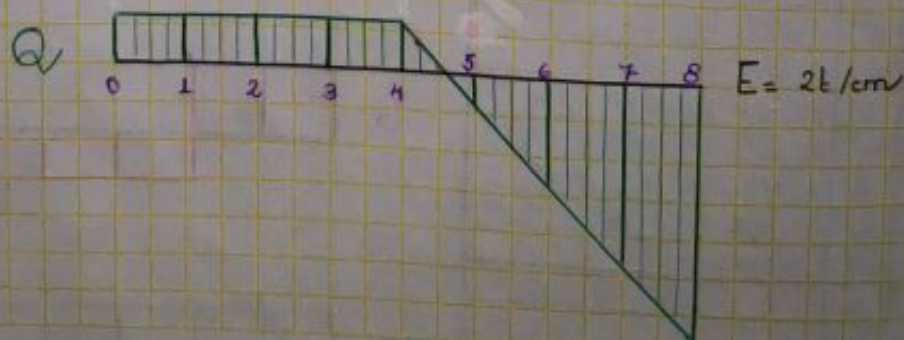
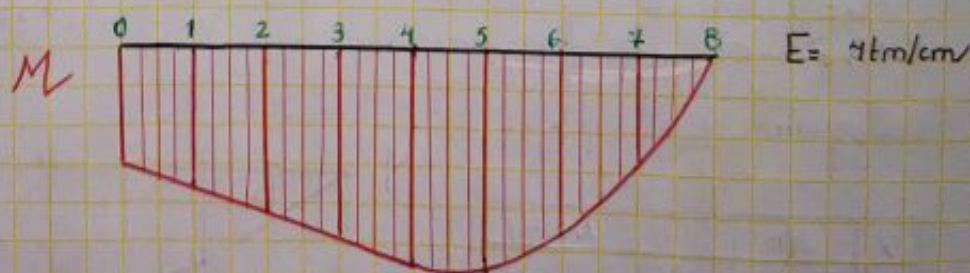
* Tramo AB $Q_x = 1,25t //$

* Tramo BC $Q_x = 1,25 - 2(x-4)$

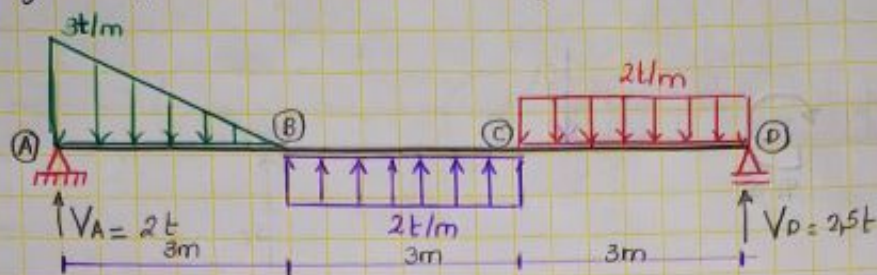
$$Q_x = 9,25 - 2x //$$

4.- Diagramas

x	Momento	Cortante
0	6	1,25
1	7,25	1,25
2	8,5	1,25
3	9,75	1,25
4	11	1,25
5	11,25	-0,75
6	9,5	-2,75
7	6,75	-4,75
8	0	-6,75



29.- Para la viga mostrada determinar los esfuerzos característicos



1.- Reacciones

$$\sum M_A = 0 \quad \curvearrowright \oplus$$

$$\frac{3(3)}{2} \cdot (1) - 2(3)(4,5) + 2(3)(7,5) - V_D(9) = 0 \quad \therefore V_D = 2,5t$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$V_A - \frac{3(3)}{2} + 2(3) - 2(3) + 2,5 = 0 \quad \therefore V_A = 2t$$

2.- Ecuaciones de Momento

* Tramo AB

$$M_x = 2 \cdot x - \left(\frac{3x^2}{2} - \frac{3x^3}{6(3)} \right)$$

$$M_x = \left[x^3 - 9x^2 + 12x \right] / 6 //$$

* Tramo BC

$$M_x = 2x - \frac{3(3)}{2} \cdot (x-1) + \frac{2(x-3)^2}{2}$$

$$M_x = \frac{2x^2 - 17x + 27}{2} //$$

* Tramo CD

$$M_x = 2,5(9-x) - \frac{2(9-x)^2}{2}$$

$$M_x = -x^2 + 15,5x - 58,5 //$$

3.- Ecuaciones de Corte

* Tramo AB

$$Q_x = 2 - \left[3 \cdot x - \frac{3x^2}{2(3)} \right]$$

$$Q_x = 0,5x^2 - 3x + 2 //$$

* Tramo BC

$$Q_x = 2 - \frac{3(3)}{2} + 2(x-3)$$

$$Q_x = 2x - 8,5 //$$

* Tramo CD

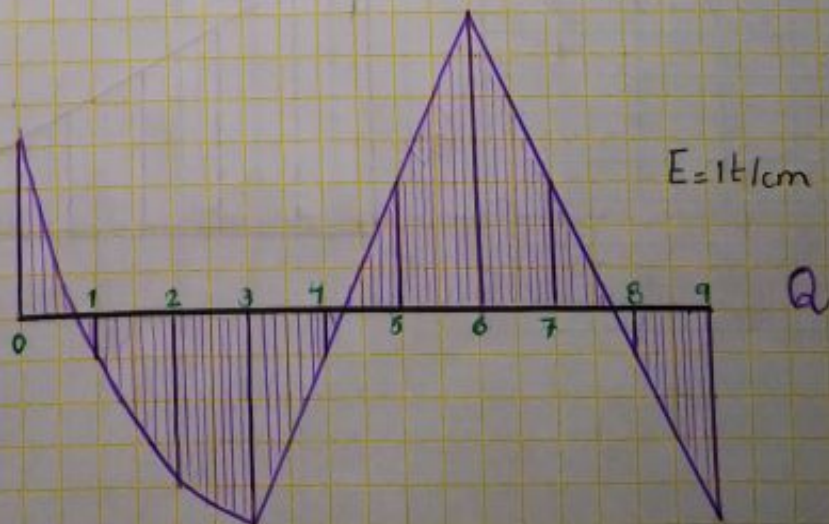
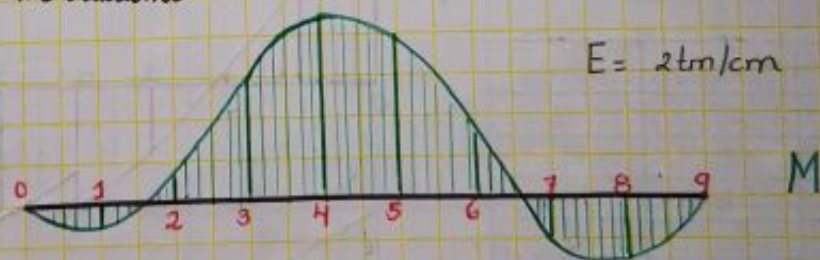
$$Q_x = -2,5 + 2(9-x)$$

$$Q_x = -2x + 15,5 //$$

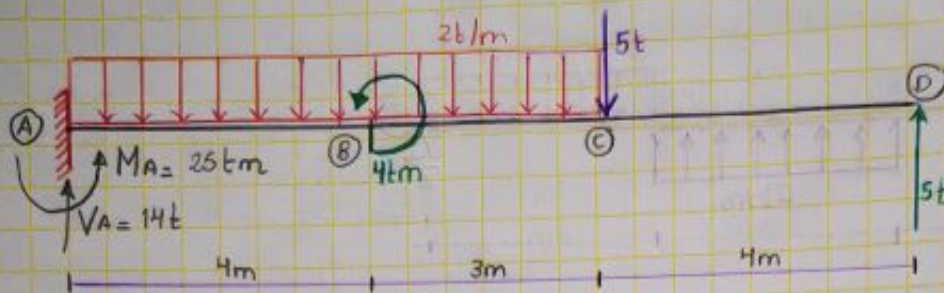
3.- Diagramas

x	Momentos	Cortantes
0	0	2
1	0,667	-0,5
2	-0,667	-2
3	-3	-2,5
4	-4,5	-0,5
5	-4	1,5
6	-1,5	8,5
7	1	1,5
8	1,5	-0,5
9	0	-2,5

1.- Reacciones



30.- Diagramar los esfuerzos característicos



1- Reacciones

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_A - 2(7) - 5 + 5 = 0 \quad \therefore V_A = 14t$$

$$\sum M_A = 0 \quad \curvearrowright \oplus$$

$$-M_A + 2(7)(3,5) + 5(7) - 4 - 5(11) = 0$$

$$M_A = 25tm$$

2- Ecuaciones de Momento

* Tramo AB

$$M_x = 14(x) - 25 - \frac{2x^2}{2}$$

$$M_x = -x^2 + 14x - 25 //$$

* Tramo BC

$$M_x = 14 \cdot x - 25 - \frac{2x^2}{2} - 4$$

$$M_x = -x^2 + 14x - 29 //$$

* Tramo CD

$$M_x = 5(11-x)$$

$$M_x = 55 - 5x //$$

3- Ecuaciones de Corte

* Tramo AC

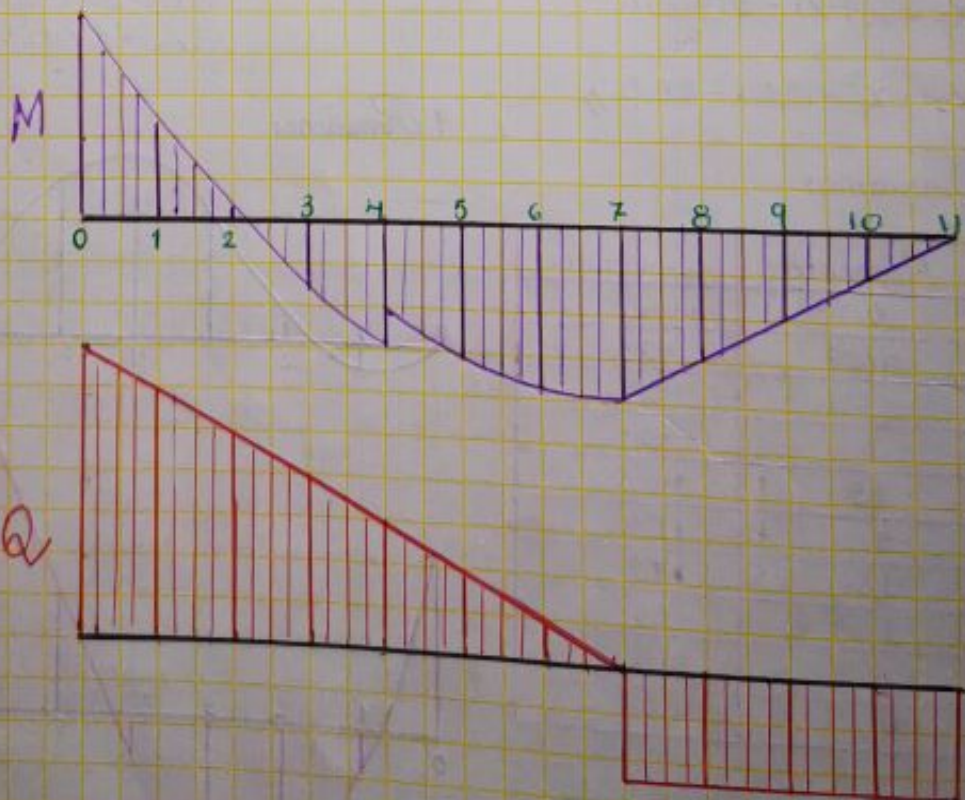
$$Q_x = 14 - 2x //$$

* Tramo CD

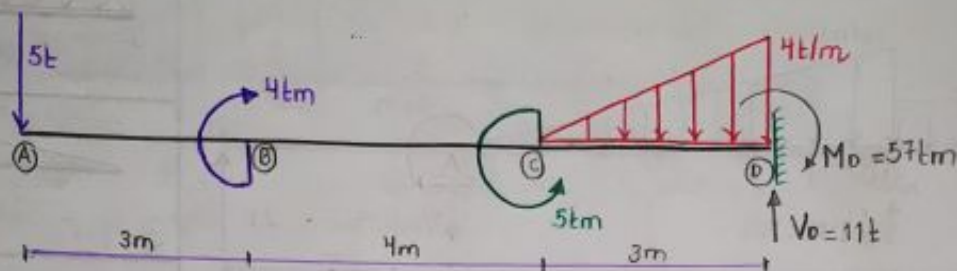
$$Q_x = -5t //$$

4- Diagramas

x	Momento	Corte
0	-25	14
1	-12	12
2	-1	10
3	8	8
4	15 / 11	6
5	16	4
6	19	2
7	20	0 / -5
8	15	-5
9	10	-5
10	5	-5
11	0	-5



31.- Diagramar los esfuerzos internos



1.- Reacciones

$$\sum F_v = 0 \uparrow \oplus$$

$$-5 - \frac{4(3)}{2} + V_D = 0 \quad \therefore V_D = 11t$$

$$\sum M_D = 0 \curvearrowright \oplus$$

$$-5(10) + 4 - 5 - \frac{4(3)}{2} \cdot (1) + M_D = 0 \quad \therefore M_D = 57tm$$

2.- Ecuaciones de Momento

* Tramo AB $M_{AB} = -5X //$

* Tramo BC $M_{BC} = -5X + 4 //$

* Tramo CD $M_{CD} = -5X + 4 - 5 - \frac{4(X-7)^3}{6(3)}$
 $M_{CD} = -\frac{2X^3 - 42X^2 + 339X - 677}{9}$

3.- Ecuaciones de Corte

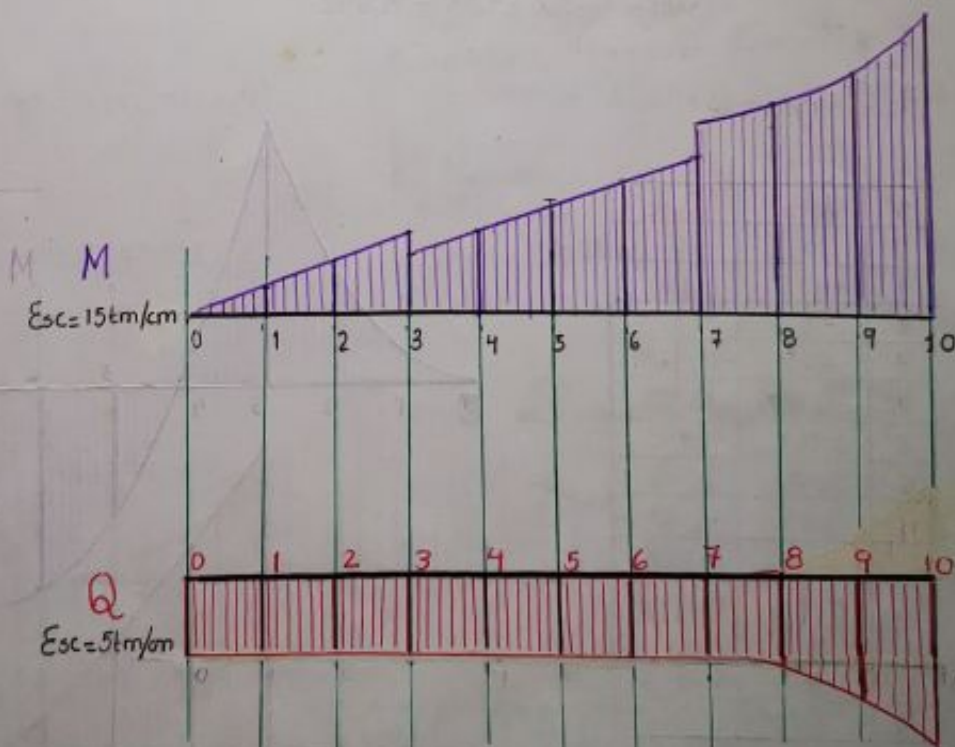
* Tramo AB $Q_{AB} = -5t$

* Tramo BC $Q_{BC} = -5t$

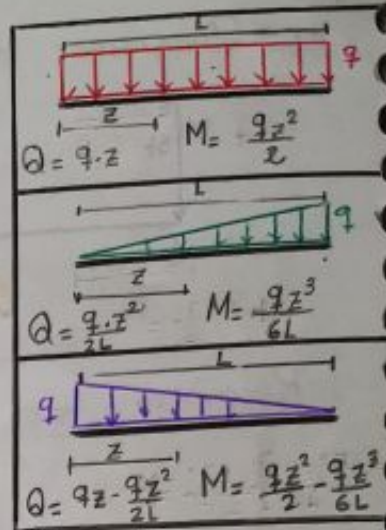
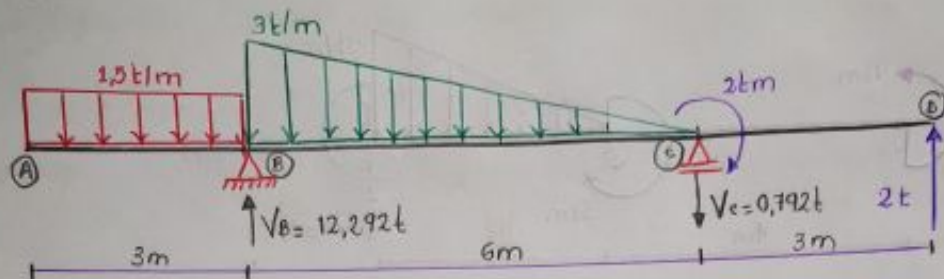
* Tramo CD $Q_{CD} = -5 - \frac{4(X-7)^2}{2 \cdot 3}$
 $Q_{CD} = \frac{-2X^2 + 28X - 113}{3}$

4.- Diagramas

x	Mto	Corte
0	0	-5
1	-5	-5
2	-10	-5
3	-15 -11	-5
4	-16	-5
5	-21	-5
6	-26	-5
7	-31 -36	-5
8	-41,222	-5,667
9	-47,778	-7,667
10	-57	-11



32. Diagramar los esfuerzos internos



1.- Reacciones

$$\sum M_B = 0 \quad \curvearrowright \oplus$$

$$-1,5(3)(1,5) + \frac{3(6)(2)}{2} + 2 - 2(9) + V_C(6) = 0 \quad \therefore V_C = 0,792t$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$-1,5(3) + V_B - \frac{3(6)}{2} - 0,792 + 2 = 0 \quad \therefore V_B = 12,292t$$

2.- Ecuaciones de Momento

* Tramo AB $M_{AB} = -\frac{1,5X^2}{2} = -0,75X^2 //$

* Tramo BC $M_{BC} = -1,5(3)(X-1,5) + 12,292(X-3) - \left[\frac{3(X-3)^2}{2} - \frac{3(X-3)^3}{6(6)} \right]$

$$M_{BC} = 0,833X^3 - 2,25X^2 + 19,042X - 45,876 //$$

* Tramo CD $M_{CD} = 2(12-X) = 24-2X$

3.- Ecuaciones de Corte

* Tramo AB $Q_{AB} = -1,5X$

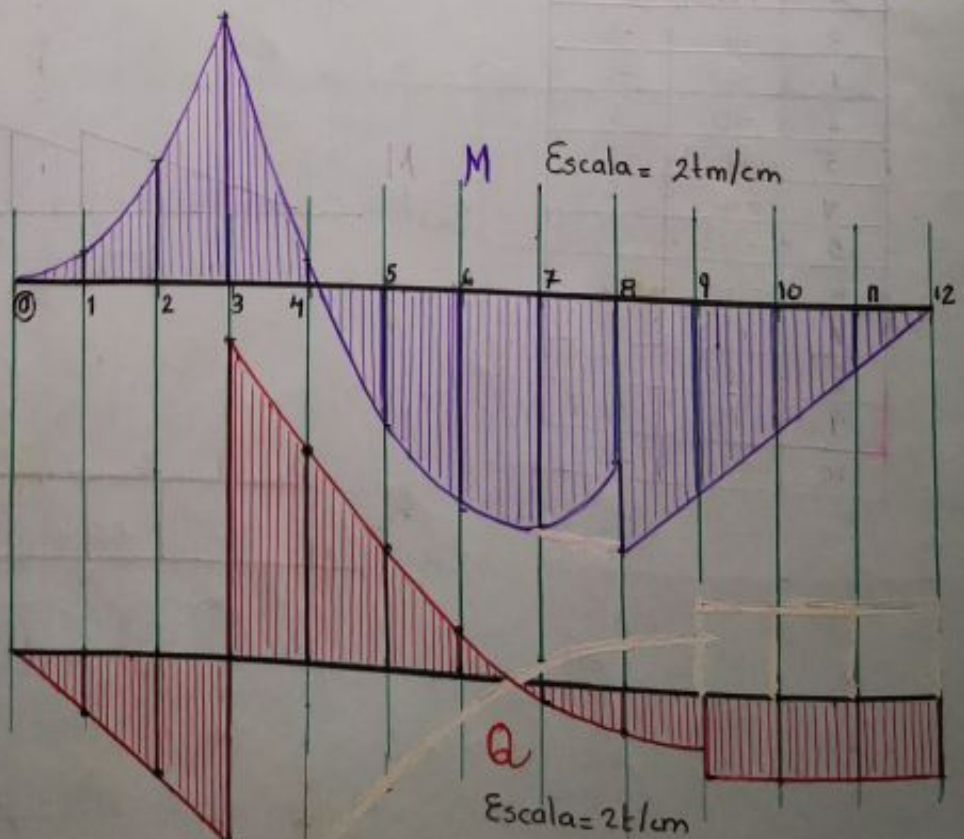
* Tramo BC $Q_{BC} = -1,5(3) + 12,292 - \left[3(X-3) - \frac{3(X-3)^2}{2(6)} \right]$

$$Q_{BC} = 0,25X^2 - 4,5X + 19,042$$

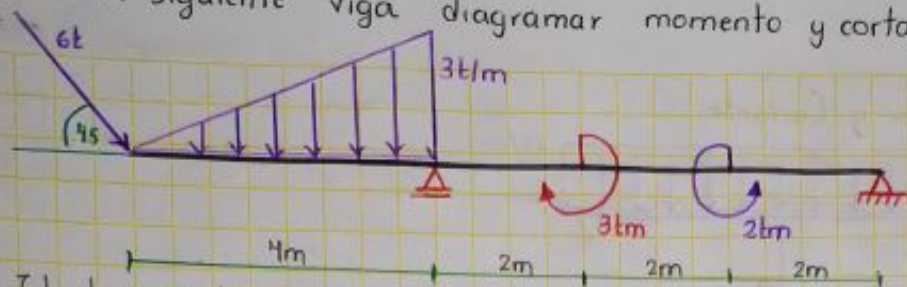
* Tramo CD $Q_{CD} = -2t$

4.- Diagramas

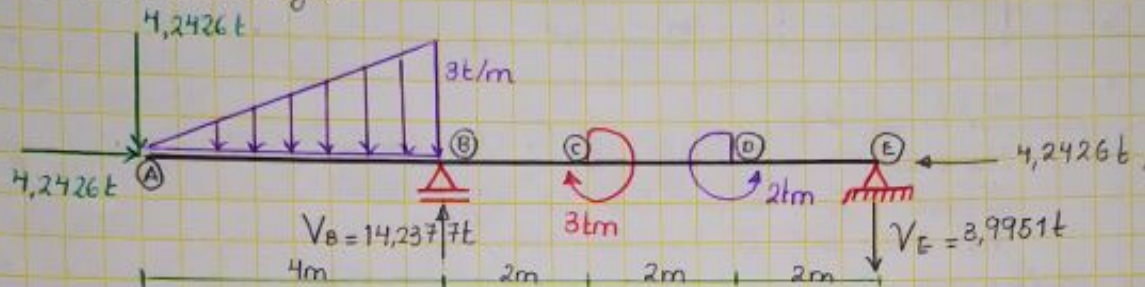
X	Mto	Corte
0	0	0
1	-0,75	-1,5
2	-3	-3
3	-6,75	-4,5
4	-0,377	5,092
5	3,497	2,792
6	5,369	1,042
7	5,740	-0,208
8	5,1096	-0,958
9	4	-1,208
10	4	-2
11	2	-2
12	0	-2



33. Para la siguiente viga diagramar momento y cortante



1 - Idealización de cargas



2 - Reacciones

$$\sum M_B = 0 \quad \curvearrowright \oplus$$

$$-4,2426(4) - 3(4) \cdot \left[\frac{1}{3} \cdot 4 \right] + 3 - 2 + V_E(6) = 0 \quad \therefore V_E = 3,9951t //$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$-4,2426 - 3(4) + V_B - 3,9951 = 0 \quad \therefore V_B = 14,2377t //$$

$$\sum F_H = 0 \quad \rightarrow \oplus \quad \therefore H_E = 4,2426t //$$

3 - Ecuaciones de Momento

* Tramo AB

$$M_x = \frac{-3x^3}{6(4)} - 4,2426 \cdot x$$

$$M_x = -0,125x^3 - 4,2426x //$$

* Tramo BC

$$M_x = -4,2426 \cdot x - 3(4) \cdot \left[x - \frac{2 \cdot 4}{3} \right] + 14,2377(x - 4)$$

$$M_x = 3,9951x - 40,9508 //$$

* Tramo CD

$$M_x = -3,9951 \cdot (10 - x) + 2 = 3,9951x - 37,951 //$$

* Tramo DE

$$M_x = -3,9951(10 - x)$$

$$M_x = 3,9951x - 39,951 //$$

4 - Diagramas

x	Momento	Corte
0	0	-4,2426
1	-4,3676	-4,6176
2	-9,4852	-5,7426
3	-16,1028	-7,6176
4	-24,9704	-10,24/3,9951
5	-20,9753	3,9951
6	-16,98/13,98	3,9951
7	-9,9858	3,9951
8	-5,99/-7,99	3,9951
9	-3,9951	3,9951
10	0	3,9951

4 - Ecuaciones de Corte

* Tramo AB

$$Q_x = -4,2426 - \frac{3x^2}{2(4)}$$

$$Q_x = -4,2426 - 0,375x^2 //$$

* Tramo BE

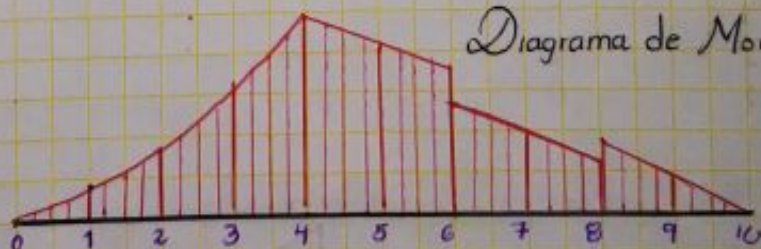
$$Q_x = -4,2426 - \frac{3(4)}{2} + 14,2377 = 3,9951t //$$

5 - Normal

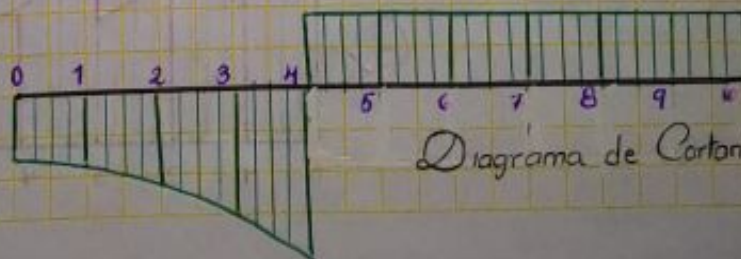
* Tramo AE

$$N = -4,2426t //$$

Diagrama de Momento



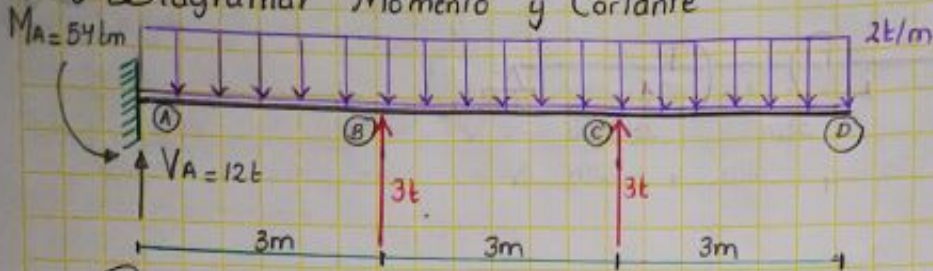
$E = 10tm/cm$



$E = 5t/cm$

Diagrama de Cortante

34- Diagramar Momento y Cortante



1- Reacciones

$$\sum M_A = 0 \quad (\rightarrow \oplus)$$

$$-M_A + 2(9)(4.5) - 3(3) - 3(6) = 0 \quad \therefore M_A = 54tm$$

$$\sum F_v = 0 \quad (\uparrow \oplus)$$

$$V_A + 3 + 3 - 2(9) = 0 \quad \therefore V_A = 12t$$

2- Ecuaciones de Momento

* Tramo AB

$$M_x = 12x - 54 - \frac{2x^2}{2}$$

$$M_x = -x^2 + 12x - 54$$

* Tramo BC

$$M_x = 12x + 3(x-3) - 54 - \frac{2x^2}{2}$$

$$M_x = -x^2 + 15x - 63$$

* Tramo CD

$$M_x = -\frac{2(9-x)^2}{2}$$

$$M_x = -x^2 + 18x - 81$$

4- Diagramas

x	Momentos	Cortantes
0	-54	12
1	-43	10
2	-34	8
3	-27	6/9
4	-19	7
5	-13	5
6	-9	3/6
7	-4	4
8	-1	2
9	0	0

3- Ecuaciones de Corte

* Tramo AB

$$Q_x = 12 - 2x$$

* Tramo BC

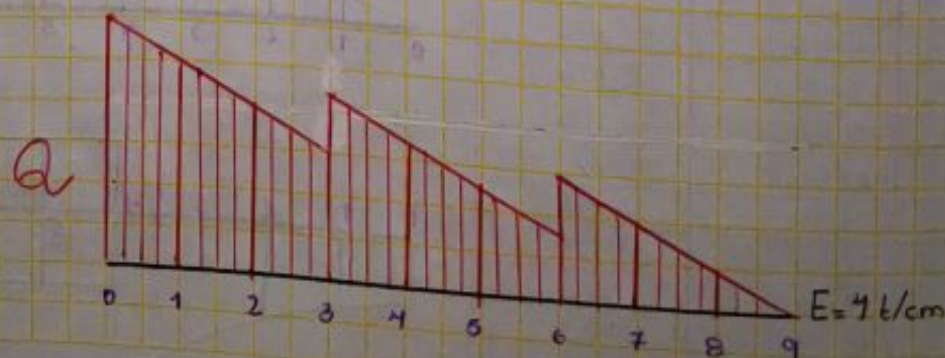
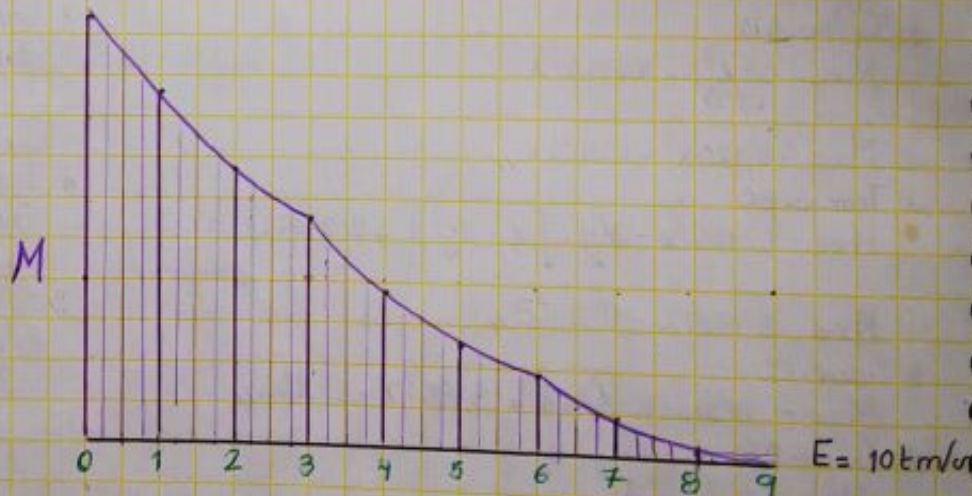
$$Q_x = 12 - 2x + 3$$

$$Q_x = 15 - 2x$$

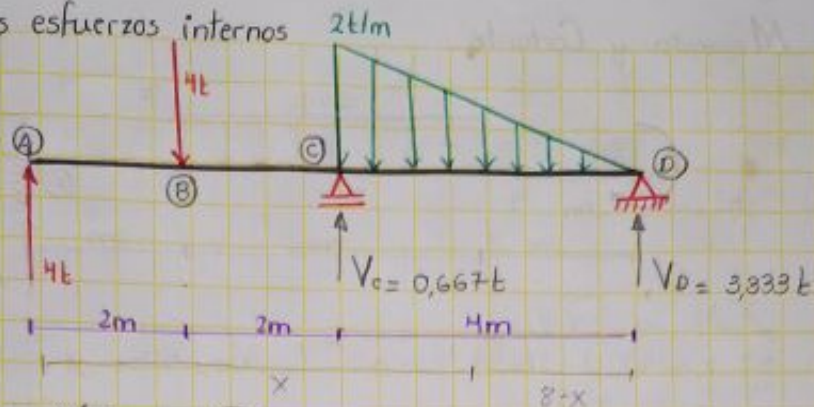
* Tramo CD

$$Q_x = 2(9-x)$$

$$Q_x = 18 - 2x$$



35. Diagramar los esfuerzos internos



1- Reacciones

$$\sum M_C = 0 \quad \curvearrowright \oplus$$

$$-V_D(4) + 2(4) \cdot \left[\frac{1}{3}(4) \right] - 4(2) + 4(4) = 0$$

$$V_D = 3,333t$$

$$\sum V = 0 \quad \uparrow \oplus$$

$$4 - 4 + V_c - \frac{2(4)}{2} + 3,333 = 0 \quad \therefore V_c = 0,667t$$

2- Ecuaciones de Momento

* Tramo AB

$$M_x = 4x$$

* Tramo BC

$$M_x = 4x - 4(x-2)$$

$$M_x = 8 //$$

* Tramo CD

$$M_x = 3,333(8-x) - \frac{2(8-x)^3}{6(4)}$$

$$M_x = 0,083x^3 - 2x^2 + 12,667x - 16,003 //$$

3- Ecuaciones de Corte

* Tramo AB

$$Q_x = 4t //$$

* Tramo BC

$$Q_x = 4 - 4 = 0t //$$

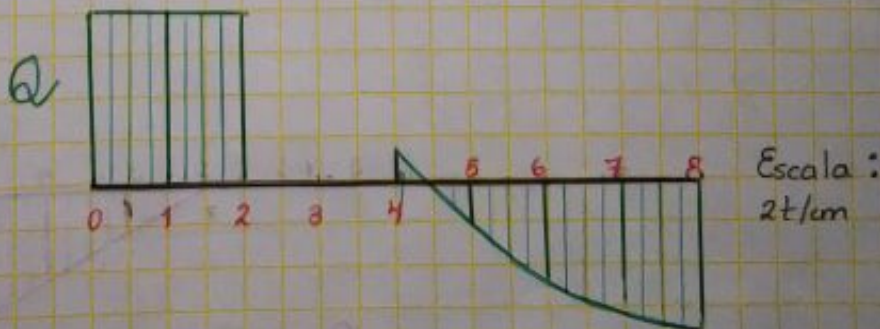
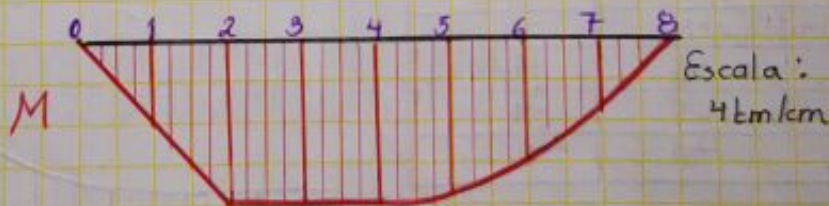
* Tramo CD

$$Q_x = -3,333 + \frac{2(8-x)^2}{2(4)}$$

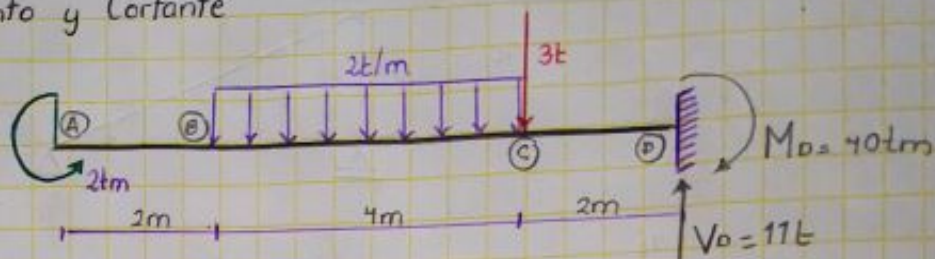
$$Q_x = 0,25x^2 - 4x + 12,667 //$$

4- Diagramas

x	Momento	Cortante
0	0	4
1	4	4
2	8	4/0
3	8	0
4	8	0/0,667
5	7,748	-1,083
6	5,998	-2,333
7	3,248	-3,083
8	0	-3,333



36- Diagramar Momento y Cortante



1- Reacciones

$$\sum M_D = 0 \quad (\uparrow \oplus)$$

$$-2 - 2(4)(4) - 3(2) + M_D = 0 \quad \therefore M_D = 40 \text{ tm} //$$

$$\sum F_v = 0 \quad (\uparrow \oplus)$$

$$-2(4) - 3 + V_D = 0 \quad \therefore V_D = 11 \text{ t} //$$

2- Ecuaciones de Momento

* Tramo AB

$$M_x = -2 \text{ tm} //$$

* Tramo BC

$$M_x = -2 - \frac{2(x-2)^2}{2}$$

$$M_x = -x^2 + 4x - 6 //$$

* Tramo CD

$$M_x = 11(8-x) - 40$$

$$M_x = -11x + 48 //$$

3- Ecuaciones de Corte

* Tramo AB

$$Q_x = 0$$

* Tramo BC

$$Q_x = -2(x-2)$$

$$Q_x = 4 - 2x //$$

* Tramo CD

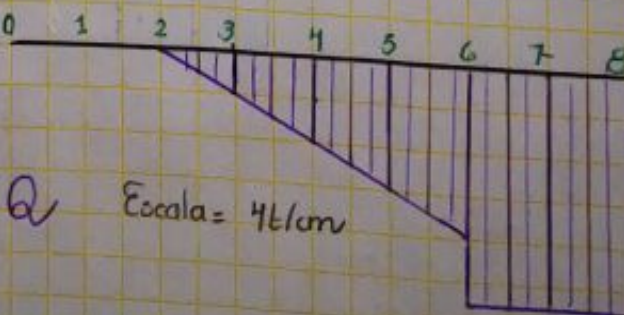
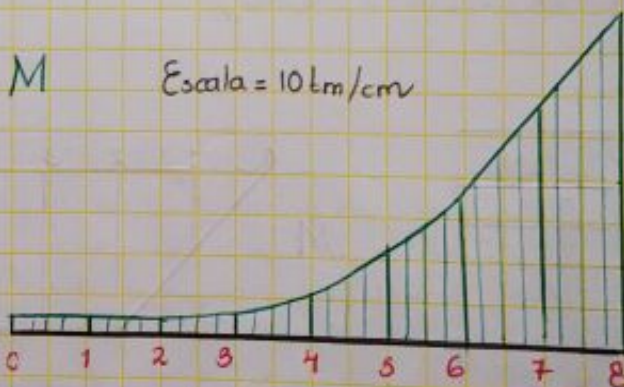
$$Q_x = -11 \text{ t} //$$

4- Diagramas

x	Momento	Cortante
0	-2	0
1	-2	0
2	-2	0
3	-3	-2
4	-6	-4
5	-11	-6
6	-18	-8/-11
7	-29	-11
8	-40	-11

M

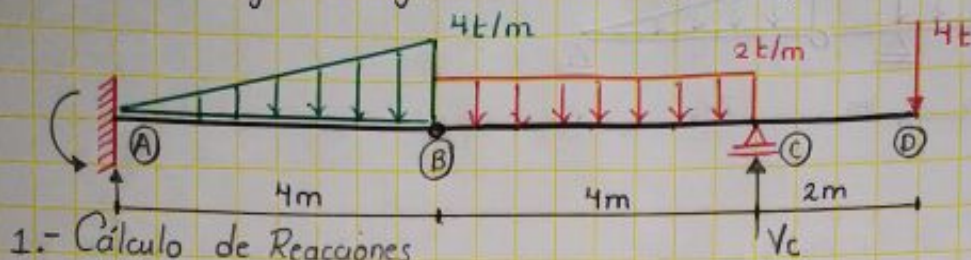
Escala = 10 tm/cm



Q

Escala = 4 t/cm

39.- Para la siguiente viga determinar los diagramas de los esfuerzos característicos



1.- Cálculo de Reacciones

$$\sum M_B = 0 \quad \curvearrowright \oplus \text{ a la derecha de la articulación}$$

$$2(4)(2) + 4(6) - V_C(4) = 0$$

$$V_C = 10t$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_A - \frac{4(4)}{2} - 2(4) + 10 - 4 = 0$$

$$V_A = 10t //$$

$$\sum M_B = 0 \quad \curvearrowright \oplus \text{ a la izquierda de la articulación}$$

$$-M_A + 10(4) - \frac{4(4)}{2} \cdot \frac{1}{3}(4) = 0$$

$$M_A = 29,333tm //$$

2.- Momentos

Barra AB $M_A = -29,333tm$

$M_B = 0tm$

Barra BC

$M_B = 0tm$

$M_C = -4(2) = -8tm$

Barra CD

$M_C = -4(2) = -8tm$

$M_D = 0tm$

3.- Cortantes

Barra AB $Q_A = 10t$

$Q_B = 10t - \frac{4(4)}{2} = 2t //$

Barra BC

$Q_B = 10 - \frac{4(4)}{2} = 2t$

$Q_C = 10 - \frac{4(4)}{2} - 2(4) = -6t$

Barra CD

$Q_C = 10 - \frac{4(4)}{2} - 2(4) + 10 = 4t$

$Q_D = 4t$

3.- Diagramas

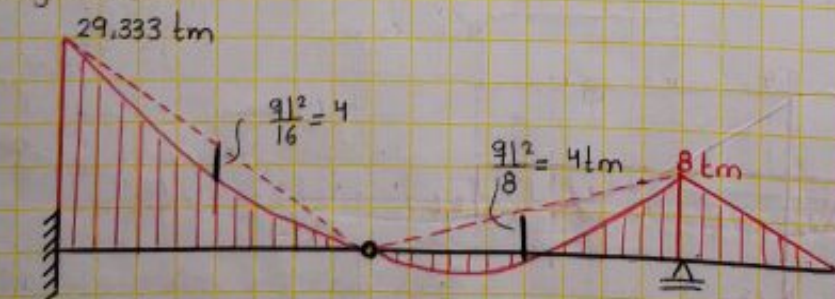


Diagrama de Momento

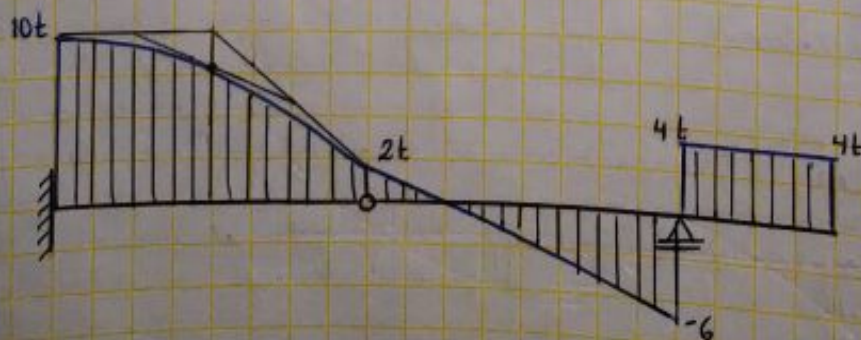
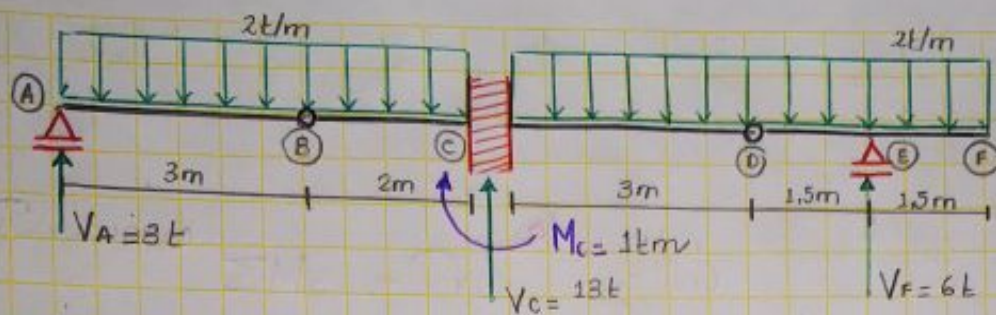


Diagrama de Cortante

40 Diagramar los esfuerzos característicos



1- Cálculo de Reacciones

$$\sum M_B = 0 \quad \curvearrowright \oplus \quad (\text{Izquierda})$$

$$V_A(3) - 2(3)(1,5) = 0$$

$$V_A = 3t //$$

$$\sum M_D = 0 \quad \curvearrowright \oplus \quad (\text{Derecho})$$

$$2(3)(1,5) - V_F(1,5) = 0$$

$$V_F = 6t$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$3 - 2(5) + V_C - 2(6) + 6 = 0$$

$$V_C = 13t$$

$$\sum M_A = 0 \quad \curvearrowright \oplus$$

$$2(5)(2,5) - 13(5) + M_C + 2(6)(8) - 6(9,5) = 0$$

$$M_C = 1tmv$$

2- Ecuaciones de Momento

* Tramo AC

$$M_x = 3 \cdot x - 2 \frac{x^2}{2} = 3x - x^2 //$$

* Tramo CE

$$M_x = 3x - 2(5)(x-2,5) + 1 + 13(x-5) - 2 \frac{(x-5)^2}{2}$$

$$M_x = -x^2 + 16x - 64 //$$

* Tramo EF

$$M_x = -2 \frac{(11-x)^2}{2}$$

$$M_x = -x^2 + 22x - 121 //$$

3- Ecuaciones de Corte

* Tramo AC

$$Q_x = 3 - 2x //$$

* Tramo CE

$$Q_x = 3 - 2x + 13$$

$$Q_x = 16 - 2x //$$

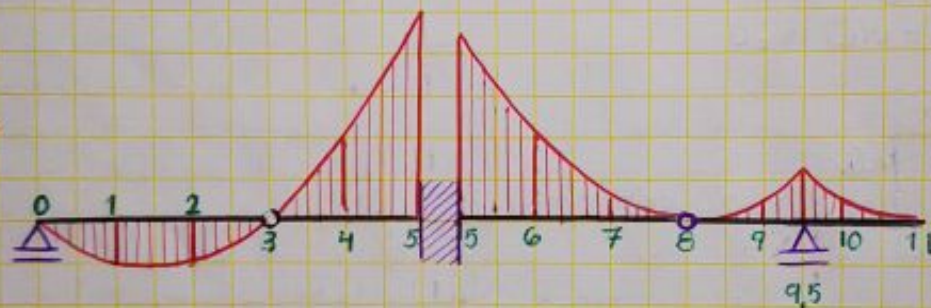
* Tramo EF

$$Q_x = 2(11-2x) = 22 - 2x //$$

4 - Diagramas

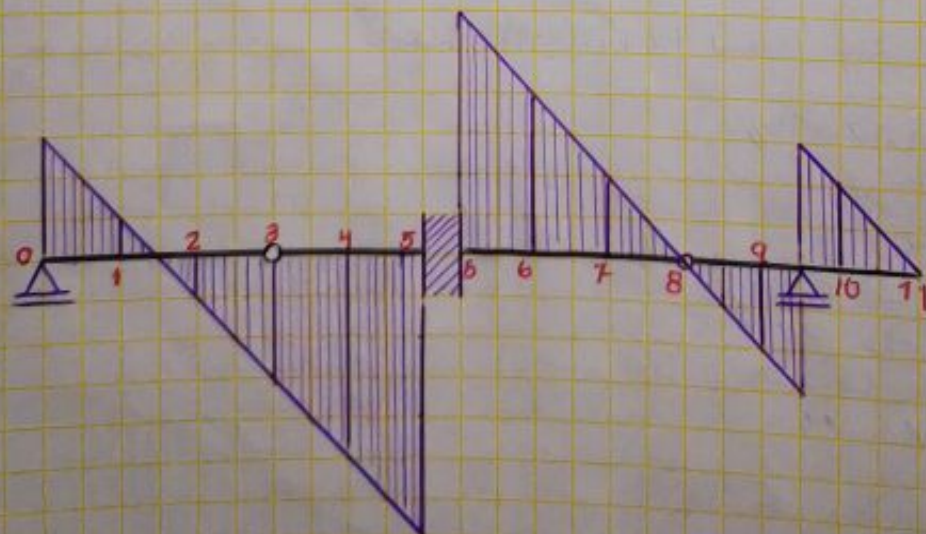
x	Momento	Cortante
0	0	3
1	2	1
2	2	-1
3	0	-3
4	-4	-5
5	-10/-9	-7/6
6	-4	4
7	-1	2
8	0	0
9	-1	-2
9,5	-2,25	-3/3
10	-1	2
11	0	0

M



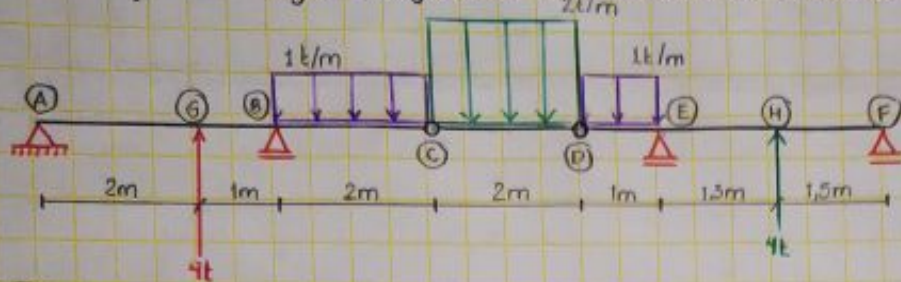
Escala = 4tm/cm

Q

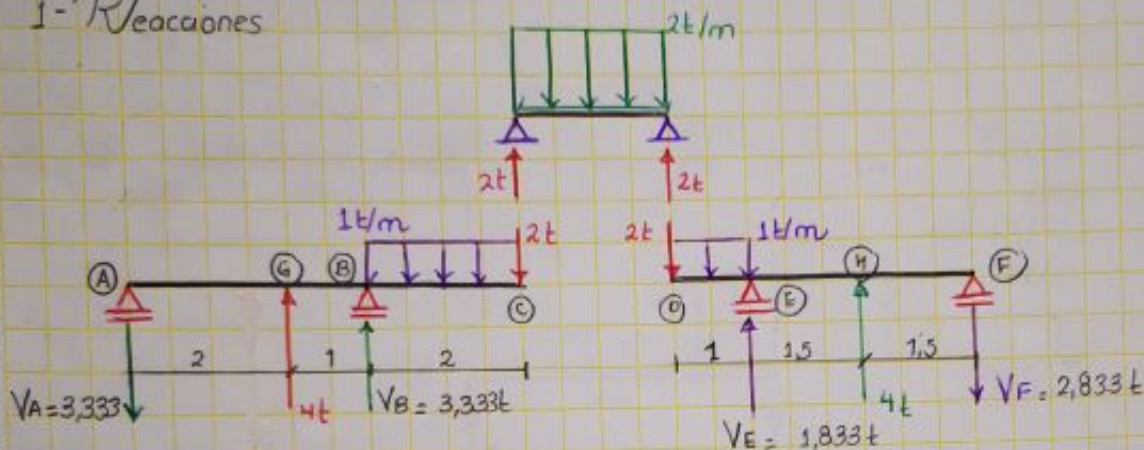


Escala = 2t/cm

41- Para la siguiente viga diagramar los esfuerzos característicos



1- Reacciones



Viga ABC $\sum M_A = 0 \curvearrowright \oplus$
 $-4(2) + 1(2)(4) + 2(5) - V_B(3) = 0 \quad \therefore V_B = 3,333t$
 $\sum F_v = 0 \uparrow \oplus$
 $-V_A + 4 + 3,333 - 2 - 1(2) = 0 \quad \therefore V_A = 3,333t$

Viga DEF $\sum M_E = 0 \curvearrowright \oplus$
 $-2(1) - 1(1)(0,5) - 4(1,5) + V_F(3) = 0 \quad \therefore V_F = 2,833t //$
 $\sum F_v = 0 \uparrow \oplus$
 $-2 - 1(1) + 4 - 2,833 + V_E = 0 \quad \therefore V_E = 1,833t //$

2- Ecuaciones de Momento

* Tramo AG

$$M_x = -3,333X //$$

* Tramo GB

$$M_x = -3,333X + 4(X-2)$$

$$M_x = 0,667X - 8 //$$

* Tramo BC

$$M_x = -3,333X + 4(X-2) + 3,333(X-3) - \frac{1(X-3)^2}{2}$$

$$M_x = -0,5X^2 + 7X - 22,499 //$$

* Tramo CD

$$M_x = 2(X-5) - \frac{2(X-5)^2}{2} = -X^2 + 12X - 35 //$$

* Tramo DE

$$M_x = -2(X-7) - \frac{1(X-7)^2}{2} = -0,5X^2 + 5X - 10,5 //$$

* Tramo EH

$$M_x = -2,833(11-X) + 4[11-X-1,5]$$

$$M_x = -1,167X + 6,837 //$$

* Tramo HF

$$M_x = -2,833(11-X)$$

3- Ecuaciones de Corte

* Tramo AG

$$Q_x = -3,333t$$

* Tramo GB

$$Q_x = -3,333 + 4 = 0,667t$$

* Tramo BC

$$Q_x = -3,333 + 4 + 3,333 - 1(X-3)$$

$$Q_x = 7 - X$$

* Tramo CD

$$Q_x = 2 - 2(X-5)$$

$$Q_x = 12 - 2X //$$

* Tramo DE

$$Q_x = -2 - 1(X-7)$$

$$Q_x = 5 - X //$$

* Tramo EH

$$Q_x = 2,833 - 4 = -1,167t //$$

* Tramo HF

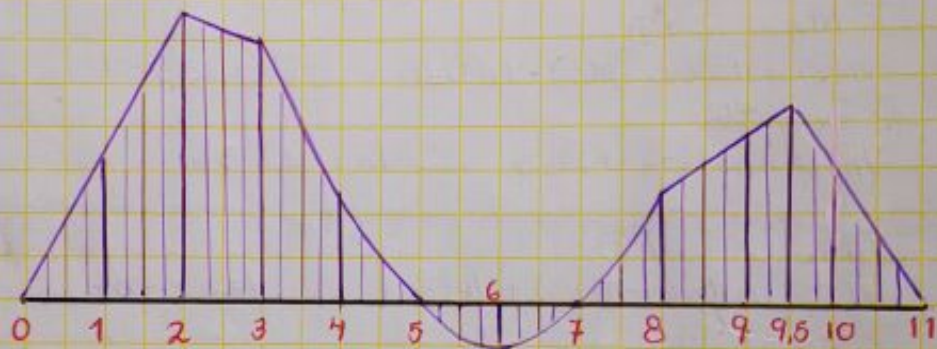
$$Q_x = 2,833t //$$

x	Momento	Cortante
0	0	-3,333
1	-3,333	-3,333
2	-6,666	$-3,333/0,667$
3	-5,999	$0,667/4$
4	-2,499	3
5	0	2
6	1	0
7	0	-2
8	-2,5	$-3/1,167$
9	-3,666	1,167
9,5	-4,2495	$1,167/2,833$
10	-2,833	2,8333
11	0	2,8333

Diagramas

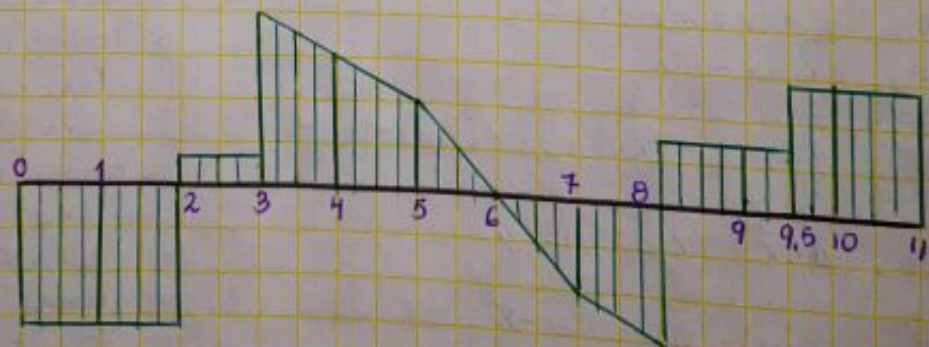
Momentos

Escala = 2tm/cm

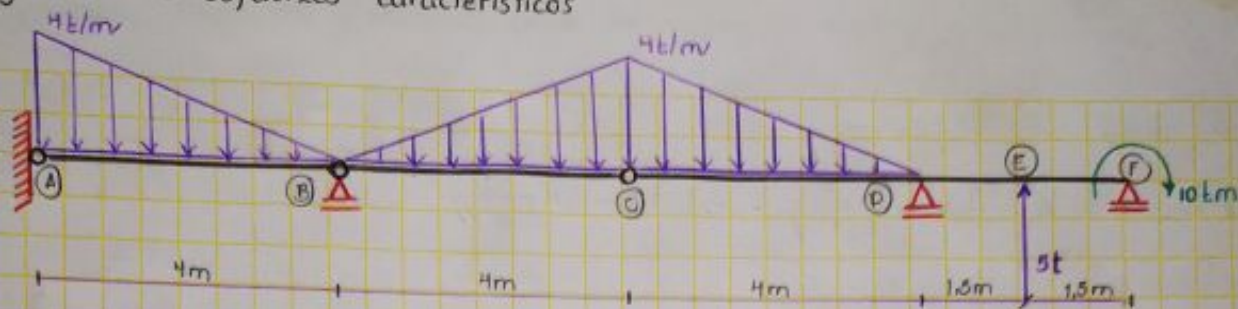


Cortante

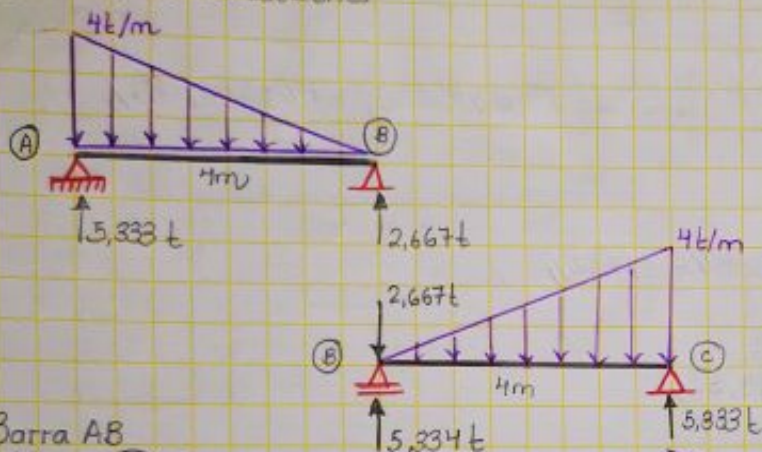
Escala = 2t/cm



42 Diagramar los esfuerzos característicos



1.- Cálculo de Reacciones



* Barra AB

$$\sum M_A = 0 \quad \curvearrowright \oplus$$

$$\frac{4(4)}{2} \cdot \left[\frac{1}{3}(4) \right] - V_B(4) = 0$$

$$V_B = 2.667 \text{ t} //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_A - \frac{4(4)}{2} + 2.667 = 0$$

$$V_A = 5.333 \text{ t} //$$

* Barra BC

$$\sum M_B = 0 \quad \curvearrowright \oplus$$

$$\frac{4(4)}{2} \left[\frac{2}{3} \cdot 4 \right] - V_C(4) = 0$$

$$V_C = 5.333 \text{ t} //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$-2.667 - \frac{4(4)}{2} + 5.333 + V_B = 0$$

$$V_B = 5.334 \text{ t} //$$

* Barra CF

$$\sum M_D = 0 \quad \curvearrowright \oplus$$

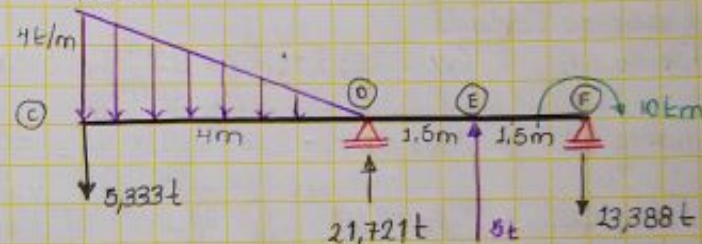
$$-5.333(4) - \frac{4(4)}{2} \cdot \left[\frac{2}{3}(4) \right] - 5(1.5) + 10 + V_F(3) = 0$$

$$V_F = 13.3884 \text{ t} //$$

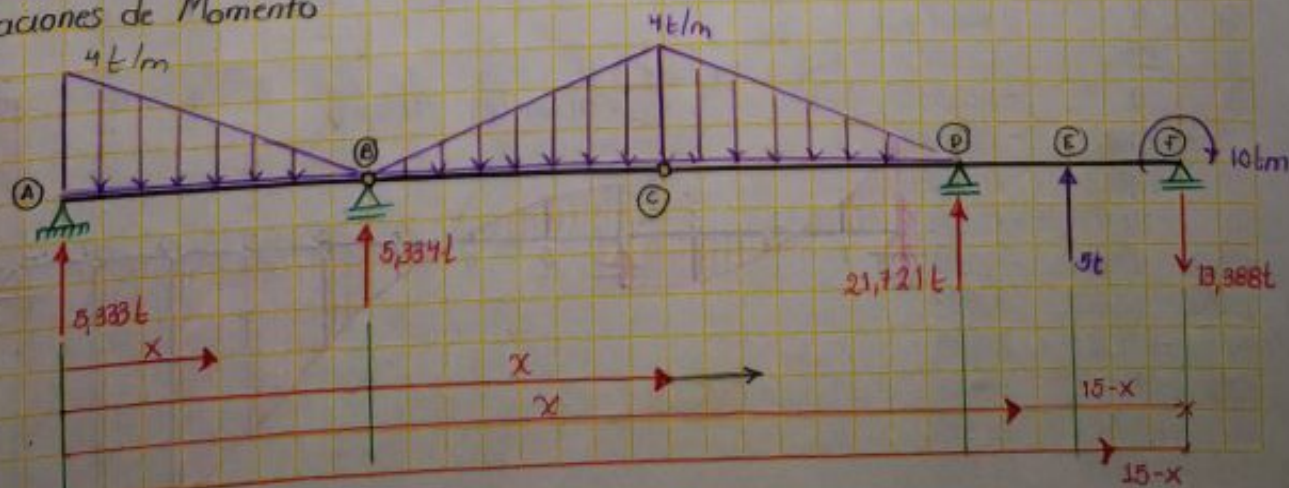
$$\sum F_v = 0 \quad \uparrow \oplus$$

$$-5.333 - \frac{4(4)}{2} + V_D + 5 - 13.388 = 0$$

$$V_D = 21.721 \text{ t} //$$



2.- Ecuaciones de Momento



* Tramo AB

$$M_x = 5,333x - \left(\frac{4x^2}{2} - \frac{4x^3}{6(4)} \right) = 5,333x - 2x^2 + \frac{x^3}{6} //$$

* Tramo BC

$$M_x = (5,333 - 2,667)(x-4) - \frac{4(x-4)^3}{6(4)} = -0,1667x^3 + 2x^2 - 5,334x + 0,003 //$$

* Tramo CD

$$M_x = -5,333(x-8) - \left[\frac{4(x-8)^2}{2} - \frac{4(x-8)^3}{6(4)} \right] = 0,167x^3 - 6x^2 + 58,667x - 170,669 //$$

* Tramo DE

$$M_x = -10 - 13,388(16-x) + 5(13,5-x) = 8,388x - 143,32 //$$

* Tramo EF

$$M_x = -10 - 13,388(15-x) = 13,388x - 210,82 //$$

3. Ecuaciones Cortante

* Tramo AB $Q_x = 5,333 - \left(4x - \frac{4x^2}{2(4)} \right) = 0,5x^2 - 4x + 5,333 //$

* Tramo BC $Q_x = 5,334 - 2,667 - \frac{4(x-4)^2}{2(4)} = -0,5x^2 + 4x - 5,333 //$

* Tramo CD $Q_x = -5,333 - \left[4(x-8) - \frac{4(x-8)^2}{2(4)} \right] = 0,5x^2 - 12x + 58,667 //$

* Tramo DE $Q_x = +13,388 - 5 = 8,388 t //$

* Tramo EF $Q_x = 13,388 t -$

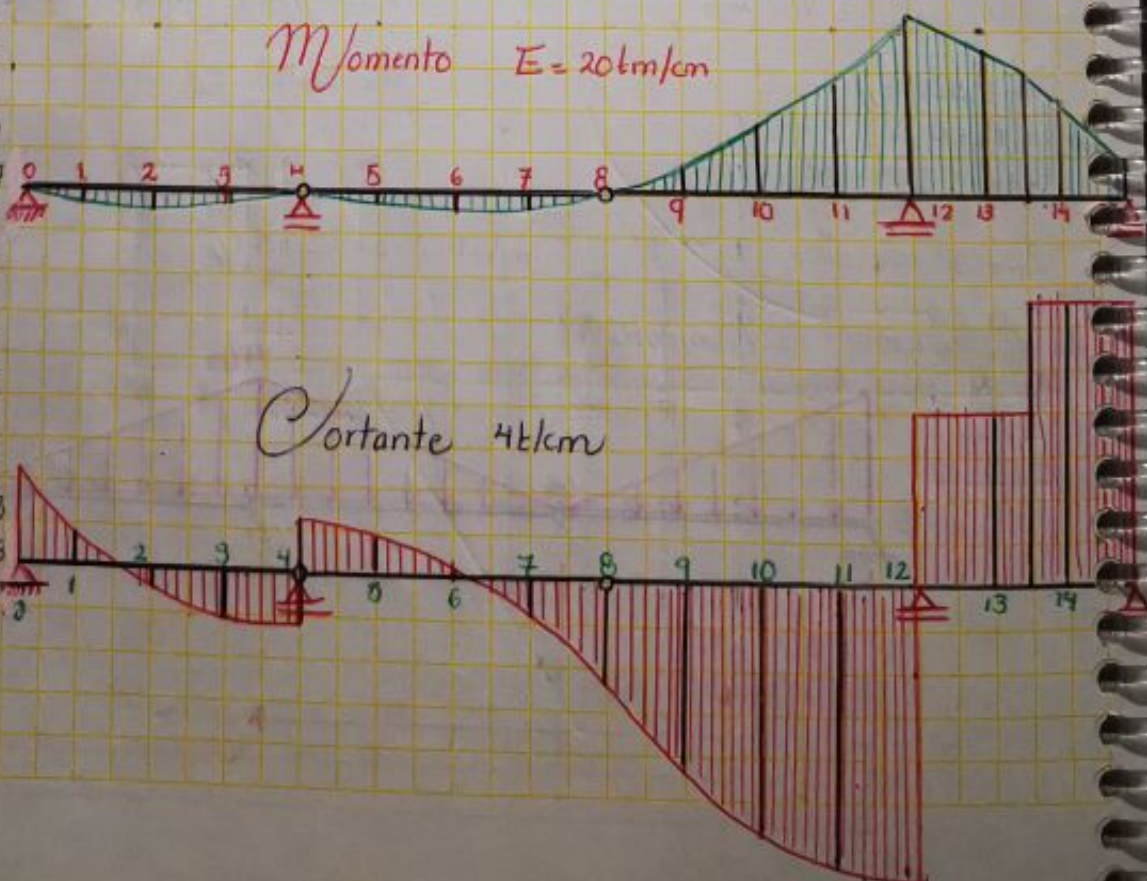
4. Diagramas

x Momento

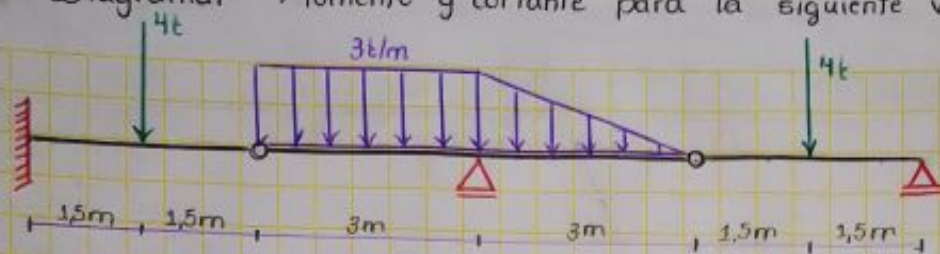
x	Momento	Q
0	0	5,3333
1	3,5	18,833
2	4	-0,6667
3	2,5	-2,1667
4(a)	0	-2,6667
4(b)	0	2,6667
5	2,5	2,1667
6	4	0,6667
7	3,5	-1,8333
8	0	-5,3333
9	-7,166	-8,8333
10	-17,333	-11,3333
11	-29,499	-12,8333
12(1)	-42,666	-13,3333
12(0)	-42,660	8,388
13	-34,266	8,388
13,5	-30,072	8,388
13,5	-30,082	13,388
14	-23,388	13,388
15	-10	13,388

Momento $E = 20 \text{ tm/cm}$

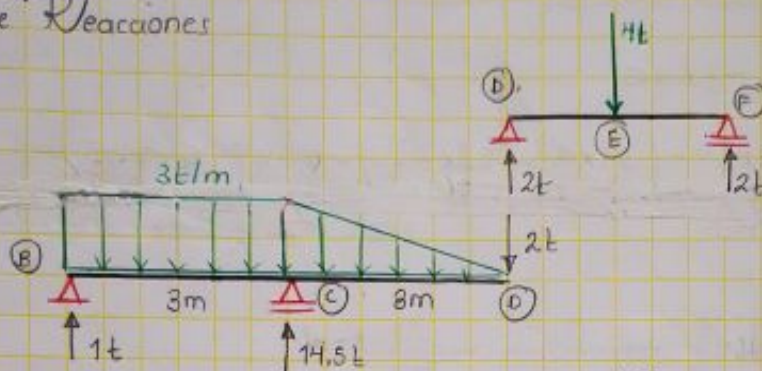
Cortante 4 t/cm



43 Diagramar Momento y cortante para la siguiente viga.



1.- Cálculo de Reacciones



* Barra BD

$$\sum M_C = 0 \quad \curvearrowright \oplus$$

$$2(3) + \frac{3(3)}{2} \cdot (1) - 3(3)(1.5) + V_B(3) = 0$$

$$V_B = 3t //$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$1 - 3(3) - \frac{3(3)}{2} - 2 + V_C = 0$$

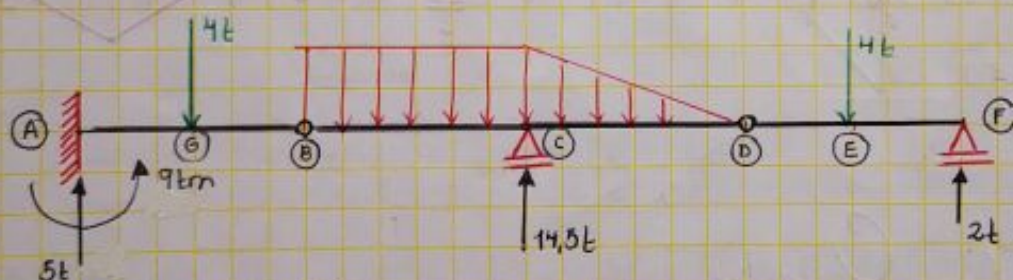
$$V_C = 14.5t //$$

$$\sum M_A = 0$$

$$M_A = 4(1.5) + 1(3) = 9tm$$

$$\sum F_V = 0$$

$$V_A = 5t //$$



2.- Ecuaciones de Momento

* Tramo AB

$$M_x = 5 \cdot x - 9 //$$

* Tramo GB

$$M_x = -1(3-x) = -3+x$$

* Tramo BC

$$M_x = 1 \cdot (x-3) - \frac{3(x-3)^2}{2} = -1.5x^2 + 10x - 16.5$$

* Tramo CD

$$M_x = 1(x-3) - 9(x-4.5) + 14.5(x-6) - \left[\frac{3 \cdot (x-6)^2}{2} - \frac{3(x-6)^3}{6(3)} \right]$$

$$M_x = 0.1667x^3 - 4.5x^2 + 42.5x - 139.5 //$$

* Tramo DE

$$M_x = 2 \cdot (x-9) = 2x-18 //$$

* Tramo EF

$$M_x = 2(12-x) = 24-2x //$$

3- Ecuaciones de Cortante

* Tramo AG

$$Q_x = 5t$$

* Tramo GB

$$Q_x = 5 - 1 = 4$$

* Tramo BC

$$Q_x = 1 - 3(x-3) = -3x + 10$$

* Tramo CD

$$Q_x = 1 - 3(3) + 14,5 - \left(3(x-6) - \frac{3(x-6)^2}{2(3)} \right) = 0,5x^2 - 9x + 42,5 //$$

* Tramo DE

$$Q_x = 2t$$

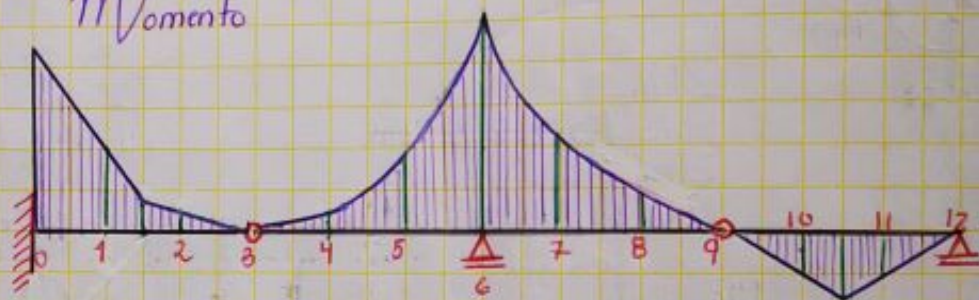
* Tramo EF

$$Q_x = -2t$$

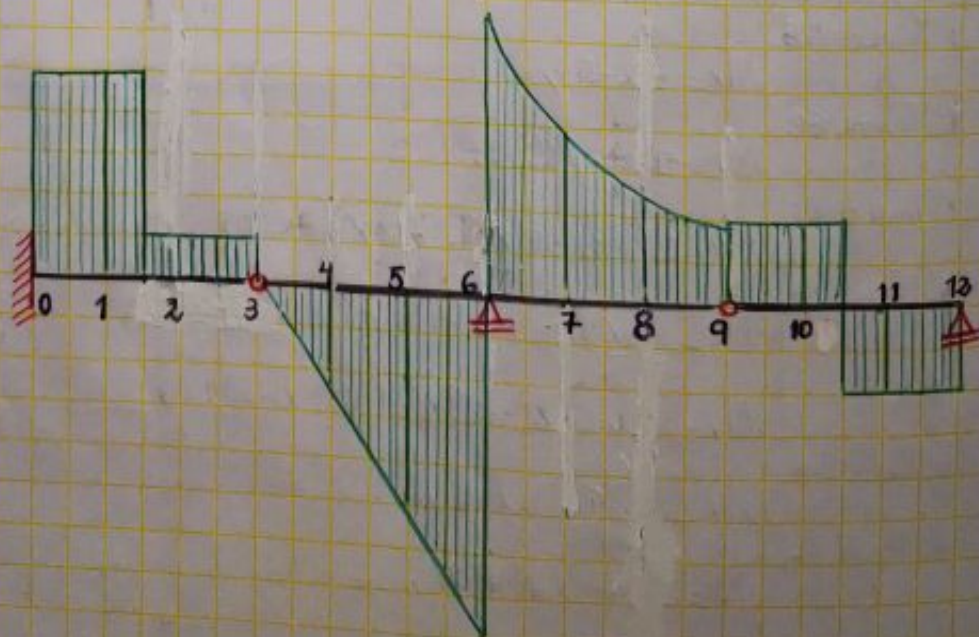
4- Diagramas

x	Momento	Cortante
0	-9	5
1	-4	5
1,5	-1,5	5/1
2	-1	1
3	0	1
4	-0,5	-2
5	-4	-5
6	-10,5	-8 / 6,5
7	-5,333	4
8	-2,166	2,5
9	0	2
10	2	2
10,5	3	2 / 2
11	2	-2
12	0	-2

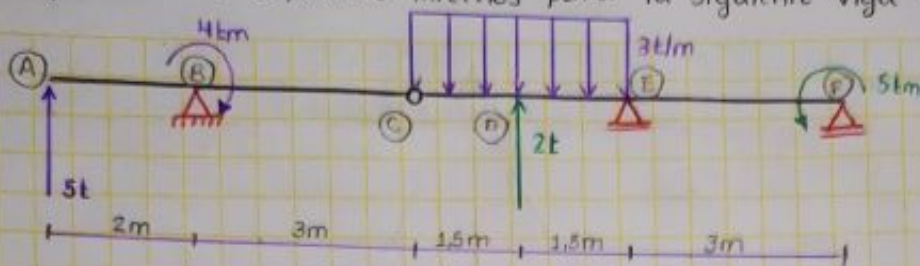
Momento



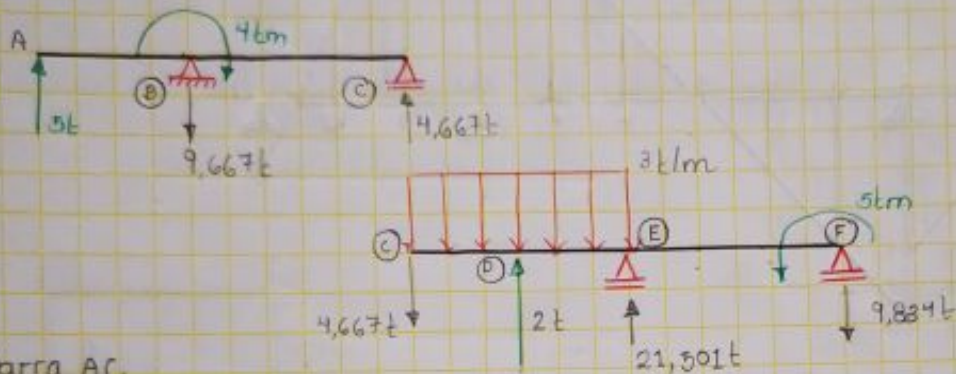
Cortante



44 Diagramar los esfuerzos internos para la siguiente viga Herber



1.- Reacciones



* Barra AC

$$\sum M_B = 0 \quad \curvearrowright \oplus$$

$$5(2) + 4 - V_C(3) = 0$$

$$V_C = 4.667t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$5 - V_B + 4.667 = 0$$

$$V_B = 9.667t //$$

* Barra CF

$$\sum M_E = 0 \quad \curvearrowright \oplus$$

$$-4.667(3) + 2(1.5) - 3(3)(1.5) - 5 + V_F(3) = 0$$

$$V_F = 9.834t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$-4.667 + 2 - 3(3) + V_E - 9.834 = 0$$

$$V_E = 21.501t //$$

2.- Ecuaciones de Momento

* Tramo AB $M_x = 5 \cdot x$

* Tramo BC $M_x = 4.667(5-x) = -4.667x + 23.335 //$

* Tramo CD $M_x = -4.667(x-5) - \frac{3(x-5)^2}{2} = -1.5x^2 + 10.333x - 14.165 //$

* Tramo DE $M_x = -4.667(x-5) - \frac{3(x-5)^2}{2} + 2(x-6.5) = -1.5x^2 + 12.333x - 27.165 //$

* Tramo EF $M_x = 5 - 9.834(11-x) = 9.834x - 103.174 //$

3.- Ecuaciones de Cortante

* Tramo AB $Q_x = 5t$

* Tramo BC $Q_x = -4.667t$

* Tramo CD $Q_x = -4.667 - 3(x-5) = -3x + 10.333 //$

* Tramo DE $Q_x = -4.667 + 2 - 3(x-5) = -3x + 12.333 //$

* Tramo EF $Q_x = 9.834t //$

4.- Diagramas

x	Momento	Cortante	x	Momento	Cortante
0	0	5	6.5	-10.373	-9.167/-7.167
1	5	5	7	-14.332	-8.667
2	10/14	5/-4.667	8	-24.498	-11.667/9.834
3	9.333	-4.667	9	-14.668	9.834
4	4.667	-4.667	10	-4.834	9.834
5	0	-4.667	11	5	9.834
6	-6.165	-7.667			

Diagrama de Momento

Esc = 5tm/km

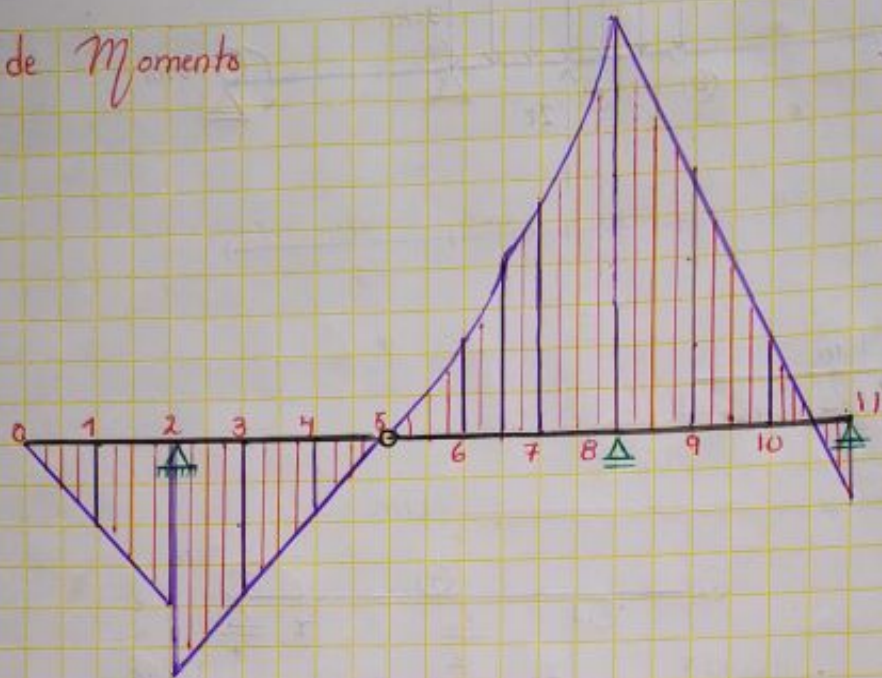
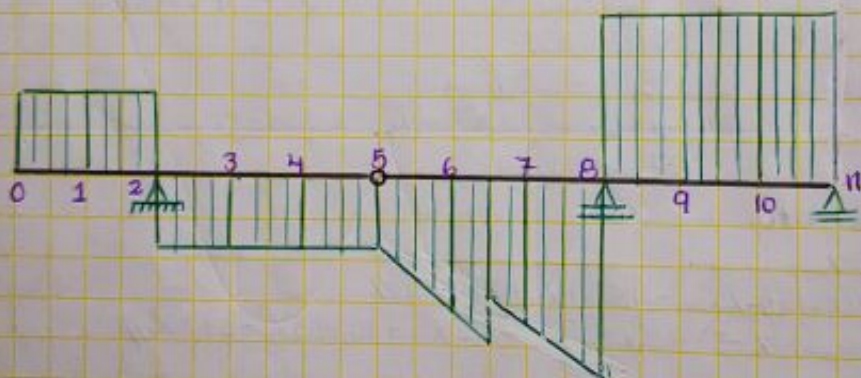
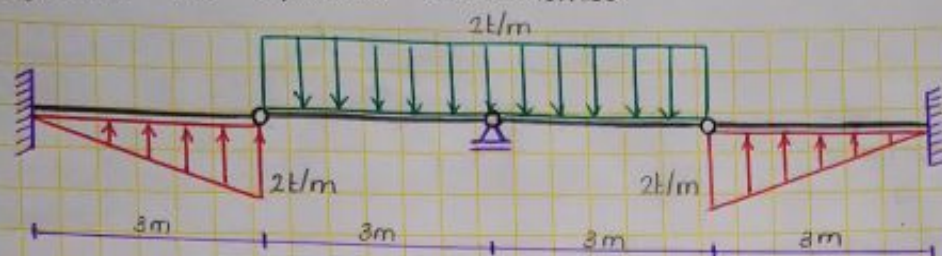


Diagrama de Cortante

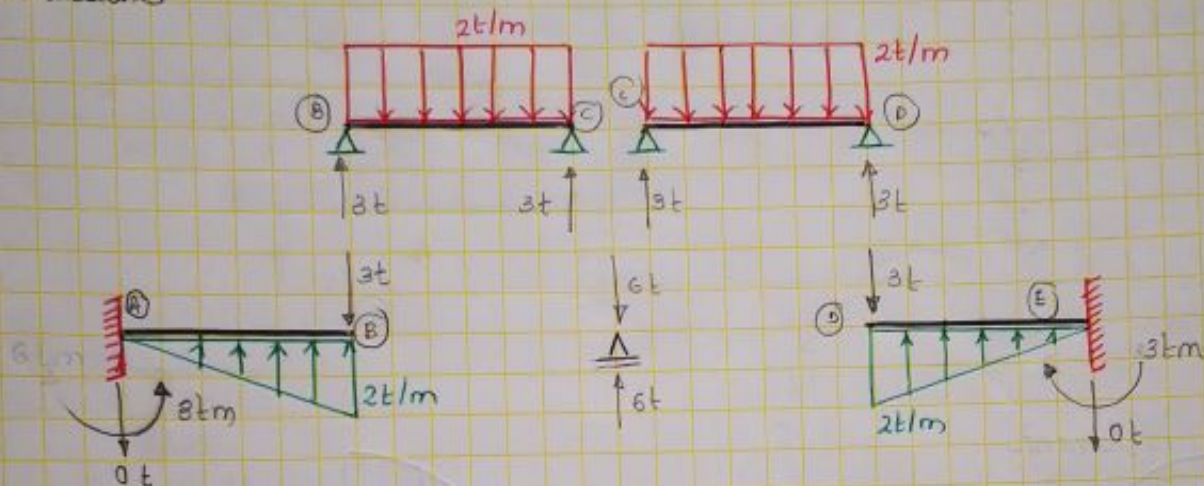
Esc = 5tm/km



45 Diagramar los esfuerzos característicos



1 Reacciones



* Tramo AB = DE

$$\sum M_A = 0 \quad \text{⤵} \oplus$$

$$-M_A - 2(3)(2) + 3(3) = 0$$

$$M_A = 3tm //$$

$$\sum F_v = 0 \quad \text{⤵} \oplus$$

$$-V_A + \frac{2(3)}{2} - 3 = 0 \quad \therefore V_A = 0t //$$

2 - Ecuaciones de Momento

* Tramo AB $M_x = -3 + \frac{2x^3}{6(3)} = -3 + \frac{x^3}{9} //$

* Tramo BC $M_x = 3(x-3) - \frac{2(x-3)^2}{2} = -x^2 + 9x - 18 //$

* Tramo CD $M_x = 3(x-6) - \frac{2(x-6)^2}{2} = -x^2 + 15x - 54 //$

* Tramo DE $M_x = -3(x-9) + \left[\frac{2(x-9)^2}{2} - \frac{2(x-9)^3}{6(3)} \right] = \frac{-x^3 + 36x^2 - 432x + 1701}{9} //$

3 - Ecuaciones de Cortante

* Tramo AB $Q_x = \frac{2x^2}{2(3)} = \frac{x^2}{3} //$

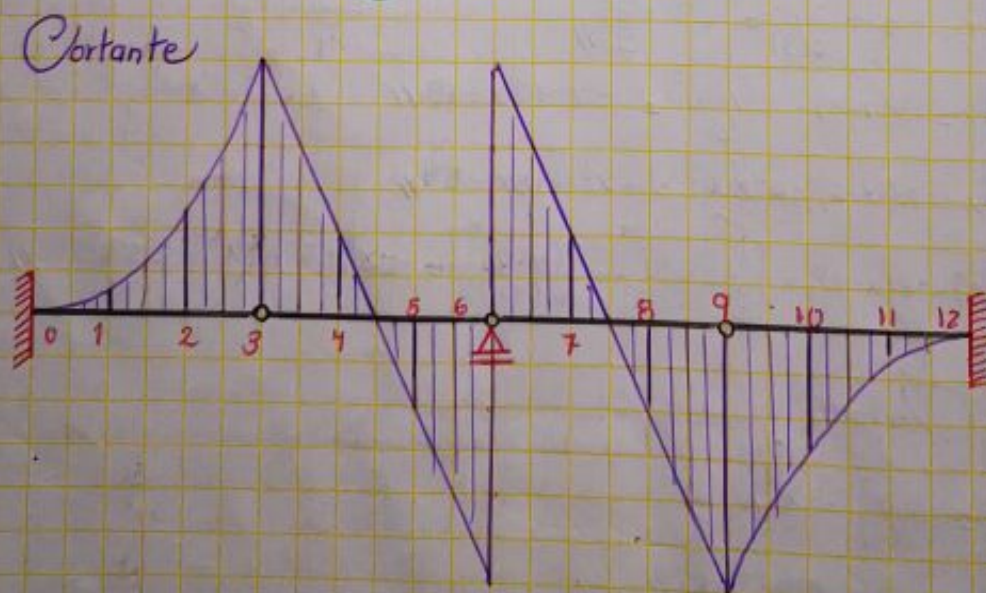
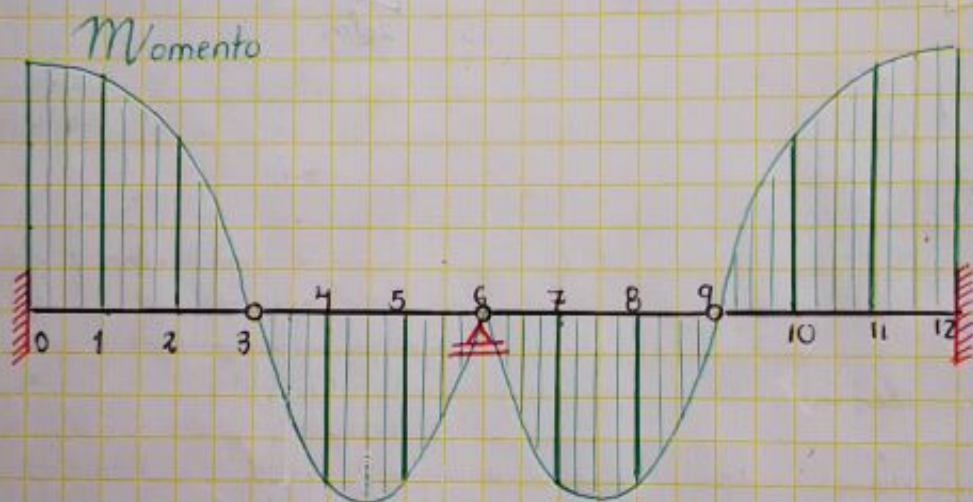
* Tramo BC $Q_x = 3 - 2(x-3) = 9 - 2x //$

* Tramo CD $Q_x = 3 - 2(x-6) = 15 - 2x //$

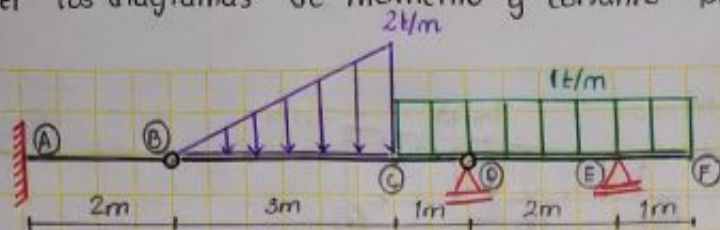
* Tramo DE $Q_x = -3 + \left[2(x-9) - \frac{2(x-9)^2}{2(3)} \right] = -0,333x^2 + 8x - 48 //$

4.-Diagramas

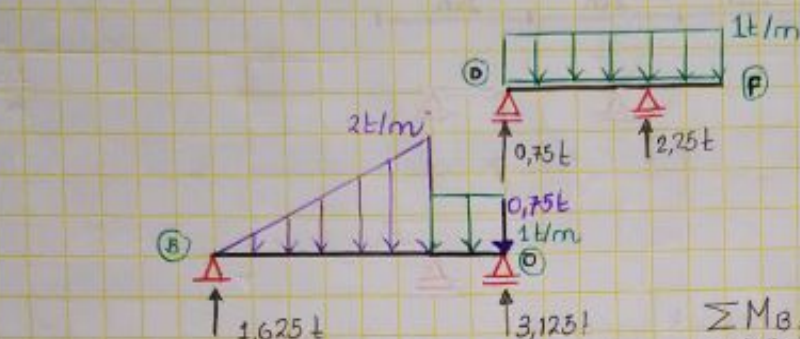
x	Momento	Cortante
0	-3	0
1	-2,889	0,333
2	-2,111	1,333
3	0	3
4	2	1
5	3	-1
6	0	-3 / 3
7	2	1
8	2	-1
9	0	-3
10	-2,117	-1,333
11	-2,888	-0,333
12	-3	0



1.6 Obtener los diagramas de momento y cortante para la siguiente viga



1.- Reacciones

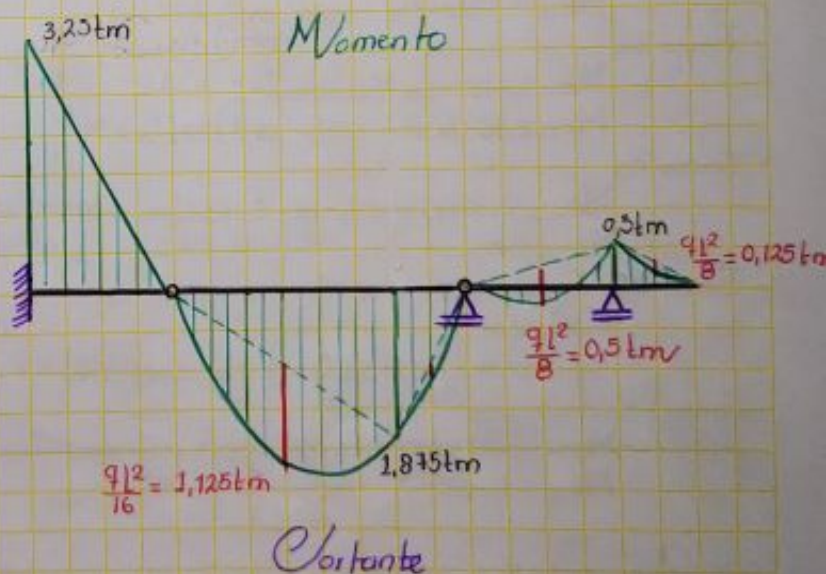


* Barra AB
 $\sum F_v = 0 \uparrow \oplus$
 $V_A = 1,625t$
 $\sum M_A = 0 \curvearrowright \oplus$
 $M_A = 1,625(2) = 3,25 \text{ tm} //$

* Barra DE
 $\sum M_D = 0 \curvearrowright \oplus$
 $1(3)(1,5) - V_E(2) = 0$
 $V_E = 2,25t //$
 $\sum F_v = 0 \uparrow \oplus$
 $V_C - 1(3) + 2,25 = 0$
 $V_C = 0,75t$

$\sum M_B = 0$ (Barra BD)
 $\frac{2(3)}{2} [2] + 1(1)(3,5) + 0,75(4) - V_D(4) = 0$
 $V_D = 3,125t$
 $\sum F_v = 0 \uparrow \oplus$
 $V_B - \frac{2(3)}{2} - 1(1) - 0,75 + 3,125 = 0$
 $V_B = 1,625t //$

4.- Diagramas



2.- Momentos

$M_A = -3,25 \text{ tm}$
 $M_B = 0 \text{ tm}$
 $M_C = 1,625(3) - \frac{2(3)}{2} [1] = 1,875 \text{ tm}$

$M_D = 0$
 $M_E = -1(1)(0,5) = -0,5 \text{ tm}$

3.- Cortantes

$Q_A = 1,625t //$
 $Q_B = 1,625t //$
 $Q_C = 1,625 - \frac{2(3)}{2} = -1,375t //$

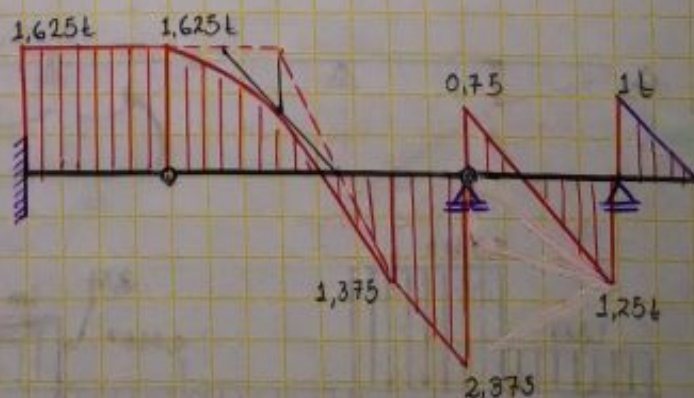
$Q_D \text{ (Izq)} = 1,625 - \frac{2(3)}{2} - 1(1) = -2,375t //$

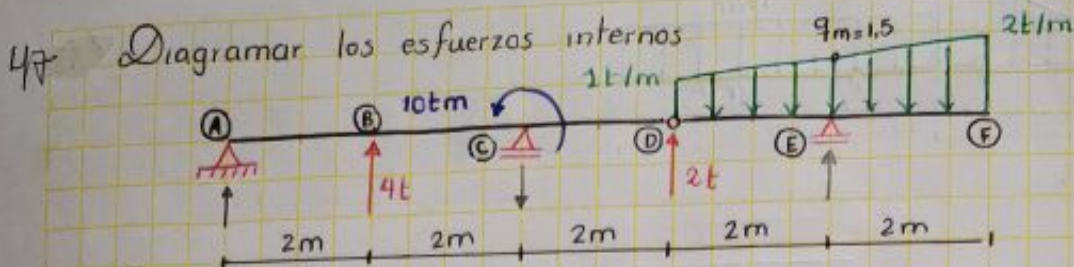
$Q_D \text{ (Der)} = 0,75t //$

$Q_E \text{ (Izq)} = 0,75 - 1(2) = -1,25t //$

$Q_E \text{ (Der)} = 1(1) = 1t //$

$Q_F = 0t //$





1.- Cálculo de Reacciones

$$\sum M_D = 0 \quad (\text{ala derecha de la articulación})$$

$$1(4)(2) + \frac{1}{2}(4) \cdot \frac{2}{3}(4) - V_E(2) = 0$$

$$V_E = 6,6667t$$

$$\sum M_A = 0 \quad (\text{ala derecha de la articulación})$$

$$-4(2) - 10 + V_C(4) - 2(6) - 6,6667(8) + 1(4)8 + \frac{1}{2}(4) \cdot \left(6 + \frac{2}{3}(4)\right) = 0$$

$$V_C = 8,5t$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_A + 4 - 8,5 + 2 + 6,6667 - \left(\frac{1}{2} \cdot 2\right) \cdot 4 = 0$$

$$V_A = 1,8333t //$$

2.- Momentos

Tramo AB

$$M_A = 0$$

$$M_B = 1,8333(2) = 3,6666tm$$

Tramo BC

$$M_B = 1,8333(2) = 3,6666tm$$

$$M_C = 1,8333(4) + 4(2) = 15,3332tm$$

Tramo CD

$$M_C = 1,8333(4) + 4(2) - 10 = 5,3332tm$$

$$M_D = 0tm$$

Tramo DE

$$M_D = 0tm$$

$$M_E = -(1,5 \cdot 2) \cdot 1 - 0,5(2) \cdot \frac{2}{3}(2) = -3,666tm$$

Tramo EF

$$M_E = -3,666tm$$

$$M_F = 0tm$$

3.- Cortante

Tramo AB

$$Q_A = 1,8333t$$

$$Q_B = 1,8333t$$

Tramo BC

$$Q_B = 1,8333 + 4 = 5,8333t$$

$$Q_C = 1,8333 + 4 = 5,8333t$$

Tramo CD

$$Q_C = 1,8333 + 4 - 8,5 = -2,6667t$$

$$Q_D = -2,6667t$$

Tramo DE

$$Q_D = 1,8333 + 4 - 8,5 + 2 = -0,6667t$$

$$Q_E = 1,8333 + 4 - 8,5 + 2 - \left(\frac{1+1,5}{2}\right) \cdot 2$$

$$Q_E = -3,1667t //$$

Tramo EF

$$Q_E = \left(\frac{1,5+2}{2}\right) \cdot 2 = 3,5t //$$

$$Q_F = 0t //$$

4.- Diagramas



Diagrama de Momento

$$\frac{(q_1 + q_2) \cdot l^2}{16} = \frac{(1,5 + 2) \cdot 2^2}{16} = 1,75tm$$

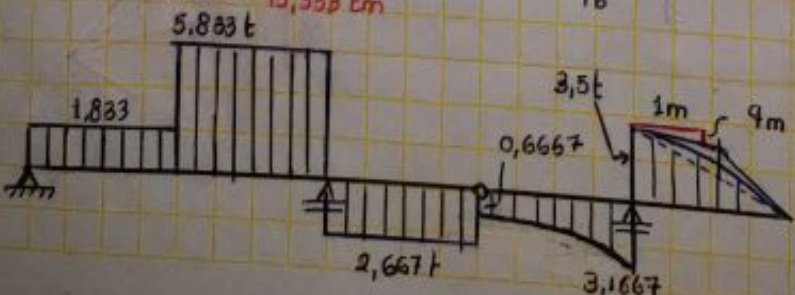
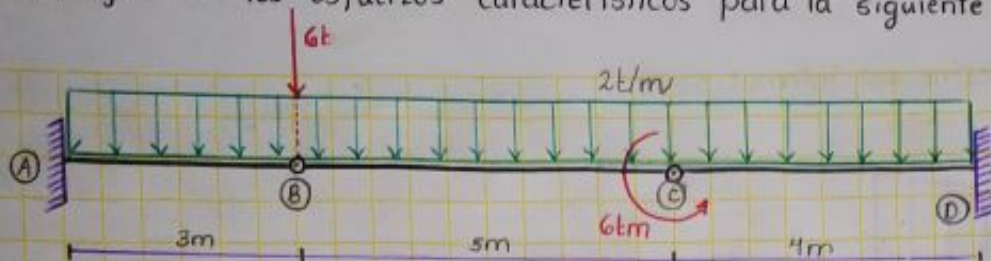
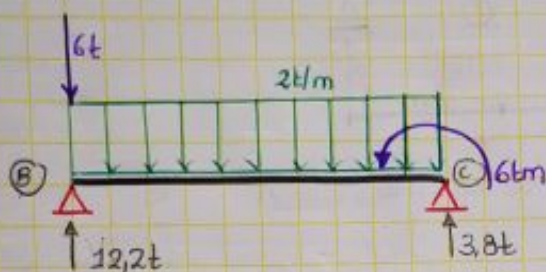


Diagrama de Cortante

48 Diagramar los esfuerzos característicos para la siguiente viga compuesta



1- Reacciones



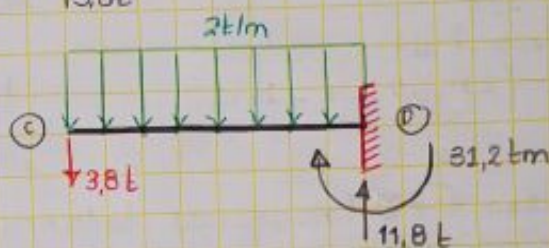
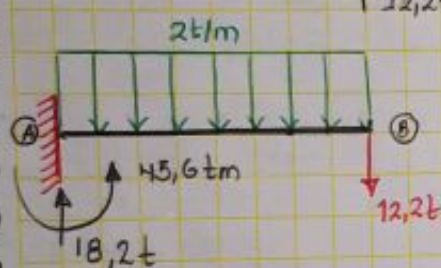
* Barra BC

$$\sum M_B = 0 \quad \curvearrowright \oplus$$

$$2(5)(2.5) - 6 - V_C(5) = 0 \quad \therefore V_C = 3.8t //$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$V_B - 6 - 2(5) + 3.8 = 0 \quad \therefore V_B = 12.2t //$$



* Barra AB

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$V_A - 2(3) - 12.2 = 0$$

$$V_A = 18.2t //$$

$$\sum M_A = 0 \quad \curvearrowright \oplus$$

$$M_A = 2(3)(1.5) + 12.2(3)$$

$$M_A = 45.6tm //$$

* Barra CD

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$-3.8 - 2(4) + V_D = 0$$

$$V_D = 11.8t //$$

$$\sum M_D = 0 \quad \curvearrowright \oplus$$

$$M_D - 3.8(4) - 2(4)(2) = 0$$

$$M_D = 31.2tm //$$

2 Momentos

$$M_A = -45.6tm$$

$$M_B = 0tm$$

$$M_C(IZQ) = 6tm$$

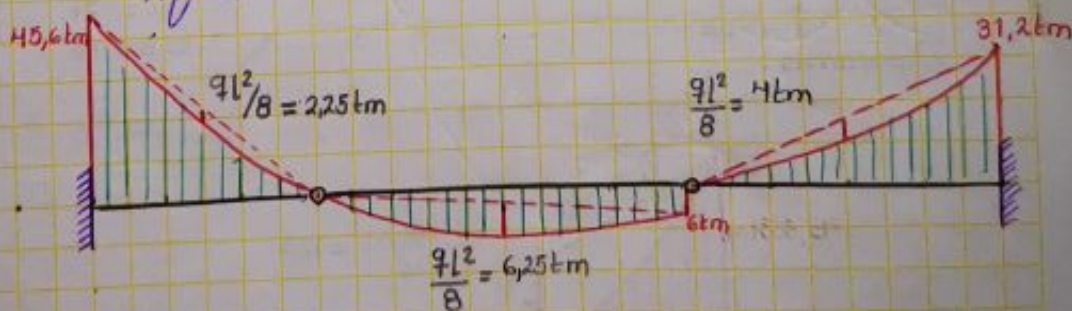
$$M_C(DER) = 0tm$$

$$M_D = -31.2tm$$

4 - Diagramas

Momento

$$E_{oc} = 20tm/cm$$



3- Cortantes

$$Q_A = 18.2t$$

$$Q_B(IZQ) = 18.2 - 2(3) = 12.2t$$

$$Q_B(DER) = 18.2 - 2(3) - 6 = 6.2t$$

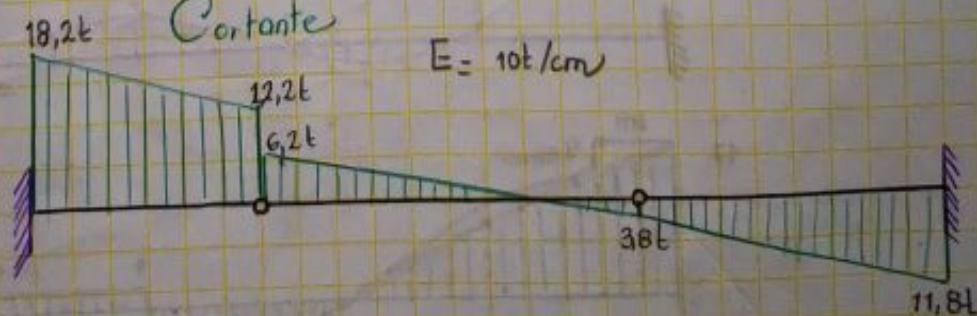
$$Q_C(IZQ) = -3.8t$$

$$Q_C(DER) = -3.8t$$

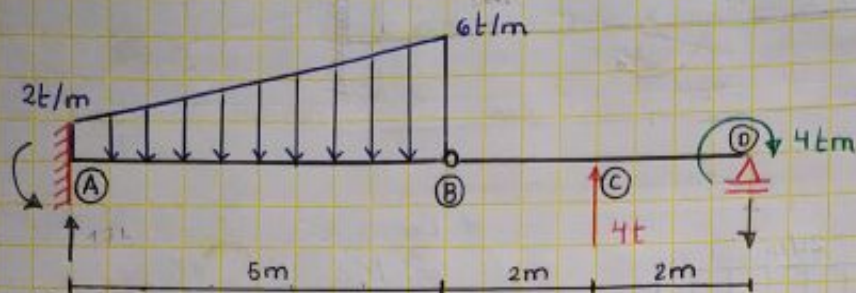
$$Q_D = -11.8t$$

Cortante

$$E = 10t/cm$$



49 Diagramar los esfuerzos internos



1.- Reacciones

$$\sum M_B = 0 \quad (\text{a la derecha de la articulación})$$

$$-4(2) + 4 + V_D(4) = 0$$

$$V_D = 1t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_A - \left(\frac{2+6}{2}\right) \cdot 5 + 4 - 1 = 0$$

$$V_A = 17t //$$

$$\sum M_B = 0 \quad (\text{a la izquierda de la articulación})$$

$$17(5) - 2(5)(2,5) - 4(5) \cdot \left[\frac{1}{3}(5)\right] - M_A = 0$$

$$M_A = 43,333tm //$$

2.- Momentos

Tramo AB

$$M_A = -43,333tm$$

$$M_B = 0tm$$

Tramo BC

$$M_B = 0tm$$

$$M_C = -4 - 1(2) = -6tm$$

Tramo CD

$$M_C = -6tm$$

$$M_D = -4tm$$

3.- Cortante

Tramo AB

$$Q_A = 17t$$

$$Q_B = 17 - \left(\frac{2+6}{2}\right) \cdot 5 = -3t$$

Tramo BC

$$Q_B = 17 - \left(\frac{2+6}{2}\right) \cdot 5 = -3t$$

$$Q_C = -3t$$

Tramo CD

$$Q_C = 1t$$

$$Q_D = 1t$$

3.- Diagramas

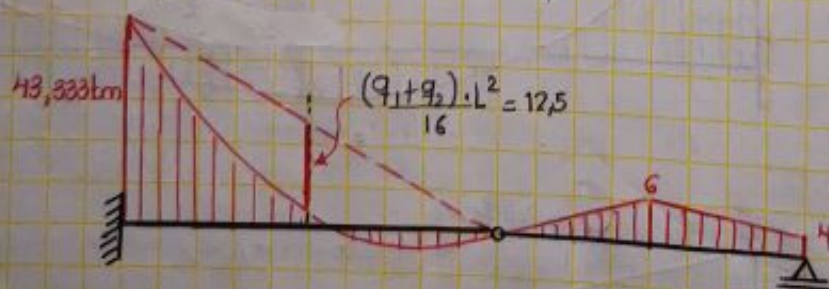
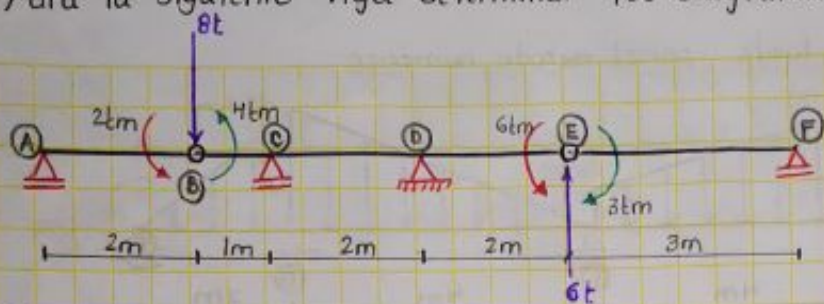


Diagrama de Momento

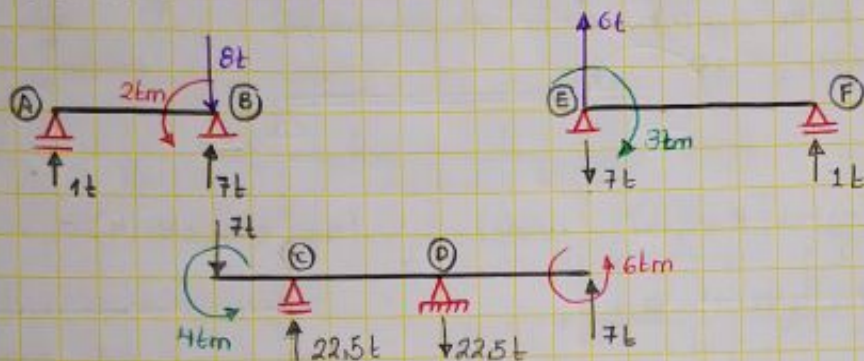


Diagrama de Cortante

50. Para la siguiente viga determinar los diagrama de Momento y Cortante



1.- Reacciones



* Barra AB

$$\begin{aligned}\sum M_A &= 0 \quad \curvearrowright \oplus \\ 8(2) - 2 - V_B(2) &= 0 \\ V_B &= 7t \\ \sum F_v &= 0 \quad \uparrow \oplus \\ V_A - 8 + 7 &= 0 \\ V_A &= 1t\end{aligned}$$

* Barra EF

$$\begin{aligned}\sum M_E &= 0 \quad \curvearrowright \oplus \\ 3 - V_F(3) &= 0 \\ V_F &= 1t \\ \sum F_v &= 0 \quad \uparrow \oplus \\ 6 + 1 - V_E &= 0 \\ V_E &= 7t\end{aligned}$$

* Barra CD

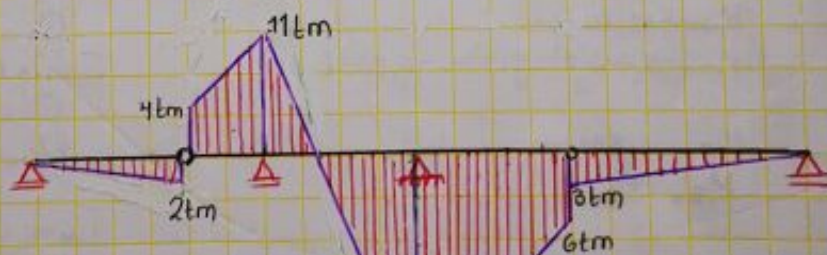
$$\begin{aligned}\sum M_C &= 0 \quad \curvearrowright \oplus \\ -4 - 6 - 7(1) - 7(4) + V_D(2) &= 0 \\ V_D &= 22.5t \\ \sum F_v &= 0 \quad \uparrow \oplus \\ V_C &= 22.5t\end{aligned}$$

2. Momentos

$$\begin{aligned}M_A &= 0tm \\ M_B(IZQ) &= 2tm \\ M_B(DER) &= -4tm \\ M_C &= -7(1) - 4 = -11tm \\ M_D &= 6 + 7(2) = 20tm \\ M_E(IZQ) &= 6tm \\ M_E(DER) &= 3tm \\ M_F &= 0tm\end{aligned}$$

4.- Diagramas.

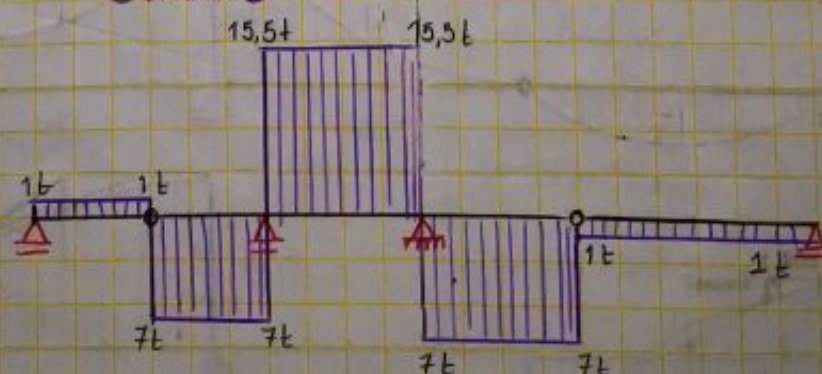
Momento



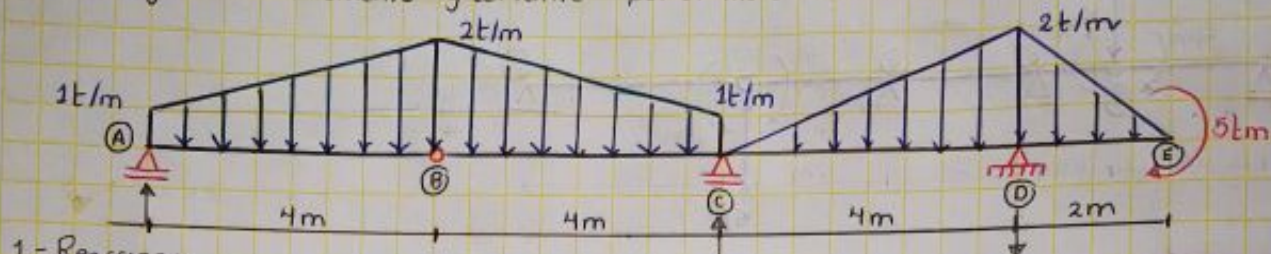
3. Cortante

$$\begin{aligned}Q_A &= 1t \\ Q_B(IZQ) &= 1t \\ Q_B(DER) &= -7t \\ Q_C(IZQ) &= -7t \\ Q_C(DER) &= -7 + 22.5 = 15.5t \\ Q_D(IZQ) &= 15.5t \\ Q_D(DER) &= 15.5 - 22.5 = -7t \\ Q_E(IZQ) &= -7t \\ Q_E(DER) &= -1t \\ Q_F &= -1t\end{aligned}$$

Cortante



51 Diagramar Momento y cortante por el método numérico



1.- Reacciones

$$\sum M_B = 0 \quad (\curvearrowright \oplus) \quad (\text{a la izquierda de la articulación})$$

$$V_A(4) - 1(4)(2) - \frac{1(4)}{2} \cdot \frac{1}{3}(4) = 0$$

$$V_A = 2,667t //$$

$$\sum M_D = 0 \quad (\curvearrowright \oplus)$$

$$2,667(12) - 1(4)(10) - \frac{1(4)}{2} \left[8 + \frac{1}{3}(4) \right] - 1(4)(6) - \frac{1(4)}{2} \left[4 + \frac{1}{3}(4) \right] + V_C(4) - \frac{2(4)}{2} \cdot \frac{1}{3}(4) + \frac{2(2)}{2} \cdot \frac{1}{3}(2) + 5 = 0$$

$$V_C = 15,749t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$2,667 - \left(\frac{1+2}{2} \right) \cdot 4 - \left(\frac{2+1}{2} \right) \cdot 4 + 15,749 - 2(6) - V_D = 0$$

$$V_D = 0,416t //$$

2.- Momento

Tramo AB $M_A = 0$

$M_B = 0$

Tramo BC $M_B = 0$

$M_C = 2,667(8) - 1(8)(4) - \frac{1(8)}{2} \cdot 4$

$M_C = -26,664tm$

Tramo CD $M_C = -26,664tm$

$M_D = -5 - \frac{2(2)}{2} \cdot \frac{1}{3}(2) = -6,333tm$

Tramo DE

$M_D = -6,333tm$

$M_E = -5tm$

3.- Cortante

Tramo AB

$Q_A = 2,667t$

$Q_B = 2,667 - \left(\frac{1+2}{2} \right) \cdot 4 = -3,333t$

Tramo BC

$Q_B = -3,333t //$

$Q_C = 2,667 - 1(8) - \frac{1(8)}{2} = -9,333t //$

Tramo CD

$Q_C = -9,333 + 15,749 = 6,416t //$

$Q_D = \frac{2(2)}{2} + 0,416 = 2,416t //$

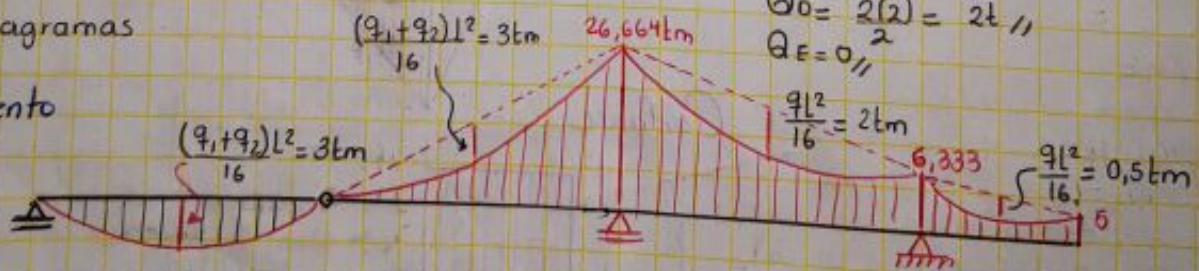
Tramo DE

$Q_D = \frac{2(2)}{2} = 2t //$

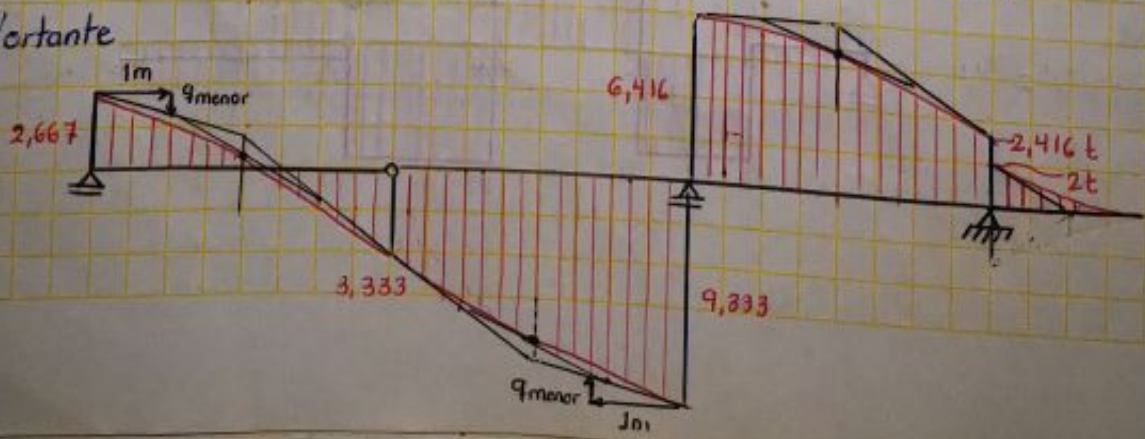
$Q_E = 0 //$

4.- Diagramas

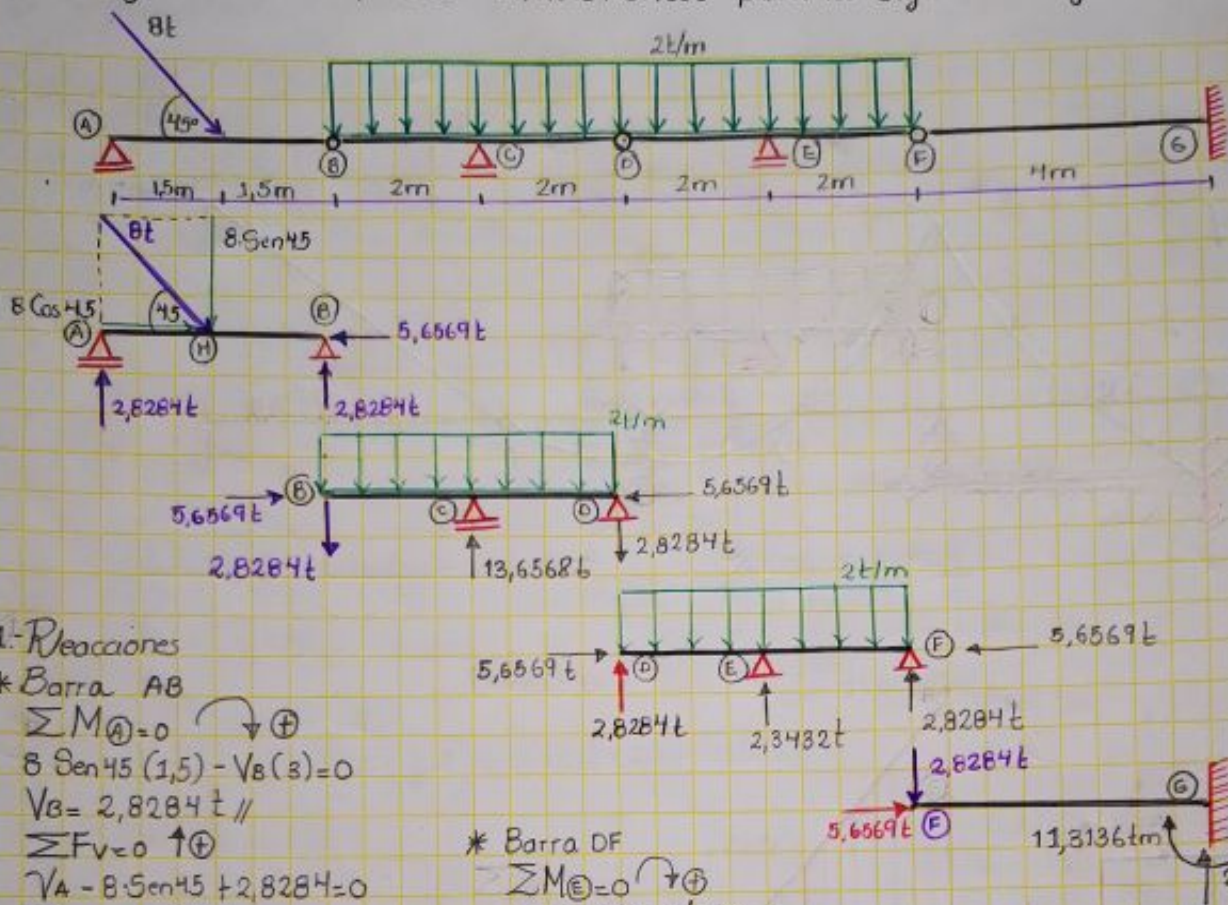
Momento



Cortante



52 Diagramar los esfuerzos característicos para la siguiente viga



1- Reacciones

* Barra AB

$$\begin{aligned}\sum M_A &= 0 \quad \curvearrowright \oplus \\ 8 \sin 45 (1,5) - V_B (3) &= 0 \\ V_B &= 2,8284 t // \\ \sum F_v &= 0 \quad \uparrow \oplus \\ V_A - 8 \sin 45 + 2,8284 &= 0 \\ V_A &= 3,8284 t // \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ H_B &= 8 \cos 45 = 5,6569 t //\end{aligned}$$

* Barra BD

$$\begin{aligned}\sum M_B &= 0 \quad \curvearrowright \oplus \\ V_D (2) - 2,8284 (2) &= 0 \\ V_D &= 2,8284 t // \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ H_D &= 5,6569 t // \\ \sum F_v &= 0 \quad \uparrow \oplus \\ -2 (4) - 2,8284 - 2,8284 + V_C &= 0 \\ V_C &= 13,6568 t\end{aligned}$$

* Barra DF

$$\begin{aligned}\sum M_D &= 0 \quad \curvearrowright \oplus \\ 2,8284 (2) - V_F (2) &= 0 \\ V_F &= 2,8284 t // \\ \sum F_v &= 0 \quad \uparrow \oplus \\ 2,8284 - 2 (4) + 2,8284 + V_E &= 0 \\ V_E &= 2,3432 t // \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ H_F &= 5,6569 t //\end{aligned}$$

* Barra FG

$$\begin{aligned}\sum F_v &= 0 \quad \uparrow \oplus \\ V_G &= 2,8284 t // \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ H_G &= 5,6569 t // \\ \sum M_G &= 0 \quad \curvearrowright \oplus \\ -2,8284 (4) + M_G &= 0 \\ M_G &= 11,3136 tm\end{aligned}$$

2- Momentos

$$\begin{aligned}M_A &= 0 tm \\ M_H &= 2,8284 (1,5) = 4,2426 t \\ M_B &= 0 tm \\ M_C &= -2,8284 (2) - 2 (2) (1) = -9,6568 tm \\ M_D &= 0 tm \\ M_E &= 2,8284 (2) - 2 (2) (1) = 1,6568 tm \\ M_F &= 0 tm \\ M_G &= -11,3136 tm\end{aligned}$$

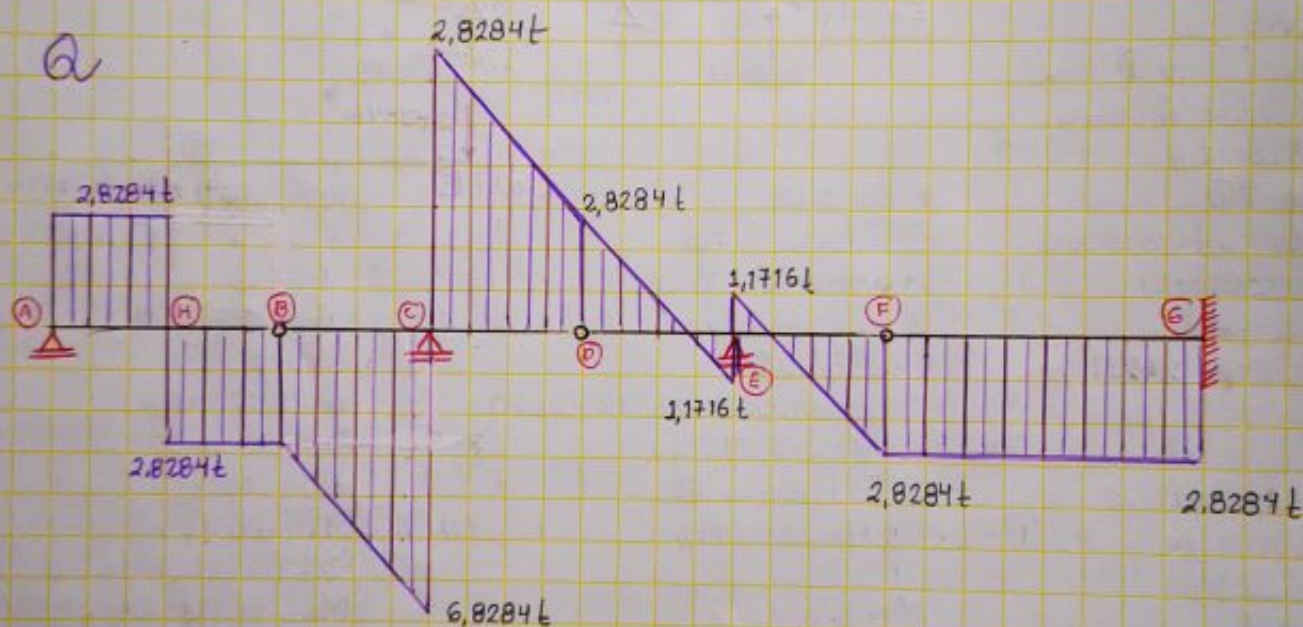
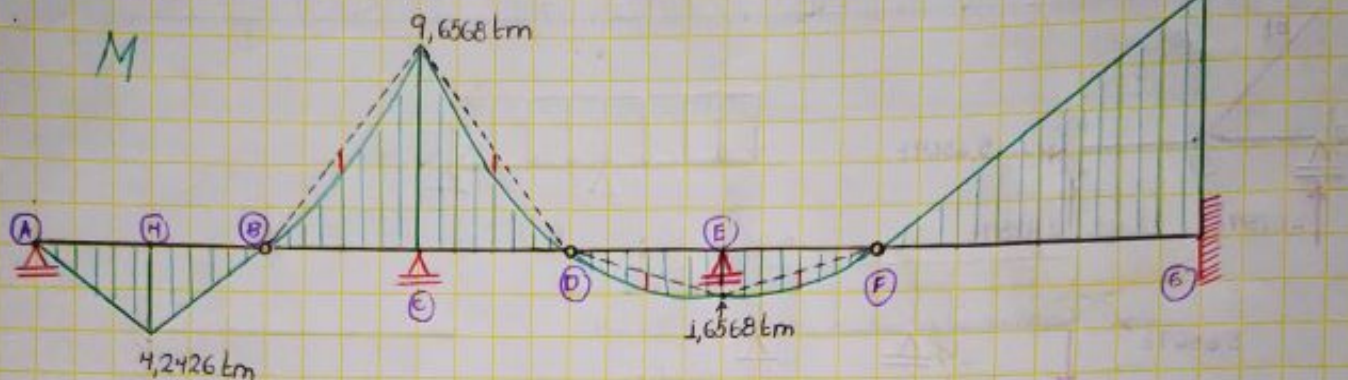
3- Cortantes

$$\begin{aligned}Q_A &= 2,8284 t // \\ Q_H (Izq) &= 2,8284 t // \\ Q_H (Der) &= 2,8284 - 8 \sin 45 = -2,8285 t // \\ Q_B &= -2,8284 t // \\ Q_C (Izq) &= -2,8284 - 2 (2) = -6,8284 t // \\ Q_C (Der) &= -2,8284 - 2 (2) + 13,6569 = 6,8285 t // \\ Q_D &= 2,8284 t // \\ Q_E (Izq) &= 2,8284 - 2 (2) = -1,1716 t // \\ Q_E (Der) &= 2,8284 - 2 (2) + 2,3432 = 1,1716 t // \\ Q_F &= -2,8284 t // \\ Q_G &= -2,8284 t //\end{aligned}$$

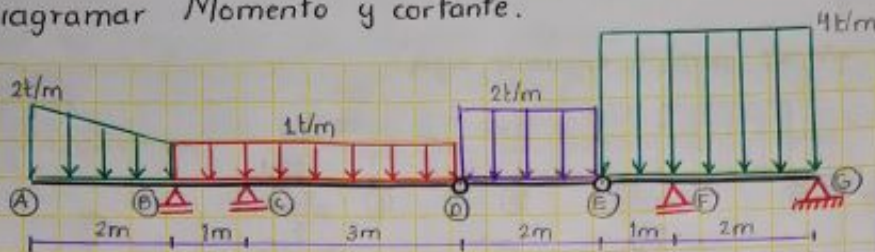
4- Normal

$$N_HG = -5,6569 t //$$

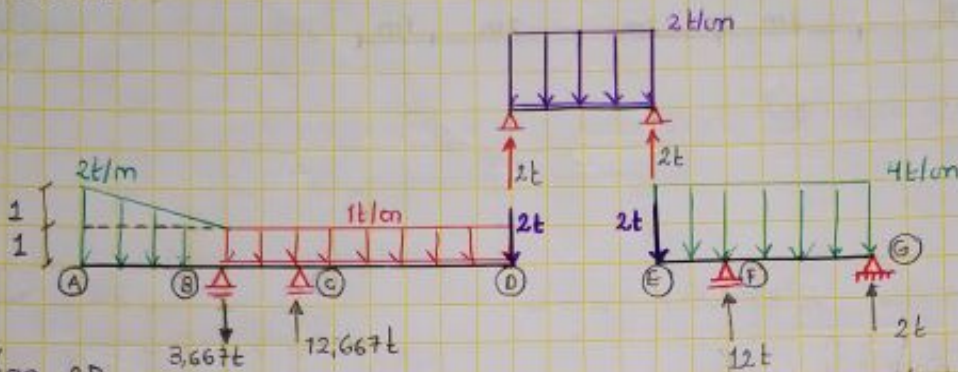
5.- Diagramas



53 Diagramar Momento y cortante.



1. Reacciones



Viga AD

$$\sum M_B = 0 \quad (\uparrow +)$$

$$-1(2)(1) - \frac{1(2)}{2} \left[\frac{2}{3} \cdot 2 \right] + 1(4)(2) + 2(4) - V_C(1) = 0$$

$$V_C = 12,667t$$

$$\sum F_v = 0 \quad (\uparrow +)$$

$$-\frac{(2+1)}{2} \cdot 2 - 1(4) - 2 + 12,667 - V_B = 0$$

$$V_B = 3,667t$$

Viga EG

$$\sum M_E = 0 \quad (\uparrow +)$$

$$-2(3) - 4(3)(1,5) + V_F(2) = 0$$

$$V_F = 12t$$

$$\sum F_v = 0 \quad (\uparrow +)$$

$$-2 - 4(3) + 12 + V_G = 0$$

$$V_G = 2t$$

2. Momentos

$$M_A = 0$$

$$M_B = -1(2)(1) - \frac{1(2)}{2} \left[\frac{2}{3} \cdot 2 \right] = -3,333tm$$

$$M_C = -2(3) - 1(3)(1,5) = -10,5tm$$

$$M_D = 0tm$$

$$M_E = 0tm$$

$$M_F = -2(1) - 4(1)(0,5) = -4tm$$

$$M_G = 0tm$$

2. Cortantes

$$Q_A = 0t$$

$$Q_B(izq) = -\frac{(2+1)}{2} \cdot 2 = -3t$$

$$Q_B(der) = -\frac{(2+1)}{2} \cdot 2 + 3,667 = -6,667t$$

$$Q_C(izq) = 2 + (1)(3) - 12,667 = -7,667t$$

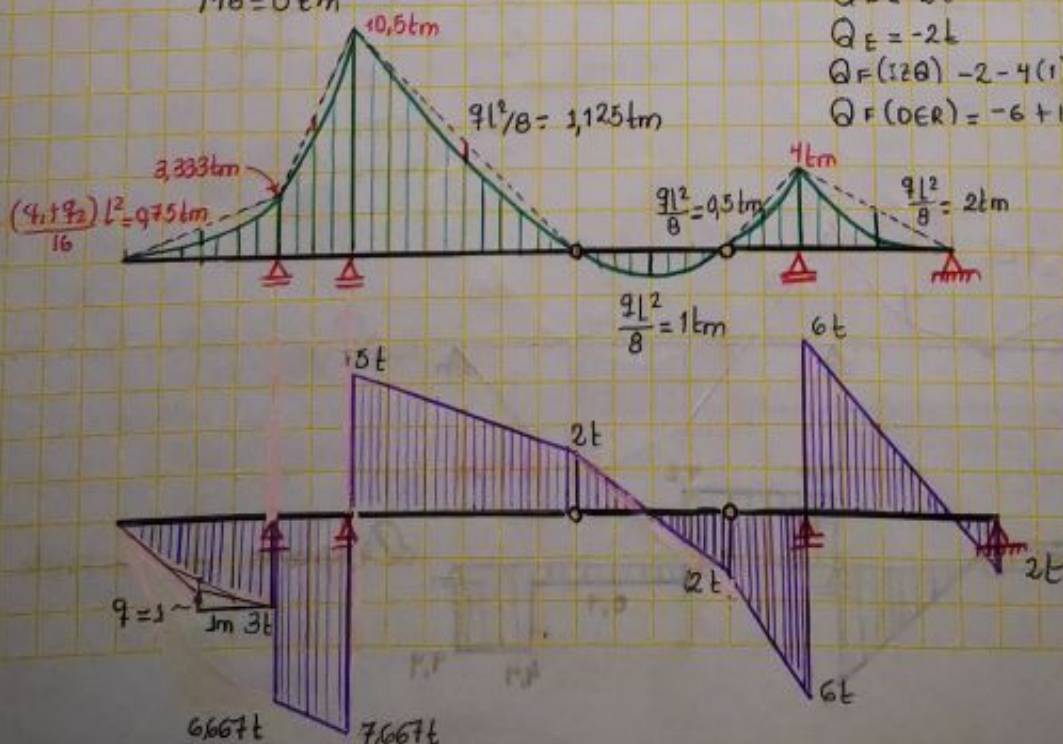
$$Q_C(der) = 2 + (1)(3) = 5t$$

$$Q_D = 2t$$

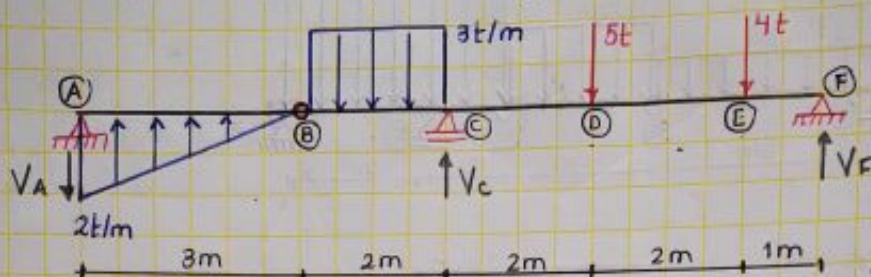
$$Q_E = -2t$$

$$Q_F(izq) = -2 - 4(1) = -6t$$

$$Q_F(der) = -6 + 12 = 6t ; Q_G = -2t$$



Obtener los esfuerzos internos para la siguiente viga



1.- Reacciones

$$\sum M_B = 0 \quad (\curvearrowright \oplus) \text{ (a la izquierda)}$$

$$\frac{2(3)(2)}{2} - V_A(3) = 0$$

$$V_A = 2t //$$

$$\sum M_F = 0 \quad (\curvearrowright \oplus)$$

$$-2(10) + \frac{2(3)(9)}{2} - 3(2)(6) + V_C(5) - 5(3) - 4(1) = 0$$

$$V_C = 9,6t //$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$-2 + \frac{2(3)}{2} - 3(2) + 9,6 - 5 - 4 + V_F = 0$$

$$V_F = 4,4t //$$

2.- Momentos

Tramo AB

$$M_A = 0$$

$$M_B = -2(3) + \frac{2(3)(2)}{2} = 0$$

Tramo BC

$$M_B = 0$$

$$M_C = -5(2) - 4(4) + 4,4(5) = -4tm$$

Tramo CD

$$M_C = -4tm$$

$$M_D = -4(2) + 4,4(3) = 5,2tm$$

Tramo DE

$$M_D = 5,2tm$$

$$M_E = 4,4(1) = 4,4tm$$

Tramo EF

$$M_E = 4,4tm$$

$$M_F = 0tm$$

4.- Diagramas

3.- Cortante

Tramo AB

$$Q_A = -2t$$

$$Q_B = -2 + \frac{2(3)}{2} = 1t$$

Tramo BC

$$Q_B = 1t$$

$$Q_C = -2 + \frac{2(3)}{2} - 3(2) = -5t$$

Tramo CD

$$Q_C = 5 + 4 - 4,4 = 4,6t$$

$$Q_D = 5 + 4 - 4,4 = 4,6t$$

Tramo DE

$$Q_D = 4 - 4,4 = -0,4t$$

$$Q_E = 4 - 4,4 = -0,4t$$

Tramo EF

$$Q_E = -4,4t$$

$$Q_F = -4,4t$$

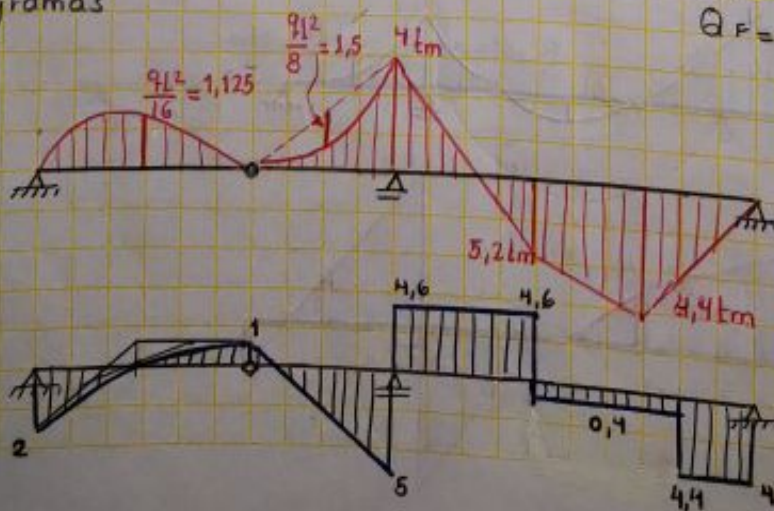
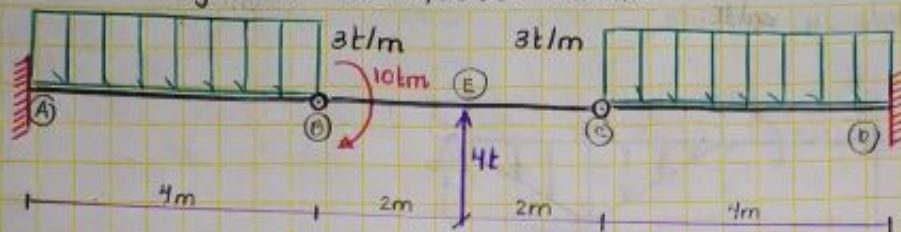


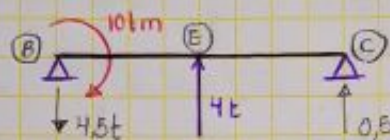
Diagrama de Momento

Diagrama de Cortante.

Obtener los diagramas de esfuerzos internos

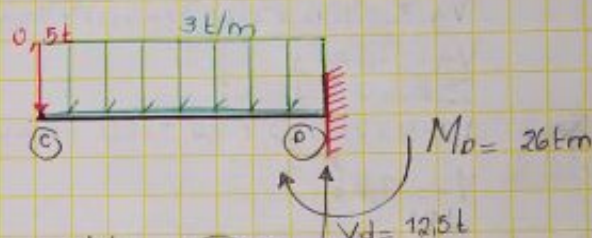
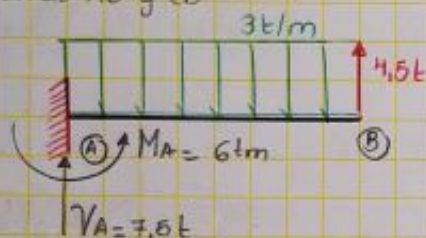


1.- Cálculo de Reacciones Barra BC



$$\begin{aligned}\sum M_B &= 0 \quad (\curvearrowright +) \\ 10 - 4(2) - V_C(4) &= 0 \quad \therefore V_C = 0.5t \\ \sum F_v &= 0 \quad \uparrow + \\ -V_B + 4 + 0.5 &= 0 \quad \therefore V_B = 4.5t\end{aligned}$$

Barra AB y CD



$$\begin{aligned}\sum M_A &= 0 \quad (\curvearrowright +) \\ -M_A - 4.5(4) + 3(4)(2) &= 0 \\ M_A &= 6tm \\ \sum F_v &= 0 \quad \uparrow + \\ V_A - 3(4) + 4.5 &= 0 \\ V_A &= 7.5t\end{aligned}$$

$$\begin{aligned}\sum M_D &= 0 \quad (\curvearrowright +) \\ M_D - 3(4)(2) - 0.5(4) &= 0 \\ M_D &= 26tm \\ \sum F_v &= 0 \quad \uparrow + \\ -0.5 - 3(4) + V_D &= 0 \\ V_D &= 12.5t\end{aligned}$$

4.- Diagramas

2.- Momentos Barra AB

$$\begin{aligned}M_A &= -6tm \\ M_B &= 0tm\end{aligned}$$

Barra BC

$$\begin{aligned}M_B &= 10tm \\ M_C &= 0tm \\ M_E &= -4.5(2) + 10 = 1tm\end{aligned}$$

Barra CD

$$\begin{aligned}M_C &= 0tm \\ M_D &= -26tm\end{aligned}$$

3.- Cortantes

Barra AB

$$\begin{aligned}Q_A &= 7.5t \\ Q_B &= -4.5t\end{aligned}$$

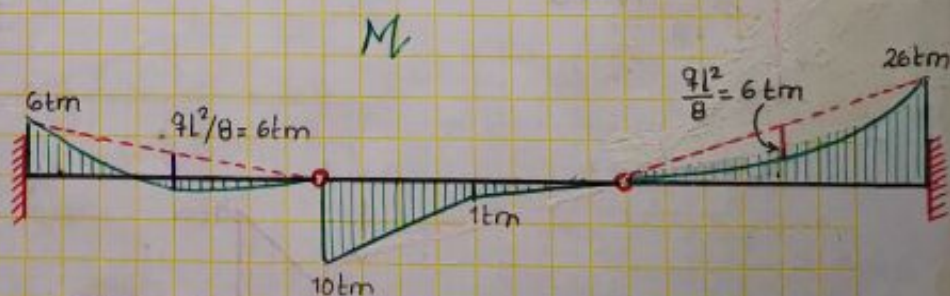
Barra BC

$$\begin{aligned}Q_B &= -4.5t \\ Q_E &= -4.5 + 4 = -0.5t \\ Q_C &= -0.5t\end{aligned}$$

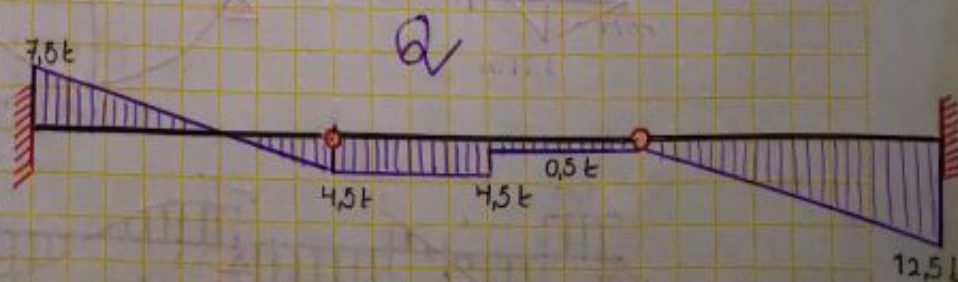
Barra CD

$$\begin{aligned}Q_C &= -0.5t \\ Q_D &= -12.5t\end{aligned}$$

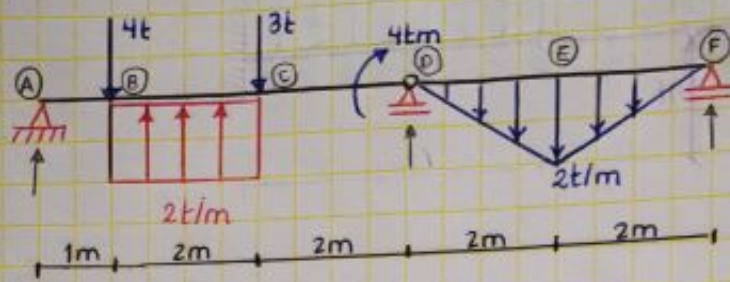
Momento: Escala = 10tm/cm



Cortante: Escala = 10t/cm



Diagramar Momento y cortante



1. Reacciones

$$\sum M_D = 0 \quad (\curvearrowright \oplus) \quad (\text{a la derecha})$$

$$\frac{2(4) \cdot 2}{2} - V_F(4) = 0$$

$$V_F = 2t$$

$$\sum M_D = 0 \quad (\curvearrowright \oplus) \quad (\text{a la izquierda})$$

$$V_A(5) - 4(4) + 2(2)(3) - 3(2) + 4 = 0$$

$$V_A = 1,2t$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$1,2 - 4 + 2(2) - 3 + V_D - \frac{2(4)}{2} + 2 = 0$$

$$V_D = 3,8t$$

2. Momentos

Tramo AB $M_A = 0$

$$M_B = 1,2(1) = 1,2tm$$

Tramo BC $M_B = 1,2tm$

$$M_C = 1,2(3) + 2(2)(1) - 4(2) = -0,4tm$$

Tramo CD $M_C = -0,4tm$

$$M_D = 4tm$$

Tramo DE $M_D = 0tm$

$$M_E = 2(2) \cdot \frac{2(2) \cdot \frac{1}{3}(2)}{2} = 2,667tm$$

Tramo EF $M_E = 2,667tm$

$$M_F = 0$$

3. Cortante

Tramo AB $Q_A = 1,2t$

$$Q_B = 1,2t$$

Tramo BC

$$Q_B = 1,2 - 4 = -2,8t$$

$$Q_C = 1,2 - 4 + 2(2) = 1,2t$$

Tramo CD

$$Q_C = 1,2 - 4 + 2(2) - 3 = -1,8t$$

$$Q_D = -1,8t$$

Tramo DE

$$Q_D = -1,8 + 3,8 = 2t$$

$$Q_E = -2 + 2(2) = 0$$

Tramo EF

$$Q_E = 0$$

$$Q_F = 0$$

4. Diagramas

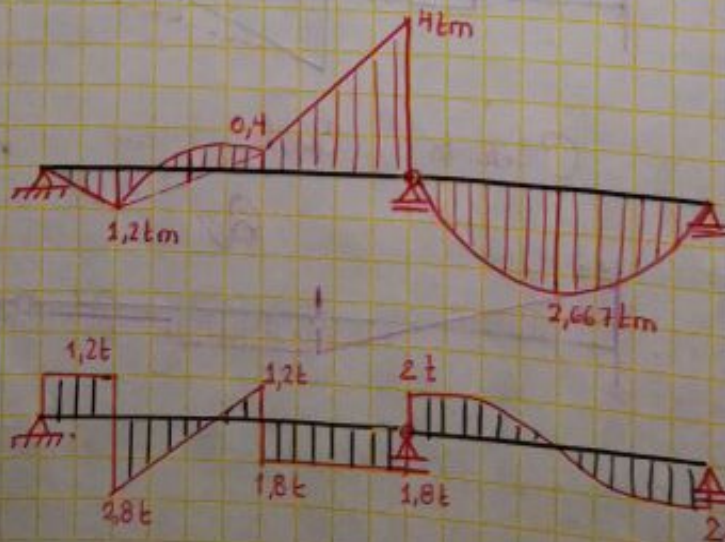
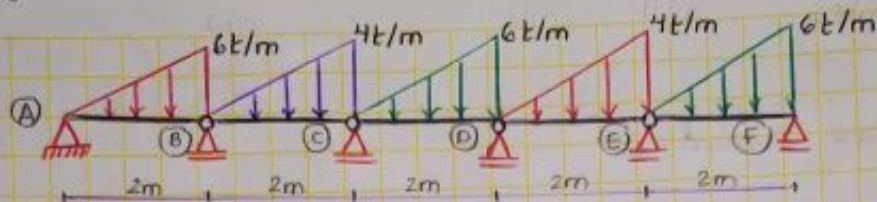


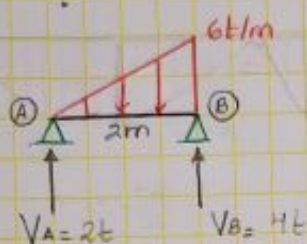
Diagrama de Momento

Diagrama de Cortante

Diagramar Momento y Cortante para la siguiente viga



1- Reacciones

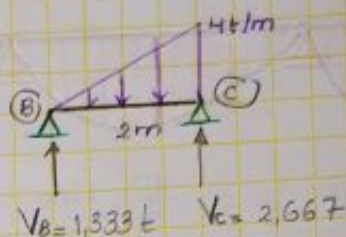


$$\sum M_A = 0 \quad \curvearrowright \oplus$$

$$\frac{6(2)}{2} \left[\frac{2}{3} \cdot 2 \right] - V_B(2) = 0 \quad \therefore V_B = 4t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_A - \frac{6(2)}{2} + 4 = 0 \quad \therefore V_A = 2t //$$



$$\sum M_B = 0 \quad \curvearrowright \oplus$$

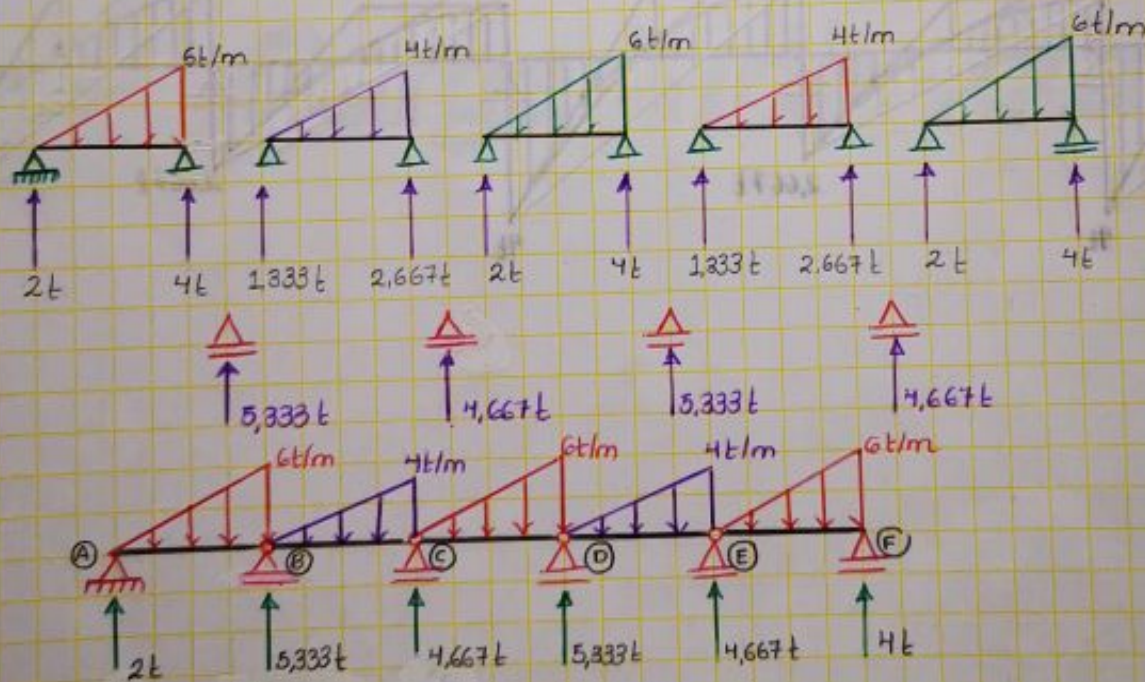
$$\frac{4(2)}{2} \left[\frac{2}{3} \cdot 2 \right] - V_C(2) = 0$$

$$V_C = 2,667t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_B - \frac{4(2)}{2} + 2,667 = 0$$

$$V_B = 1,333t //$$



2- Cortantes

Barra AB

$$Q_A = 2t$$

$$Q_B = -4t$$

Barra BC

$$Q_B = 1,333t$$

$$Q_C = -2,667t$$

Barra CD

$$Q_C = 2t$$

$$Q_D = 4t$$

Barra DE

$$Q_D = 1,333t$$

$$Q_E = -2,667t$$

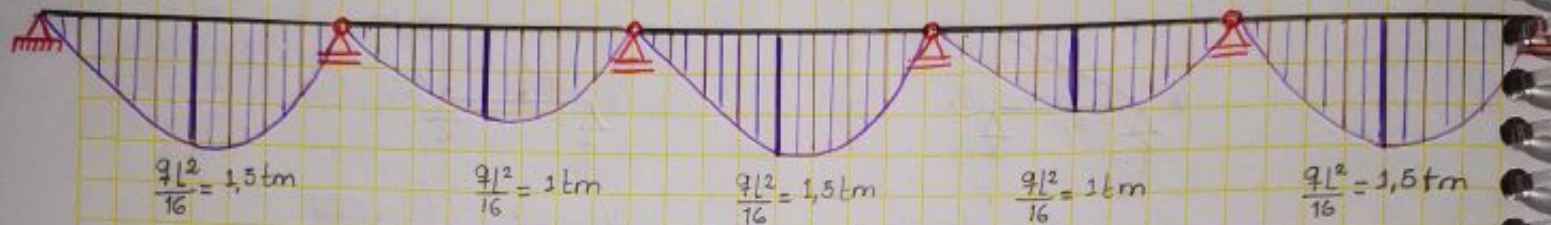
Barra EF

$$Q_E = 2t$$

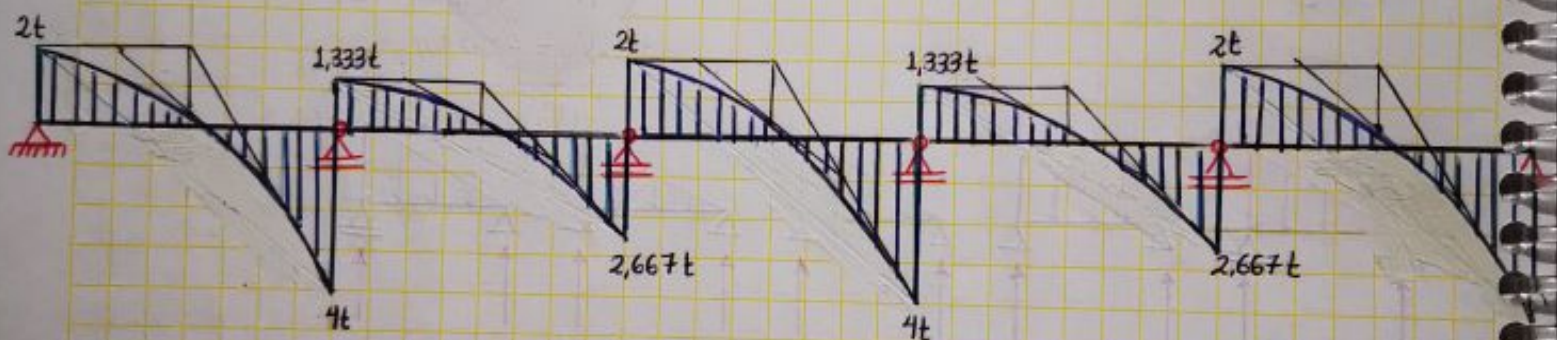
$$Q_F = -4t$$

3-Diagramas

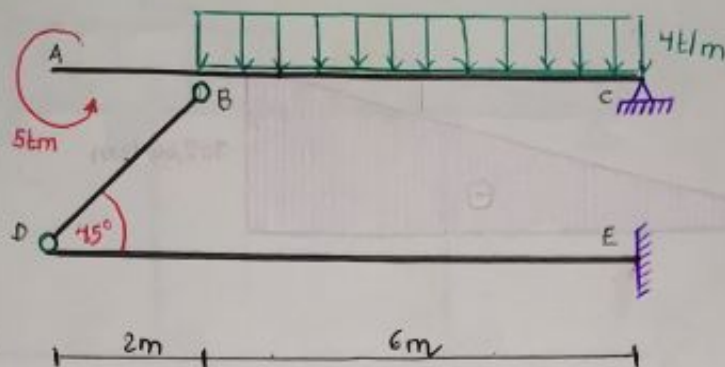
Momento Escala = 1 t·m/cm



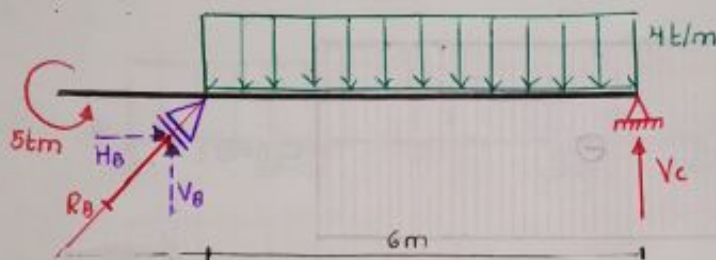
Cortante
Escala = 2 t/cm



Diagramar los esfuerzos característicos para la viga D-E



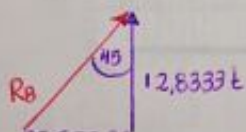
1.- Barra AC



$$\sum M_C = 0 \quad \curvearrowright \oplus$$

$$-5 + V_B(6) - 4(6)(3) = 0$$

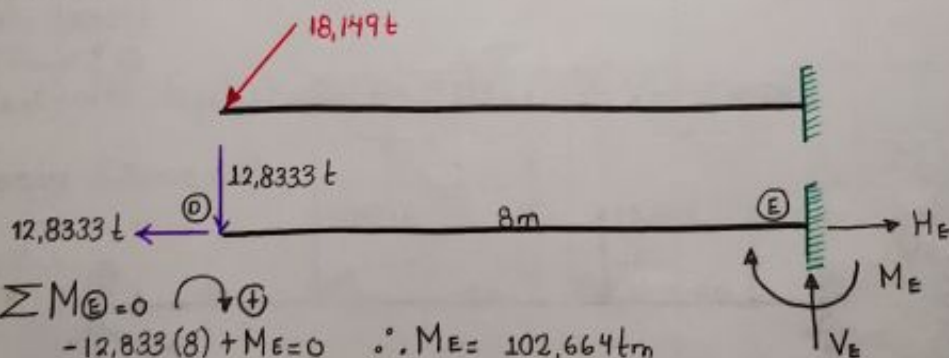
$$V_B = 12,8333 \text{ t}$$



$$\cos 45 = \frac{12,8333}{R_B}$$

$$R_B = \frac{12,8333}{\cos 45} = 18,149 \text{ t}$$

2.- Reacciones barra DE



$$\sum M_E = 0 \quad \curvearrowright \oplus$$

$$-12,833(8) + M_E = 0 \quad \therefore M_E = 102,664 \text{ tm}$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$V_E = 12,8333 \text{ t}$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$H_E = 12,8333 \text{ t}$$

3.- Ecuaciones

- Momento $M_x = -12,8333 \cdot x$

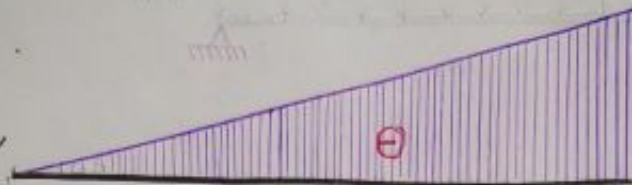
- Cortante $Q_x = -12,8333 \text{ t}$

- Normal $N_x = 12,8333 \text{ t}$

4.- Diagramas

M

$$E_{sc} = 50 \text{ tm/cm}$$



$$102,664 \text{ tm}$$

Q-

$$E_{sc} = 5 \text{ t/cm}$$



$$12,833 \text{ t}$$

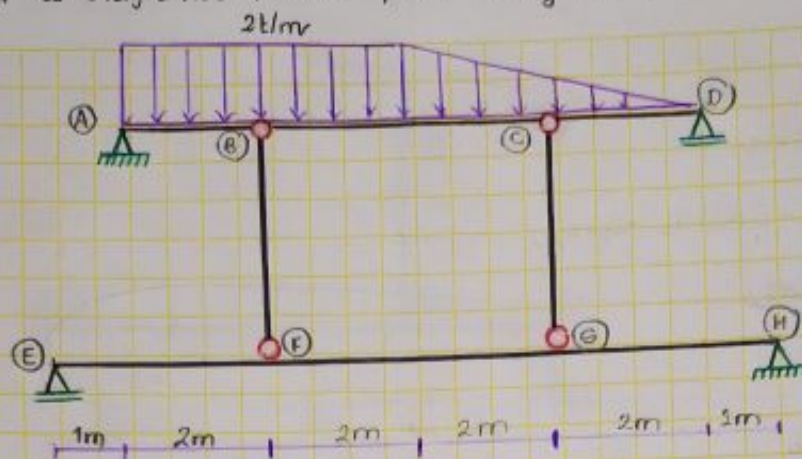
N

$$E_{sc} = 5 \text{ t/cm}$$



$$12,833 \text{ t}$$

Obtener los diagramas internos para la viga E-H



1 Barra AD



* Barra AB

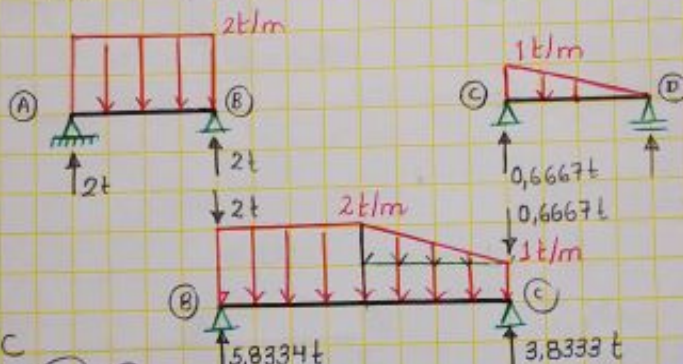
$$V_A = V_B = \frac{2(2)}{2} = 2t //$$

* Barra CD

$$\sum M_D = 0 \quad \curvearrowright \oplus$$

$$- \frac{1(2)}{2} \left(\frac{2 \cdot 2}{3} \right) + V_C(2) = 0$$

$$V_C = 0,6667 t //$$



* Barra BC

$$\sum M_B = 0 \quad \curvearrowright \oplus$$

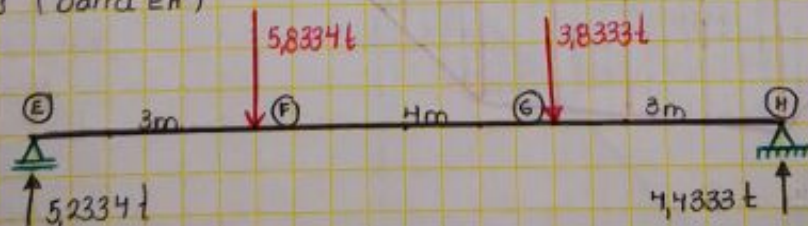
$$2(2)(1) + 1(2)(3) + 0,6667(4) + \frac{1(2)}{2} \cdot \left[2 + \frac{1}{3}(2) \right] - V_C(4) = 0$$

$$V_C = 3,8333 t$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$V_B - 2 - 2(2) - \frac{(2+1)(2)}{2} - 0,6667 + 3,8333 = 0 \quad \therefore V_B = 5,8334 t$$

2.- Reacciones (Barra EH)



$$\sum M_E = 0 \quad \curvearrowright \oplus$$

$$5,8334(3) + 3,8333(7) - V_H(10) = 0$$

$$V_H = 4,4333 t //$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$V_E - 5,8334 - 3,8333 + 4,4333 = 0$$

$$V_E = 5,2334 t //$$

3.- Ecuaciones de Momento

* Tramo EF

$$M_x = 5,2334 \cdot x //$$

* Tramo FG

$$M_x = 5,2334x - 5,8334(x-3) = -0,6x + 17,5 //$$

* Tramo GH

$$M_x = 4,4333(10-x) = 44,333 - 4,4333x //$$

4.- Cortantes

* Tramo EF

$$Q = 5,2334 \text{ t //}$$

* Tramo FG

$$Q = 5,2334 - 5,8334 = -0,6 \text{ t //}$$

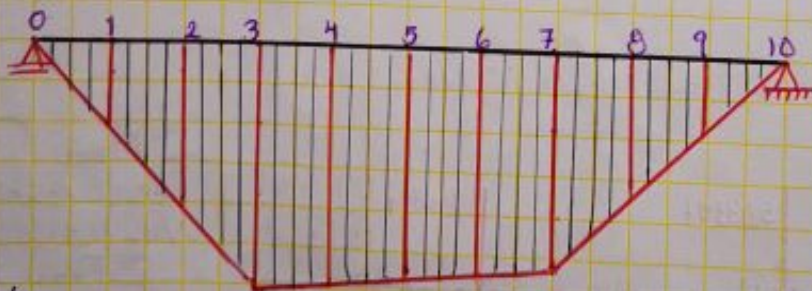
* Tramo GH

$$Q = -4,4333 \text{ t //}$$

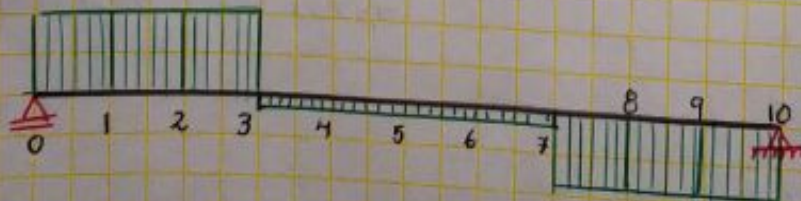
5.- Diagramas

x	Momento	Cortante
0	0	5,2334
1	5,2334	5,2334
2	10,4668	5,2334
3	15,7002	5,2334 / -0,6
4	15,1	-0,6
5	14,5	-0,6
6	13,9	-0,6
7	13,3	-0,6 / -4,4333
8	8,866	-4,4333
9	4,4333	-4,4333
10	0	-4,4333

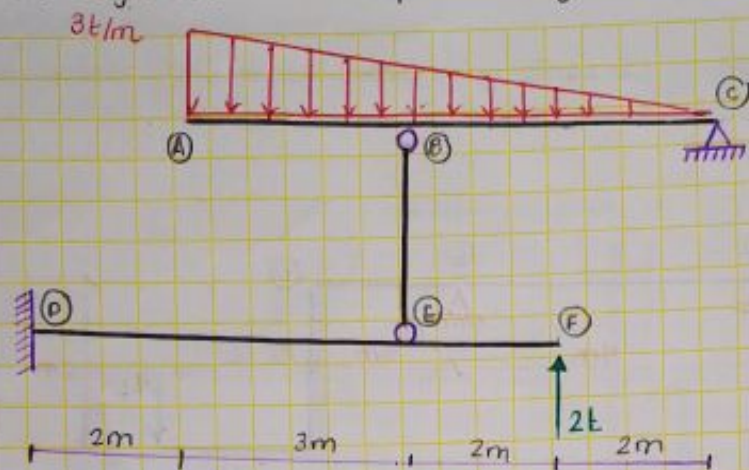
Momento



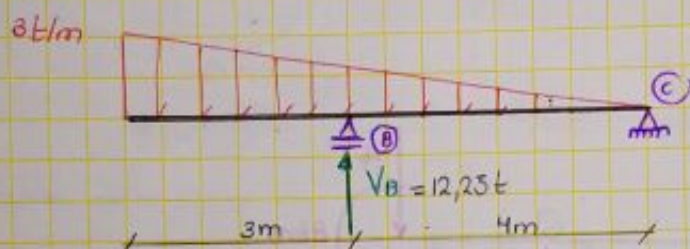
Cortante



Obtener los diagramas internos para la viga D-F



1.- Barra AC

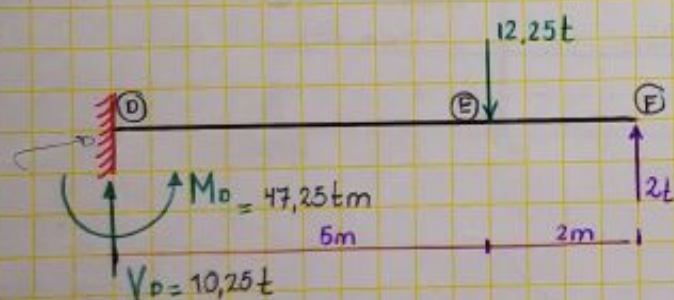


$$\sum M_O = 0 \quad (\curvearrowright \oplus)$$

$$V_B(4) - \frac{3(7)}{2} \left[\frac{2}{3}(7) \right] = 0$$

$$V_B = 12,25t$$

2.- Barra DF



$$\sum M_O = 0 \quad (\curvearrowright \oplus)$$

$$-M_D + 12,25(5) - 2(7) = 0$$

$$M_D = 47,25tm$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_D - 12,25 + 2 = 0$$

$$V_D = 10,25t$$

3.- Momentos

$$M_D = -47,25tm$$

$$M_E = 2(2) = 4tm$$

$$M_F = 0tm$$

4.- Cortantes

$$Q_D = 10,25t$$

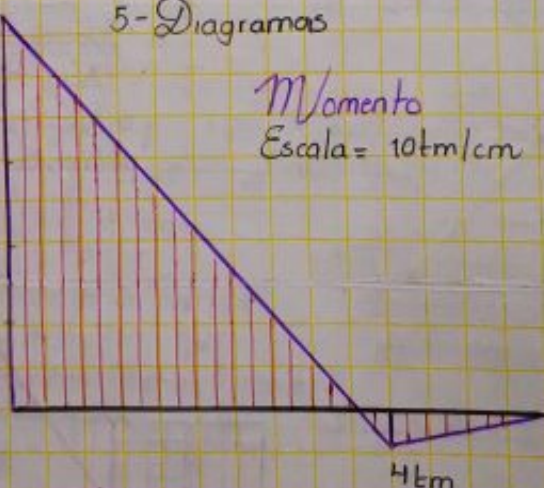
$$Q_E = 10,25t \text{ (Izq.)}$$

$$Q_E = 10,25 - 12,25$$

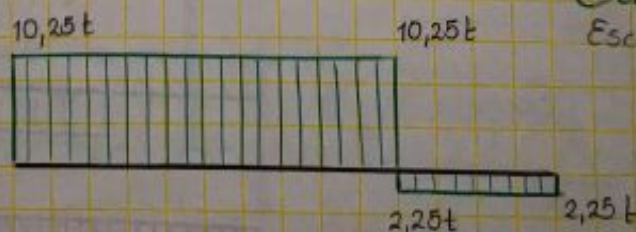
$$Q_E = -2,25t \text{ (Der.)}$$

5.- Diagramas

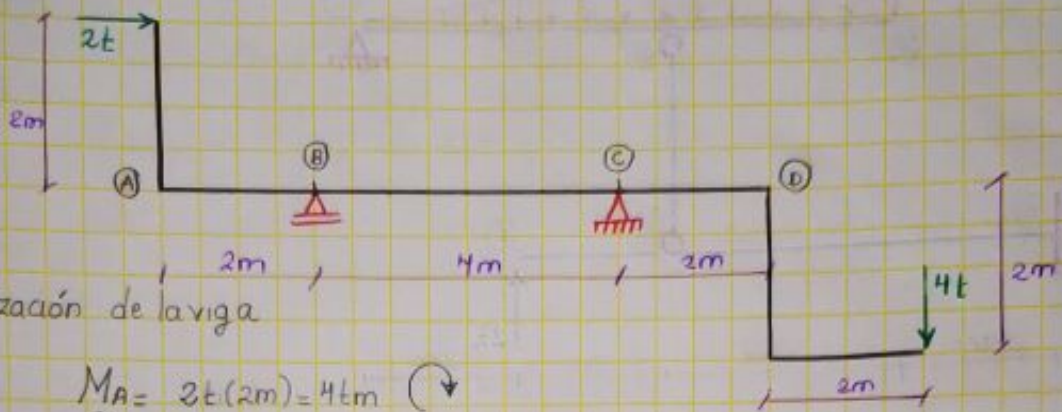
Momento
Escala = 10tm/cm



Cortante
Escala = 10t/cm



Obtener los diagramas de Momento, Cortante y Normal de la viga AD



1 - Idealización de la viga

$$M_A = 2t(2m) = 4tm \quad (\curvearrowright)$$

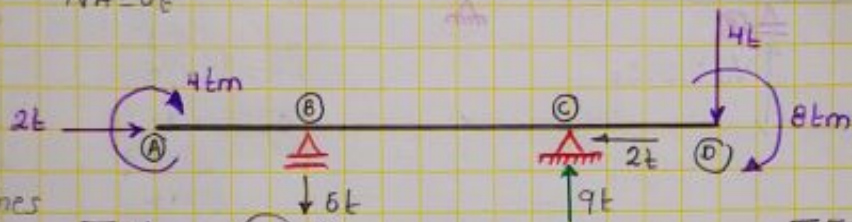
$$Q_A = 0t$$

$$N_A = -2t$$

$$M_D = -4t(2m) = -8tm$$

$$Q_D = 4t$$

$$N_D = 0t$$



2 Reacciones

$$\sum M_B = 0 \quad (\curvearrowright \oplus)$$

$$4 + 8 + 4(6) - V_C(4) = 0 \quad \therefore V_C = 9t //$$

$$\sum F_V = 0 \quad \uparrow \oplus$$

$$-V_B + 9 - 4 = 0 \quad \therefore V_B = 5t //$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$H_C = 2t //$$

3 - Momentos

$$M_A = 4tm$$

$$M_B = 4tm$$

$$M_C = -8tm - 4(2) = -16tm$$

$$M_D = -8tm$$

4 - Cortante

$$Q_A = 0$$

$$Q_B = -5t ; Q_C = -5 \text{ (Izg)}$$

$$Q_C = -5 + 9 = 4t \text{ (Der)}$$

$$Q_D = 4t$$

5 - Normal

$$N_A = -2t$$

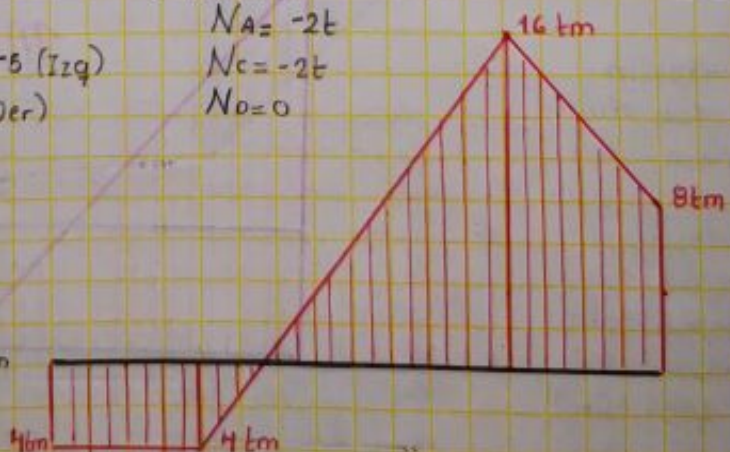
$$N_C = -2t$$

$$N_D = 0$$

6 - Diagramas

Momento

Escala = 4tm/cm



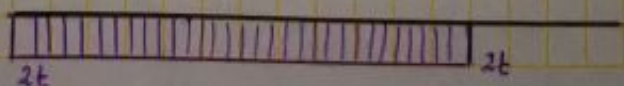
Cortante

Escala = 10t/cm

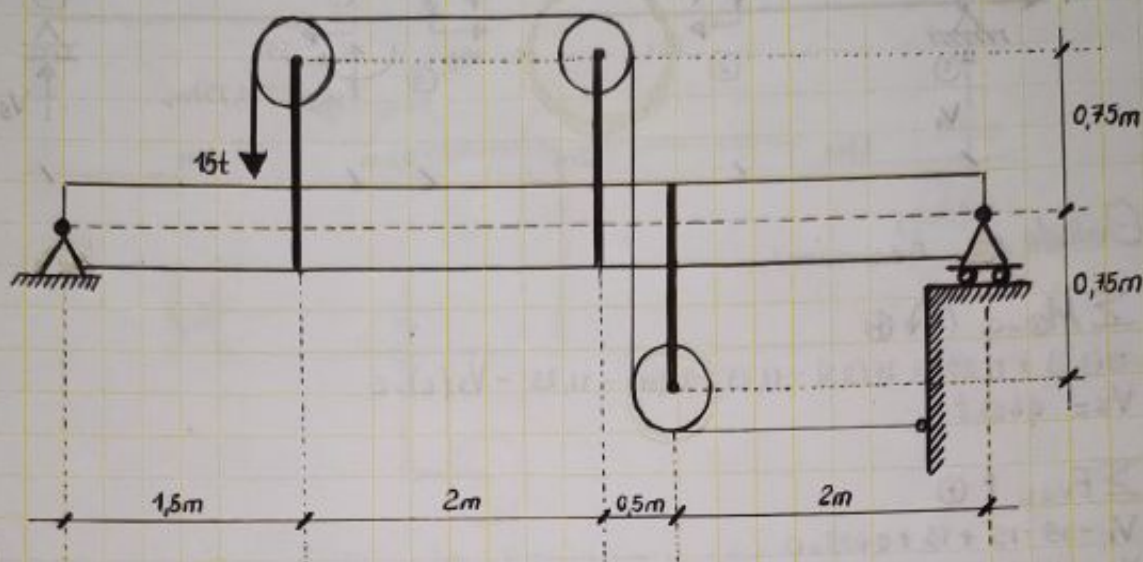


Normal

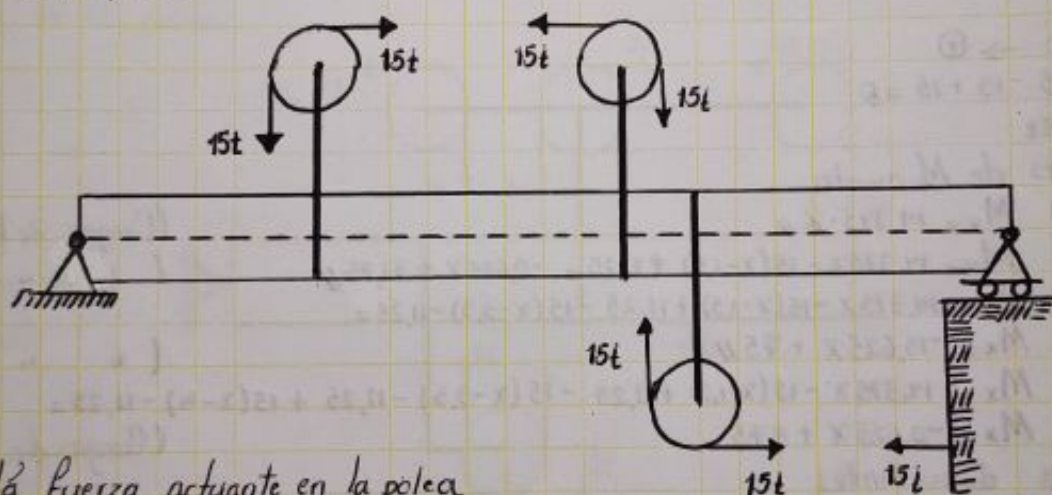
Escala = 4t/cm



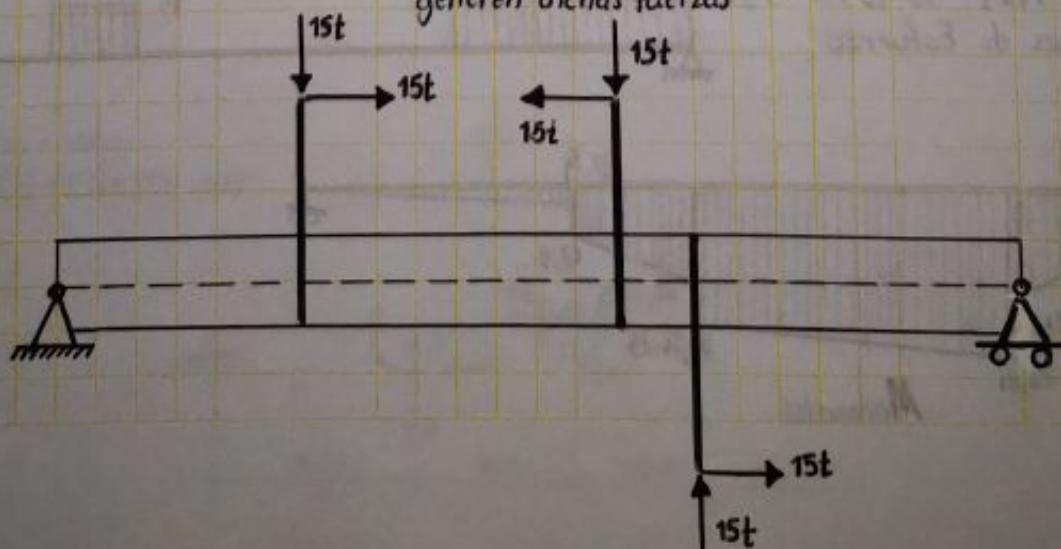
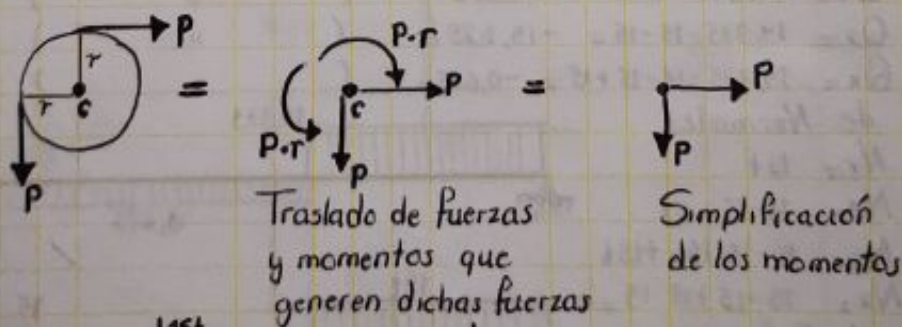
1.- Calcular reacciones y diagramar Momento



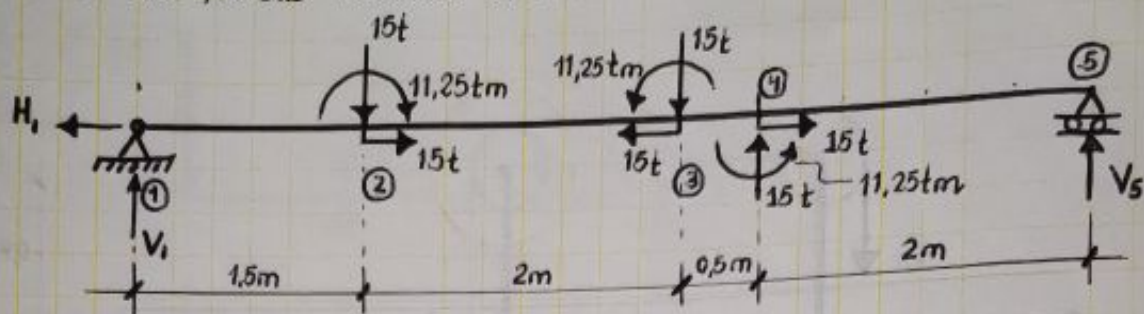
1.- Análisis en las poleas



Cada fuerza actuante en la polea deberá trasladarse al centro de la misma



Trasladamos las fuerzas ubicadas en el centro de las poleas al baricentro de la viga



2.- Cálculo de Reacciones

$$\sum M_0 = 0 \quad \curvearrowright \oplus$$

$$15(1.5) + 11.25 + 15(3.5) - 11.25 - 15(4) - 11.25 - V_5(6) = 0$$

$$V_5 = 0.625 \text{ t}$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_1 - 15 - 15 + 15 + 0.625 = 0$$

$$V_1 = 14.375 \text{ t}$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$-H_1 + 15 - 15 + 15 = 0$$

$$H_1 = 15 \text{ t}$$

3.- Ecuaciones de Momento

Tramo 1-2 $M_x = 14.375 - x //$

Tramo 2-3 $M_x = 14.375x - 15(x-1.5) + 11.25 = -0.625x + 33.75 //$

Tramo 3-4 $M_x = 14.375x - 15(x-1.5) + 11.25 - 15(x-3.5) - 11.25 =$

$$M_x = -15.625x + 75 //$$

Tramo 4-5 $M_x = 14.375x - 15(x-1.5) + 11.25 - 15(x-3.5) - 11.25 + 15(x-4) - 11.25 =$

$$M_x = -0.625x + 3.75$$

(Cargas de la izquierda)

(" " " ")

(" " " ")

(Cargas de la izquierda)

4.- Ecuaciones de cortantes

Tramo 1-2 $Q_x = 14.375 \text{ t}$

Tramo 2-3 $Q_x = 14.375 - 15 = -0.625 \text{ t}$

Tramo 3-4 $Q_x = 14.375 - 15 - 15 = -15.625 \text{ t}$

Tramo 4-5 $Q_x = 14.375 - 15 - 15 + 15 = -0.625 \text{ t}$

(cargas de la izquierda)

(" ")

(" ")

(" ")

5.- Ecuaciones de Normales

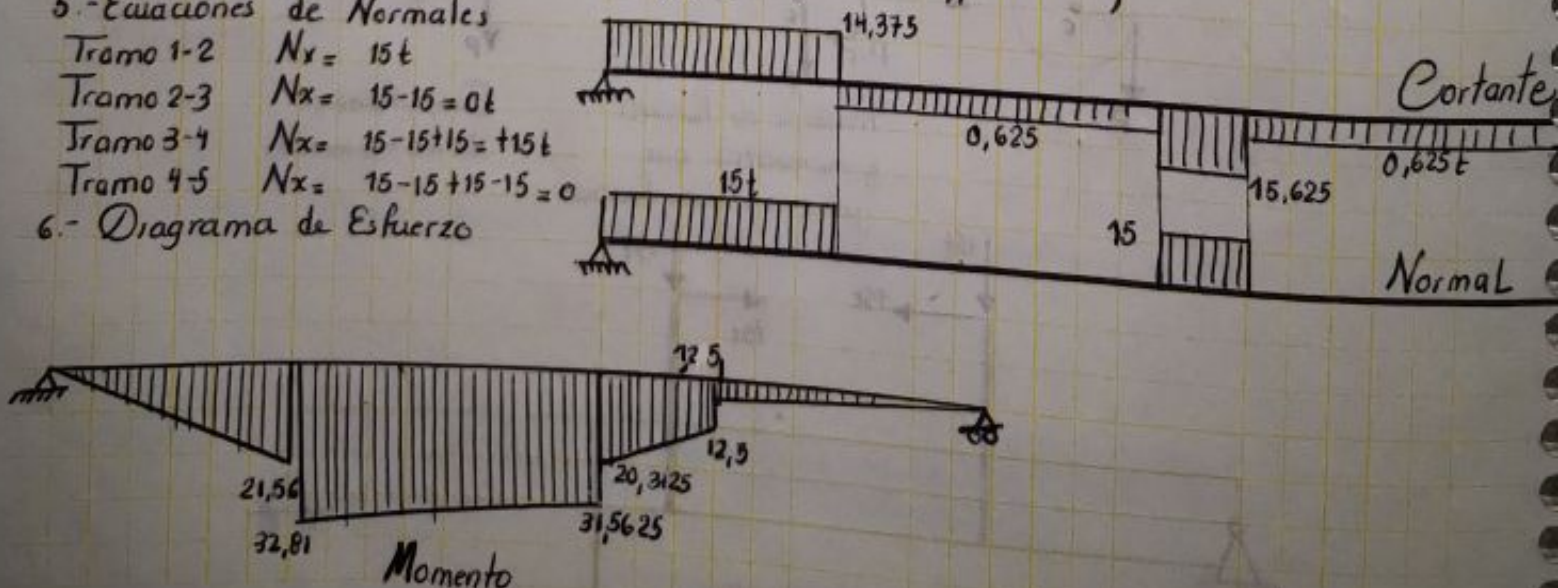
Tramo 1-2 $N_x = 15 \text{ t}$

Tramo 2-3 $N_x = 15 - 15 = 0 \text{ t}$

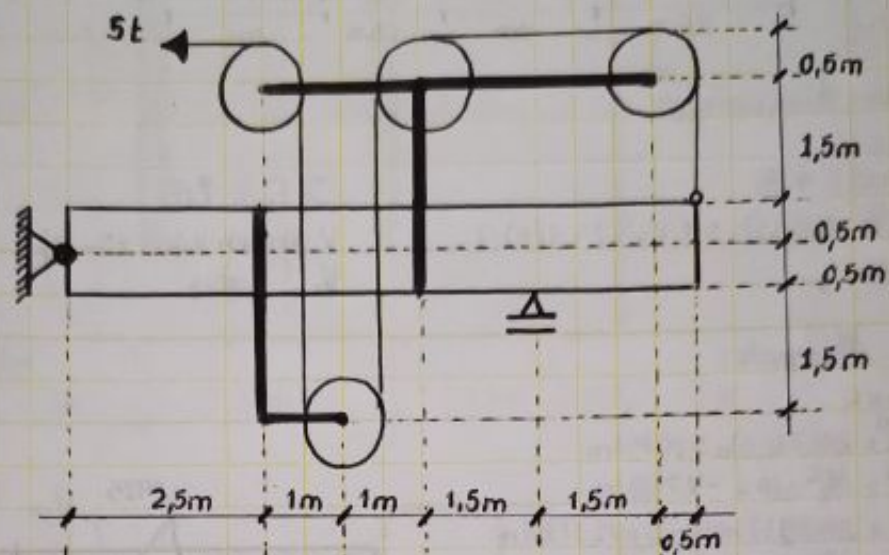
Tramo 3-4 $N_x = 15 - 15 + 15 = 15 \text{ t}$

Tramo 4-5 $N_x = 15 - 15 + 15 - 15 = 0$

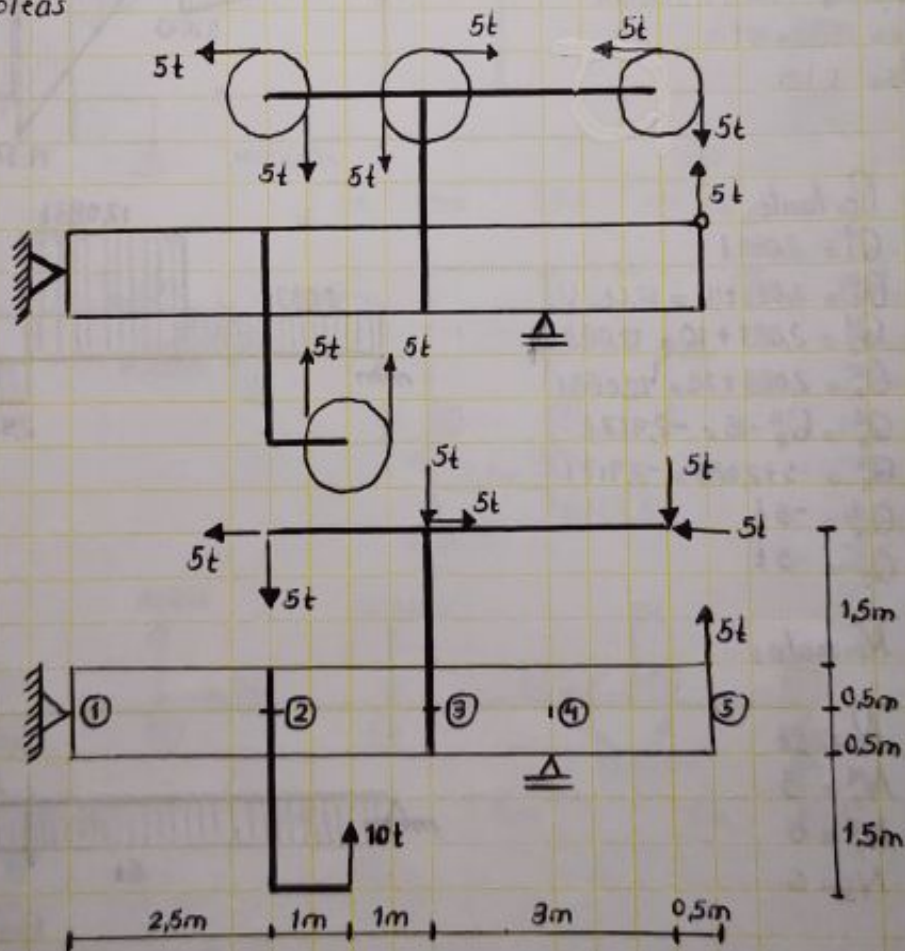
6.- Diagrama de Esfuerzo



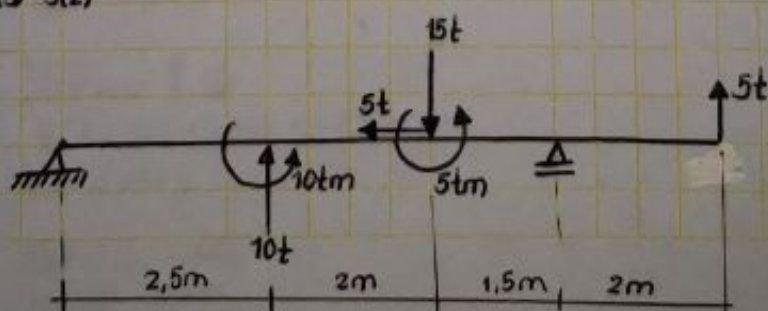
2. Calcular reacciones y diagramar momento, normal y cortante

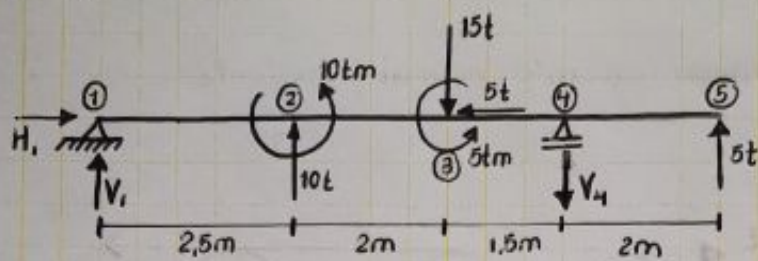


1. Análisis de las poleas



$$\begin{aligned} M_2 &= -10t \cdot 1m = -10tm \\ M_3 &= -5(2) - 5(2) + 5(2) + 5(3) - 5(2) \\ M_3 &= -5tm \end{aligned}$$





Cálculo de Reacciones

$$\sum M_1 = 0 \curvearrowright \oplus$$

$$-10 - 10(2.5) + 15(4.5) - 5 + V_4(6) - 5(8) = 0$$

$$V_4 = 2.083t$$

$$\sum F_V = 0 \uparrow \oplus$$

$$V_1 + 10 - 15 - 2.083 + 5 = 0$$

$$V_1 = 2.083t$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$H_1 - 5 = 0$$

$$H_1 = 5t //$$

Cálculo de Momentos

$$(CI) \quad M_1 = 0$$

$$(CI) \quad M_2^a = 2.083(2.5) = 5.2075tm$$

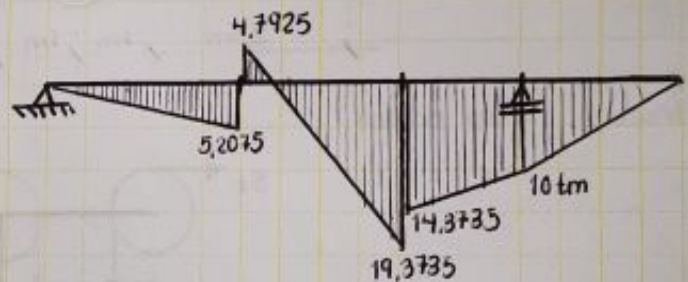
$$(CI) \quad M_2^d = M_2^a - 10 = -4.7925tm$$

$$(CI) \quad M_3^a = 2.083(4.5) + 10(2) - 10 = 19.3735tm$$

$$(CI) \quad M_3^d = M_3^a - 5 = 14.3735tm$$

$$(CD) \quad M_4 = 5(2) = 10tm$$

$$(CD) \quad M_5 = 0tm$$



Cálculo de Cortante

$$CI \quad Q_1^d = 2.083t$$

$$CI \quad Q_2^a = 2.083t$$

$$CI \quad Q_2^d = 2.083 + 10 = 12.083t$$

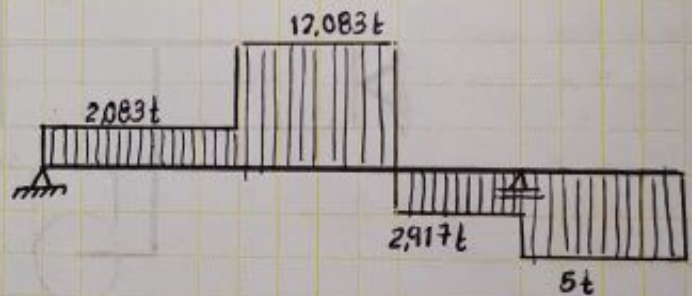
$$CI \quad Q_3^a = 2.083 + 10 = 12.083t$$

$$CI \quad Q_3^d = Q_3^a - 15 = -2.917t$$

$$CD \quad Q_4^a = -5 + 2.083 = -2.917t$$

$$CD \quad Q_4^d = -5t$$

$$CD \quad Q_5^a = -5t$$



Cálculo de Normales

$$N_1 = -5t$$

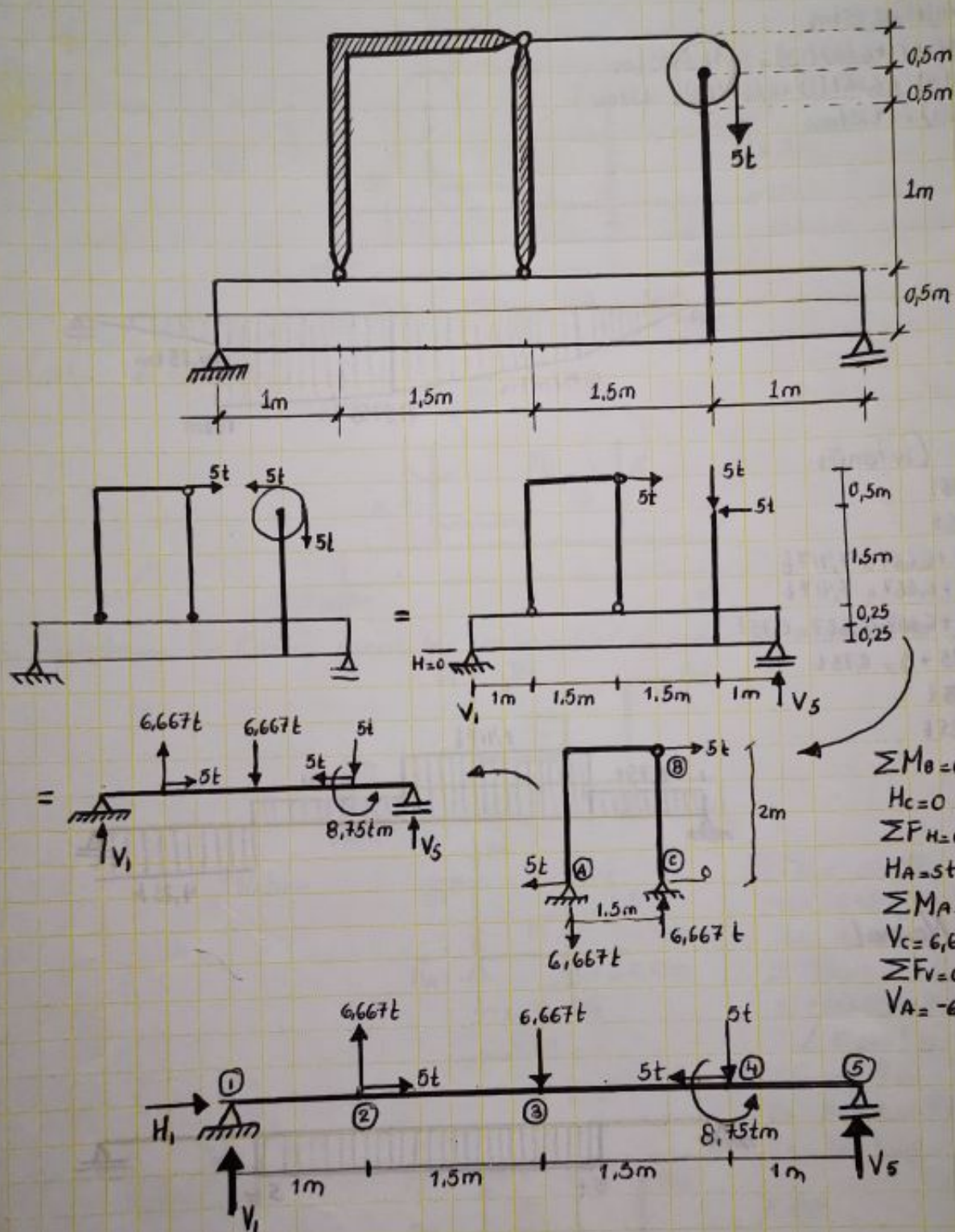
$$N_2^a = -5$$

$$N_3^a = 0$$

$$N_5 = 0$$



3 Analizar la siguiente estructura horizontal



$$\sum M_B = 0 \quad (\curvearrowright \oplus) \quad (B-C)$$

$$H_C = 0$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$H_A = 5t$$

$$\sum M_A = 0 \quad (\curvearrowright \oplus)$$

$$V_C = 6.667t$$

$$\sum F_V = 0 \uparrow \oplus$$

$$V_A = -6.667t$$

Calculo de Reacciones

$$\sum M_O = 0 \quad (\curvearrowright \oplus)$$

$$-6.667(1) + 6.667(2.5) + 5(4) - 8.75 - V_5(5) = 0$$

$$V_5 = 4.25t //$$

$$\sum F_V = 0 \uparrow \oplus$$

$$V_1 + 6.667 - 6.667 - 5 + 4.25 = 0$$

$$V_1 = 0.75t //$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$H_1 + 5 - 5 = 0$$

$$H_1 = 0 //$$

Cálculo de Momentos

$$CI \ M_1 = 0$$

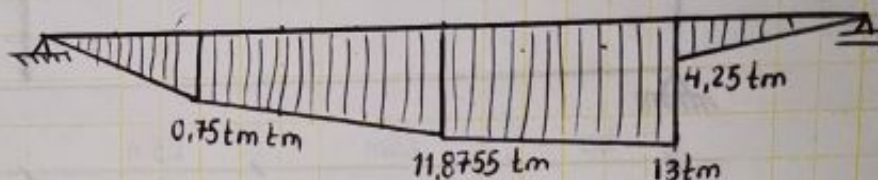
$$CI \ M_2 = 0,75(1) = 0,75 \text{ tm}$$

$$CI \ M_3 = 0,75(2,5) + 6,667(1,5) = 11,875 \text{ tm}$$

$$CI \ M_4^a = 0,75(4) + 6,667(3) - 6,667(1,5) = 13 \text{ tm}$$

$$CD \ M_4^d = 4,25(1) = 4,25 \text{ tm}$$

$$CD \ M_5 = (0)$$



Cálculo de Cortantes

$$CI \ Q_1^d = 0,75 \text{ t}$$

$$CI \ Q_2^a = 0,75 \text{ t}$$

$$CI \ Q_2^d = 0,75 + 6,667 = 7,417 \text{ t}$$

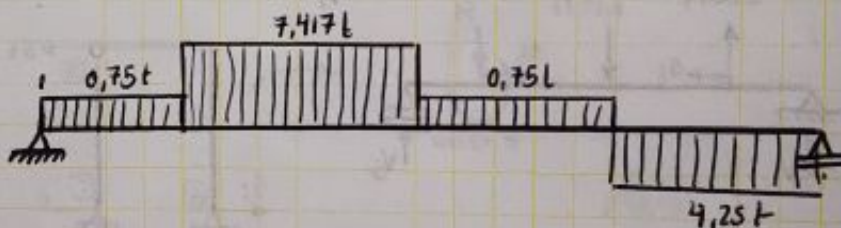
$$CI \ Q_3^a = 0,75 + 6,667 = 7,417 \text{ t}$$

$$CI \ Q_3^d = 0,75 + 6,667 - 6,667 = 0,75 \text{ t}$$

$$CD \ Q_4^a = -4,25 + 5 = 0,75 \text{ t}$$

$$CD \ Q_4^d = -4,25 \text{ t}$$

$$CD \ Q_5^a = -4,25 \text{ t}$$



Cálculo de Normal

$$N_1 = 0$$

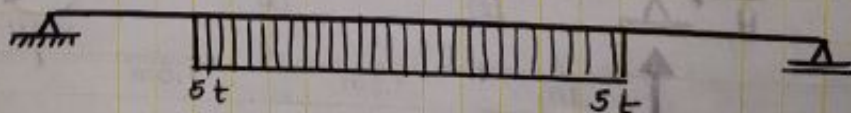
$$N_2^a = 0$$

$$N_2^d = -5$$

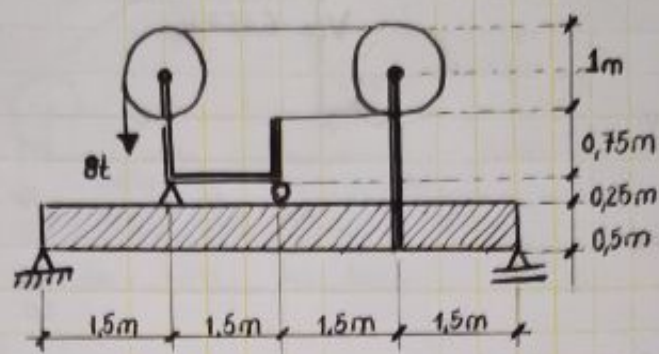
$$N_4^a = -5$$

$$N_4^d = 0$$

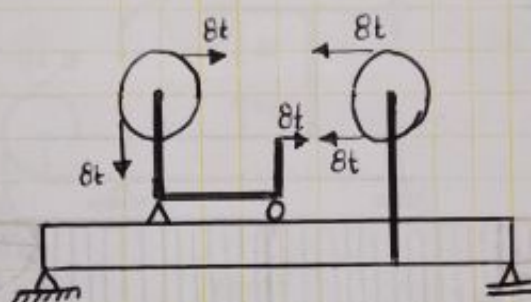
$$N_5 = 0$$



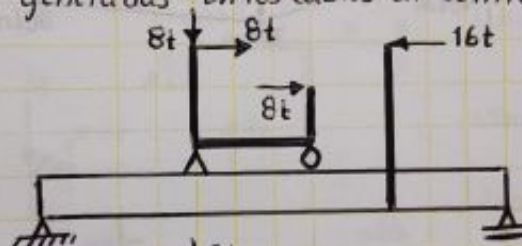
4. Analizar los diagramas de esfuerzos en la sgte viga



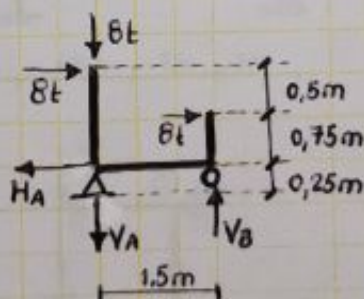
1.- Análisis de las poleas



Trasladamos las fuerzas generadas en los cables al centro de cada polea

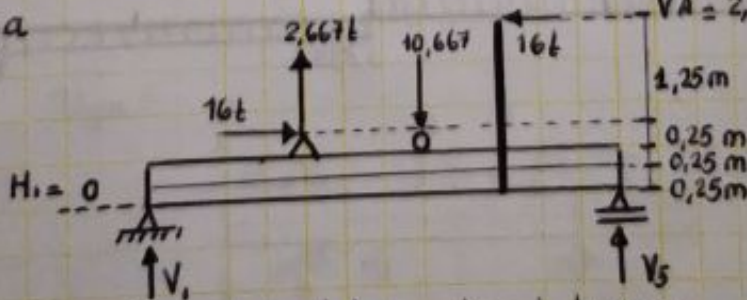


Pórtico

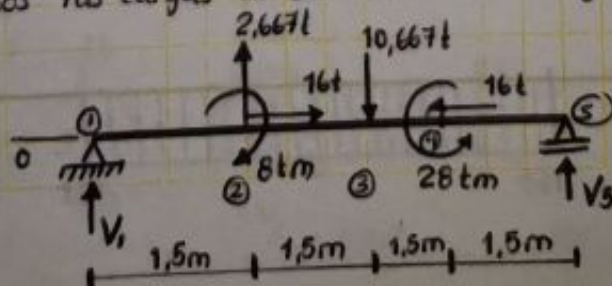


$$\begin{aligned} \sum F_H &= 0 \rightarrow \oplus \\ -H_A + 8 + 8 &= 0 \\ H_A &= 16t // \\ \sum M_A &= 0 \curvearrowright \oplus \\ 8(1,25) + 8(0,75) - V_B(1,5) &= 0 \\ V_B &= 10,667t // \\ \sum F_V &= 0 \uparrow \oplus \\ -V_A - 8 + 10,667 &= 0 \\ V_A &= 2,667t // \end{aligned}$$

Viga



Trasladamos las cargas al baricentro de la viga



Matrices de Rigidez para elementos curvos

Cálculo de Reacciones

$$\sum M_0 = 0 \quad \curvearrowright \oplus$$

$$-2,667(1,5) + 8 + 10,667(3) - 28 - V_5(6) = 0$$

$$V_5 = 1,333t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_1 + 2,667 - 10,667 + 1,333 = 0$$

$$V_1 = 6,667t //$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$H_1 + 16 - 16 = 0$$

$$H_1 = 0 //$$

Cálculo de Momento

$$CI \quad M_1 = 0$$

$$CI \quad M_{2a} = 6,667(1,5) = 10tm$$

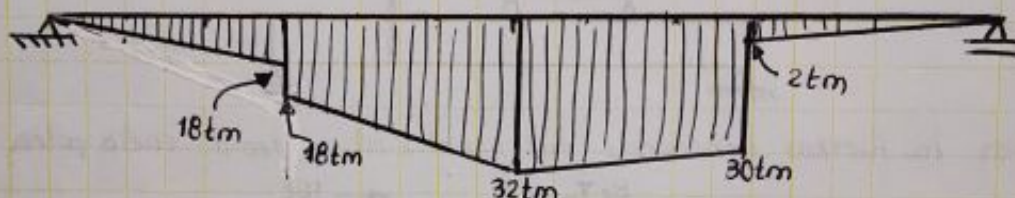
$$CI \quad M_{2d} = M_{2a} + 8 = 18tm$$

$$CI \quad M_3 = 6,667(3) + 2,667(1,5) + 8 = 32tm$$

$$CD \quad M_{4a} = 28 + 1,333(1,5) = 30tm$$

$$CD \quad M_{4d} = 1,333(1,5) = 2tm$$

$$CD \quad M_5 = 0$$



Cálculo de Portante

$$CI \quad Q_{1d} = 6,667t$$

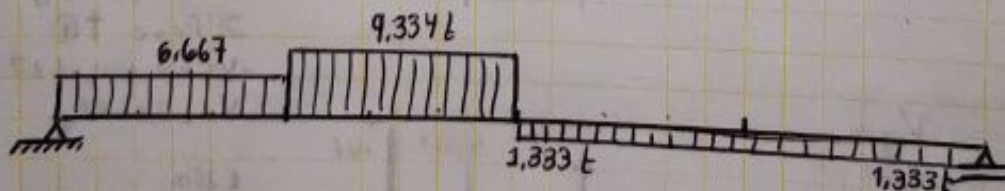
$$CI \quad Q_{2a} = 6,667t$$

$$CI \quad Q_{2d} = 6,667 + 2,667 = 9,334t$$

$$CI \quad Q_{3a} = 6,667 + 2,667 = 9,334t$$

$$CD \quad Q_{3d} = -1,333t$$

$$CD \quad Q_{5a} = -1,333t$$



Cálculo de Normales

$$CI \quad N_1 = 0$$

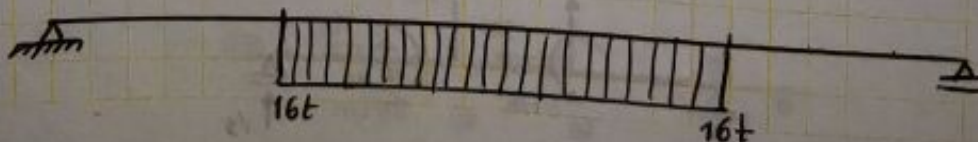
$$CI \quad N_{2a} = 0$$

$$CI \quad N_{2d} = -16t$$

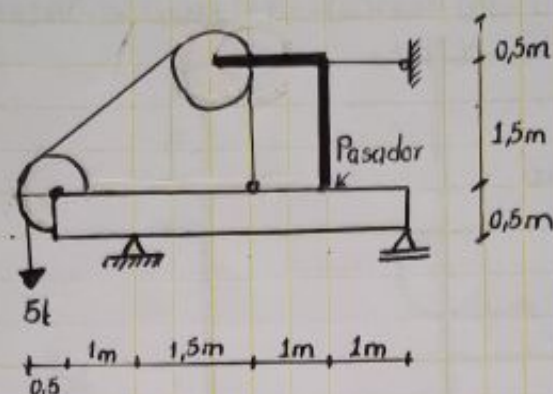
$$CI \quad N_{4a} = -16t$$

$$CD \quad N_{4d} = 0$$

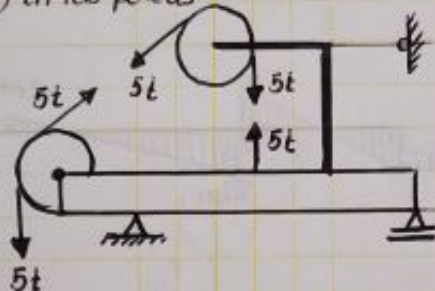
$$CD \quad N_5 = 0$$



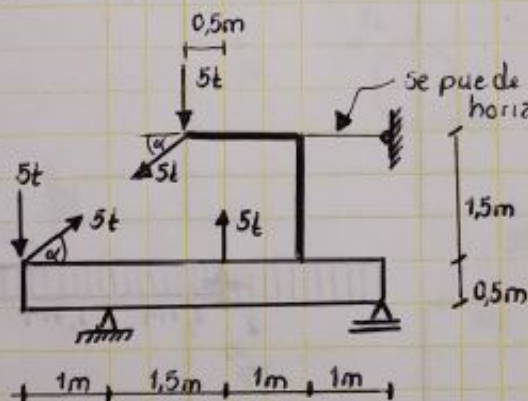
5.- Analizar la Viga determinando sus diagramas de esfuerzos internos



1.- Análisis (en el aparejo) en las poleas

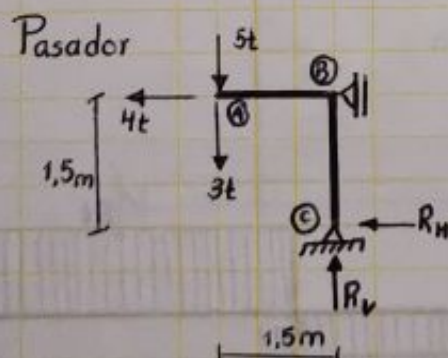


2.- Análisis del pasador



Se puede reemplazar por un apoyo móvil horizontal solo para calcular las reacciones

$$\alpha = \arctg\left(\frac{1.5}{2}\right) = 36,87^\circ$$



$$\sum F_v = 0 \uparrow \oplus$$

$$-5 - 3 + R_v = 0$$

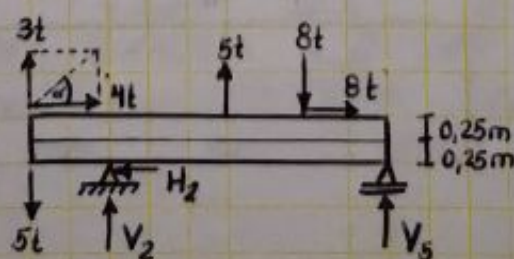
$$R_v = 8t //$$

$$\sum M_B = 0 \curvearrowright \oplus$$

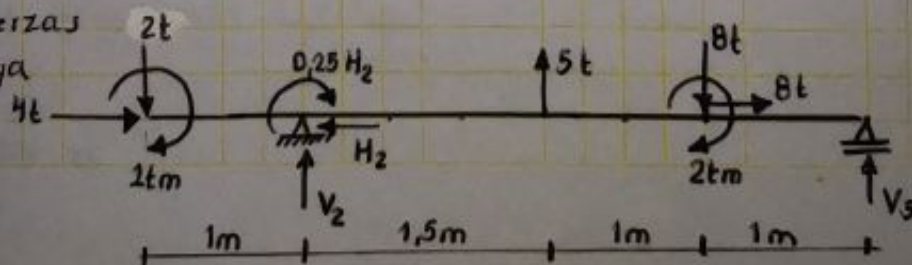
$$-(5+3)(1.5) + R_H(1.5) = 0$$

$$R_H = 8t //$$

Viga



Trasladamos las fuerzas al baricentro de la viga



Reacciones

$$\sum F_H = 0 \rightarrow \oplus$$

$$+4 - H_2 + 8 = 0$$

$$H_2 = 12t //$$

$$\sum M_2 = 0 \curvearrowright \oplus$$

$$1 - 2(1) + 0,25(12) - 5(1,5) + 8(2,5) + 2 - V_5(3,5) = 0$$

$$V_5 = 4,714t //$$

$$\sum F_V = 0 \uparrow \oplus$$

$$-2 + V_2 + 5 - 8 + 4,714 = 0$$

$$V_2 = 0,286t //$$

Cálculo de Momentos

$$CI \ M_{1d} = 1tm$$

$$CI \ M_{2a} = 1 - 2(1) = -1tm$$

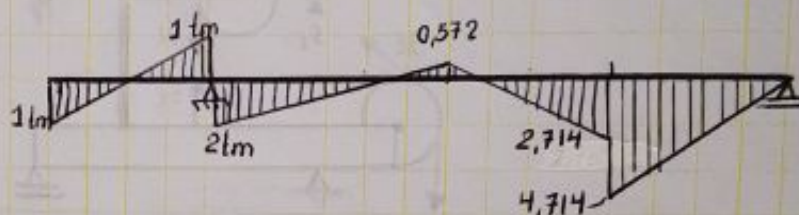
$$CI \ M_{2d} = M_{2a} + 0,25(12) = 2tm$$

$$CD \ M_3 = -2 - 8(1) + 4,714(2) = -0,572tm$$

$$CD \ M_{4a} = -2 + 4,714(1) = 2,714tm$$

$$CD \ M_{4d} = 4,714(1) = 4,714tm$$

$$CD \ M_5 = 0 //$$



Cálculo de Cortante

$$CI \ Q_{1d} = -2t$$

$$CI \ Q_{2a} = -2t$$

$$CI \ Q_{2d} = -2 + 0,286 = -1,714t //$$

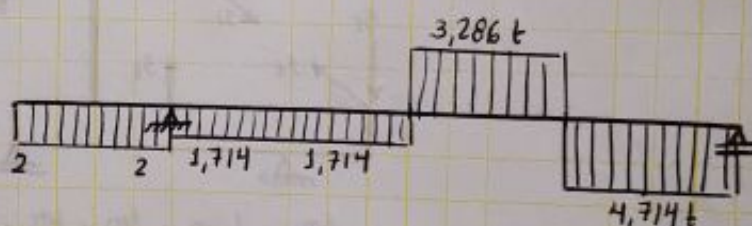
$$CI \ Q_{3a} = -2 + 0,286 = -1,714t$$

$$CI \ Q_{3d} = -2 + 0,286 + 5 = 3,286t$$

$$CD \ Q_{4a} = 8 - 4,714 = 3,286t$$

$$CD \ Q_{4d} = -4,714t //$$

$$CD \ Q_{5a} = -4,714t //$$



Cálculo de Normales

$$CI \ N_{1d} = -4t$$

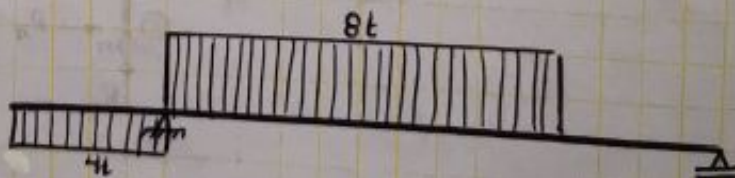
$$CI \ N_{2a} = -4t$$

$$CI \ N_{2d} = -4 + 12 = 8t$$

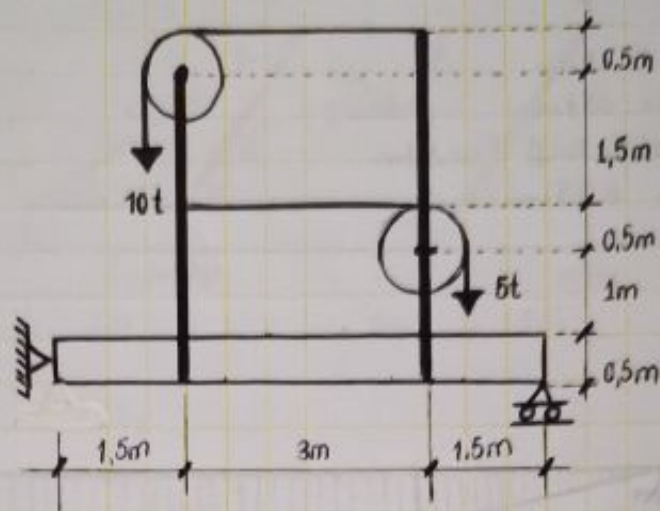
$$CD \ N_{4a} = 8t$$

$$CD \ N_{4d} = 0$$

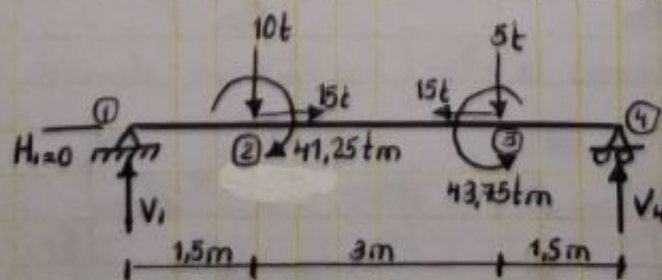
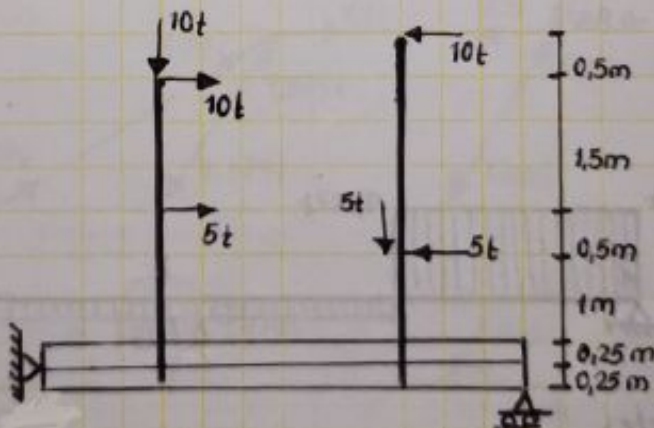
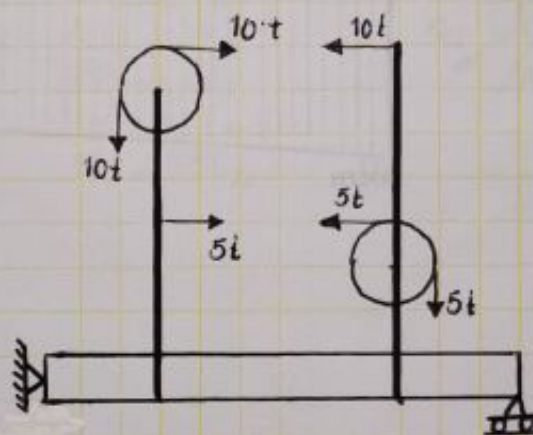
$$CD \ N_5 = 0$$



6 Calcular reacciones, diagramar momento



1. Análisis en las poleas



Cálculo de Reacciones

$$\sum M_0 = 0 \quad \curvearrowright \oplus$$

$$10(1.5) + 41.25 + 5(4.5) - 43.75 - V_4(6) = 0$$

$$V_4 = 5.833t //$$

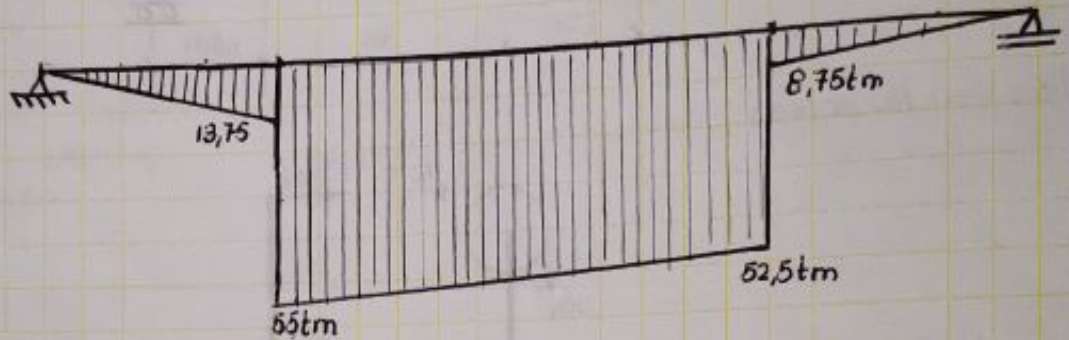
$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_1 - 10 - 5 + 5.833 = 0$$

$$V_1 = 9.167t //$$

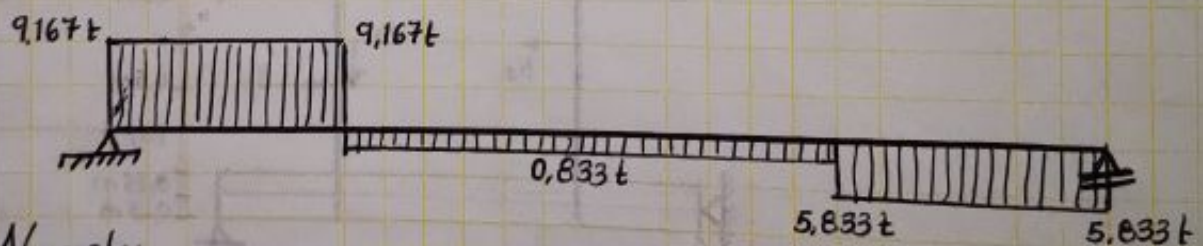
Cálculo de Momento

$$\begin{aligned}
 \text{CI } M_1 &= 0 \\
 \text{CI } M_{2a} &= 9,167(1,5) = 13,75 \text{ tm} \\
 \text{CI } M_{2d} &= M_{2a} + 41,25 = 55 \text{ tm} \\
 \text{CI } M_{3a} &= 9,167(4,5) + 41,25 - 10(3) = 52,5 \text{ tm} \\
 \text{CI } M_{3d} &= M_{3a} - 43,75 = 8,75 \text{ tm} \\
 \text{CD } M_4 &= 0
 \end{aligned}$$



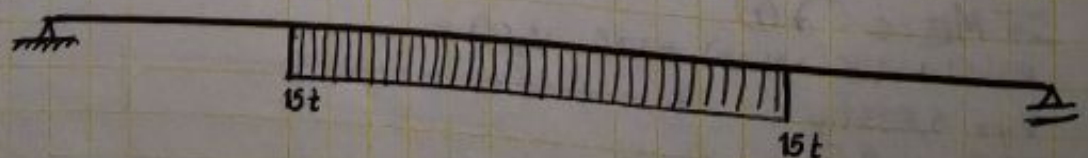
Cálculo de Cortante

$$\begin{aligned}
 \text{CI } Q_{1d} &= 9,167 \text{ t} \\
 \text{CI } Q_{2a} &= 9,167 \text{ t} \\
 \text{CI } Q_{2d} &= 9,167 - 10 = -0,833 \text{ t} \\
 \text{CI } Q_{3a} &= 9,167 - 10 = -0,833 \text{ t} \\
 \text{CD } Q_{3d} &= -5,833 \text{ t} \\
 \text{CD } Q_{4a} &= -5,833 \text{ t}
 \end{aligned}$$

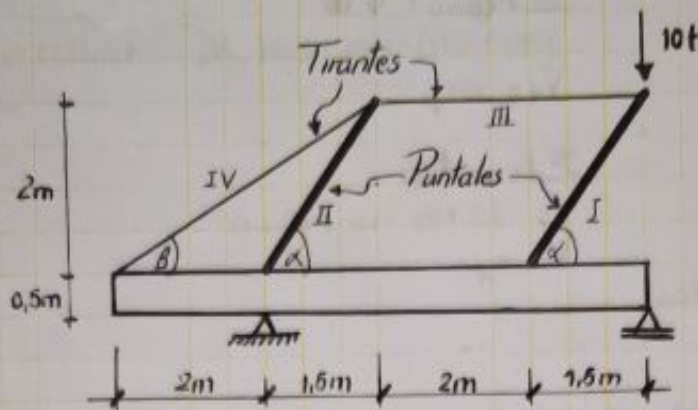


Cálculo de Normais

$$\begin{aligned}
 \text{CI } N_1 &= 0 \\
 \text{CI } N_{2a} &= 0 \\
 \text{CI } N_{2d} &= -15 \text{ t} \\
 \text{CI } N_{3a} &= -15 \text{ t} \\
 \text{CD } N_{3d} &= 0 \\
 \text{CD } N_4 &= 0
 \end{aligned}$$

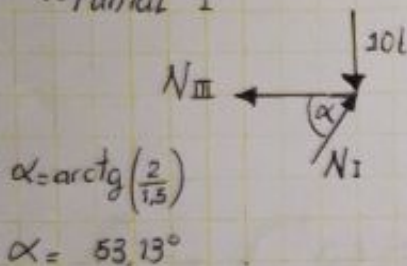


7.- Diagramar esfuerzos en el elemento horizontal



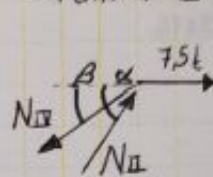
1. Análisis de puntales y tirantes

-Puntal I



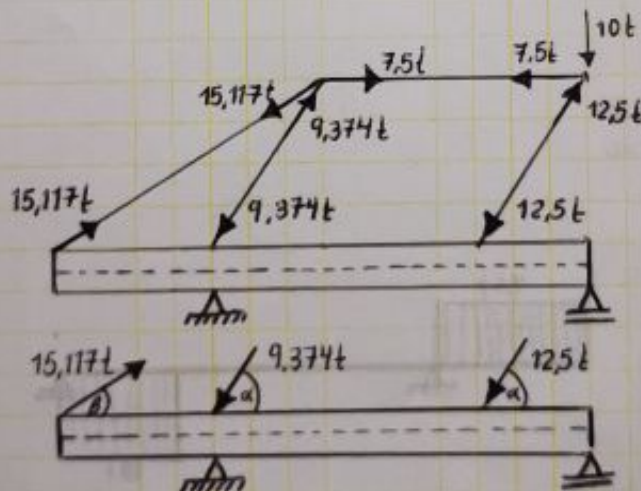
$$\begin{aligned}\sum F_v = 0 \uparrow (+) \\ N_I \text{ Sen } \alpha - 10 = 0 \\ N_I = 12.5 \text{ t} \parallel \\ \sum F_h = 0 \rightarrow (+) \\ -N_{III} + 12.5 \text{ Cos } \alpha = 0 \\ N_{III} = 7.5 \text{ t} \parallel\end{aligned}$$

-Puntal II



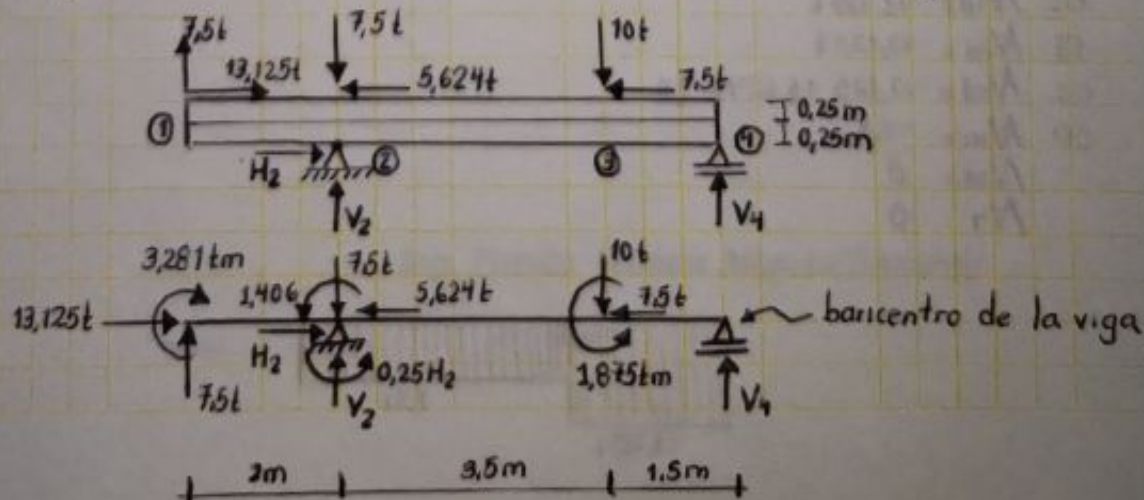
$$\begin{aligned}\beta = \arctg\left(\frac{2}{3.5}\right) = 29.745^\circ \\ \sum F_h = 0 \rightarrow (+) \\ -N_{IV} \text{ Cos } \beta + N_{II} \text{ Cos } \alpha + 7.5 = 0 \\ -0.8682 N_{IV} + 0.600 N_{II} = -7.5\end{aligned}$$

$$\begin{aligned}\sum F_v = 0 \uparrow (+) \\ -N_{IV} \text{ Sen } \beta + N_{II} \text{ Sen } \alpha = 0 \\ -0.4961 N_{IV} + 0.8 N_{II} = 0 \parallel \\ N_{IV} = 15.117 \text{ t} \parallel \\ N_{II} = 9.374 \text{ t} \parallel\end{aligned}$$



$$\alpha = 53.13^\circ \\ \beta = 29.745^\circ$$

Las cargas y los apoyos deberán trasladarse hasta el baricentro de la viga.



Cálculo de Reacciones

$$\sum F_H = 0 \rightarrow \oplus$$

$$13,125 + H_2 - 5,625 - 7,5 = 0$$

$$H_2 = 0 //$$

$$\sum M_2 = 0 \curvearrowright \oplus$$

$$3,281 + 7,5(2) - 1,406 - 0,25(H_2^0) + 10(3,5) - 1,875 - V_4(5) = 0$$

$$V_4 = 10t //$$

$$\sum F_V = 0 \uparrow \oplus$$

$$7,5 - 7,5 + V_2 - 10 + 10 = 0$$

$$V_2 = 0 //$$

Cálculo de Momentos

$$CI \quad M_{1d} = 3,281tm$$

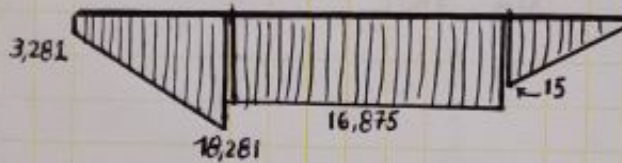
$$CI \quad M_{2a} = 3,281 + 7,5(2) = 18,281tm$$

$$CI \quad M_{2d} = M_{2a} - 1,406 = 16,875tm$$

$$CD \quad M_{3a} = 10(1,5) + 1,875 = 16,875tm$$

$$CD \quad M_{3d} = 10(1,5) = 15tm$$

$$M_4 = 0$$



Cálculo de Cortantes

$$CI \quad Q_{1d} = 7,5$$

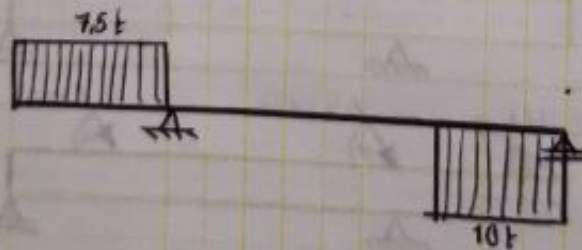
$$CI \quad Q_{2a} = 7,5$$

$$CI \quad Q_{2d} = 7,5 - 7,5 = 0$$

$$CI \quad Q_{3a} = +7,5 - 7,5 = 0$$

$$CD \quad Q_{3d} = -10t$$

$$CD \quad Q_4 = -10t$$



Cálculo de Normal

$$CI \quad N_{1d} = -13,125t$$

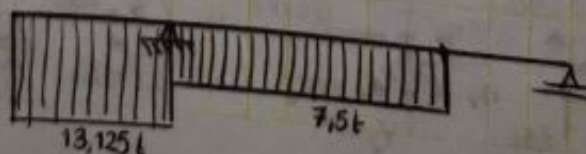
$$CI \quad N_{2a} = -13,125t$$

$$CI \quad N_{2d} = -13,125 + 5,624 = -7,5$$

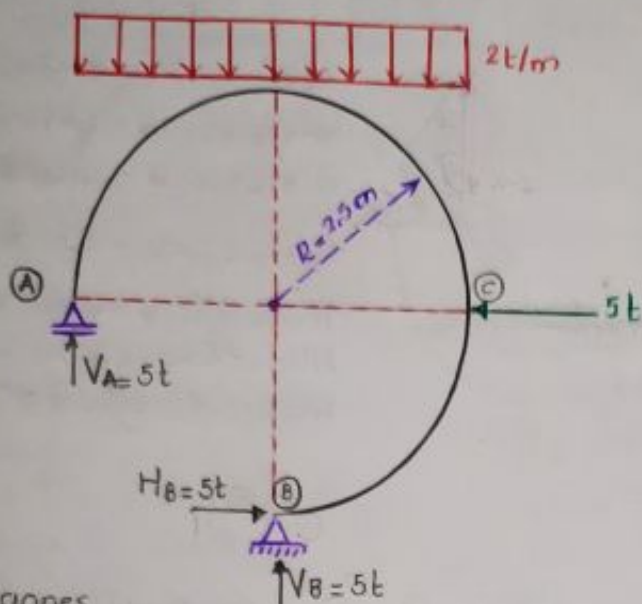
$$CD \quad N_{3a} = -7,5$$

$$N_{3d} = 0$$

$$N_4 = 0$$



Diagramar Momento, Normal y Cortante para la siguiente estructura

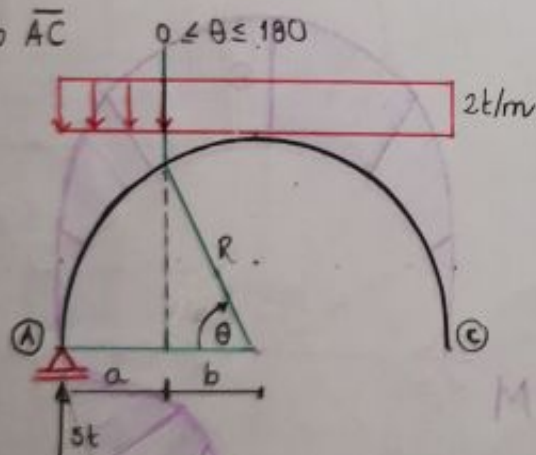


1.- Reacciones

$$\begin{aligned}\sum M_B &= 0 \quad \curvearrowright \oplus \\ -5(2,5) + V_A(2,5) &= 0 \quad \therefore V_A = 5t \\ \sum F_V &= 0 \quad \uparrow \oplus \\ 5 - 2(5) + V_B &= 0 \quad \therefore V_B = 5t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ H_B &= 5t\end{aligned}$$

2.- Momentos

* Tramo AC



$$\begin{aligned}a &= R - b \\ b &= R \cos \theta \\ a &= R - R \cos \theta = R(1 - \cos \theta)\end{aligned}$$

$$M(\theta) = 5 \cdot a - \frac{2(a)^2}{2} = 5R(1 - \cos \theta) - R^2(1 - \cos \theta)^2$$

$$M(\theta) = 5R - 5R \cos \theta - R^2(1 - 2 \cos \theta + \cos^2 \theta)$$

$$M(\theta) = 5R - 5R \cos \theta - R^2 + 2R^2 \cos \theta - R^2 \cos^2 \theta$$

$$M(\theta) = 5(2,5) - 5(2,5) \cos \theta - (2,5)^2 + 2(2,5)^2 \cos \theta - (2,5)^2 \cos^2 \theta$$

$$M(\theta) = -6,25 \cos^2 \theta + 6,25$$

$$c = R \cos \alpha \quad ; \quad e = R \cdot \sin \alpha$$

$$d = R - c = R - R \cos \alpha = R(1 - \cos \alpha)$$

$$M_2(\theta) = 5R(1 - \cos \alpha) - 5(R \cdot \sin \alpha)$$

$$M_2(\theta) = 5(2,5)(1 - \cos \alpha) - 5(2,5 \cdot \sin \alpha)$$

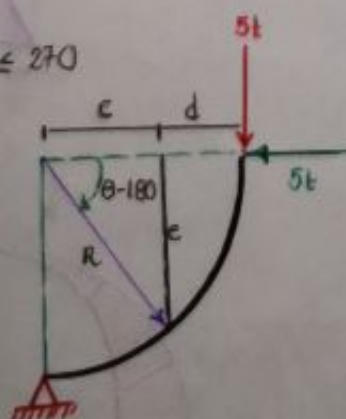
$$M_2(\theta) = -12,5 \cos \alpha - 12,5 \sin \alpha + 12,5$$

$$\alpha = \theta - 180$$

$$M_2(\theta) = -12,5 \cos(\theta - 180) - 12,5 \sin(\theta - 180) + 12,5$$

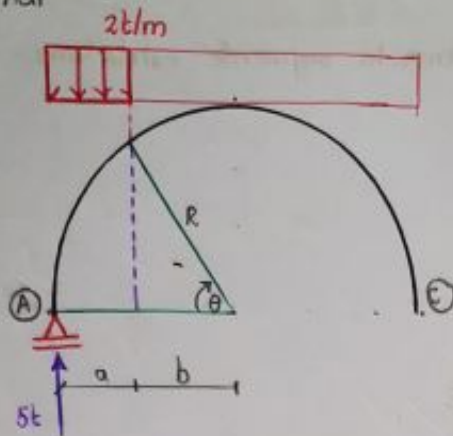
* Tramo CB

$$180 \leq \theta \leq 270$$

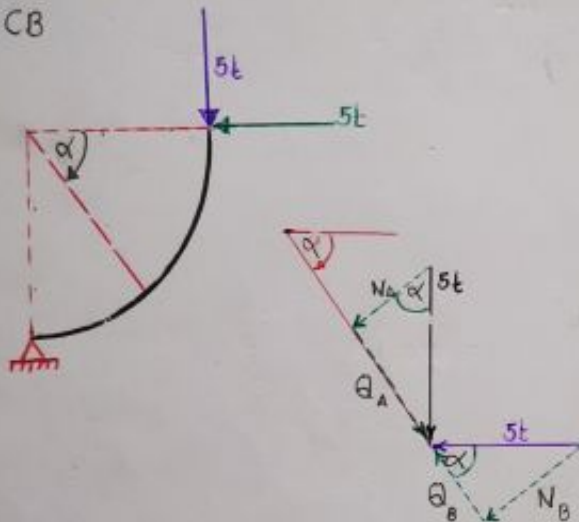


3.- Corte - Normal

* Tramo AC

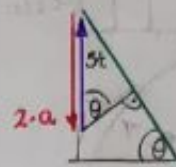


* Tramo CB



$$a = R - b \quad ; \quad b = R \cos \theta$$

$$a = R(1 - \cos \theta)$$



$$Q(\theta) = 5 \cdot \text{Sen} \theta - 2a \text{Sen} \theta$$

$$Q(\theta) = 5 \text{Sen} \theta - 2R(1 - \cos \theta) \text{Sen} \theta$$

$$Q(\theta) = 5 \text{Sen} \theta - 2R \text{Sen} \theta + 2R \text{Sen} \theta \cos \theta$$

$$Q(\theta) = 5 \text{Sen} \theta - 5 \text{Sen} \theta + 5 \text{Sen} \theta \cos \theta$$

$$N(\theta) = -5 \cos \theta + 2a \cos \theta$$

$$N(\theta) = -5 \cos \theta + 2(25)(1 - \cos \theta) \cos \theta$$

$$N(\theta) = -5 \cos \theta + 5 \cos \theta + 5 \cos^2 \theta$$

$$Q(\theta) = -Q_A + Q_B = -5 \text{Sen} \alpha + 5 \cos \alpha$$

$$Q(\theta) = -5 \text{Sen} \alpha + 5 \cos \alpha$$

$$N(\theta) = -N_A + N_B$$

$$N(\theta) = -5 \cos \alpha - 5 \text{Sen} \alpha$$

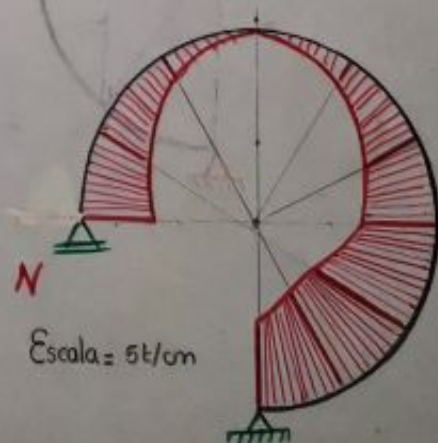
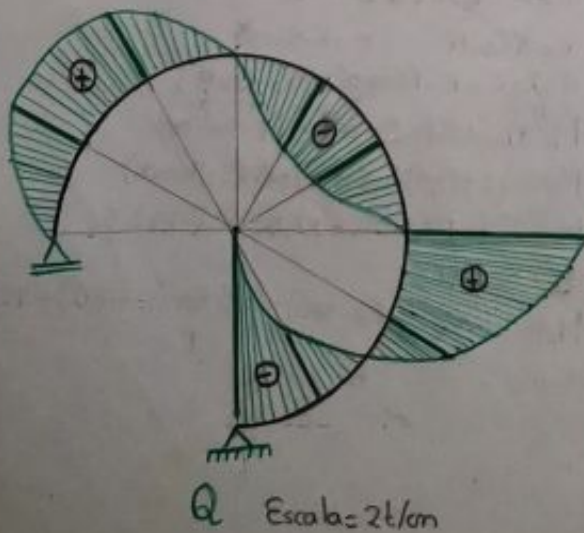
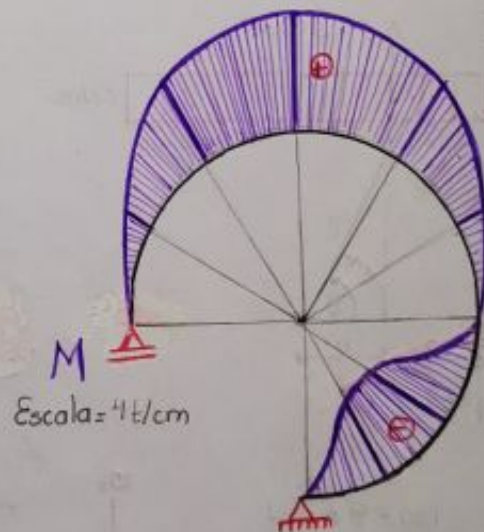
$$\text{don } \alpha = \theta - 180$$

$$Q(\theta) = -5 \text{Sen}(\theta - 180) + 5 \cos(\theta - 180)$$

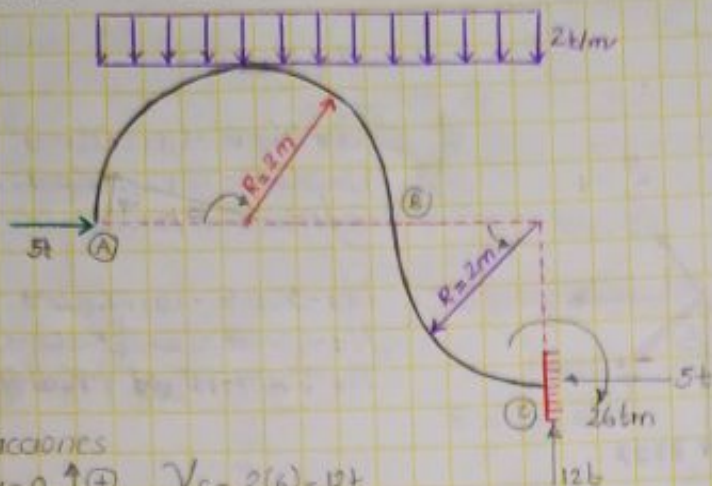
$$N(\theta) = -5 \cos(\theta - 180) - 5 \text{Sen}(\theta - 180)$$

4.- Diagramas

θ	Mto	Cortante	Normal
0	0	0	-5
30	1,563	2,165	-3,75
60	4,688	2,165	-1,25
90	6,25	0	0
120	4,688	-2,165	-1,25
150	1,563	-2,165	-3,75
180	0	0/5	-5
210	-4,575	1,830	-6,830
240	-4,575	-1,830	-6,830
270	0	-5	-5



Diagramar los esfuerzos internos



1.- Reacciones

$$\sum F_v = 0 \uparrow \oplus \quad V_c = 2(6) = 12t$$

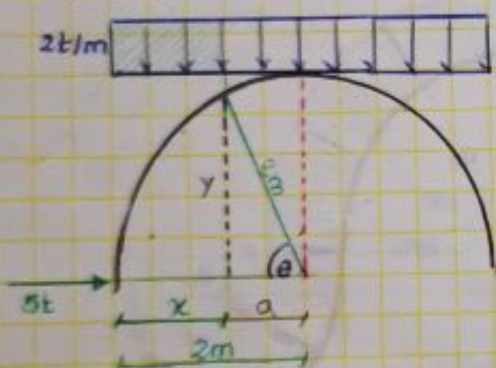
$$\sum F_H = 0 \rightarrow \oplus \quad H_c = 5t$$

$$\sum M_c = 0 \curvearrowright \oplus$$

$$5(2) - 2(6)(3) + M_c = 0 \quad \therefore M_c = 26tm$$

2.- Ecuaciones de Momento

* Tramo AB



$$x = 2 - a \quad ; \quad a = 2 \cos \theta$$

$$x = 2 - 2 \cos \theta$$

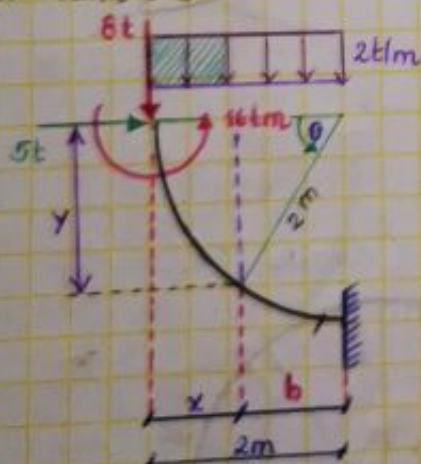
$$x = 2(1 - \cos \theta)$$

$$y = 2 \sin \theta$$

$$M = -5 \cdot y - 2 \frac{x^2}{2} = -5(2 \sin \theta) - [2(1 - \cos \theta)]^2$$

$$M = -10 \sin x - 4 \cos^2 x + 8 \cos x - 4 //$$

* Tramo BC



$$M = -16 - 8 \cdot x + 5y - 2 \frac{x^2}{2}$$

$$x = 2 - b \quad ; \quad b = 2 \cos \theta$$

$$x = 2 - 2 \cos \theta$$

$$x = 2(1 - \cos \theta)$$

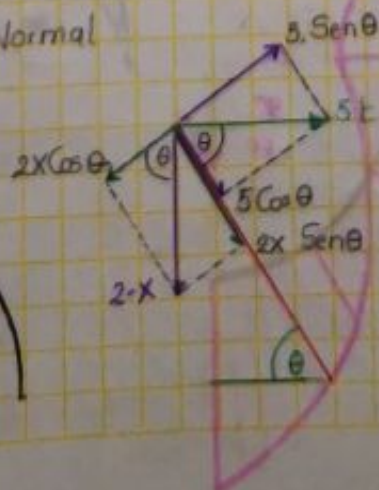
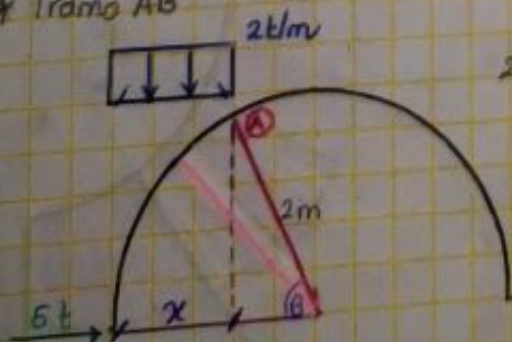
$$y = 2 \sin \theta$$

$$M = -16 - 8[2(1 - \cos \theta)] + 5(2 \sin \theta) - [2(1 - \cos \theta)]^2$$

$$M = 10 \sin(x) - 4 \cos^2(x) + 24 \cos(x) - 36 //$$

3.- Ecuaciones de Cortes y Normal

* Tramo AB



$$Q = -5 \cos \theta - 2x \sin \theta$$

$$Q = -5 \cos \theta - 2[2(1 - \cos \theta)] \sin \theta$$

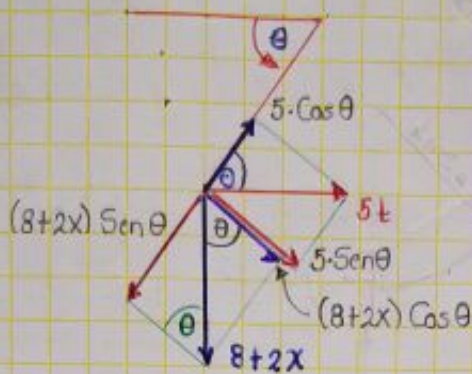
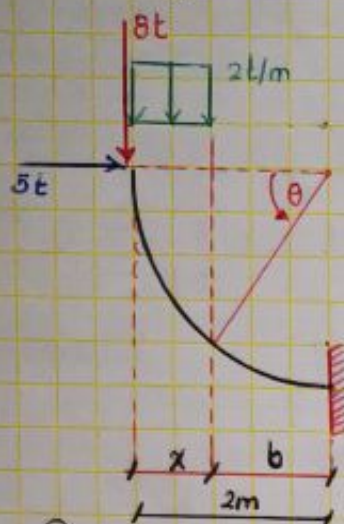
$$Q = (4 \cos(x) - 4) \sin(x) - 5 \cos(x) //$$

$$N = 2x \cos \theta - 5 \sin \theta$$

$$N = 2[2(1 - \cos \theta)] \cos \theta - 5 \sin \theta$$

$$N = -5 \sin(x) - 4 \cos^2(x) + 4 \cos(x) //$$

* Tramo BC



$$x = 2(1 - \cos \theta)$$

$$Q = 5 \cos \theta - (8 + 2x) \sin \theta$$

$$Q = 5 \cos \theta - (8 + 2[2 - 2 \cos \theta]) \sin \theta$$

$$Q = (4 \cos x - 12) \sin(x) + 5 \cos(x)$$

$$N = -5 \sin \theta - (8 + 2x) \cos \theta$$

$$N = -5 \sin \theta - (8 + 2[2 - 2 \cos \theta]) \cos \theta$$

$$N = -5 \sin \theta - 12 \cos \theta + 4 \cos^2 \theta$$

4.- Diagramas

Tramo AB

x Momento Cortante Normal

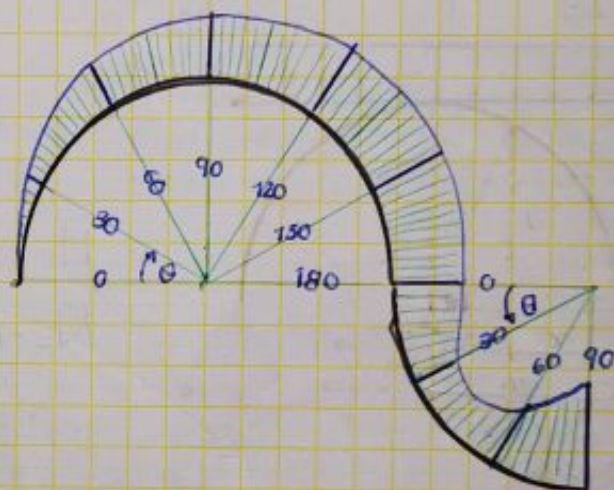
0	0	-5	0
30	-5,072	-4,598	-2,036
60	-9,66	-4,232	-3,33
90	-14	-4	-5
120	-17,66	-2,696	-7,33
150	-18,93	0,598	-8,964
180	-16	5	-8

Tramo BC

x Momento Cortante Normal

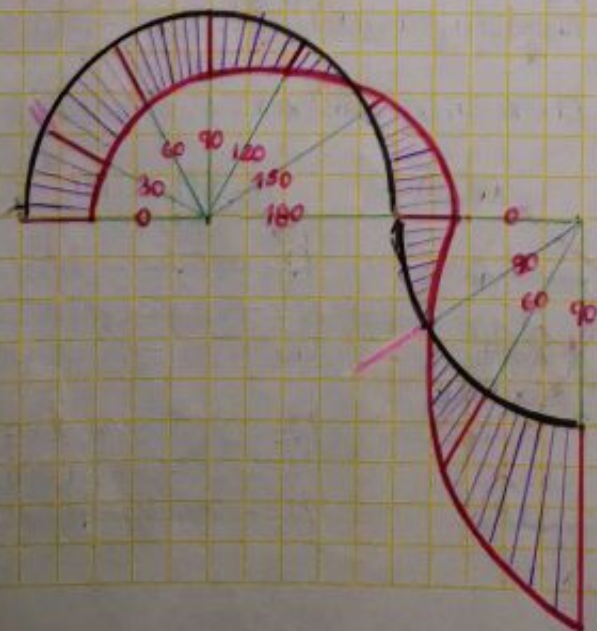
0	-16	5	-8
30	-13,215	0,062	-9,89
60	-16,34	-6,160	-9,333
90	-26	-12,000	-5

Momento Escala = 20 tm/cm

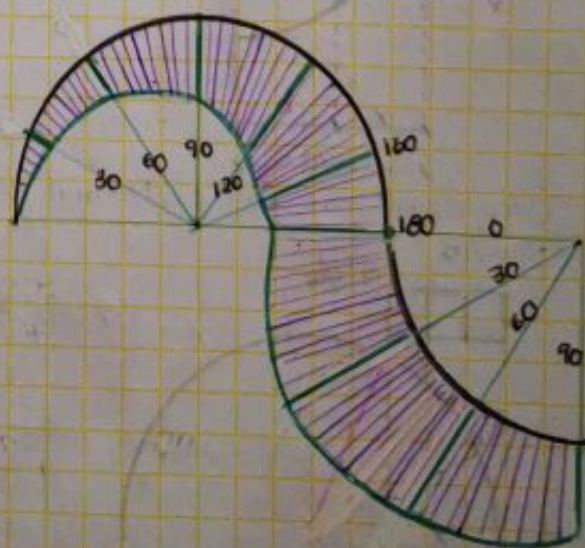


E = 20 tm/cm

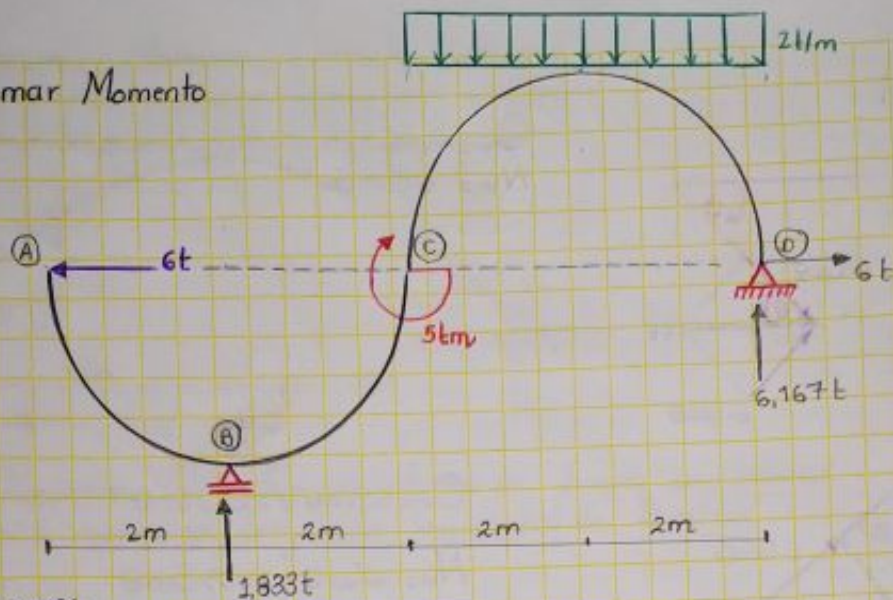
Cortante Escala = 5 t/cm



Normal Escala = 5 t/cm



Diagramar Momento



1.- Reacciones

$$\sum M_{\odot} = 0 \quad \curvearrowright \oplus$$

$$V_B(6) + 5 - 2(4)(2) = 0 \quad \therefore V_B = 1,833t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

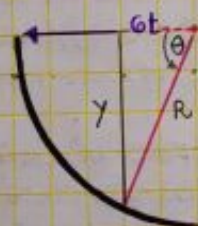
$$1,833 - 2(4) + V_D = 0 \quad \therefore V_D = 6,167t //$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$H_D = 6t //$$

2.- Momentas

* Tramo AB

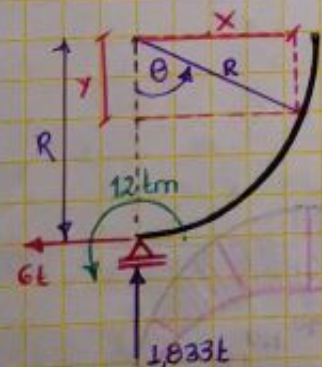


$$M_{\theta} = -6 \cdot y \quad \text{donde } y = R \cdot \text{Sen} \theta$$

$$M_{\theta} = -6R \text{Sen} \theta$$

$$M_{\theta} = -12 \text{Sen} \theta$$

* Tramo BC



$$M_{\theta} = -12 + 1,833 \cdot X + 6(R - y)$$

$$R = 2m$$

$$y = R \cos \theta = 2 \cos \theta$$

$$x = R \text{Sen} \theta = 2 \text{Sen} \theta$$

$$M_{\theta} = -12 + 1,833(2 \text{Sen} \theta) + 6(2 - 2 \cos \theta)$$

$$M_{\theta} = 3,666 \text{Sen}(\theta) - 12 \cdot \cos(\theta)$$

* Tramo CD



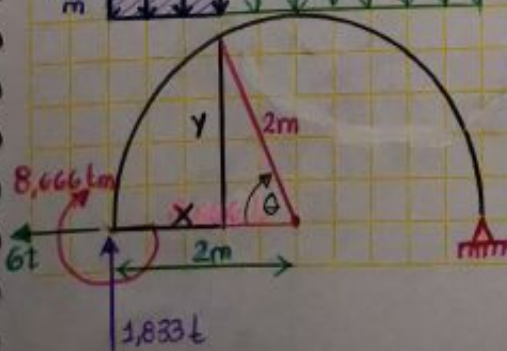
$$y = 2 \cdot \text{Sen} \theta$$

$$x = 2 - 2 \cos \theta$$

$$M_{\theta} = 8,666 + 1,833x + 6 \cdot y - 2 \frac{x^2}{2}$$

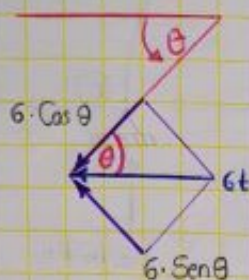
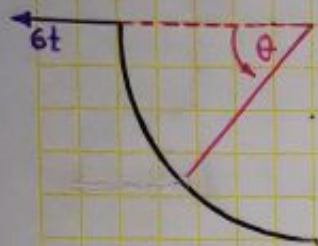
$$M_{\theta} = 8,666 + 1,833 \cdot (2 - 2 \cos \theta) + 6(2 \text{Sen} \theta) - (2 - 2 \cos \theta)^2$$

$$M_{\theta} = 12 \text{Sen} \theta - 4 \cos^2 \theta + 4,334 \cos \theta + 8,332 //$$



3.- Cortante y Normal

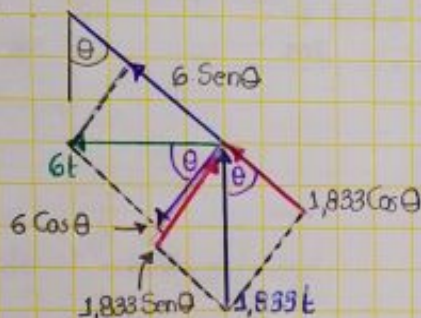
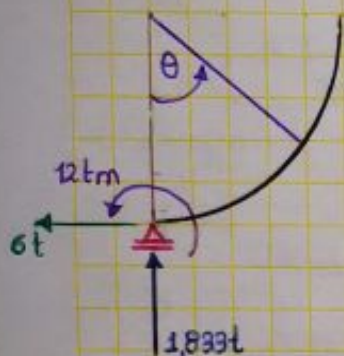
* Tramo AB



$$Q_{\theta} = -6 \cos \theta$$

$$N_{\theta} = 6 \sin \theta$$

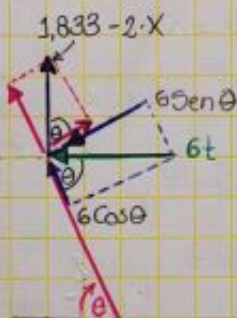
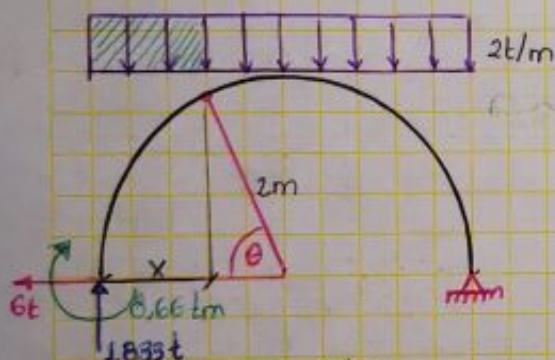
* Tramo BC



$$Q_{\theta} = 6 \sin \theta + 1,833 \cos \theta$$

$$N_{\theta} = 6 \cos \theta - 1,833 \sin \theta$$

* Tramo CD



$$x = 2 - 2 \cos \theta$$

$$Q_{\theta} = (1,833 - 2x) \sin \theta + 6 \cos \theta$$

$$Q_{\theta} = (1,833 - 2[2 - 2 \cos \theta]) \sin \theta + 6 \cos \theta$$

$$Q_{\theta} = (4 \cos \theta - 2,167) \sin \theta + 6 \cos \theta //$$

$$N_{\theta} = 6 \sin \theta - (1,833 - 2x) \cos \theta$$

$$N_{\theta} = 6 \sin \theta - (1,833 - 2[2 - 2 \cos \theta]) \cos \theta$$

$$N_{\theta} = 6 \sin \theta - 4 \cos^2 \theta + 2,167 \cos \theta //$$

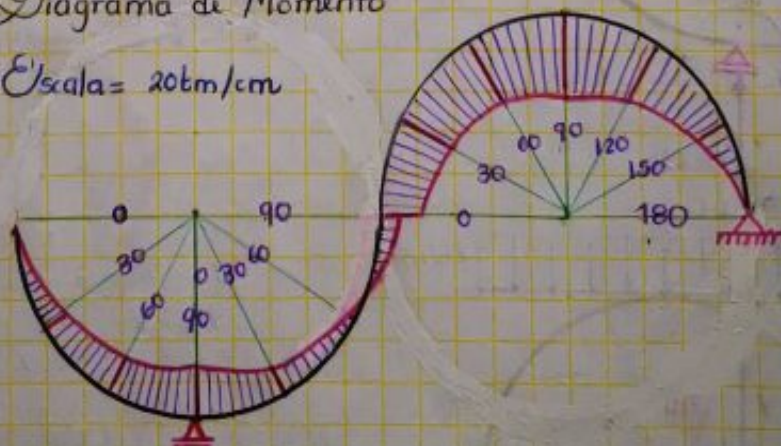
4.- Diagrama de Mto

x Momento

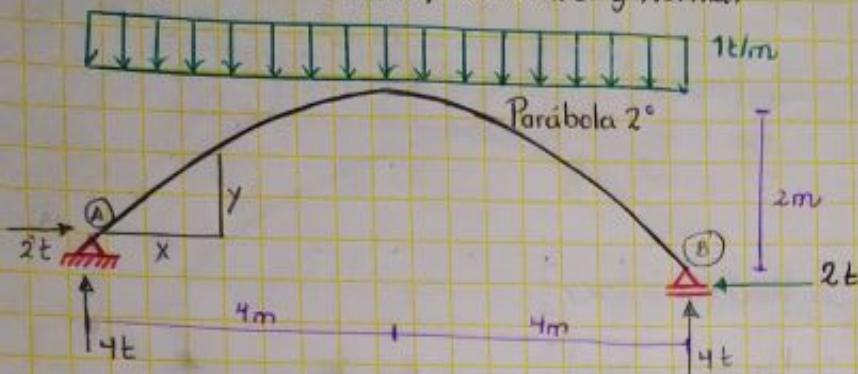
0°	0
30°	-6
60°	-10,892
90°	-12
0°	-12
30°	-8,56
60°	-2,825
90°	3,666
0°	8,666
30°	15,085
60°	19,891
90°	20,332
120°	15,557
150°	7,58
180°	0

Diagrama de Momento

Escala = 20tm/cm



Diagramar Momento, Cortante y Normal



Las diagramas de esfuerzos internos en barras parabólicas se realiza con respecto a una recta, que es la proyección horizontal de la curva

1- Ecuación de la curva

Datos

Origen = Pto A

Pto Conocido $x=0$
 $y=0$

Vertice $h=4$
 $k=2$

$$(x-h)^2 = -4a(y-k)$$

$$(0-4)^2 = -4a(0-2)$$

$$16 = +8a$$

$$a = 2 //$$

Ecuación $\rightarrow (x-4)^2 = -4(2)(y-2)$

despejamos y

$$x^2 - 8x + 16 = -8(y-2)$$

$$x^2 - 8x + 16 = -8y + 16$$

$$y = \frac{8x - x^2}{8}$$

$$y = x - \frac{x^2}{8}$$

2. Reacciones

$$\sum M_A = 0 \curvearrowright \oplus$$

$$1(8)(4) - V_B(8) = 0 \quad \therefore V_B = 4t //$$

$$\sum F_v = 0 \uparrow \oplus$$

$$V_A = 8 - 4 = 4t //$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$H_A = 2t$$

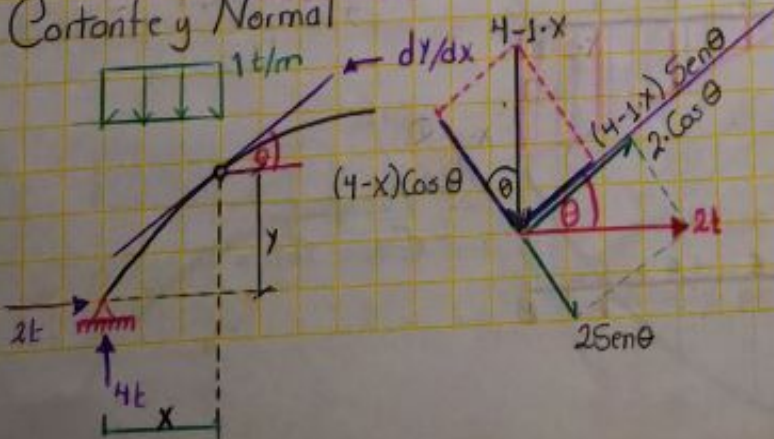
3- Ecuación de Momento

$$M_x = 4 \cdot x - 2 \cdot y - \frac{1 \cdot x^2}{2} = 4x - 2\left(x - \frac{x^2}{8}\right) - \frac{x^2}{2}$$

$$M_x = 4x - 2x + \frac{x^2}{4} - \frac{x^2}{2}$$

$$M_x = 2x - \frac{x^2}{4}$$

4.- Cortante y Normal



$$\frac{dy}{dx} = \tan \theta = 1 - \frac{2x}{8} = 1 - \frac{x}{4} //$$

$$\theta = \arctg(1 - 0,25x)$$

$$Q_x = (4-x) \cos \theta - 2 \sin \theta$$

$$N_x = (4-x) \sin \theta - 2 \cos \theta$$

$$\theta = \arctg(1-0,25x)$$

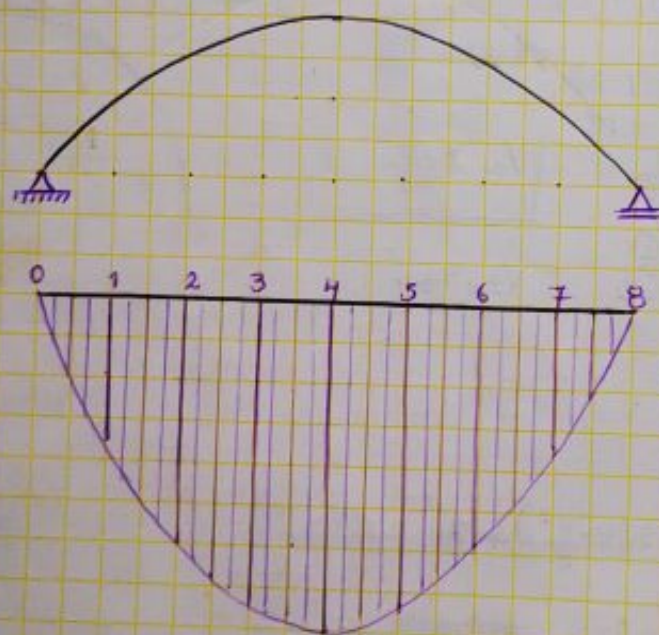
Reemplazamos θ en Q_x y N_x

$$Q_x = (4-x) \cos [\arctg(1-0,25x)] - 2 \sin [\arctg(1-0,25x)]$$

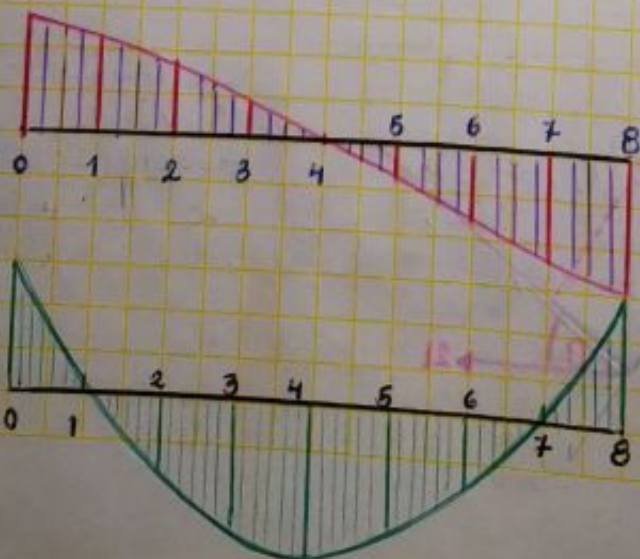
$$N_x = (4-x) \sin [\arctg(1-0,25x)] - 2 \cos [\arctg(1-0,25x)]$$

x	Momento	Cortante	Normal
0	0	1,4142	1,4142
1	1,75	1,2	0,2
2	3	0,894	-0,894
3	3,75	0,485	-1,698
4	4	0	-2
5	3,75	-0,485	-1,698
6	3	-0,894	-0,894
7	1,75	-1,2	0,2
8	0	-1,414	1,4142

Diagramas

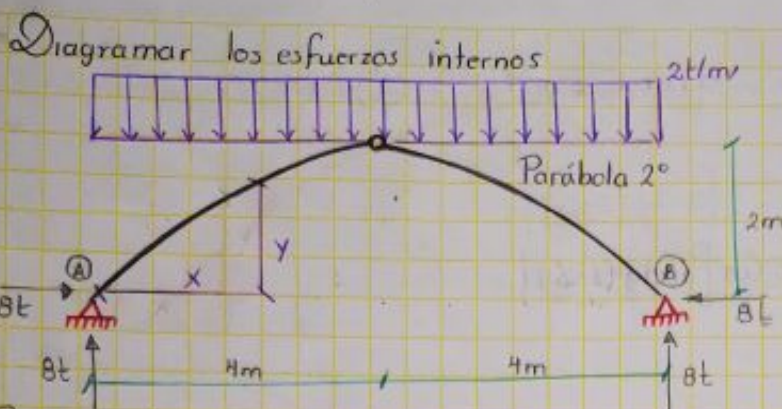


Momento
Esc = 1 ton/cm



Cortante
Esc = 1 t/cm

Normal
Esc = 1 t/cm



1.- Ecuación de la barra parabólica

Datos

Origen del eje cartesiano = pto A = (0,0)
 Vertice = (4,2)

$$(x-h)^2 = -4a(y-k)$$

$$(0-4)^2 = -4a(0-2)$$

$$16 = -4a(-2)$$

$$16 = 8a$$

$$a = 2 //$$

$$(x-4)^2 = -4(2)(y-2)$$

$$x^2 - 8x + 16 = -8(y-2)$$

$$x^2 - 8x + 16 = -8y + 16$$

$$y = \frac{8x - x^2}{8}$$

$$y = x - \frac{x^2}{8} //$$

2.- Reacciones

$$\sum M_A = 0 \curvearrowright \oplus$$

$$2(8)(4) - V_B(8) = 0 \therefore V_B = 8t //$$

$$\sum F_V = 0 \uparrow \oplus$$

$$V_A - 2(8) + 8 = 0 \therefore V_A = 8t //$$

$$\sum M_B = 0 \curvearrowright \oplus \text{ (Izg)}$$

$$8(4) - 2(4)(2) - H_A(2) = 0 \therefore H_A = 8t$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$H_B = 8t$$

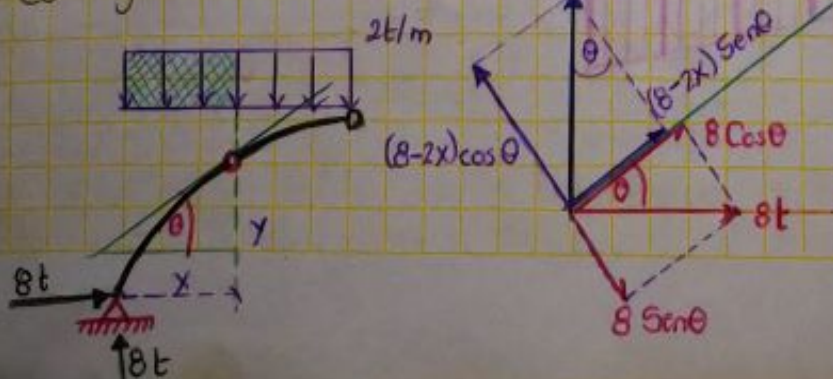
4.- Ecuación de Momento

$$M = 8 \cdot x - 8y - \frac{2x^2}{2} = 8x - 8y - x^2$$

$$M = 8x - \frac{8(8x - x^2)}{8} - x^2 = 8x - 8x + x^2 - x^2 = 0$$

No presenta diagrama de Mto flector

5. Corte y Normal



$$\frac{dy}{dx} = \tan \theta = 1 - \frac{x}{4} //$$

$$\theta = \arctan \left(1 - \frac{x}{4} \right) //$$

$$Q = (8-2x) \cos \theta - 8 \sin \theta$$

$$N = -(8-2x) \sin \theta - 8 \cos \theta$$

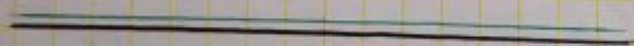
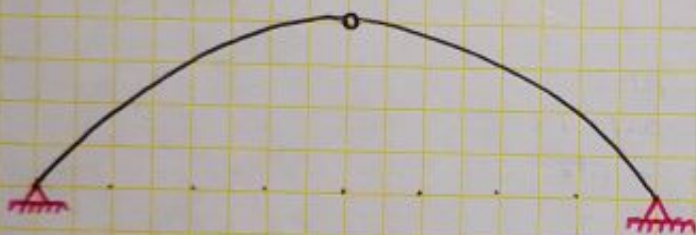
donde $\theta = \arctg \left(1 - \frac{x}{4} \right)$

$$Q = (8-2x) \cos \left[\arctg \left(1 - \frac{x}{4} \right) \right] - 8 \sin \left[\arctg \left(1 - \frac{x}{4} \right) \right]$$

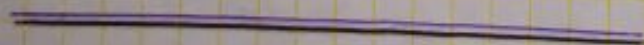
$$N = -(8-2x) \sin \left[\arctg \left(1 - \frac{x}{4} \right) \right] - 8 \cos \left[\arctg \left(1 - \frac{x}{4} \right) \right]$$

x	Cortante	Normal
0	0	-11,31
1	0	-10
2	0	-8,944
3	0	-8,246
4	0	-8

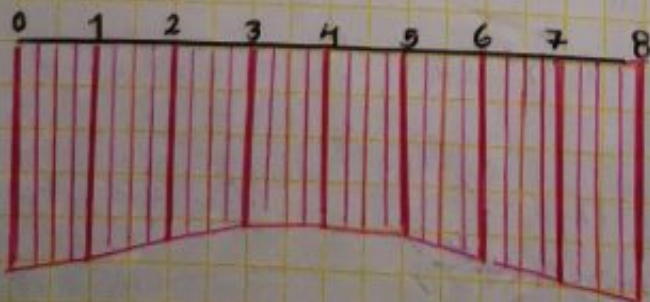
6 - Diagramas



Momento



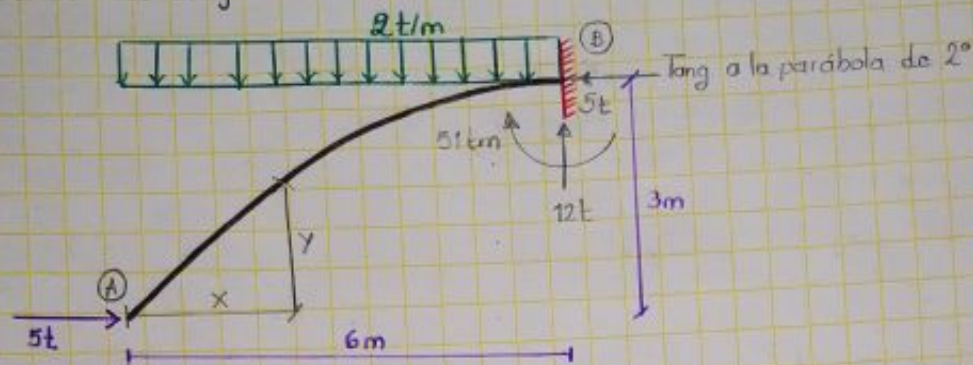
Cortante



Normal

$$Esc = 4t/cm$$

Obtener los diagramas característicos



1.- Ecuación de la barra parabólica

Datos

Eje cartesiano de referencia = Pto A = (0,0)

Vértice = pto B = (6,3)

$$(x-h)^2 = -4a(y-k)$$

$$(0-6)^2 = -4a(0-3)$$

$$36 = 12a$$

$$a = 3$$

$$(x-6)^2 = -4(3)(y-3)$$

$$y = x - \frac{x^2}{12} //$$

2.- Reacciones

$$\sum F_v = 0 \uparrow \oplus$$

$$-2(6) + V_B = 0 \therefore V_B = 12t$$

$$\sum F_H = 0 \rightarrow \oplus \therefore H_B = 5t$$

$$\sum M_B = 0 \curvearrowright \oplus$$

$$-2(6)(3) - 5(3) + M_B = 0 \therefore M_B = 51tm$$

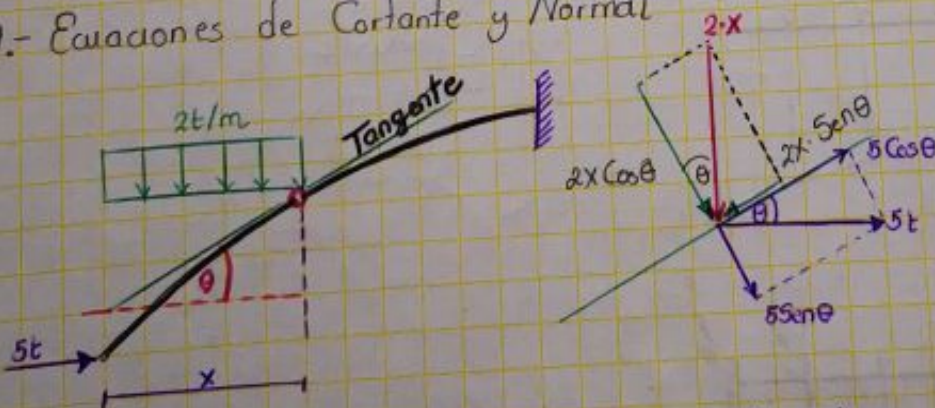
3.- Ecuación de Momento

$$M_x = -5y - 2\frac{x^2}{2} = -5y - x^2$$

$$M_x = -5\left(x - \frac{x^2}{12}\right) - x^2 = -5x + \frac{5x^2}{12} - x^2$$

$$M_x = -5x - \frac{7x^2}{12} //$$

4.- Ecuaciones de Cortante y Normal



$$\frac{dy}{dx} = \text{Tang } \theta = 1 - \frac{x}{6} \Rightarrow \theta = \arctg\left(1 - \frac{x}{6}\right)$$

$$Q_x = -2x \cos \theta - 5 \sin \theta$$

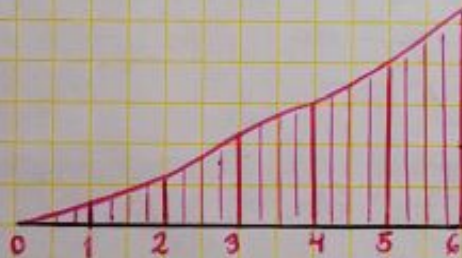
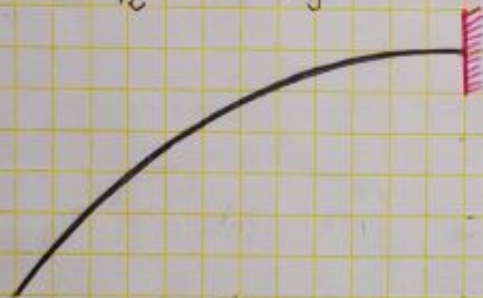
$$Q_x = -2x \cos [\arctg(1 - \frac{x}{6})] - 5 \sin [\arctg(1 - \frac{x}{6})] //$$

$$N_x = 2x \sin \theta - 5 \cos \theta$$

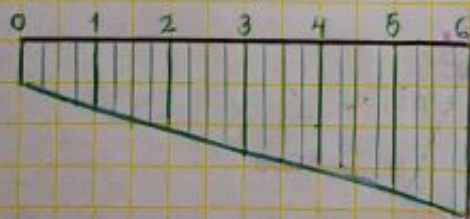
$$N_x = 2x \sin [\arctg(1 - \frac{x}{6})] - 5 \cos [\arctg(1 - \frac{x}{6})]$$

5.- Diagramas

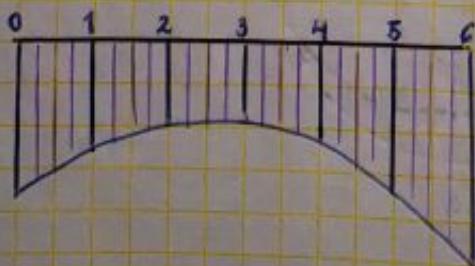
x	Momento	Cortante	Normal
0	0	-3,535	-3,535
1	-5,5833	-4,787	-2,561
2	-12,333	-6,102	-1,941
3	-20,25	-7,603	-1,789
4	-29,33	-9,171	-2,21
5	-39,583	-10,686	-3,288
6	-51	-12	-5



Momento
Esc = 20 kN/cm

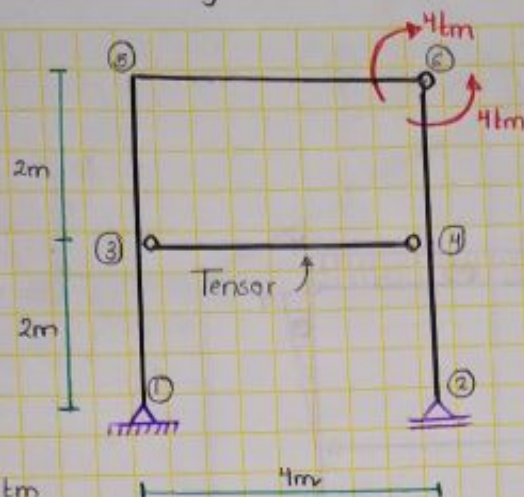


Cortante
Esc = 6 kN/cm

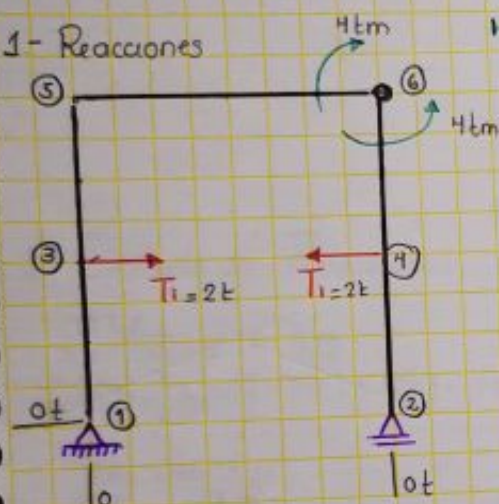


Normal
Esc = 2 kN/cm

Diagramar Momento, Cortante y Normal



1- Reacciones



$$\begin{aligned}\sum M_6 &= 0 \quad \curvearrowright \text{ @ } 6 \\ -4 + T_1(2) &= 0 \quad \therefore T_1 = 2t \\ \sum M_1 &= 0 \quad \curvearrowright \text{ @ } 1 \\ V_2 &= 0t \\ \sum F_v &= 0 \quad \uparrow \text{ @ } 4 \\ V_1 &= 0t \\ \sum F_H &= 0 \quad \rightarrow \text{ @ } 1 \\ H_1 &= 0t\end{aligned}$$

2- Momentos

Barra 1-3	$M_1 = 0$ $M_3 = 0$
Barra 3-5	$M_3 = 0$ $M_5 = -2(2) = -4tm$
Barra 5-6	$M_5 = -4tm$ $M_6 = -4tm$
Barra 2-4	$M_2 = 0$ $M_4 = 0$
Barra 4-6	$M_4 = 0$ $M_6 = -2(2) = -4tm$

3 Cortantes

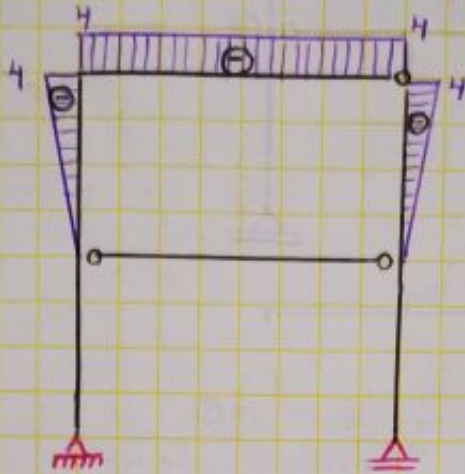
Barra 1-3	$Q_1 = 0$ $Q_3 = 0$
Barra 3-5	$Q_{3-5} = -2t$
Barra 5-6	$Q_{5-6} = 0t$
Barra 2-4	$Q_{2-4} = 0t$
Barra 4-6	$Q_{4-6} = 2t$

4- Normales

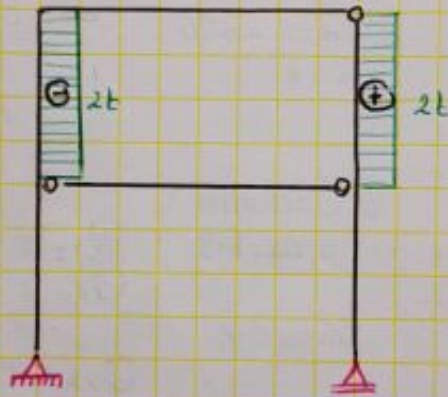
Barra 1-3	$N_{1-3} = 0t$
Barra 3-5	$N_{3-5} = 0t$
Barra 5-6	$N_{5-6} = -2t$
Barra 2-4	$N_{2-4} = 0t$
Barra 4-6	$N_{4-6} = 0t$
Barra 3-4	$N_{3-4} = 2t$

5- Diagramas

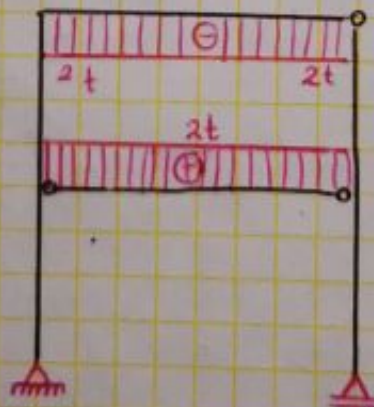
Momentos



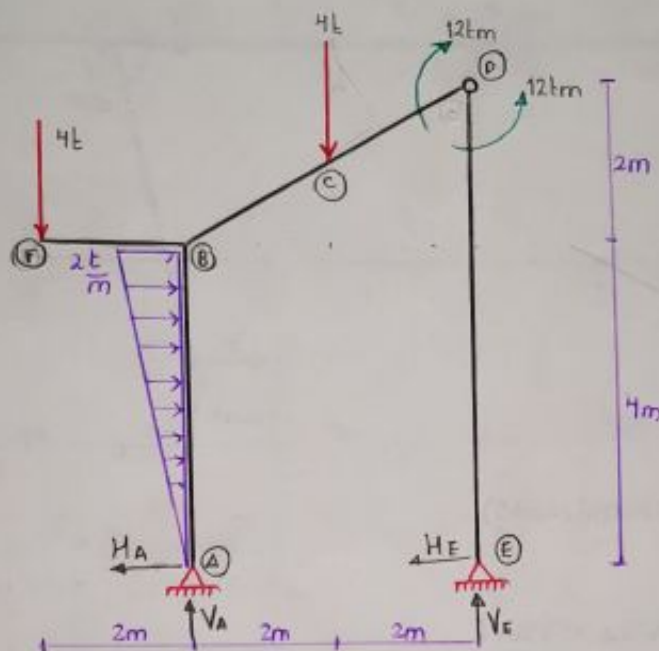
Cortante



Normal



Obtener los diagramas de esfuerzos internos para la siguiente estructura



1.- Cálculo de Reacciones

$$\begin{aligned}\sum M_A &= 0 \quad \curvearrowright \oplus \\ \frac{2(4)}{2} \cdot \left[\frac{2}{3} \cdot (4) \right] - 4(2) + 4(2) - V_E(4) &= 0 \\ V_E &= 2,667t // \\ \sum F_v &= 0 \quad \uparrow \oplus \\ V_A - 4 - 4 + 2,667 &= 0 \\ V_A &= 5,333t //\end{aligned}$$

$$\begin{aligned}\sum M_D &= 0 \quad \curvearrowright \oplus \quad (\text{barra D-E}) \\ -12 + H_E(6) &= 0 \\ H_E &= 2t // \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ -H_A + \frac{2(4)}{2} - 2 &= 0 \\ H_A &= 2t //\end{aligned}$$

2.- Momentos

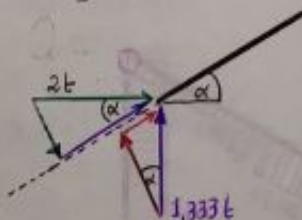
$$\begin{aligned}M_A &= 0tm \\ M_{BA} &= 2(4) - \frac{2(4)}{2} \cdot \left[\frac{1}{3} \cdot 4 \right] \\ M_{BA} &= 2,667tm \\ M_F &= 0tm \\ M_{BF} &= -4(2) = -8tm \\ M_{BC} &= 2(4) - \frac{2(4)}{2} \cdot \left[\frac{1}{3} \cdot 4 \right] - 4(2) = -5,333tm \\ M_C &= 5,333(2) + 2(5) - \frac{2(4)}{2} \cdot \left[\frac{1}{3} \cdot 4 + 1 \right] - 4(4) = -4,667tm \\ M_D &= 0tm \\ M_{DE} &= -12tm \\ M_E &= 0tm\end{aligned}$$

Cortantes

$$\begin{aligned}Q_{BC} &= 1,333 \cos(26,565) - 2 \sin(26,565) \\ Q_{BC} &= 0,2981t \\ Q_{CD} &= 0,2981 - 4 \cos(26,565) \\ Q_{CD} &= -3,2796 \\ Q_{DE} &= -3,2796 \\ Q_{DE} &= 2t \\ Q_E &= 2t\end{aligned}$$

3.- Cortantes

$$\begin{aligned}Q_A &= 2t \\ Q_{BA} &= 2 - \frac{2(4)}{2} = -2t \\ Q_F &= -4t \\ Q_{BF} &= -4t\end{aligned}$$



4 - Normales

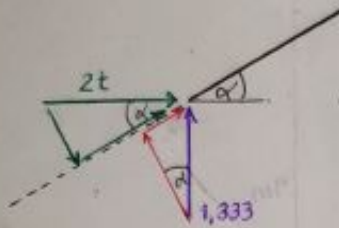
$$N_A = -5,333t$$

$$N_{BA} = -5,333t$$

$$N_F = 0$$

$$N_{BF} = 0$$

$$N_{BC} = ?$$



$$N_{BC} = -2 \cos(26,565) - 1,333 \sin(26,565)$$

$$N_{BC} = -2,385t$$

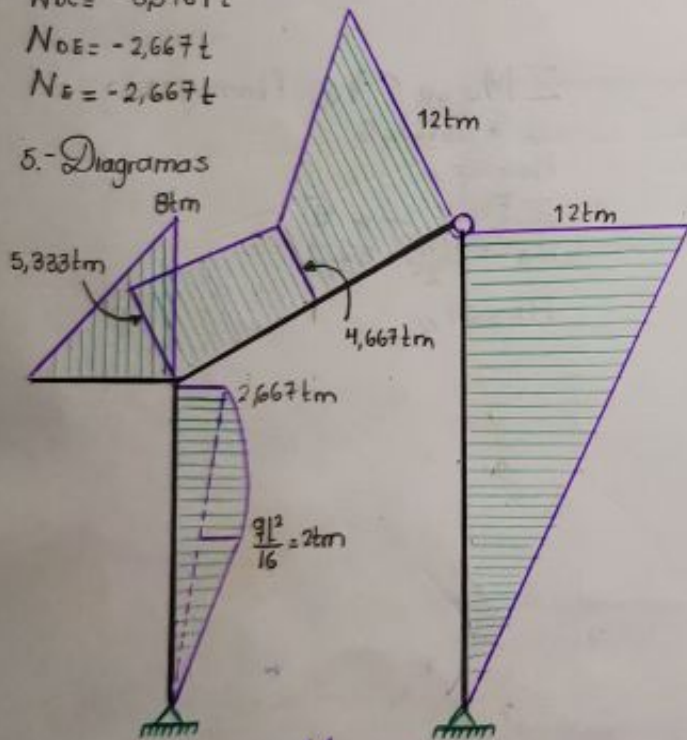
$$N_{CD} = -2,385 + 4 \cdot \sin(26,565) = -0,5961t$$

$$N_{DC} = -0,5961t$$

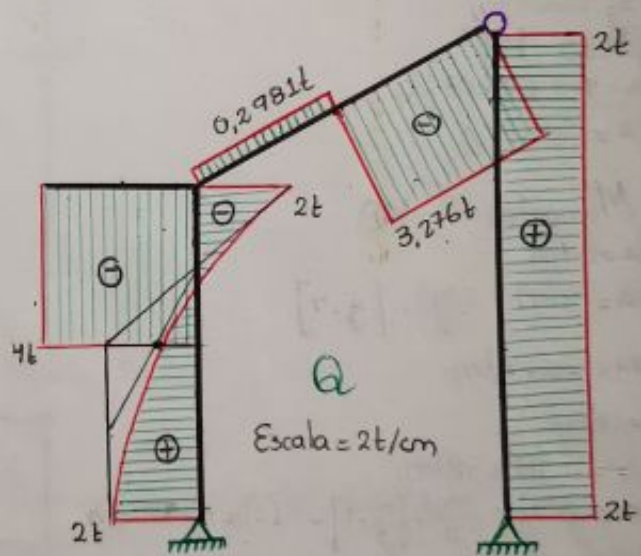
$$N_{DE} = -2,667t$$

$$N_E = -2,667t$$

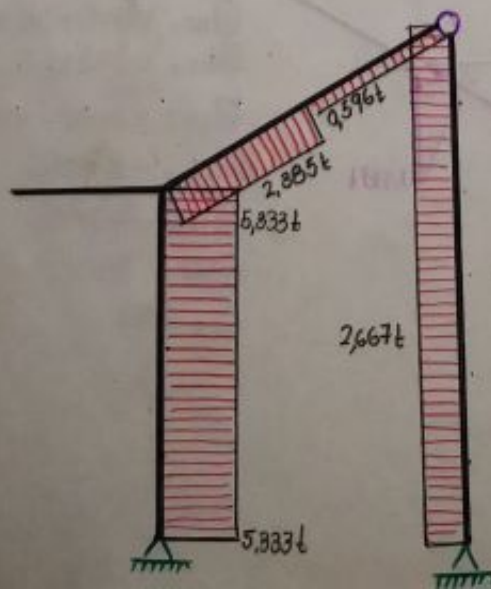
5 - Diagramas



M
Escala = 4tm/cm

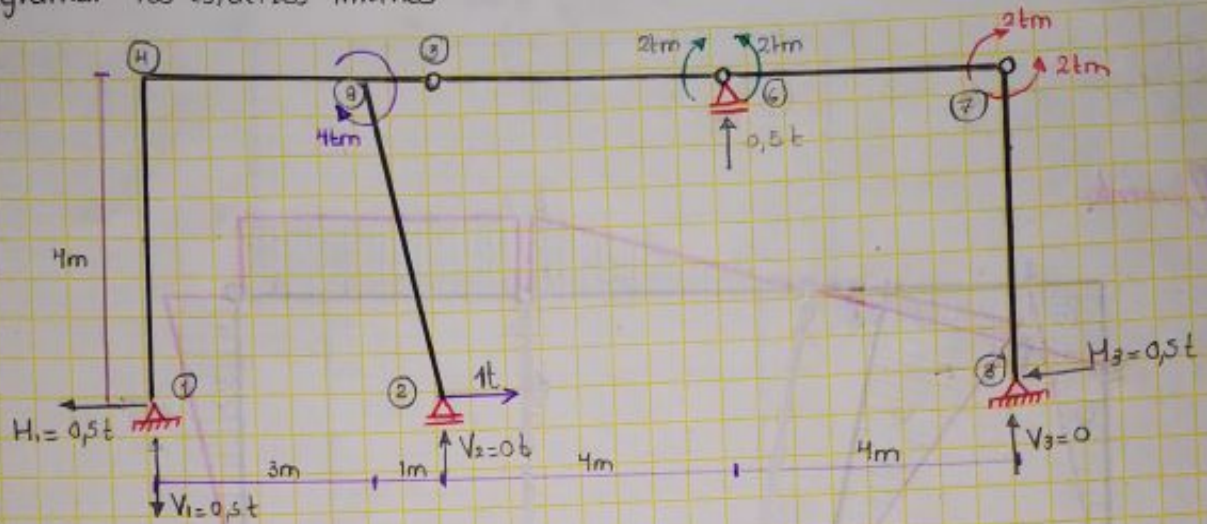


Q
Escala = 2t/cm



N
Escala = 5t/cm

Diagramar los esfuerzos internos



1.- Reacciones

$$\begin{aligned} \sum M_7 &= 0 \quad (\curvearrowright) \quad (\text{abajo}) \\ -2 + H_3(4) &= 0 \quad \therefore H_3 = 0.5t // \\ \sum F_H &= 0 \rightarrow \oplus \\ -H_1 + 1 - 0.5 &= 0 \quad \therefore H_1 = 0.5t // \\ \sum M_6 &= 0 \quad (\curvearrowright) \quad (\text{Derecha}) \\ -2 + 0.5(4) - V_3(4) &= 0 \quad \therefore V_3 = 0 \\ \sum M_5 &= 0 \quad (\curvearrowright) \quad (\text{Derecha}) \\ 0.5(4) - V_6(4) &= 0 \quad \therefore V_6 = 0.5t // \\ \sum M_2 &= 0 \quad (\curvearrowright) \quad (\text{Izquierda}) \\ 4 + 0.5(4) - 1(4) + V_1(4) &= 0 \quad \therefore V_1 = 0.5t \\ \sum F_V &= 0 \uparrow \oplus \\ -0.5 + V_2 + 0.5 &= 0 \quad \therefore V_2 = 0 \end{aligned}$$

2.- Momentos

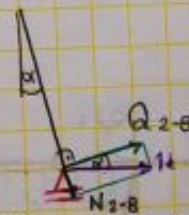
- * Barra 1-4 $M_1 = 0$
 $M_4 = 0.5(4) = 2tm$
- * Barra 4-8 $M_4 = 2tm$
 $M_8 = 0.5(4) - 0.5(3) = 0.5tm$
- * Barra 2-8 $M_2 = 0$
 $M_8 = -1(4) = -4tm$
- * Barra 3-7 $M_3 = 0tm$
 $M_7 = -0.5(4) = -2tm //$
- * Barra 6-7 $M_6 = -0.5(4) = -2tm$
 $M_7 = -0.5(4) = -2tm //$
- * Barra 5-6 $M_5 = +0.5(4) - 0.5(4) = 0tm //$
 $M_6 = 2tm //$
- * Barra 5-8 $M_5 = 0tm //$
 $M_8 = +0.5(5) - 0.5(4) = +0.5tm //$

4.- Normales

- * Barra 1-4 $N_{1-4} = 0.5t$
- * Barra 4-8 $N_{4-8} = 0.5t$
- * Barra 2-8 $N_{2-8} = -1 \cdot \text{Sen}(14.036) = -0.243t //$
- * Barra 3-7 $N_{3-7} = 0t$
- * Barra 6-7 $N_{6-7} = -0.5t$
- * Barra 5-6 $N_{5-6} = -0.5t$
- * Barra 5-8 $N_{5-8} = -0.5t$

3.- Cortantes

Barra 1-4 $Q_{1-4} = 0.5t$
Barra 4-8 $Q_{4-8} = -0.5t$
Barra 2-8

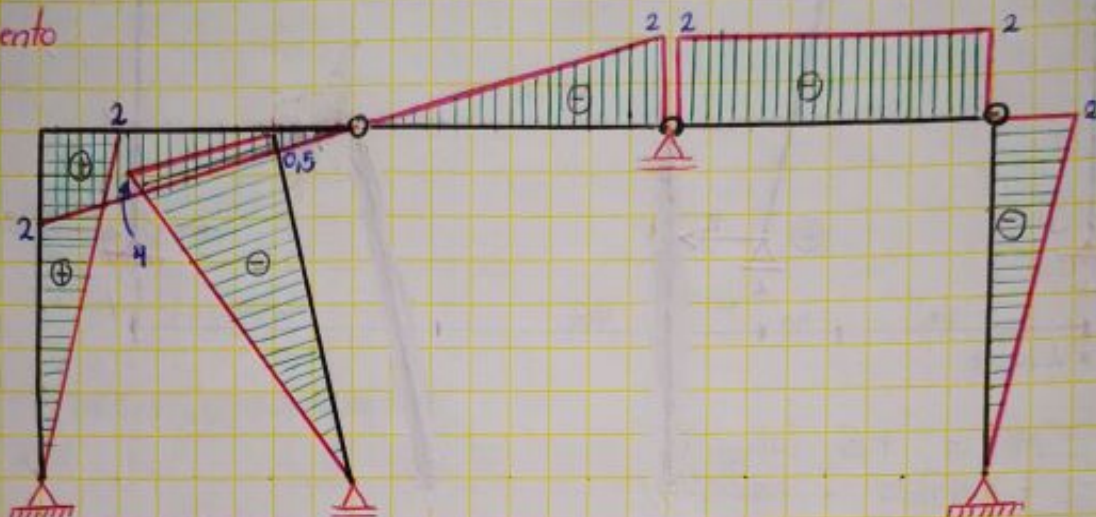


$$\alpha = \arctg\left(\frac{1}{4}\right) = 14.036$$

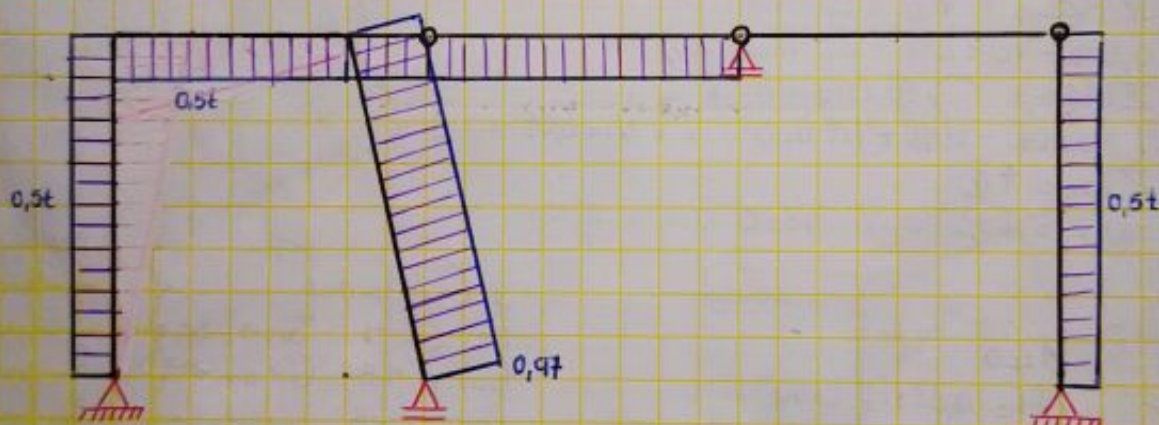
$Q_{2-8} = -1 \cos(14.036) = -0.97t //$
Barra 3-7 $Q_{3-7} = 0.5t$
Barra 6-7 $Q_{6-7} = 0t$
Barra 5-6 $Q_{5-6} = -0.5t$
Barra 8-5 $Q_{8-5} = -0.5t$

5 - Diagramas

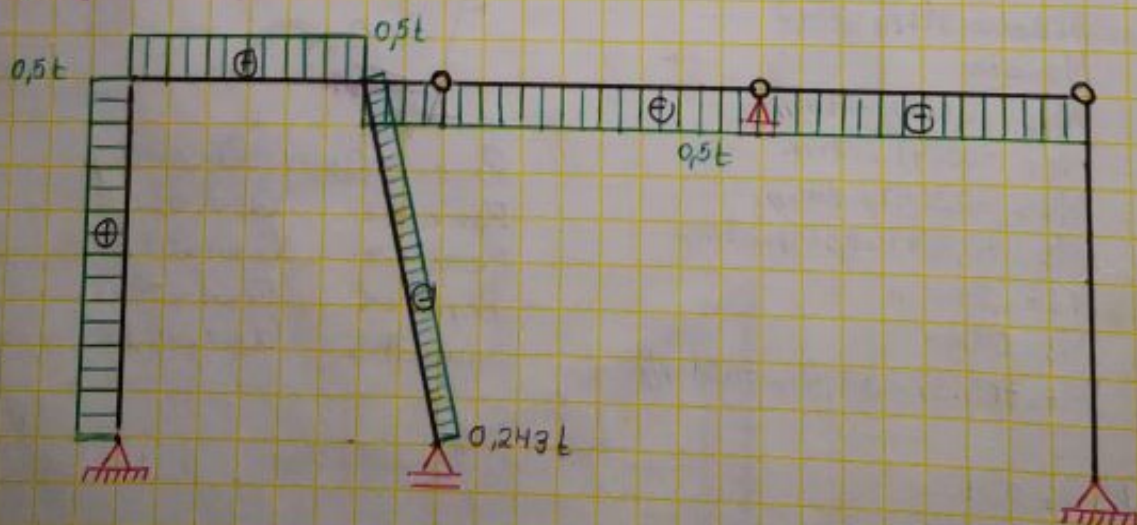
Momento



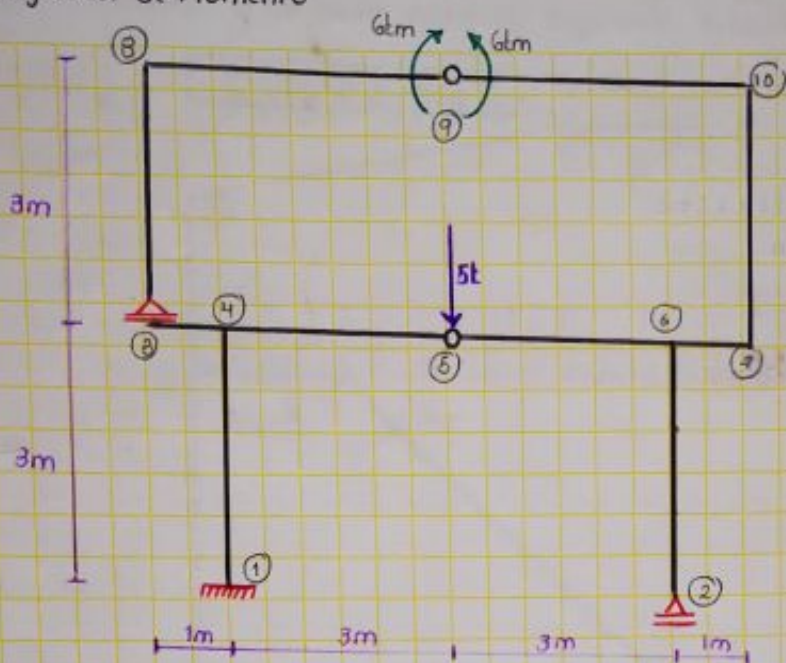
Cortante



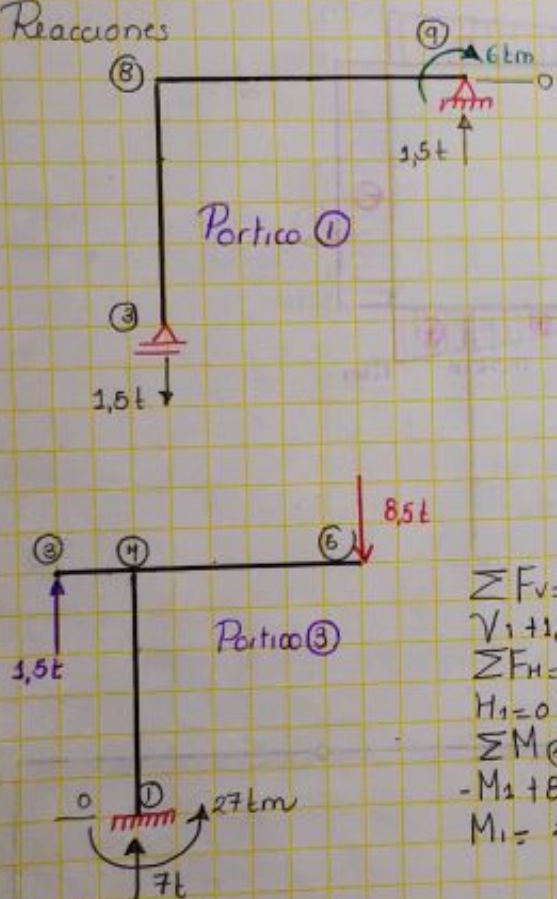
Normal



Obtener el diagrama de Momento



1. Reacciones



$$\left. \begin{aligned} \sum M_9 &= 0 \quad (\curvearrowright +) \\ 6 - V_3(4) &= 0 \\ V_3 &= 1,5t \\ \sum F_v &= 0 \quad \uparrow (+) \\ V_9 &= 1,5t \end{aligned} \right\}$$

$$\left. \begin{aligned} \sum M_5 &= 0 \quad (\curvearrowright +) \\ -6 + V_2(3) &= 0 \\ V_2 &= 2t \\ \sum F_v &= 0 \quad \uparrow (+) \\ V_5 - 5 - 1,5 - 2 &= 0 \\ V_5 &= 8,5t \end{aligned} \right\}$$

$$\left. \begin{aligned} \sum F_v &= 0 \quad \uparrow (+) \\ V_1 + 1,5 - 8,5 &= 0 \quad \therefore V_1 = 7t \\ \sum F_H &= 0 \quad \rightarrow (+) \\ H_1 &= 0t \\ \sum M_1 &= 0 \quad (\curvearrowright +) \\ -M_1 + 8,5(3) + 1,5(1) &= 0 \\ M_1 &= 27tm \end{aligned} \right\}$$

2. Momentos

- Portico 1

* Barra 3-8 $M_3 = 0$
 $M_8 = 0$
 * Barra 8-9 $M_8 = 0$
 $M_9 = -1,5(4) = -6tm$

- Portico 2

* Barra 9-10 $M_9 = -6tm$
 $M_{10} = -6 - 1,5(4) = -12tm$
 * Barra 10-7 $M_{10} = -12tm$
 $M_7 = -12tm$

* Barra 5-6

$M_5 = 0tm$
 $M_6 = (8,5 - 5)(3) = 10,5tm$
 * Barra 2-6 $M_2 = M_6 = 0$
 * Barra 6-7 $M_6 = (8,5 - 5)(3) = 10,5tm$
 $M_7 = (8,5 - 5)(4) - 2(1) = 12tm$

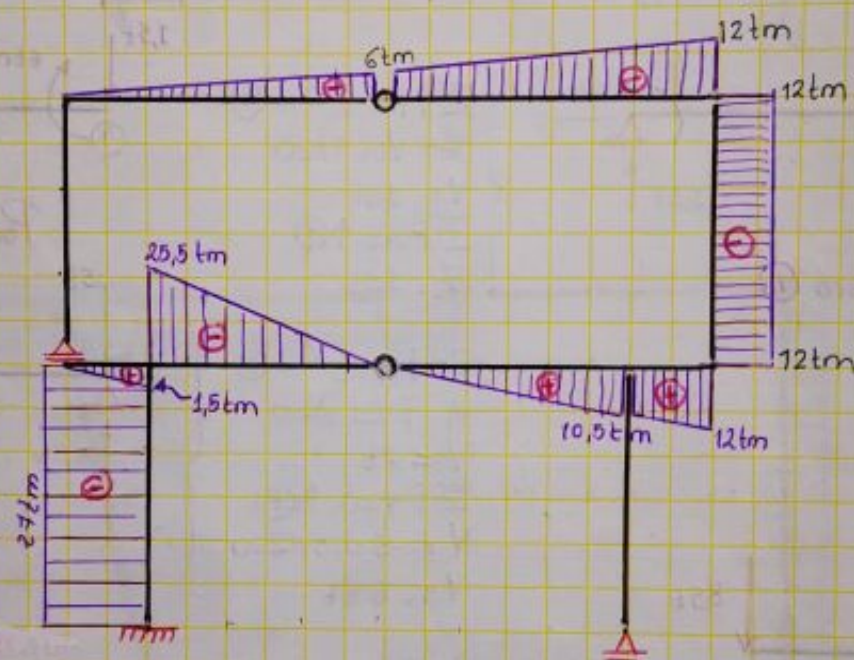
- Portico ③

* Barra 3-4 $M_3 = 0$
 $M_4 = 1,5(1) = 1,5 \text{ tm}$

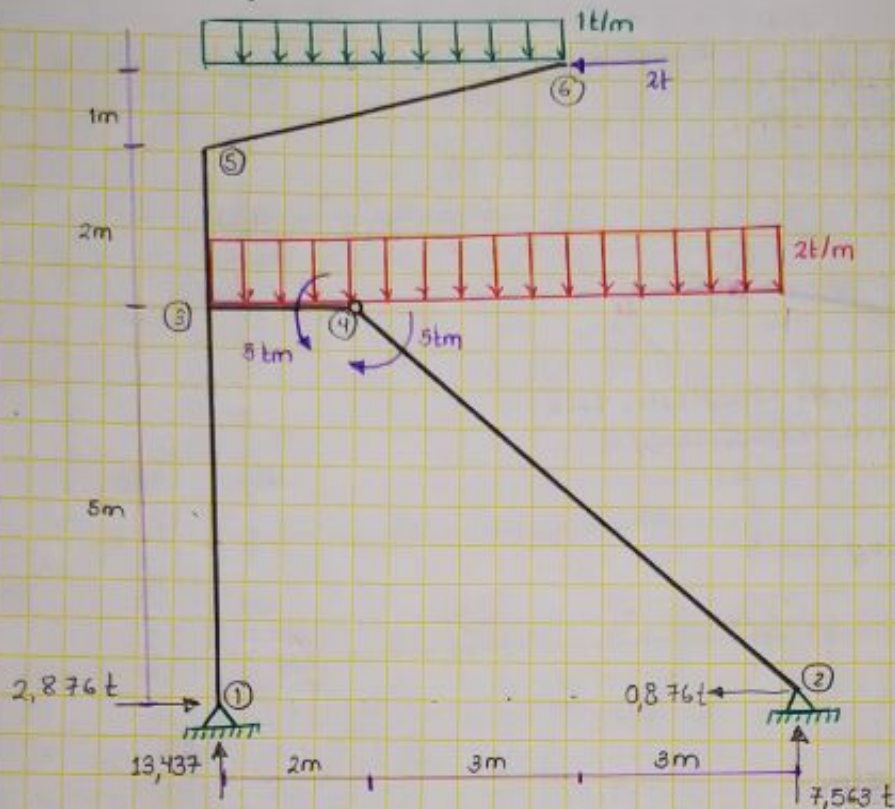
* Barra 1-4 $M_1 = -27 \text{ tm}$
 $M_4 = -27 \text{ tm}$

* Barra 4-5 $M_5 = 0 \text{ tm}$
 $M_4 = -8,5(3) = -25,5 \text{ tm}$

3. Diagrama



Diagramar Momento y cortante para la siguiente estructura



1.- Cálculo de Reacciones

$$\begin{aligned}\sum M_{(1)} &= 0 \quad (\curvearrowright) \\ 2(8)(4) + 1(5)(2.5) - 2(8) - V_2(8) &= 0 \quad \therefore V_2 = 7.563 \text{ t} \\ \sum F_v &= 0 \quad (\uparrow) \\ V_1 - 1(5) - 2(8) + 7.563 &= 0 \quad \therefore V_1 = 13.437 \text{ t} \\ \sum M_{(4)} &= 0 \quad (\curvearrowright) \quad (\text{Derecho}) \\ 5 + 2(6)(3) - 7.563(6) + H_2(5) &= 0 \quad \therefore H_2 = 0.876 \text{ t} \\ \sum F_h &= 0 \quad (\rightarrow) \\ H_1 - 2 - 0.876 &= 0 \quad \therefore H_1 = 2.876 \text{ t}\end{aligned}$$

2.- Momento

Barra 1-3 $M_1 = 0$

$$M_3 = -2.876(5) = -14.38 \text{ tm}$$

Barra 2-4 $M_2 = 0$

$$M_4 = -0.876(5) + 7.563(6) - 2(6)(3) = 5 \text{ tm}$$

Barra 3-4

$$M_3 = -2.876(5) - 2(3) + 1(5)(2.5) = -7.88$$

$$M_4 = 5 \text{ tm}$$

Barra 5-6 $M_6 = 0$

$$M_5 = 2(1) - 1(5)(2.5) = -10.5 \text{ tm}$$

Barra 3-5 $M_5 = -10.5 \text{ tm}$

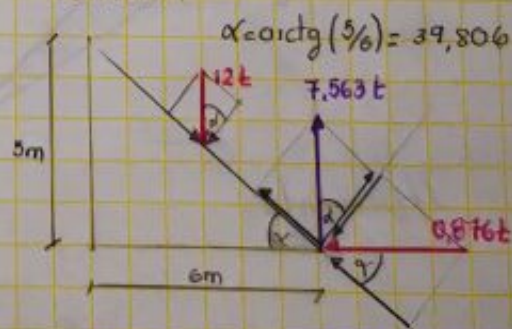
$$M_3 = 2(3) - 1(5)(2.5) = -6.5 \text{ tm}$$

3.- Cortantes

Barra 1-3

$$Q_{1-3} = -2.876 \text{ t}$$

Barra 2-4



$$Q_2 = 0.876 \text{ Sen}(39.806) - 7.563 \text{ Cos}(39.806)$$

$$Q_2 = -5.249 \text{ t}$$

$$Q_4 = -5.249 + 12 \cdot \text{Cos}(39.806)$$

$$Q_4 = 3.969 \text{ t}$$

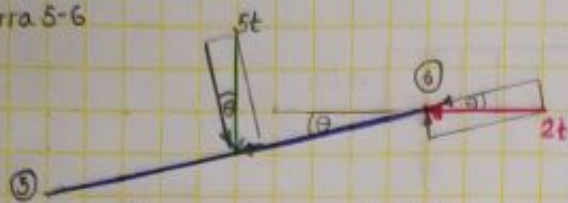
Barra 3-4

$$Q_4 = -7,563 + 2(6) = 4,437 \text{ t}$$

$$Q_3 = 4,437 + 2(2) = 8,437 \text{ t}$$

Barra 5-6

$$\theta = \arctg\left(\frac{1}{5}\right) = 11,31^\circ$$



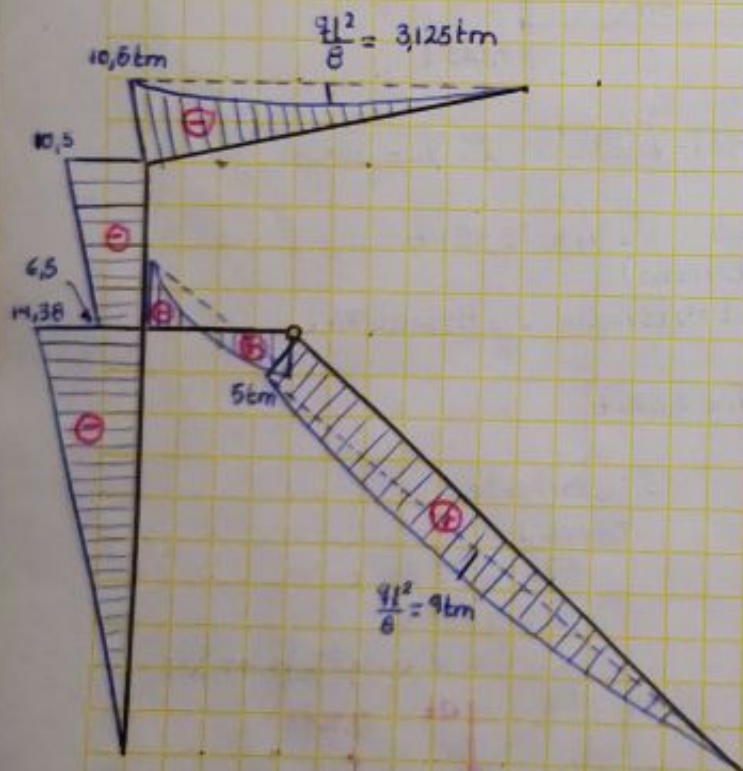
$$Q_5 = -2 \cdot \text{Sen}(11,31) + 5 \cdot \text{Cos}(11,31) = 4,51 \text{ t}$$

$$Q_6 = -2 \cdot \text{Sen}(11,31) = -0,392 \text{ t}$$

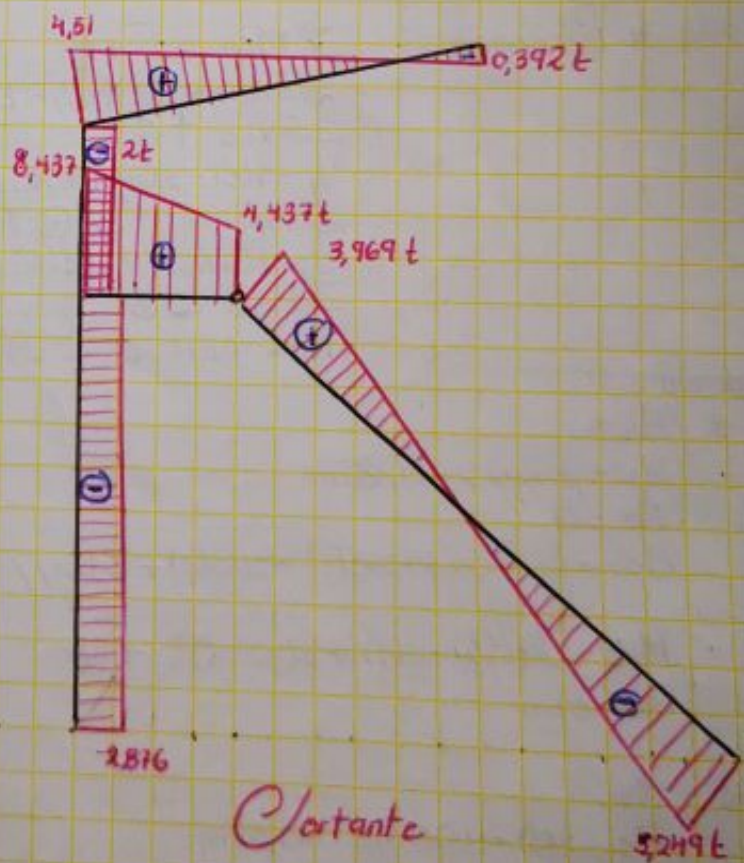
Barra 3-5

$$Q_{35} = -2 \text{ t}$$

4 - Diagramas

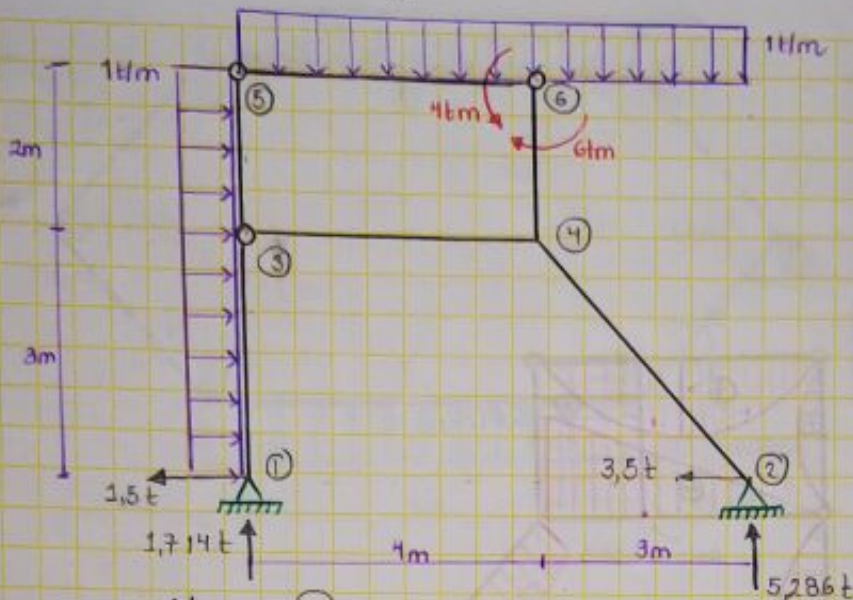


Memento



Cortante

Diagramar Momento en la siguiente estructura



1- Reacciones

$$\sum M_{\odot} = 0 \quad \curvearrowright \oplus$$

$$1(5)(2,5) + 1(7)(3,5) - V_2(7) = 0 \quad \therefore V_2 = 5,286t,,$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_1 - 1(7) + 5,286t = 0 \quad \therefore V_1 = 1,714t,,$$

$$\sum M_{\odot} = 0 \quad \curvearrowright \oplus$$

$$H_1(3) - 1(3)(1,5) = 0 \quad \therefore H_1 = 1,5t,,$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$1(5) - 1,5 - H_2 = 0 \quad \therefore H_2 = 3,5t,,$$

2- Momentos

Barra 1-3

$$M_1 = 0$$

$$M_3 = 1,5(3) - 1(3)(1,5) = 0$$

$$\bar{M}_{13} = \frac{qL^2}{8} = \frac{1(3)^2}{8} = 1,125tm$$

Barra 2-4

$$M_2 = 0$$

$$M_4 = 5,286(3) - 3,5(3) - 1(3)(1,5) = 0,858tm$$

Barra 4-6

$$M_6 = 6tm$$

$$M_4 = 6tm$$

Barra 3-4

$$M_3 = 0$$

$$M_4 = -6 - 3,5(3) + 5,286(3) - 1(3)(1,5) = -5,142$$

Barra 3-5

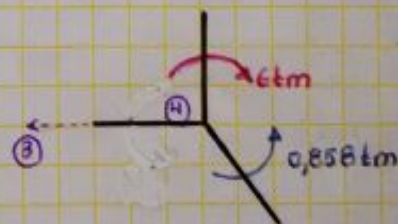
$$\bar{M}_{35} = \frac{qL^2}{8} = \frac{1(2)^2}{8}$$

$$\bar{M}_{35} = 0,5tm$$

Barra 5-6

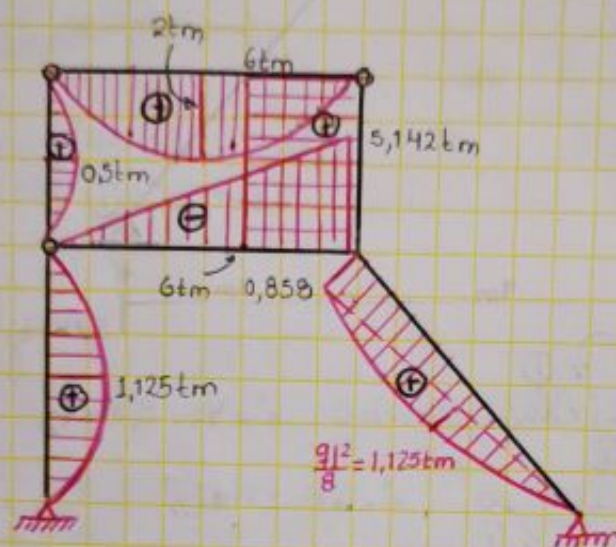
$$\bar{M}_{56} = \frac{1(4)^2}{8}$$

$$\bar{M}_{56} = 2tm$$

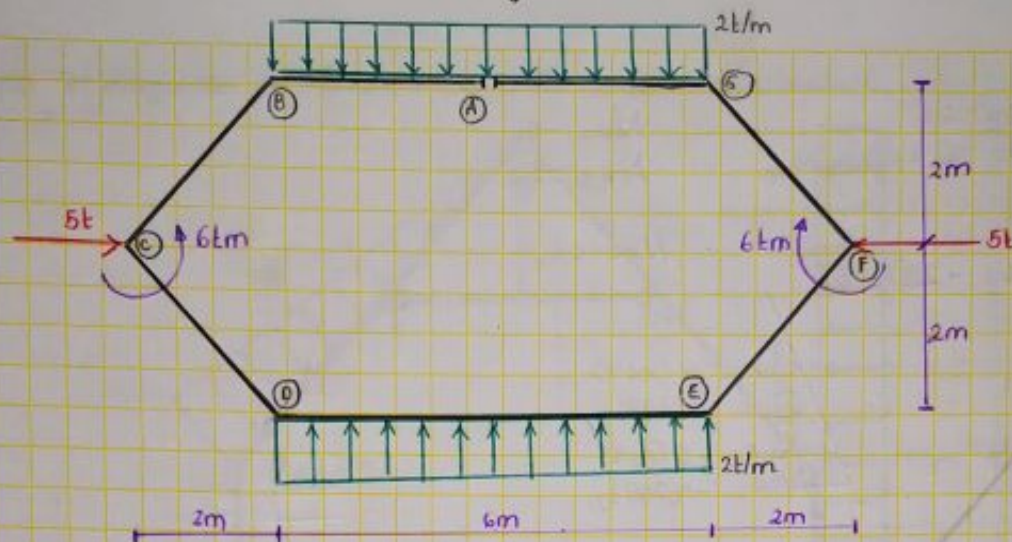


$$M_4 = -6 + 0,858 = -5,142tm$$

3. Diagrama



Para la estructura autosustentada diagramar los esfuerzos internos



1.- Momento

Barra AB $M_A = 0 \text{ tm}$
 $M_B = -2(3)(1,5) = -9 \text{ tm}$

Barra BC $M_B = -2(3)(1,5) = -9 \text{ tm}$
 $M_C = -21 \text{ tm}$

Barra CD $M_C = -6 + 2(3)(3,5) = 15 \text{ tm}$
 $M_D = 2(3)(1,5) + 5(2) - 6 = 13 \text{ tm}$

Barra DE $M_D = 13 \text{ tm}$
 $M_E = 2(3)(1,5) + 5(2) - 6 = 13 \text{ tm}$

Barra AG $M_A = 0$
 $M_G = -2(3)(1,5) = -9 \text{ tm}$

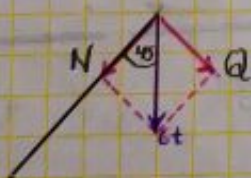
Barra GF $M_G = -9 \text{ tm}$
 $M_F = -2(3)(3,5) = -21 \text{ tm}$

Barra FE $M_F = 2(3)(3,5) - 6 = 15 \text{ tm}$
 $M_E = 2(3)(1,5) + 5(2) - 6 = 13 \text{ tm}$

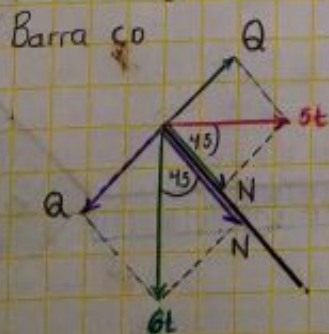
2.- Cortante y Normal

Barra AB $Q_A = 0$ $N_A = 0$
 $Q_B = 2(3) = 6 \text{ tm}$ $N_B = 0$

Barra BC



$Q_{BC} = 6 \cdot \text{Sen } 45 = 4,243 \text{ t}$
 $N_{BC} = -6 \cdot \text{Cos } 45 = -4,243 \text{ t}$



$Q_{CD} = 5 \text{ Sen } 45 - 6 \text{ Sen } 45 =$
 $Q_{CD} = -0,707 \text{ t}$

$N_{CD} = -5 \text{ Cos } 45 - 6 \text{ Cos } 45$
 $N_{CD} = -7,778 \text{ t}$

Barra DE

$$Q_D = -6t$$

$$Q_E = -6 + 2(6) = 6t_{//}$$

$$N_D = -5t$$

$$N_E = -5t$$

Barra AG

$$Q_A = 0t$$

$$Q_G = -2(3) = -6t$$

$$N_{AG} = 0t$$

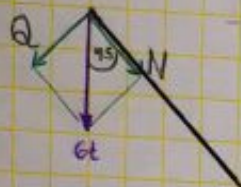
Barra GF

$$Q_{GF} = -6 \cdot \text{Sen } 45$$

$$Q_{GF} = -4,243t_{//}$$

$$N_{GF} = -6 \text{ Cos } 45$$

$$N_{GF} = -4,243t_{//}$$



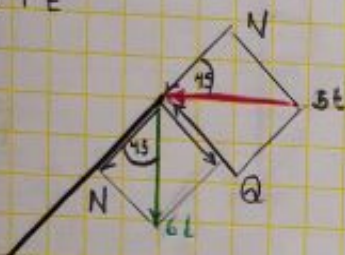
Barra FE

$$Q_{FE} = -5 \text{ Sen } 45 + 6 \cdot \text{Sen } 45$$

$$Q_{FE} = 0,707t_{//}$$

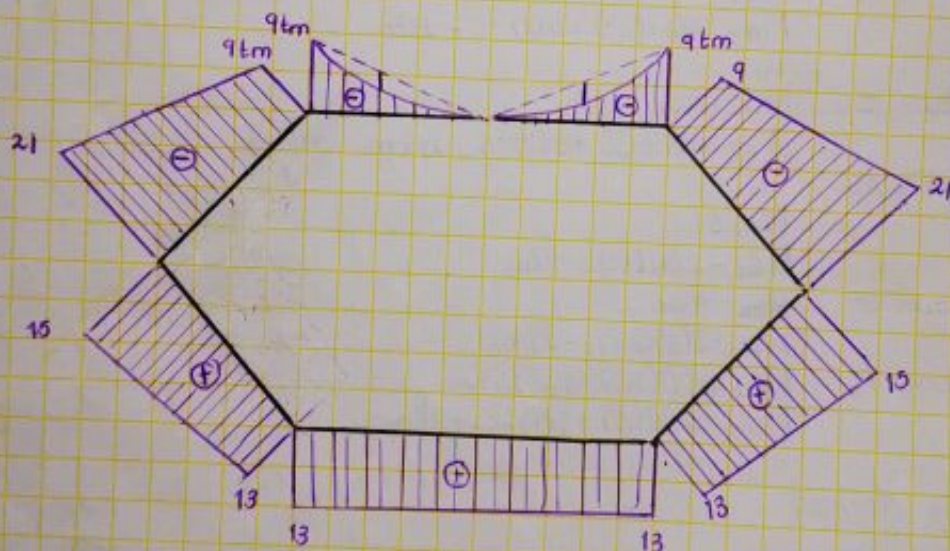
$$N_{FE} = -5 \text{ Cos } 45 - 6 \text{ Cos } 45$$

$$N_{FE} = -7,778t_{//}$$

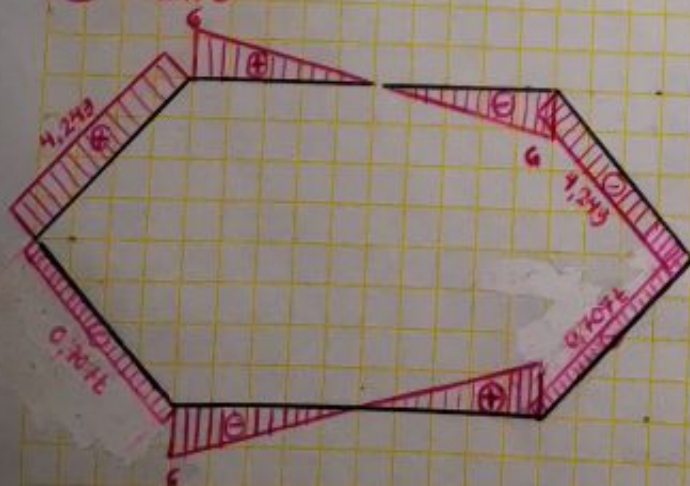


3. Diagramas

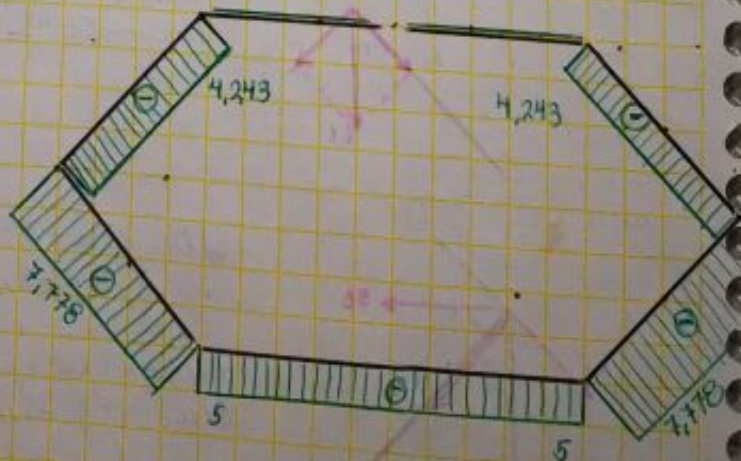
Momento



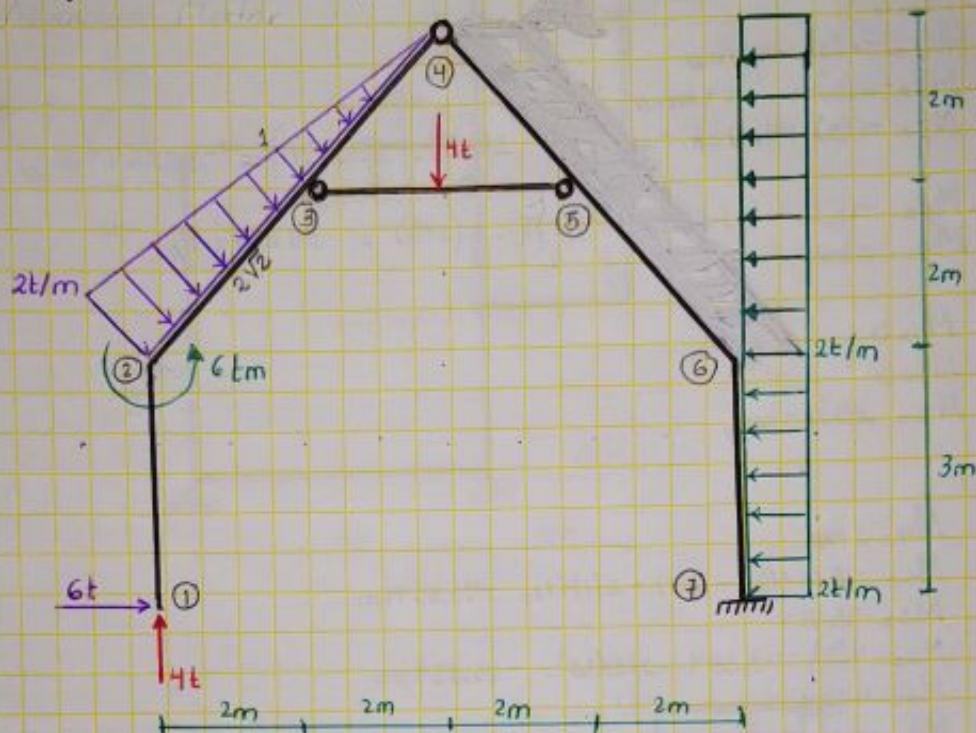
Constante



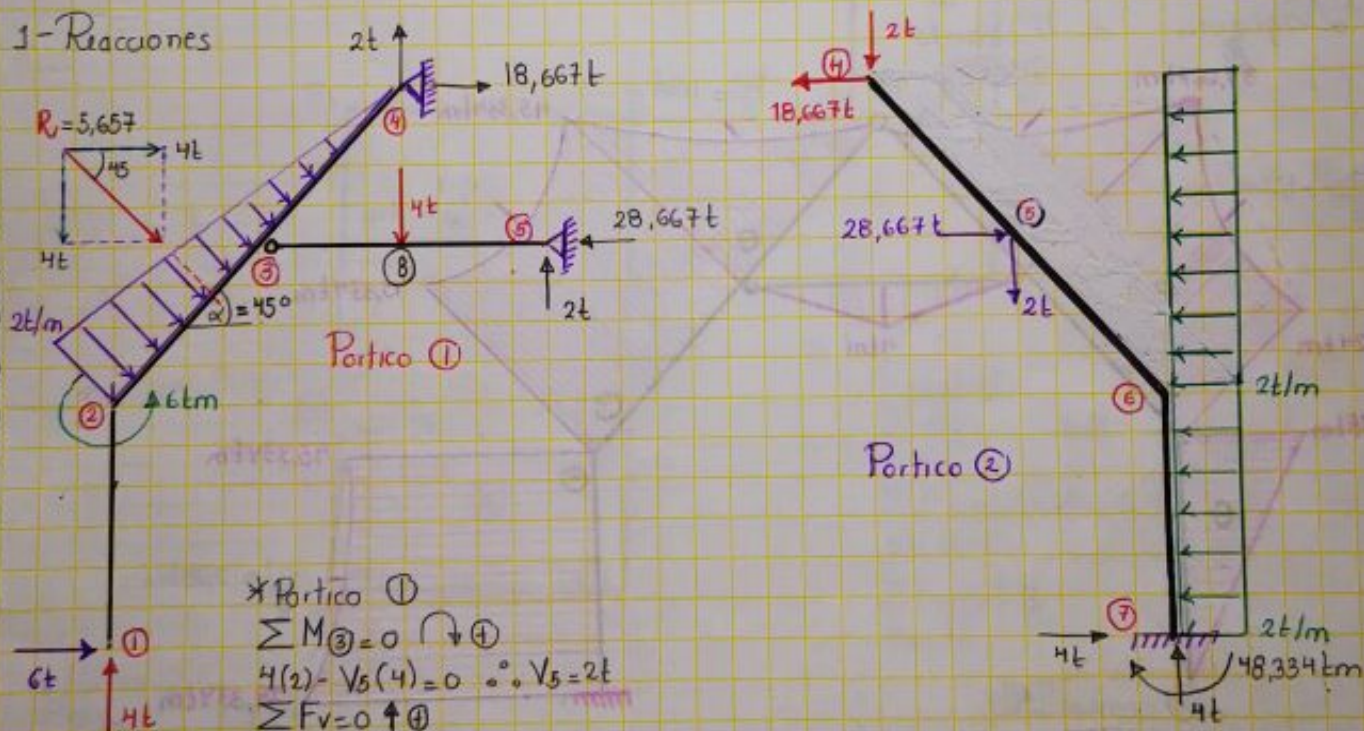
Normal



Obtener el diagrama de momento para la siguiente estructura



1-Reacciones



*Portico ①

$$\sum M_{(3)} = 0 \curvearrowright \textcircled{4}$$

$$4(2) - V_3(4) = 0 \quad \therefore V_3 = 2t$$

$$\sum F_v = 0 \uparrow \textcircled{4}$$

$$4 - 4 - 4 + 2 + V_4 = 0 \quad \therefore V_4 = 2t$$

$$\sum M_{(4)} = 0 \curvearrowright \textcircled{3}$$

$$-2(2) - 5,657 \cdot \left[\frac{2}{3} (4\sqrt{2}) \right] - 6 - 6(7) + 4(4) + H_5(2) = 0$$

$$H_5 = 28,667t$$

$$\sum F_H = 0 \rightarrow \textcircled{3}$$

$$+6 + 4 - 28,667 + H_4 = 0$$

$$H_4 = 18,667t //$$

*Portico ②

$$\sum F_v = 0 \uparrow \textcircled{6}$$

$$-2 - 2 + V_7 = 0 \quad \therefore V_7 = 4t$$

$$\sum F_H = 0 \rightarrow \textcircled{6}$$

$$-18,667 + 28,667 - 2(7) + H_7 = 0 \quad \therefore H_7 = 4t$$

$$\sum M_{(7)} = 0 \curvearrowright \textcircled{6}$$

$$-18,667(7) - 2(4) + 28,667(5) - 2(2) - 2(7)(3,5) + M_7 = 0$$

$$M_7 = 48,334tm //$$

2.- Momentos

Barra 1-2 $M_1 = 0$

$$M_2 = -6(3) = -18 \text{ tm} //$$

Barra 2-3 $M_2 = -6(3) - 6 = -24 \text{ tm} //$

$$M_3 = -6(5) + 4(2) - 6 - \frac{1(2\sqrt{2})^2}{2} - \frac{1(2\sqrt{2})}{2} \cdot \left[\frac{2}{3}(2\sqrt{2}) \right] = -34,667 \text{ tm} //$$

Barra 3-4 $M_3 = -18,667(2) + 2(2) - \frac{1(2\sqrt{2})}{2} \cdot \frac{1}{3}(2\sqrt{2}) = -34,667 \text{ tm} //$

$$M_4 = 0 \text{ tm}$$

Barra 3-5 $M_3 = 0$

$$M_5 = 0$$

$$M_8 = 2(2) = 4 \text{ tm} //$$

Pórtico 2

Barra 4-5

$$M_4 = 0 \text{ tm}$$

$$M_5 = -18,667(2) - 2(2) - 2(2)(1) = -45,334 \text{ tm}$$

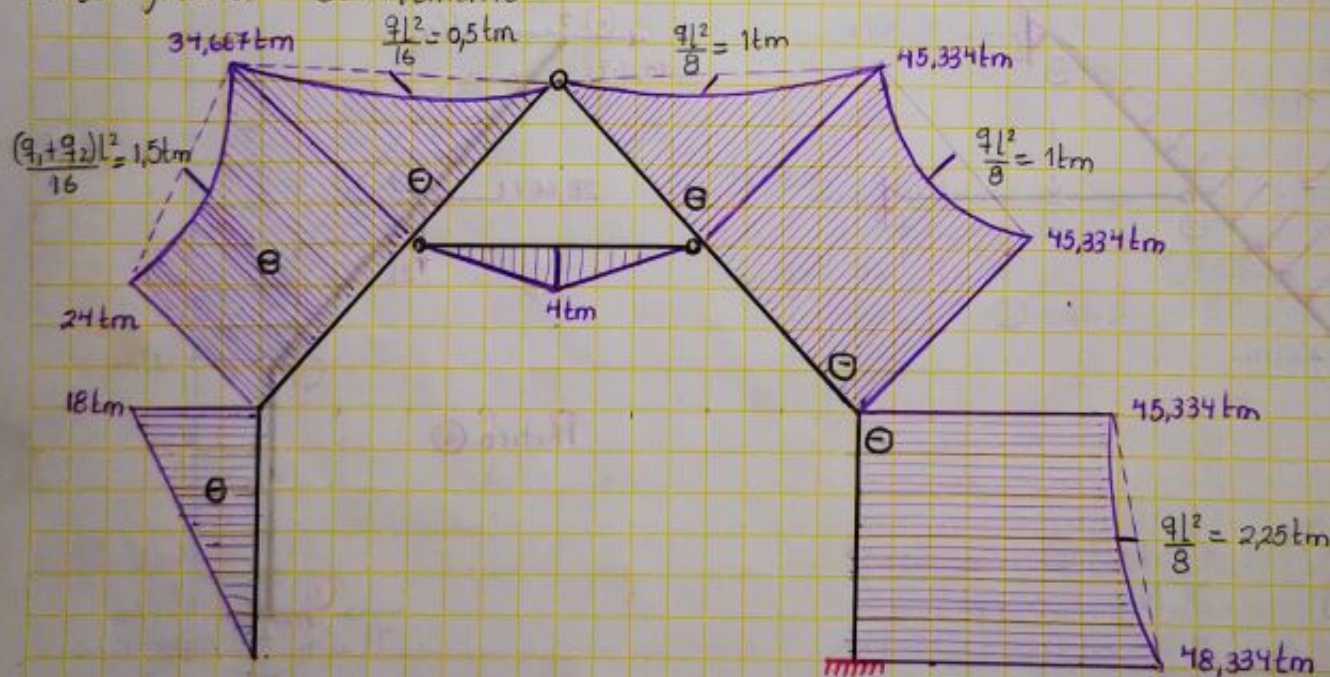
Barra 5-6 $M_5 = -45,334$

$$M_6 = 4(3) - 48,334 - 2(3)(1,5) = -45,334 \text{ tm}$$

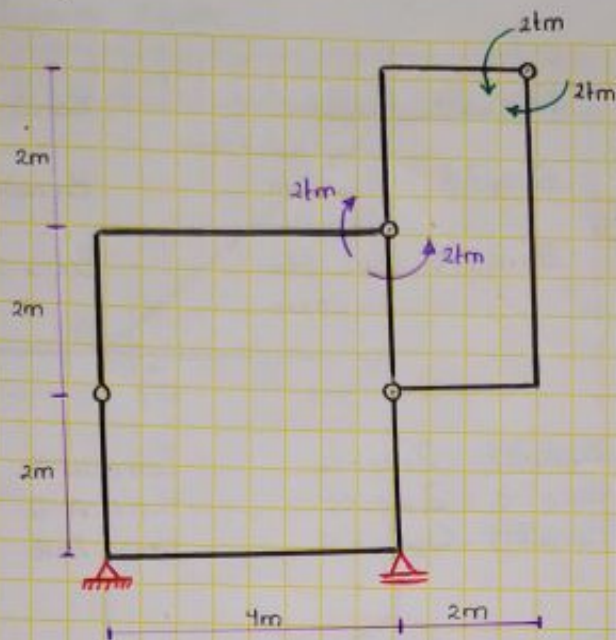
Barra 6-7 $M_6 = -45,334 \text{ tm}$

$$M_7 = -48,334 \text{ tm}$$

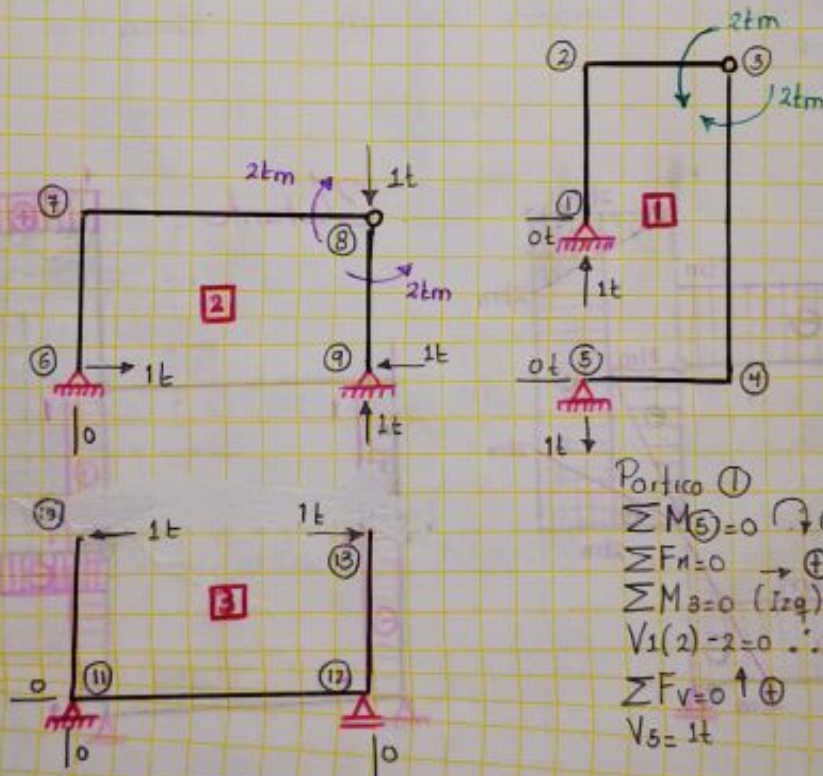
3.- Diagrama de Momento



Diagramar los esfuerzos internos



1. Reacciones



Portico 1

$$\begin{aligned} \sum M_5 &= 0 \quad \curvearrowright \oplus \Rightarrow H_1 = 0 \\ \sum F_H &= 0 \quad \rightarrow \oplus \Rightarrow H_5 = 0t \\ \sum M_3 &= 0 \quad (\text{Izq}) \quad \curvearrowright \oplus \\ V_1(2) - 2 &= 0 \quad \therefore V_1 = 1t // \\ \sum F_V &= 0 \quad \uparrow \oplus \\ V_5 &= 1t \end{aligned}$$

*Portico 2

$$\begin{aligned} \sum M_4 &= 0 \quad \curvearrowright \oplus \\ 1(4) - V_9(4) &= 0 \quad \therefore V_9 = 1t \\ \sum F_V &= 0 \quad \uparrow \oplus \\ V_6 &= 0t \\ \sum M_8 &= 0 \quad \curvearrowright \oplus \quad (\text{abajo}) \\ -2 + H_9(2) &= 0 \quad \therefore H_9 = 1t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ H_6 &= 1t // \end{aligned}$$

2- Momentos

Barra 1-2 $M_1 = 0 \text{ tm}$
 $M_2 = 0 \text{ tm}$

Barra 2-3 $M_2 = 0 \text{ tm}$
 $M_3 = 1(2) = 2 \text{ tm}$

Barra 3-4 $M_3 = 2 \text{ tm}$
 $M_4 = 1(2) = 2 \text{ tm}$

Barra 4-5 $M_4 = -1(2) = -2 \text{ tm}$
 $M_5 = 0 \text{ tm}$

Barra 6-7 $M_6 = 0 \text{ tm}$
 $M_7 = -1(2) = -2 \text{ tm}$

Barra 7-8 $M_7 = -2 \text{ tm}$
 $M_8 = -2 \text{ tm}$

Barra 8-9 $M_8 = -2 \text{ tm}$
 $M_9 = 0 \text{ tm}$

Barra 10-11 $M_{10} = 0 \text{ tm}$
 $M_{11} = -1(2) = -2 \text{ tm}$

Barra 11-12 $M_{11} = -2 \text{ tm}$
 $M_{12} = -2 \text{ tm}$

Barra 12-13 $M_{12} = -2 \text{ tm}$
 $M_{13} = 0 \text{ tm}$

3- Cortantes

Barra 1-2 $Q_{1-2} = 0 \text{ t}$

Barra 2-3 $Q_{2-3} = 1 \text{ t}$

Barra 3-4 $Q_{3-4} = 0 \text{ t}$

Barra 5-4 $Q_{5-4} = -1 \text{ t}$

Barra 6-7 $Q_{6-7} = -1 \text{ t}$

Barra 7-8 $Q_{7-8} = 0 \text{ t}$

Barra 8-9 $Q_{8-9} = +1 \text{ t}$

Barra 10-11 $Q_{10-11} = -1 \text{ t}$

Barra 11-12 $Q_{11-12} = 0 \text{ t}$

Barra 12-13 $Q_{12-13} = 1 \text{ t}$

4- Normales

Barra 1-2 $N_{1-2} = -1 \text{ t}$

Barra 2-3 $N_{2-3} = 0 \text{ t}$

Barra 3-4 $N_{3-4} = 1 \text{ t}$

Barra 4-5 $N_{4-5} = 0 \text{ t}$

Barra 6-7 $N_{6-7} = 0 \text{ t}$

Barra 7-8 $N_{7-8} = -1 \text{ t}$

Barra 8-9 $N_{8-9} = -1 \text{ t}$

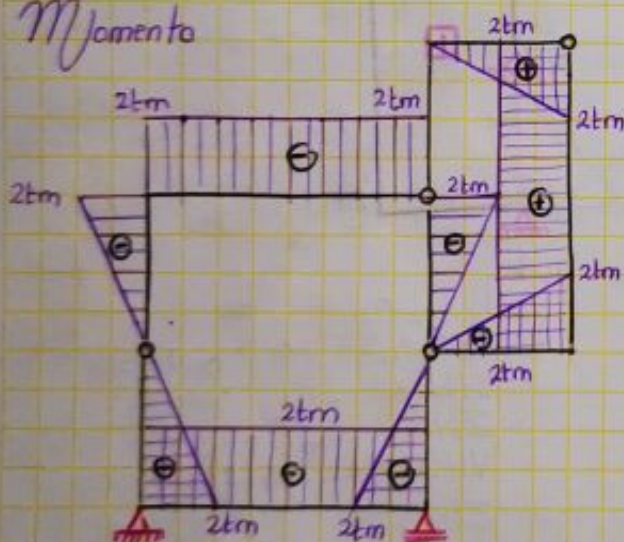
Barra 10-11 $N_{10-11} = 0 \text{ t}$

Barra 11-12 $N_{11-12} = 1 \text{ t}$

Barra 12-13 $N_{12-13} = 0 \text{ t}$

6- Diagramas

Momento



Cortante



Normal

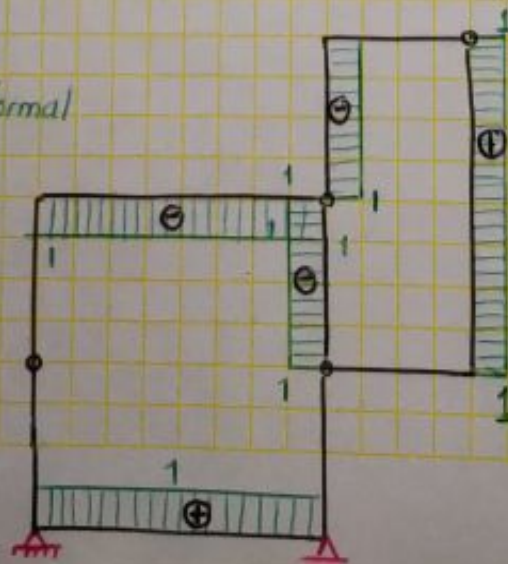
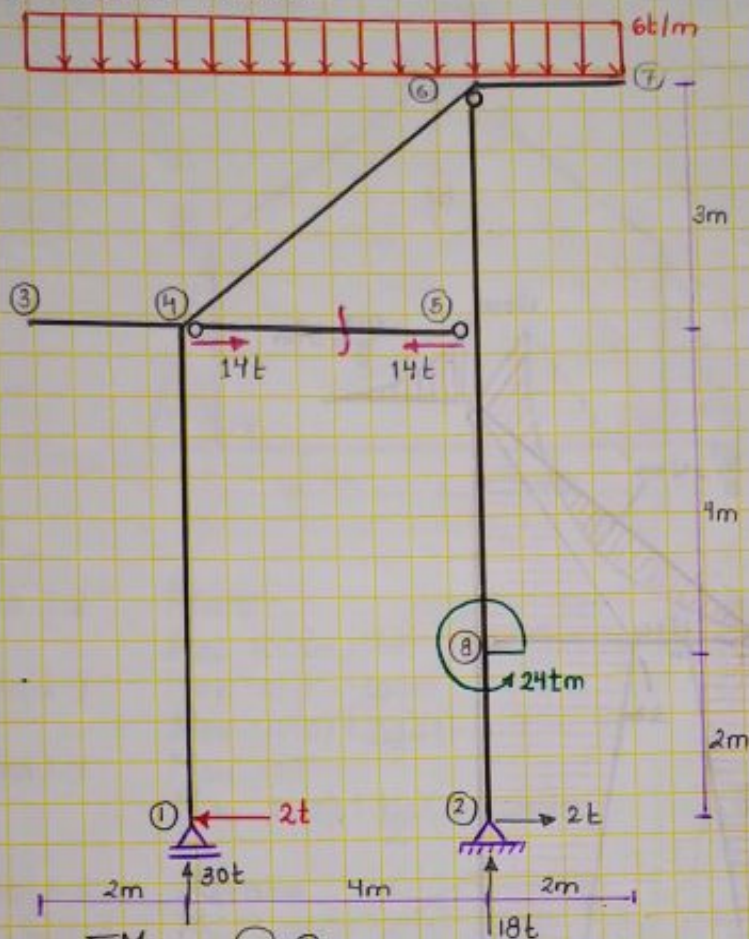


Diagrama Momento flector



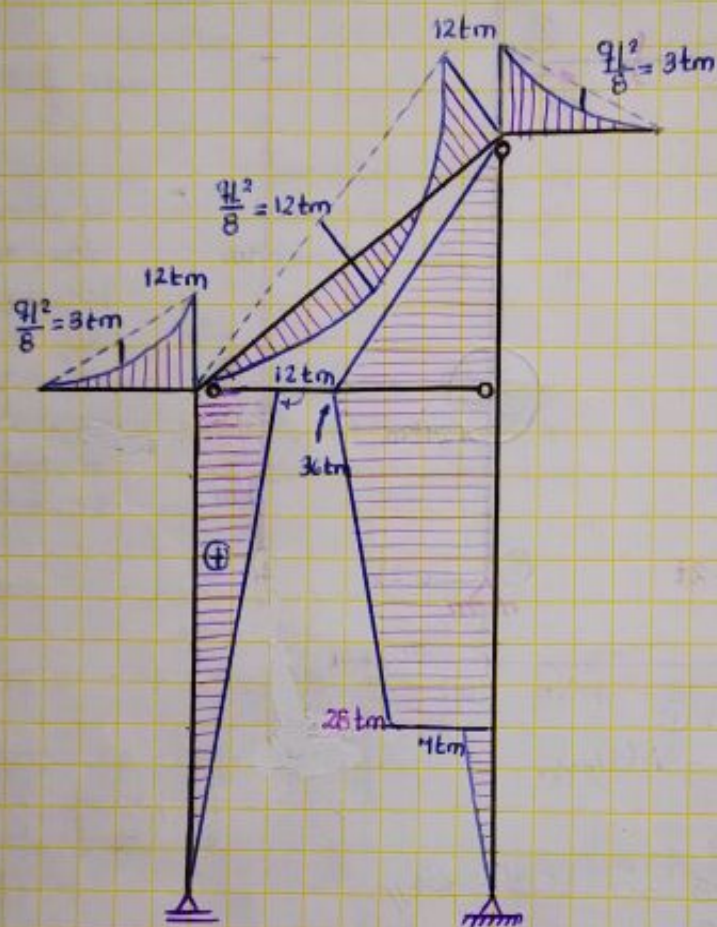
1.- Reacciones

$$\begin{aligned} \sum M_{\odot} &= 0 \quad \curvearrowright \oplus \\ 6(8)(2) - 24 - V_2(4) &= 0 \\ V_2 &= 18t // \\ \sum F_v &= 0 \quad \uparrow \oplus \\ V_1 - 6(8) + 18 &= 0 \quad \therefore V_1 = 30t // \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ H_2 &= 2t // \\ \sum M_{\odot} &= 0 \quad \curvearrowright \oplus \text{ (abajo)} \\ N_{5-6}(3) - 24 - 2(9) &= 0 \quad \therefore N_{5-6} = 14t \end{aligned}$$

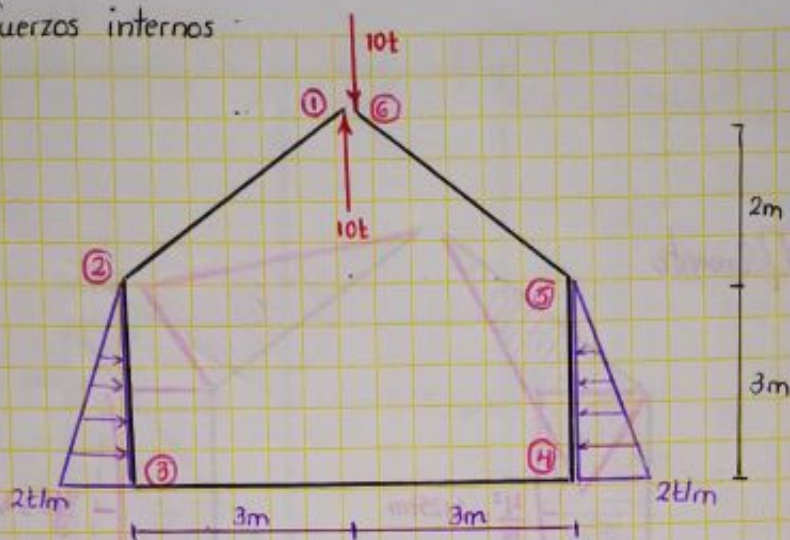
2.- Momento

Barra 1-4	$M_1 = 0$ $M_4 = 2(6) = 12tm$
Barra 3-4	$M_3 = 0$ $M_4 = -6(2)(1) = -12tm$
Barra 4-6	$M_4 = 2(6) - 6(2)(1) = 0tm$ $M_6 = 30(4) + 2(9) - 14(3) - 6(6)(3) = -12tm$
Barra 6-7	$M_6 = -6(2)(1) = -12tm$ $M_7 = 0tm$
Barra 2-8	$M_2 = 0tm$ $M_8 = 2(2) = 4tm$
Barra 8-5	$M_8 = 2(2) + 24 = 28tm$ $M_5 = 2(6) + 24 = 36tm$
Barra 5-6	$M_5 = 2(9) + 24 - 14(3) = 0tm$

3.- Diagrama de Momento



Diagramar los esfuerzos internos

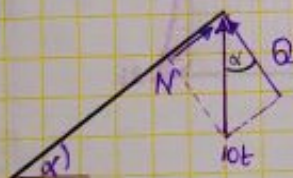


- Momentos

Barra 1-2 $M_1 = 0$
 $M_2 = 10(3) = 30 \text{ tm}$
 Barra 2-3 $M_2 = -10(3) = -30 \text{ tm}$
 $M_3 = -10(3) + \frac{2(3)(1)}{2} = -27 \text{ tm}$
 Barra 3-4 $M_3 = -27 \text{ tm}$
 $M_4 = 10(3) + \frac{2(3)(1)}{2} = 33 \text{ tm}$
 Barra 5-6 $M_6 = 0 \text{ tm}$
 $M_5 = -10(3) = -30 \text{ tm}$
 Barra 5-4 $M_5 = 10(3) = 30 \text{ tm}$
 $M_4 = 10(3) + \frac{2(3)(1)}{2} = 33 \text{ tm}$

- Cortante y Normal

Barra 1-2



$$\alpha = \arctg\left(\frac{2}{3}\right) = 33,69^\circ$$

$$Q_{1-2} = -10 \cdot \cos 33,69 = -8,321 \text{ t}$$

$$N_{1-2} = 10 \cdot \sin 33,69 = 5,547 \text{ t}$$

Barra 2-3

$$Q_2 = 0 \text{ t}$$

$$Q_3 = \frac{2(3)}{2} = 3 \text{ t}$$

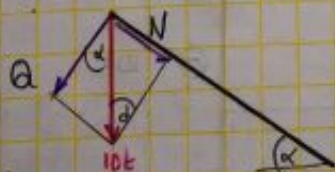
$$N_{2-3} = 10 \text{ t}$$

Barra 3-4

$$Q_{3-4} = 10 \text{ t}$$

$$N_{3-4} = -\frac{2(3)}{2} = -3 \text{ t}$$

Barra 6-5



$$Q_{6-5} = -10 \cdot \cos 33,69 = -8,32 \text{ t}$$

$$N_{6-5} = -10 \cdot \sin 33,69 = -5,55 \text{ t}$$

Barra 5-4

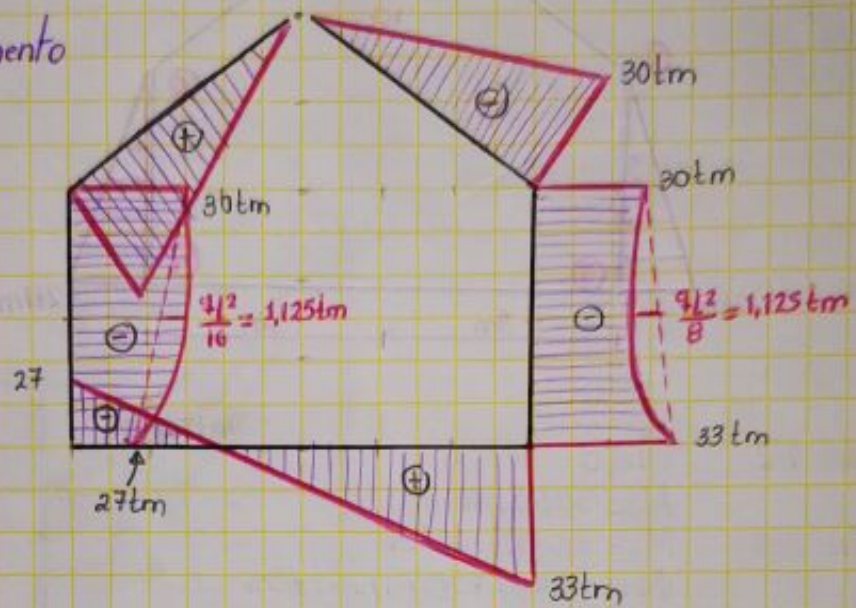
$$Q_5 = 0 \text{ t}$$

$$Q_4 = -\frac{2(3)}{2} = -3 \text{ t}$$

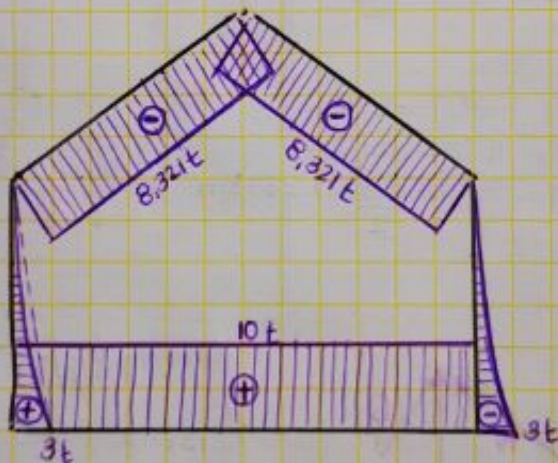
$$N_{5-4} = -10 \text{ t}$$

3. Diagramas

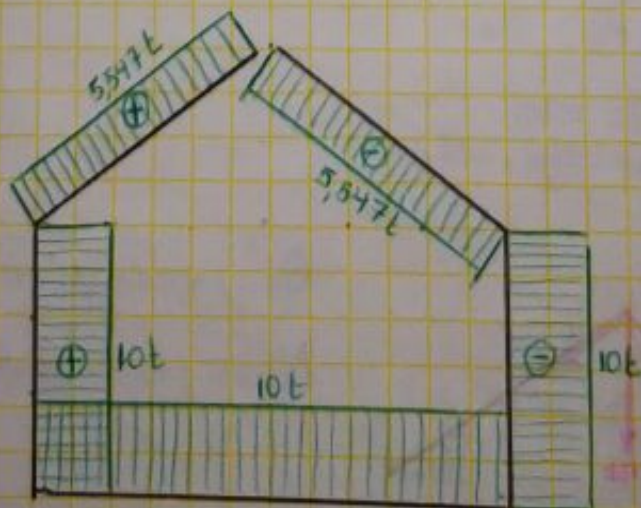
Momento



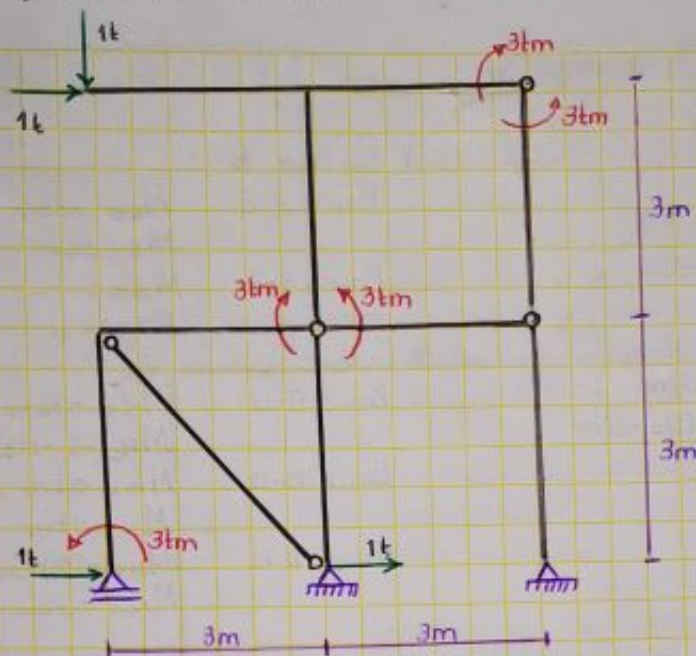
Cortante



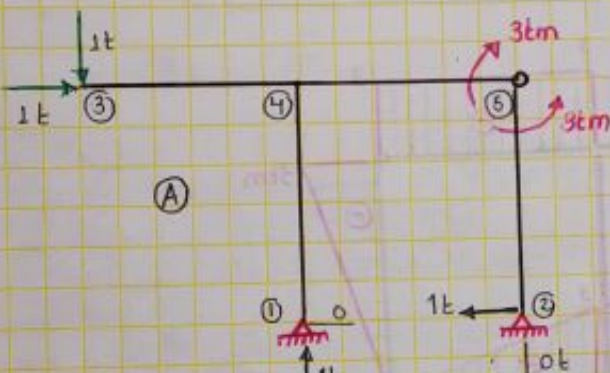
Normal



Obtener el diagrama de Momento



1. Reacciones



* Portico A

$$\sum M_{\odot} = 0 \quad \curvearrowright \oplus$$

$$1(3) - 1(3) + V_2(3) = 0 \quad \therefore V_2 = 0t$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_1 = 1t$$

$$\sum M_{\odot} = 0 \quad \curvearrowright \oplus \text{ (abajo)}$$

$$-3 + H_2(3) = 0 \quad \therefore H_2 = 1t$$

$$\sum F_H = 0 \quad \rightarrow \oplus \quad \therefore H_1 = 0$$

* Portico B

$$\sum M_{\odot} = 0 \quad \curvearrowright \oplus \text{ (abajo)}$$

$$H_2 = 0t$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

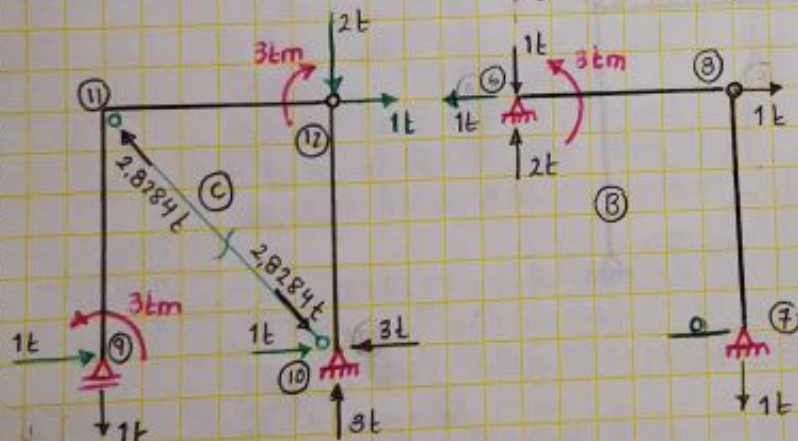
$$H_6 = 1t$$

$$\sum M_{\odot} = 0 \quad \curvearrowright \oplus$$

$$-3 + V_7(3) = 0 \quad \therefore V_7 = 1t$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_6 = 2t$$



* Portico C

$$\sum M_{\odot} = 0 \quad \curvearrowright \oplus$$

$$-3 + 3 + 2(3) + 1(3) - V_{10}(3) = 0$$

$$V_{10} = 3t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$-V_9 - 2 + 3 = 0 \quad \therefore V_9 = 1t$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$H_{10} = 3t$$

$$\sum M_{\odot} = 0 \quad \curvearrowright \oplus \text{ (abajo)}$$

$$2(3) - N_{10-11} \cdot (1.5\sqrt{2}) = 0$$

$$N_{10-11} = 2.8284t //$$

3.- Momentos

* Portico A

Barra 1-4 $M_{1-4} = 0 \text{ tm}$

Barra 2-5 $M_{2-5} = 0 \text{ tm}$

$M_5 = -1(3) = -3 \text{ tm}$

Barra 3-4 $M_{3-4} = 0 \text{ tm}$

$M_4 = -1(3) = -3 \text{ tm}$

Barra 4-5 $M_{4-5} = -1(3) = -3 \text{ tm}$

$M_3 = -1(6) + 1(3) = -3 \text{ tm}$

* Portico B

Barra 6-8 $M_{6-8} = -3 \text{ tm}$

$M_8 = 0 \text{ tm}$

Barra 8-7 $M_{8-7} = 0$

$M_7 = 0 \text{ tm}$

* Portico C

Barra 9-11 $M_{9-11} = -3 \text{ tm}$

$M_{11} = -3 - 1(3) = -6 \text{ tm}$

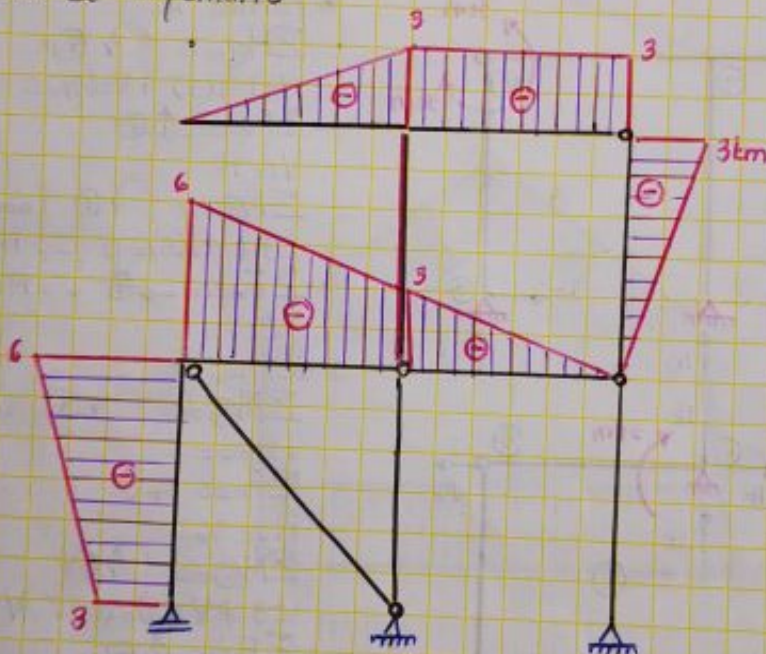
Barra 10-12 $M_{10-12} = 0 \text{ tm}$

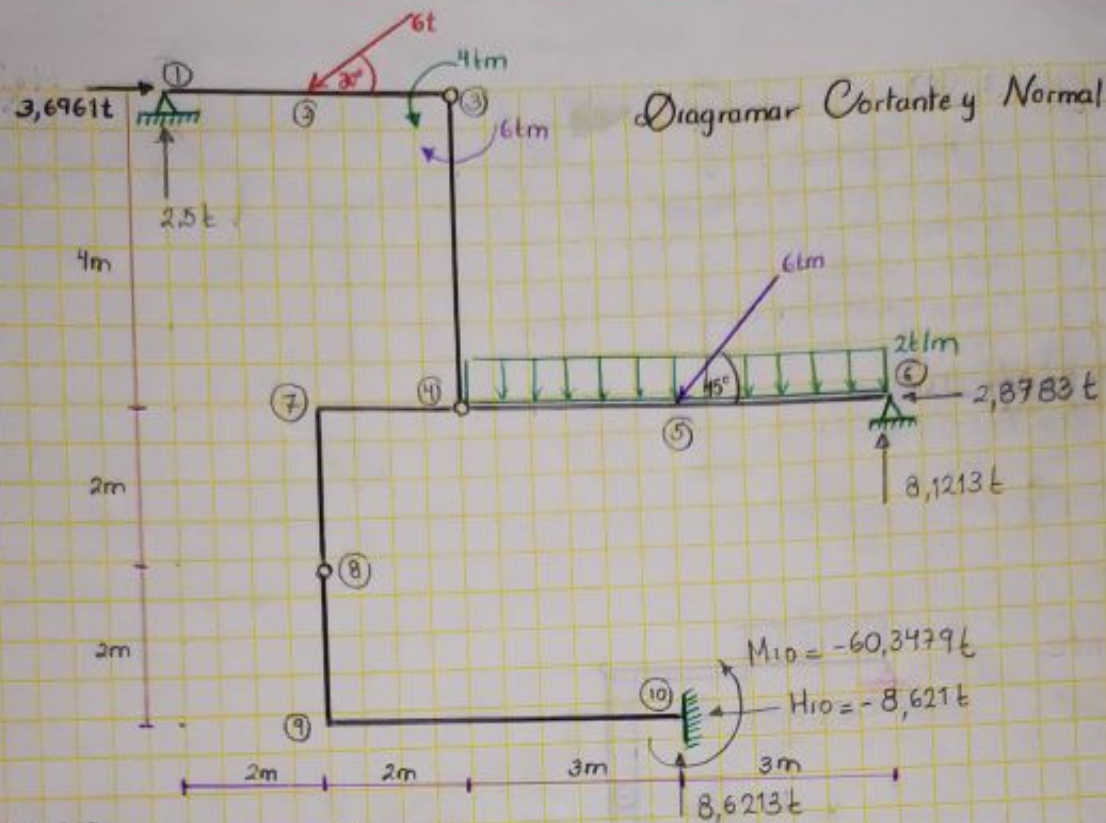
$M_{12} = 0 \text{ tm}$

Barra 11-12 $M_{11-12} = -1(3) - 3 = -6 \text{ tm}$

$M_{12} = -3 \text{ tm}$

4.- Diagrama de Momento





1 Reacciones

$$\begin{aligned} \sum M_{(3)} &= 0 \quad \curvearrowright \text{ (Barra 1-3)} \\ +V_1(4) - 4 - 6 \cdot \text{Sen} 30 \cdot 2 &= 0 \quad \therefore V_1 = 2.5t // \\ \sum M_{(4)} &= 0 \quad \curvearrowright \text{ (Barra 4-6)} \\ 6 \cdot \text{Sen} 45 (3) + 2(6)(3) - V_6(6) &= 0 \quad \therefore V_6 = 8.1213t // \\ \sum F_v &= 0 \quad \uparrow \text{ (4)} \\ 2.5 - 6 \cdot \text{Sen} 30 - 2(6) - 6 \cdot \text{Sen} 45 + 8.1213 + V_{10} &= 0 \quad \therefore V_{10} = 8.6213t // \\ \sum M_{(8)} &= 0 \quad \curvearrowright \text{ (abajo)} \\ -M_{10} - 8.6213(5) + H_{10}(2) &= 0 \quad \text{①} \\ \sum M_{(7)} &= 0 \quad \curvearrowright \text{ (izda)} \\ -8.6213(3) + H_{10}(4) - M_{10} &= 0 \\ -M_{10} &= 25.8639 - 4H_{10} \quad \text{(en ①)} \\ 25.8639 - 4H_{10} - 8.6213(5) + 2H_{10} &= 0 \quad \therefore H_{10} = -8.621t // \\ -M_{10} &= 25.8639 - 4(-8.621) \quad \therefore M_{10} = -60.3479tm \\ \sum M_{(10)} &= 0 \quad \curvearrowright \text{ (arriba)} \\ H_1(4) + 2.5(4) - 6 \cdot \text{Sen} 30(2) - 6 \cdot \text{Cos} 30(4) - 4 + 6 &= 0 \quad \therefore H_1 = 3.6961t \\ \sum F_H &= 0 \quad \rightarrow \text{ (4)} \\ 3.6961 - 6 \cdot \text{Cos} 30 - 6 \cdot \text{Cos} 45 + H_6 + 8.621 &= 0 \quad \therefore H_6 = 2.8783t \end{aligned}$$

2 - Cortante

Barra 1-2 $Q_1 = 2.5 //$

$Q_2 = 2.5 //$

Barra 2-3 $Q_2 = 2.5 - 6 \cdot \text{Sen} 30 = -0.5t$

$Q_3 = -0.5t //$

Barra 3-4 $Q_3 = 3.6961 - 6 \cdot \text{Cos} 30 = -1.5t //$

$Q_4 = -1.5t //$

Barra 5-6 $Q_6 = -8.1213t$

$Q_5 = -8.1213 + 2(3) = -2.1213t //$

Barra 5-4 $Q_5 = -8.1213 + 2(3) + 6 \cdot \text{Sen} 45 = 2.1213t //$

$Q_4 = 2.1213 + 2(3) = 8.1213t //$

Barra 9-10

$Q_{10} = -8.6213t //$

$Q_9 = -8.6213t //$

Barra 9-7 $Q_9 = -8.621t //$

$Q_7 = -8.621t //$

Barra 7-4 $Q_7 = 8.6213t //$

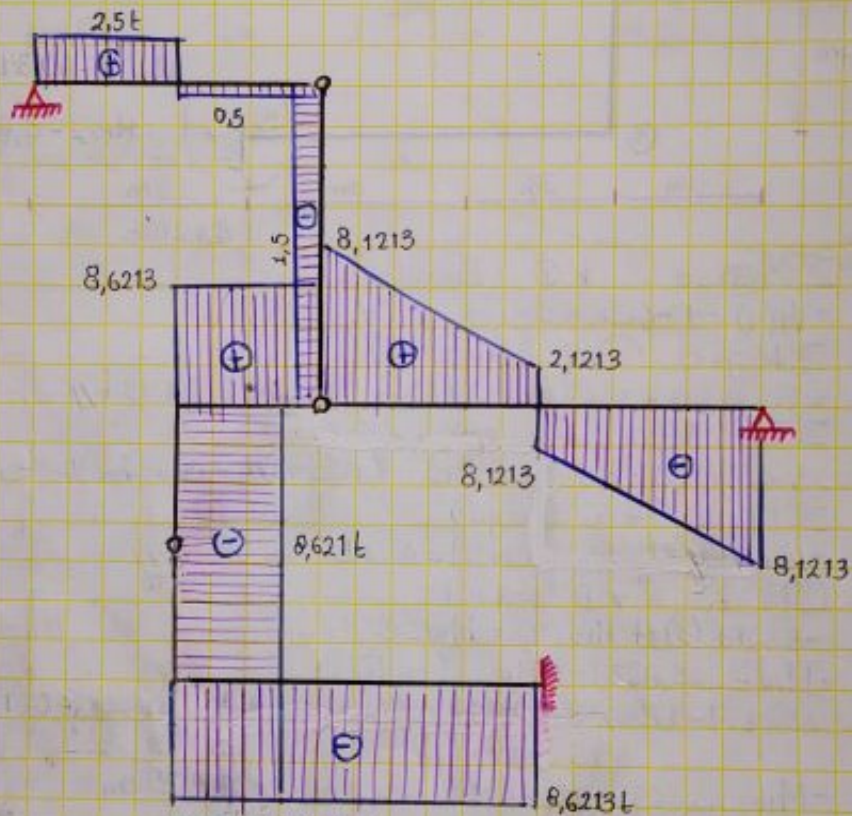
$Q_4 = 8.6213t //$

3 - Normales

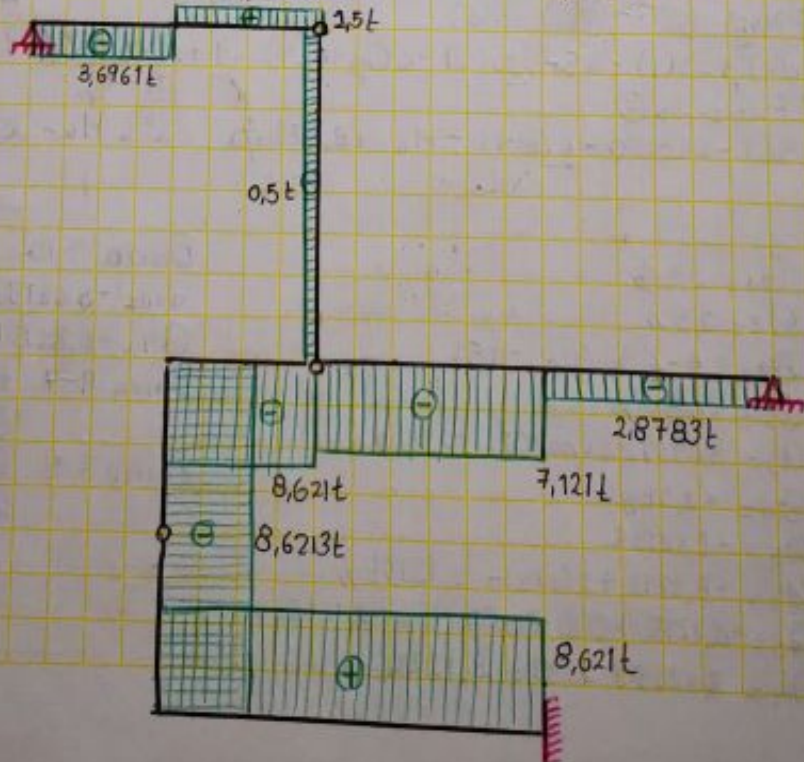
Barra 1-2 $N_{1-2} = -3,6961t //$
 Barra 2-3 $N_{2-3} = -3,6961 + 6 \cos 30 = 1,5t //$
 Barra 3-4 $N_{3-4} = 2,5 - 6 \sin 30 = -0,5t //$
 Barra 5-6 $N_{5-6} = -2,8783t //$
 Barra 4-5 $N_{4-5} = -2,8783 - 6 \cos 45 = -7,121t //$
 Barra 9-10 $N_{9-10} = 8,621t //$
 Barra 7-9 $N_{7-9} = -8,6213t //$
 Barra 7-4 $N_{7-4} = -8,621t //$

4 - Diagramas

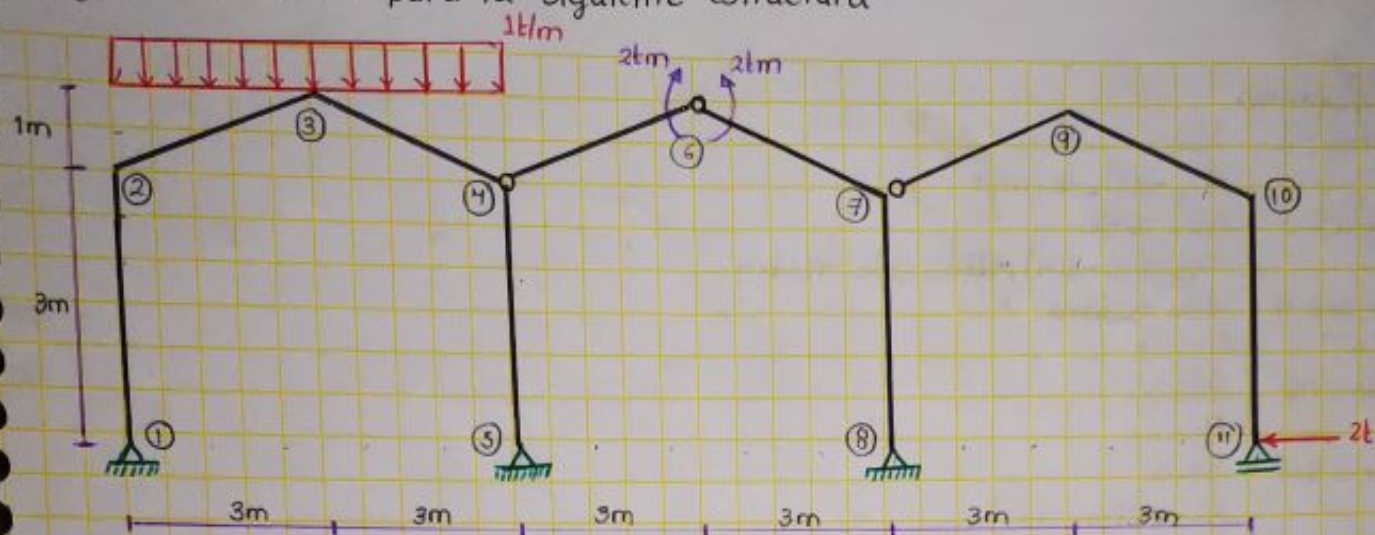
Cortante



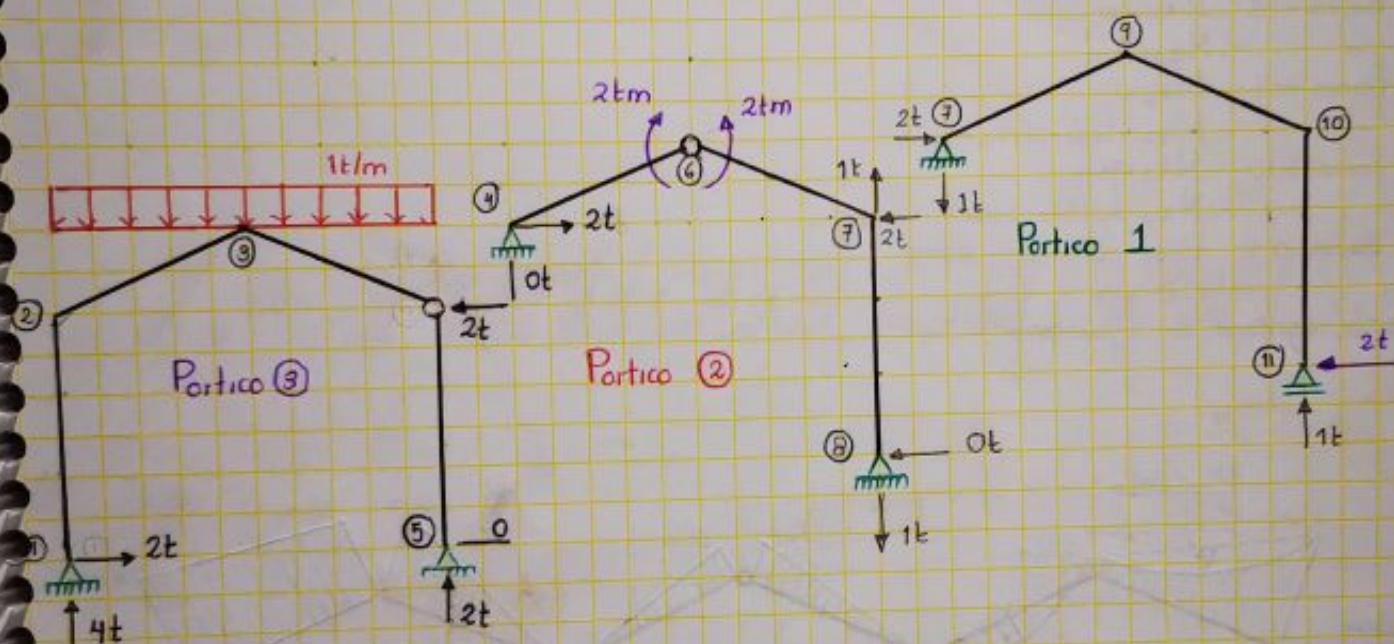
Normal



Diagramar Momento para la siguiente estructura



1.- Separar la estructura



* Portico ①

$$\sum M_7 = 0 \quad (\curvearrowright \oplus)$$

$$2(3) - V_{11}(6) = 0 \quad \therefore V_{11} = 1t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_7 = 1t$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$H_7 = 2t$$

* Portico ③

$$\sum M_{10} = 0 \quad (\curvearrowright \oplus)$$

$$1(6)(3) - 2(3) - V_5(6) = 0 \quad \therefore V_5 = 2t //$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_1 - 1(6) + 2 = 0 \quad \therefore V_1 = 4t //$$

$$\sum M_{11} = 0 \quad (\curvearrowright \oplus) \text{ (abajo)}$$

$$H_5 = 0t$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$H_7 = 2t$$

* Portico ②

$$\sum M_6 = 0 \quad (\curvearrowright \oplus) \text{ (22Q)}$$

$$-H_4(1) - V_4(3) + 2 = 0$$

$$H_4 = 2 - 3V_4 \quad \text{①}$$

$$\sum M_8 = 0 \quad (\curvearrowright \oplus)$$

$$H_4(3) - V_4(6) - 2(3) = 0$$

$$3H_4 - 6V_4 - 6 = 0 \quad \text{②}$$

Reemplazamos ① en ②

$$3(2 - 3V_4) - 6V_4 - 6 = 0$$

$$6 - 9V_4 - 6V_4 - 6 = 0$$

$$V_4 = 0t //$$

$$H_4 = 2 - 3(0) = 2t //$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$H_8 - 2 + 2 = 0 \quad \therefore H_8 = 0t$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_8 = 1t$$

2 - Momentos

Barra 1-2 $M_1 = 0$

$M_2 = -2(3) = -6 \text{ tm}$

Barra 2-3 $M_2 = -6 \text{ tm}$

$M_3 = 4(3) - 2(4) - \frac{1}{2}(3)(1,5) = -0,5 \text{ tm}$

Barra 3-4 $M_3 = -0,5 \text{ tm}$

$M_4 = 0 \text{ t}$

Barra 4-5 $M_4 = 0$

$M_5 = 0$

Barra 4-6 $M_4 = 0$

$M_6 = -2 \text{ tm}$

Barra 6-7 $M_6 = -2 \text{ tm}$

$M_7 = 0 \text{ tm}$

Barra 7-8 $M_7 = 0 \text{ tm}$

$M_8 = 0 \text{ tm}$

Barra 7-9 $M_7 = 0$

$M_9 = -1(3) - 2(1) = -5 \text{ tm}$

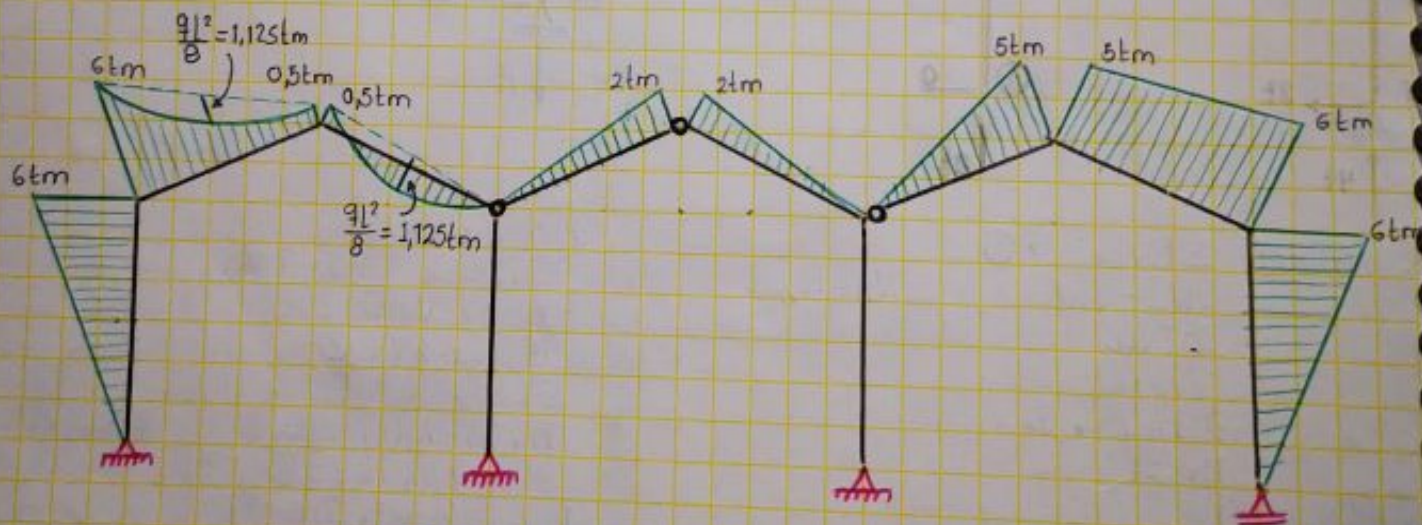
Barra 9-10 $M_9 = -5 \text{ tm}$

$M_{10} = -2(3) = -6 \text{ tm}$

Barra 10-11 $M_{10} = -6 \text{ tm}$

$M_{11} = 0 \text{ tm}$

3 - Diagrama de Momento



Para la siguiente estructura obtener el diagrama de Momento

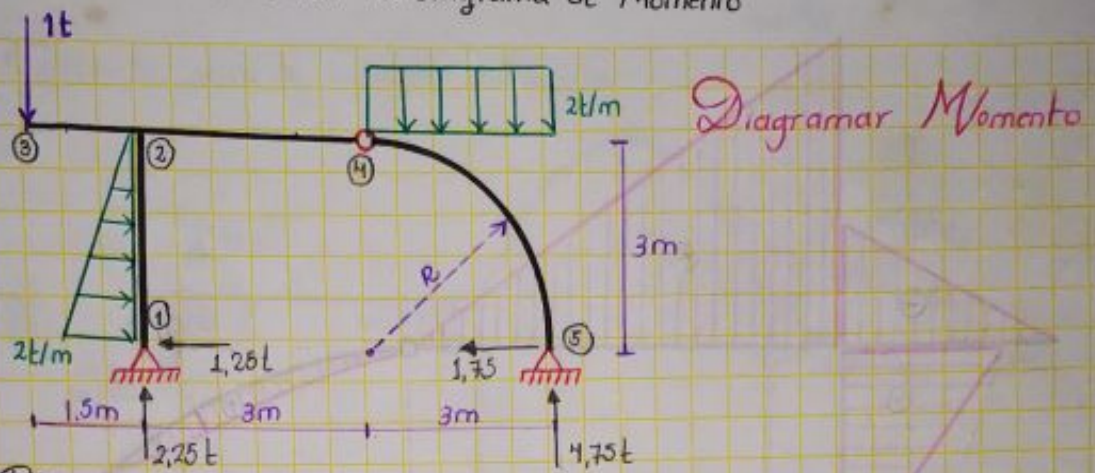


Diagrama Momento

1- Cálculo de Reacciones

$$\begin{aligned}\sum M_O &= 0 \quad \curvearrowright \oplus \\ \frac{2(3)(1)}{2} - 1(1.5) + 2(3)(4.5) - V_5(6) &= 0 \\ V_5 &= 4.75t \\ \sum F_v &= 0 \quad \uparrow \oplus \\ -1 + V_1 - 2(3) + 4.75 &= 0 \\ V_1 &= 2.25t \\ \sum M_H \text{ (DER)} &= 0 \quad \curvearrowright \oplus \\ 2(3)(1.5) - 4.75(3) + H_5(3) &= 0 \\ H_5 &= 1.75t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ \frac{2(3)}{2} - H_1 - 1.75 &= 0 \\ H_1 &= 1.25t\end{aligned}$$

2- Momentos.

Barra 1-2

$$\begin{aligned}M_1 &= 0 \\ M_2 &= 1.25(3) - \frac{2(3) \cdot 2}{2} = -2.25tm\end{aligned}$$

Barra 3-2

$$\begin{aligned}M_3 &= 0 \\ M_2 &= -1(1.5) = -1.5tm\end{aligned}$$

Barra 2-4

$$M_2 = 1.25(3) - \frac{2(3) \cdot 2}{2} - 1(1.5) = -3.75tm$$

$$M_4 = 0tm$$

$$M_{\theta=30} = 1.25 \cdot d_1 - 2 \cdot d_1 \cdot \frac{d_1}{2} + 1.75(R - R \cos \alpha)$$

$$M_{\theta=30} = 1.25(1.5) - (1.5)^2 + 1.75(3 - 3 \cos 30)$$

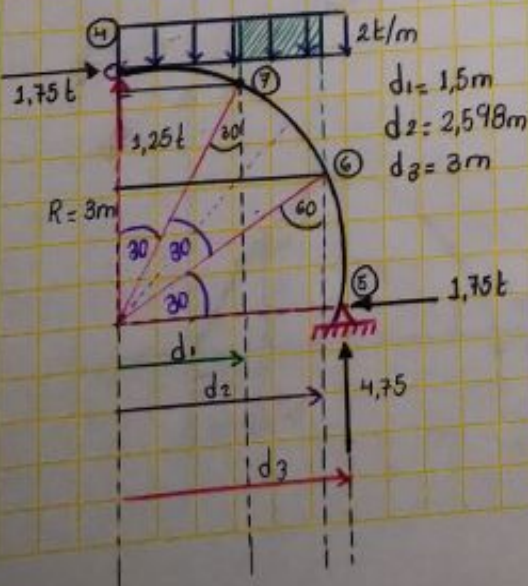
$$M_{\theta=30} = 0.3285tm$$

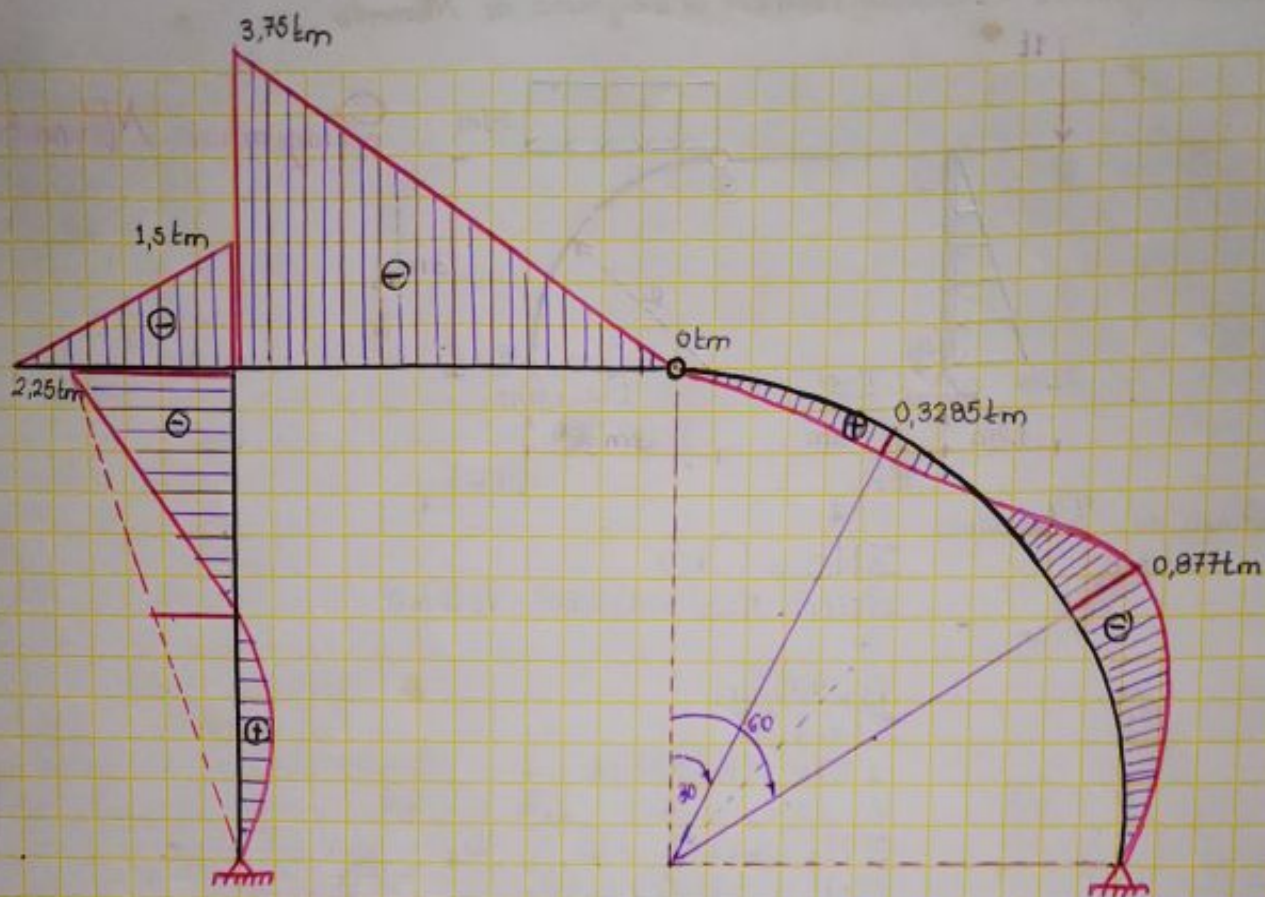
$$M_{\theta=60} = 1.25 \cdot d_2 - 2 \cdot d_2 \cdot \frac{d_2}{2} + 1.75(R - R \cos \alpha)$$

$$M_{\theta=60} = 1.25(2.598) - (2.598)^2 + 1.75(3 - 3 \cos 60)$$

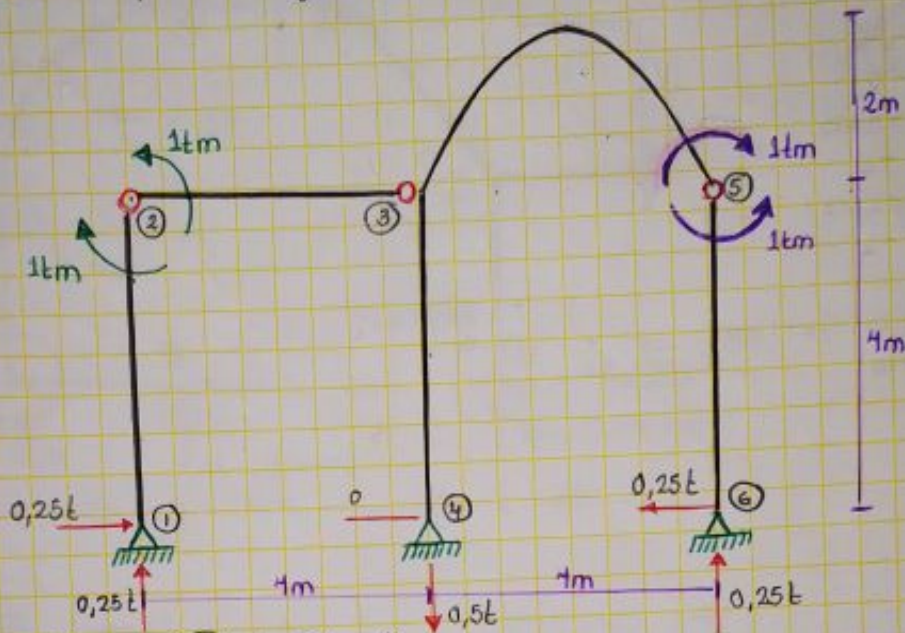
$$M_{\theta=60} = -0.877tm$$

$$M_{\theta=90} = 0 //$$





Diagramar Momento para la siguiente estructura



1- Reacciones

$$\begin{aligned} \sum M_{(2)} &= 0 \quad (\rightarrow \oplus) \text{ (abajo)} \\ 1 - H_1(4) &= 0 \quad \therefore H_1 = 0,25t \\ \sum M_{(5)} &= 0 \quad (\rightarrow \oplus) \text{ (abajo)} \\ -1 + H_6(4) &= 0 \quad \therefore H_6 = 0,25t \\ \sum F_H &= 0 \rightarrow \oplus \\ H_4 &= 0t \\ \sum M_{(3)} &= 0 \quad (\rightarrow \oplus) \text{ (izquierda)} \\ V_1(4) - 0,25(4) &= 0 \quad \therefore V_1 = 0,25t \\ \sum M_{(3)} &= 0 \quad (\rightarrow \oplus) \text{ (Derecho)} \\ 0,25(4) - V_6(4) &= 0 \quad \therefore V_6 = 0,25t \\ \sum F_V &= 0 \uparrow \oplus \\ 0,25 - V_4 + 0,25 &= 0 \quad \therefore V_4 = 0,5t \end{aligned}$$

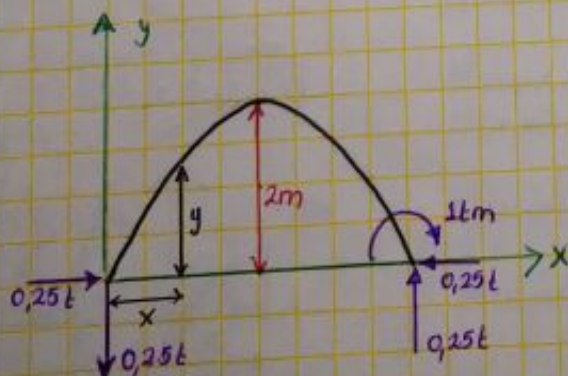
2- Momentos

Barra 1-2 $M_1 = 0$
 $M_2 = -0,25(4) = -1tm$

Barra 2-3 $M_2 = -1tm$
 $M_3 = 0tm$

Barra 3-4 $M_3 = 0$
 $M_4 = 0$

Barra 3-5



$$y = f(x) \text{ Parabol} \\ (x-h)^2 = -4a(y-k)$$

$$P(0,0) ; V(h,k) = (2,2)$$

$$(0-2)^2 = -4a(0-2)$$

$$4 = +8a$$

$$a = 0,5$$

$$(x-2)^2 = -4(0,5)(y-2)$$

$$Y = -0,5X^2 + 2X$$

$$M = -0,25X - 0,25Y = -0,25X - 0,25(-0,5X^2 + 2X)$$

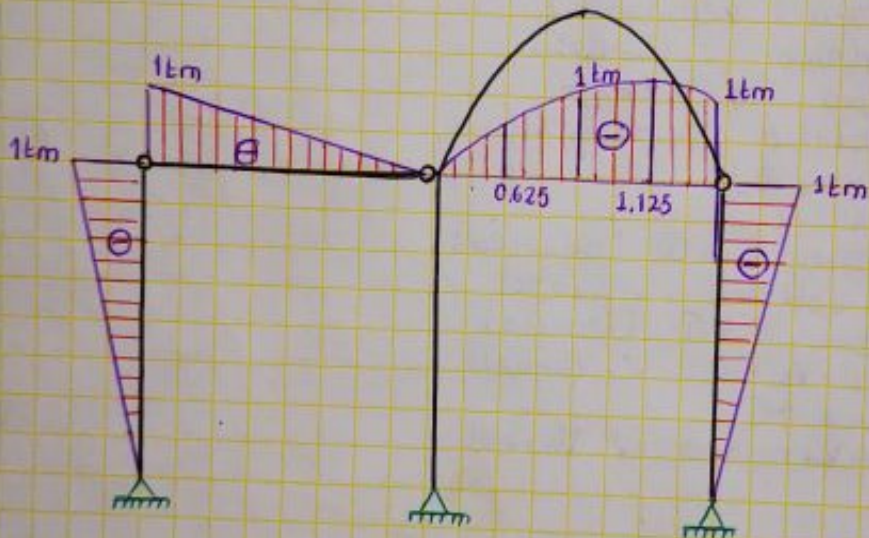
$$M_x = 0,125x^2 - 0,75x$$

x	M
0	0
1	-0,625
2	-1
3	-1,125
4	-1

Barra 5-6

$$M_5 = -0,25(4) = -1 \text{ tm}$$

$$M_6 = 0 \text{ tm}$$



Obtener el diagrama de Momento para la siguiente estructura

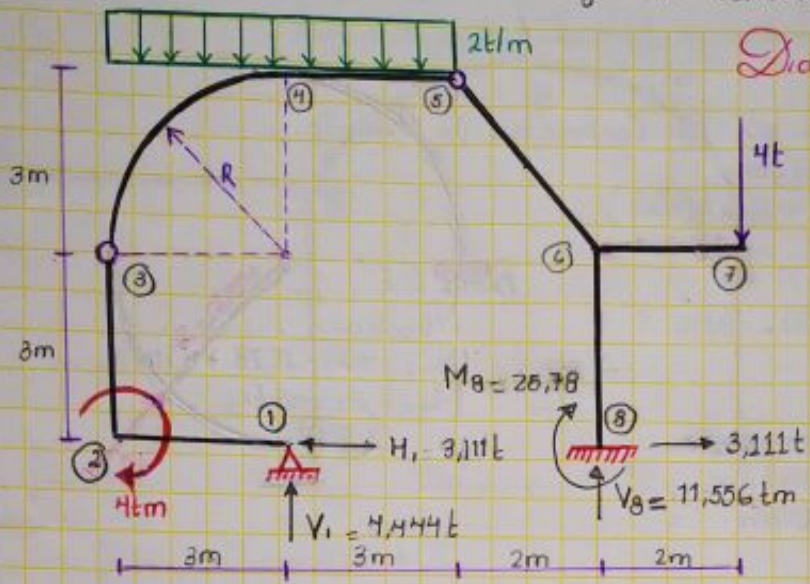


Diagrama Momento

1.- Reacciones

$$\begin{aligned} \sum M_8 = 0 \quad \curvearrowright \oplus \quad (\text{abajo}) \\ 4 - V_1(3) + H_1(3) = 0 \\ \sum M_5 = 0 \quad \curvearrowright \oplus \quad (\text{Izquierda}) \\ V_1(3) + H_1(6) + 4 - 2(6)(3) = 0 \end{aligned}$$

$$\begin{aligned} -3V_1 + 3H_1 &= -4 \\ 3V_1 + 6H_1 &= +32 \\ \hline 9H_1 &= +36 \end{aligned}$$

$$\begin{aligned} H_1 &= +3,111t \\ 4 - V_1(3) + 3,111(3) &= 0 \\ V_1 &= 4,444t \end{aligned}$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$4,444 - 2(6) - 4 + V_8 = 0 \quad V_8 = 11,556t$$

$$\sum F_H = 0 \quad \rightarrow \oplus$$

$$-3,111 + H_8 = 0 \quad \therefore H_8 = 3,111t$$

$$\sum M_1 = 0 \quad \curvearrowright \oplus$$

$$4 + 4(7) - 11,556(5) + M_8 = 0 \quad \therefore M_8 = 25,78t$$

2.- Momentos

Barra 1-2 $M_1 = 0$

$$M_2 = 4,444(3) = 13,332t$$

Barra 2-3 $M_2 = -4,444(3) + 4 = -9,332t$

$$M_2 = -9,332t$$

$$M_3 = 0t$$

Barra 3-4

$$d_1 = R \cos 60 = 3 \cos 60 = 1,5m$$

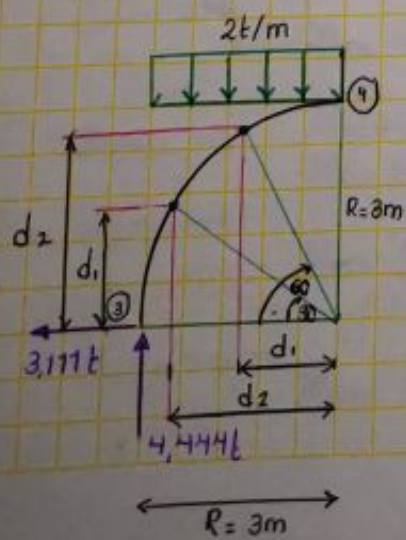
$$d_2 = R \cos 30 = 3 \cos 30 = 2,598m$$

$$M(\theta=0) = 0$$

$$M(\theta=30) = 4,444(3 - 2,598) + 3,111(1,5) - \frac{2(3 - 2,598)^2}{2} = 6,291t$$

$$M(\theta=60) = 4,444(3 - 1,5) + 3,111(2,598) - \frac{2(3 - 1,5)^2}{2} = 12,498t$$

$$M(\theta=90) = 4,444(3) + 3,111(3) - 2(3)(1,5) = 13,665t$$



Barra 4-5

$M_4 = M_{(0-90)}$ Por Continuidad de Momento

$$M_4 = 13,665 \text{ tm}$$

$$M_5 = 0 \text{ (articulación)}$$

Barra 6-7

$$M_6 = -4(2) = -8 \text{ tm}$$

$$M_7 = 0 \text{ tm}$$

Barra 6-8

$$M_6 = 3,111(3) - 25,78$$

$$M_6 = -16,447 \text{ tm}$$

$$M_8 = -25,78 \text{ tm}$$

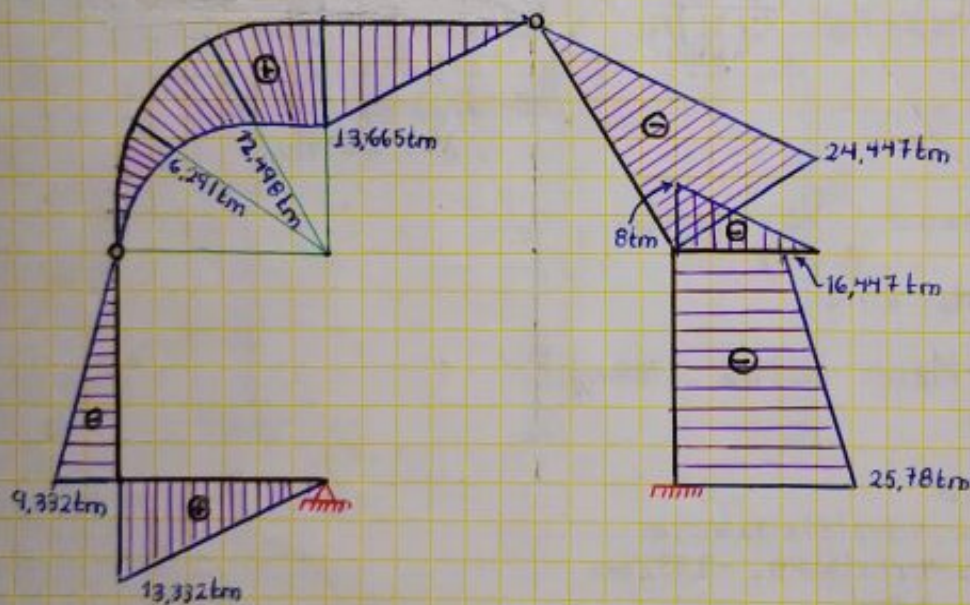
Barra 5-6

$$M_5 = 0 \text{ tm}$$

$$M_6 = -4(2) - 25,78 + 3,111(3)$$

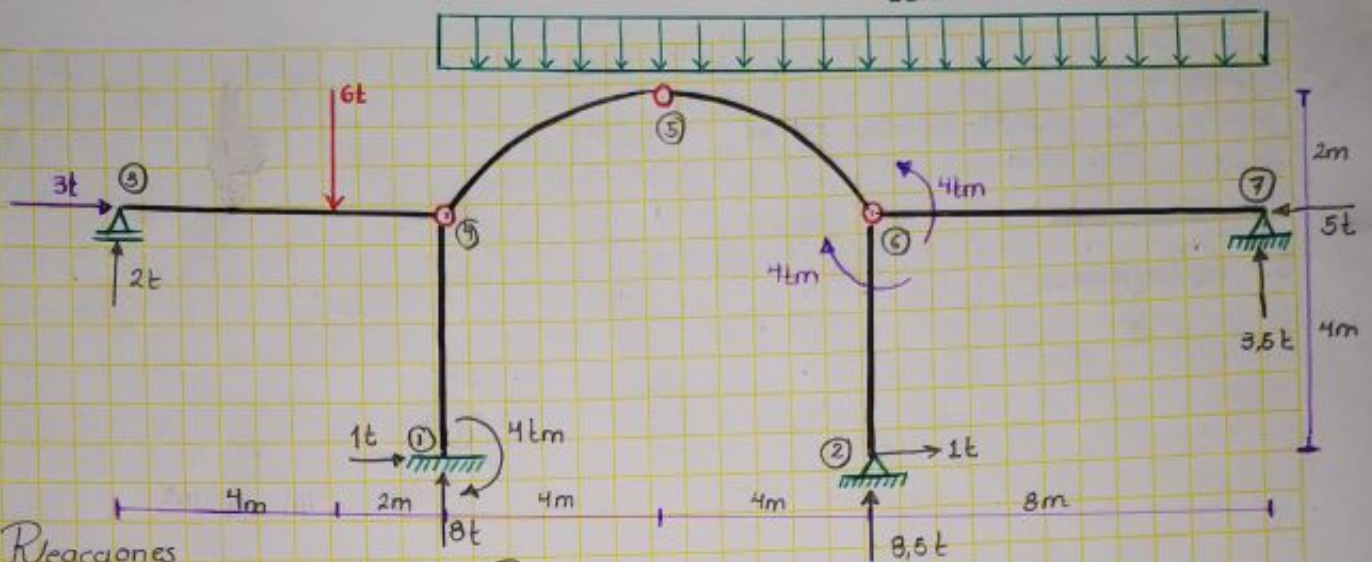
$$M_6 = -24,447 \text{ tm}$$

3.- Diagramas



Diagramar Momento para la siguiente estructura

1t/m



1- Reacciones

$$\sum M_4 = 0 \text{ (Izquierda)} \curvearrowright \oplus$$

$$V_3(6) - 6(2) = 0$$

$$V_3 = 2t //$$

$$\sum M_6 = 0 \curvearrowright \oplus \text{ (derecha)}$$

$$-4 + 1(8)(4) - V_7(8) = 0$$

$$V_7 = 3.5t$$

$$\sum M_6 = 0 \curvearrowright \oplus \text{ (abajo)}$$

$$4 - H_2(4) = 0 \quad \therefore H_2 = 1t$$

$$\sum M_4 = 0 \curvearrowright \oplus \text{ (Derecha)}$$

$$1(16)(8) - 3.5(16) - 1(4) - V_2(8) = 0$$

$$V_2 = 8.5t$$

$$\sum F_v = 0 \uparrow \oplus$$

$$2 - 6 + V_1 - 1(16) + 8.5 + 3.5 = 0$$

$$V_1 = 8t$$

$$\sum M_5 = 0 \curvearrowright \oplus \text{ (derecha)}$$

$$1(12)(6) - 1(6) - 8.5(4) - 3.5(12) + H_7(2) = 0$$

$$H_7 = 5t //$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$3 + 1 - 5 + H_1 = 0 \quad \therefore H_1 = 1t //$$

$$\sum M_4 = 0 \curvearrowright \oplus \text{ (abajo)}$$

$$-1(4) + M_1 = 0 \quad \therefore M_1 = 4tm$$

2. Momentas

Barra 1-4

$$M_1 = 4tm$$

$$M_4 = 0tm$$

Barra 3-4

$$M_p = P_{ab} = \frac{6(4)(2)}{6} = 8tm$$

Barra 2-6

$$M_2 = 0tm$$

$$M_6 = 1(4) = 4tm$$

Barra 4-5

$$M_m = \frac{q l^2}{8} = \frac{1(4)^2}{8} = 2tm$$

Barra 5-6

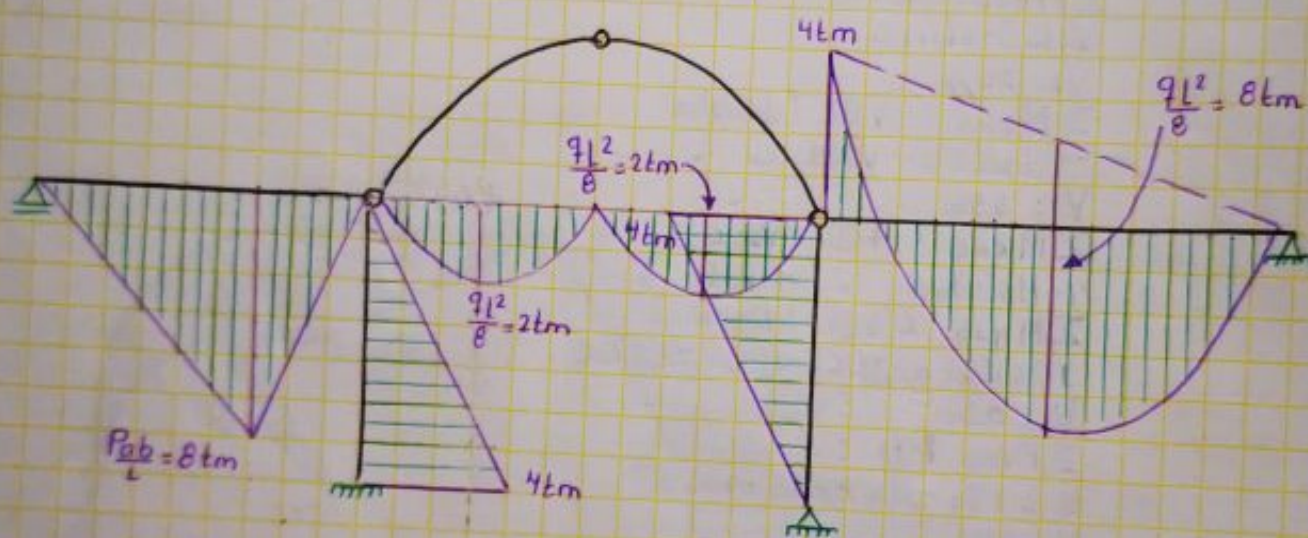
$$M_m = \frac{q l^2}{8} = \frac{1(4)^2}{8} = 2tm$$

Barra 6-7

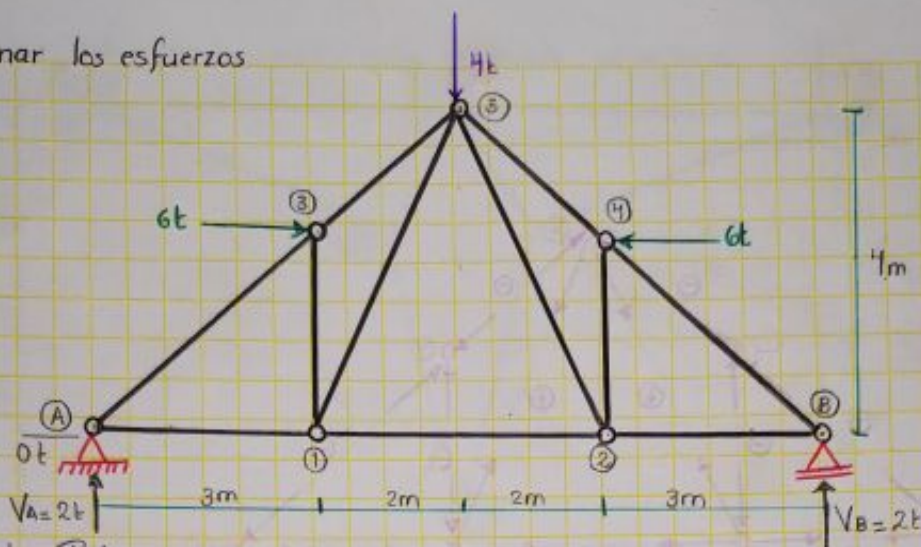
$$M_6 = 3,5(8) - 1(8)(4) = -4 \text{ tm}$$

$$M_7 = 0 \text{ tm}$$

3- Diagramas



Determinar los esfuerzos



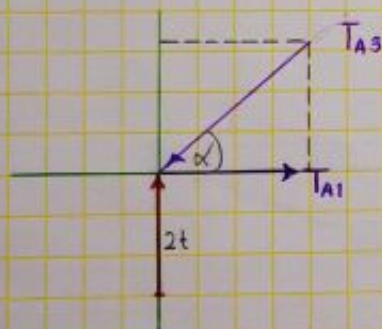
1- Cálculo de Reacciones

$$\begin{aligned}\sum M_A &= 0 \quad \curvearrowright \oplus \\ 4(5) - V_B(10) &= 0 \quad \therefore V_B = 2t \\ \sum F_V &= 0 \quad \uparrow \oplus \\ V_A - 4 + 2 &= 0 \quad \therefore V_A = 2t\end{aligned}$$

2 Tensiones

Siendo la Estructura Simétrica y la carga Simétrica solo analizaremos los nudos A, 1 y 3

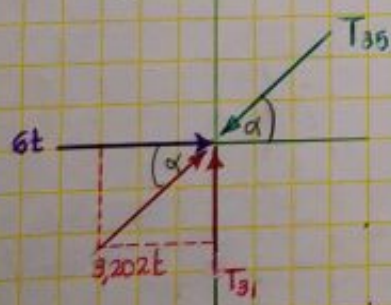
* Nudo A



$$\alpha = \arctg\left(\frac{4}{3}\right) = 38,66^\circ$$

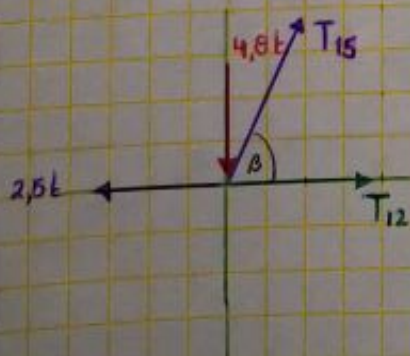
$$\begin{aligned}\sum F_V &= 0 \quad \uparrow \oplus \\ 2 - T_{A3} \sin(38,66) &= 0 \quad \therefore T_{A3} = 3,202t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ T_{A1} - 3,202 \cos(38,66) &= 0 \quad \therefore T_{A1} = 2,5t\end{aligned}$$

* Nudo 3



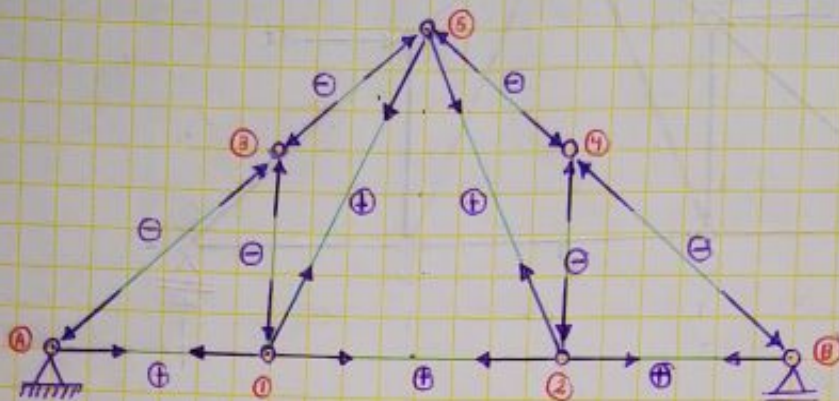
$$\begin{aligned}\sum F_V &= 0 \quad \uparrow \oplus \\ T_{31} - T_{35} \sin(38,66) + 3,202 \sin(38,66) &= 0 \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ -T_{35} \cos(38,66) + 6 + 3,202 \cos(38,66) &= 0 \\ T_{35} &= 10,886t \\ \text{Reemplazamos en } \textcircled{1} \\ T_{31} - 10,886 \sin(38,66) + 3,202 \sin(38,66) &= 0 \\ T_{31} &= 4,8t\end{aligned}$$

* Nudo 1



$$\begin{aligned}\sum F_V &= 0 \quad \uparrow \oplus \\ T_{15} \sin(63,435) - 4,8 &= 0 \\ T_{15} &= 5,367t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ -2,5 + T_{12} + 5,367 \cos(63,435) &= 0 \\ T_{12} &= -0,1t\end{aligned}$$

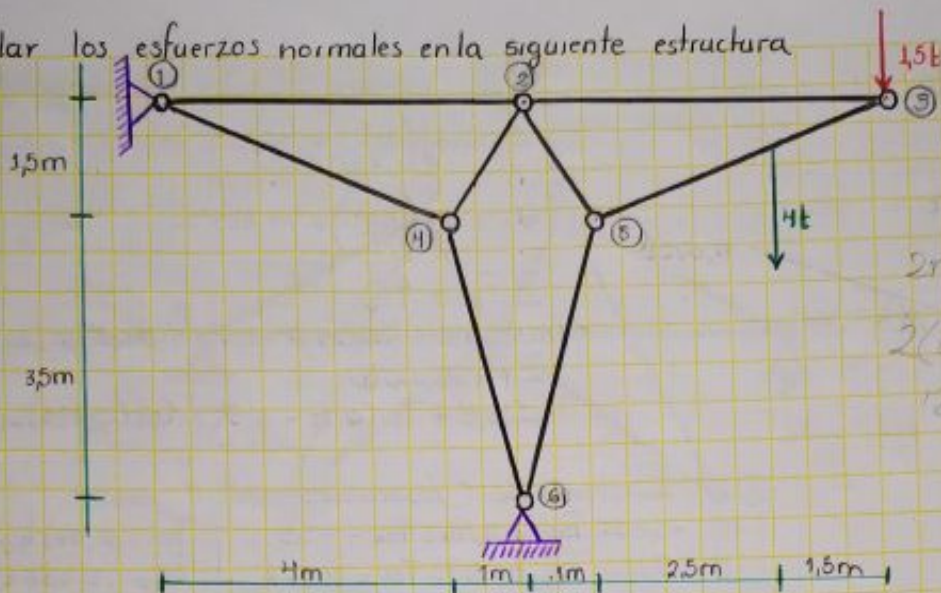
3- Diagrama de Tensiones



4- Tabla Final

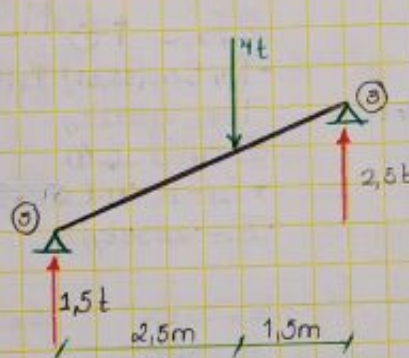
Barra	Tensión	Esfuerzos
A-1	2,5 t	Tiración
A-3	3,202 t	Compresión
1-2	0,1 t	Tiración
1-3	4,8 t	Compresión
1-5	5,367 t	Tiración
2-8	2,5 t	Tiración
2-4	4,8 t	Compresión
2-5	5,367 t	Tiración
3-4	3,202 t	Compresión
3-5	10,886 t	Compresión
4-5	10,886 t	Compresión

Calcular los esfuerzos normales en la siguiente estructura

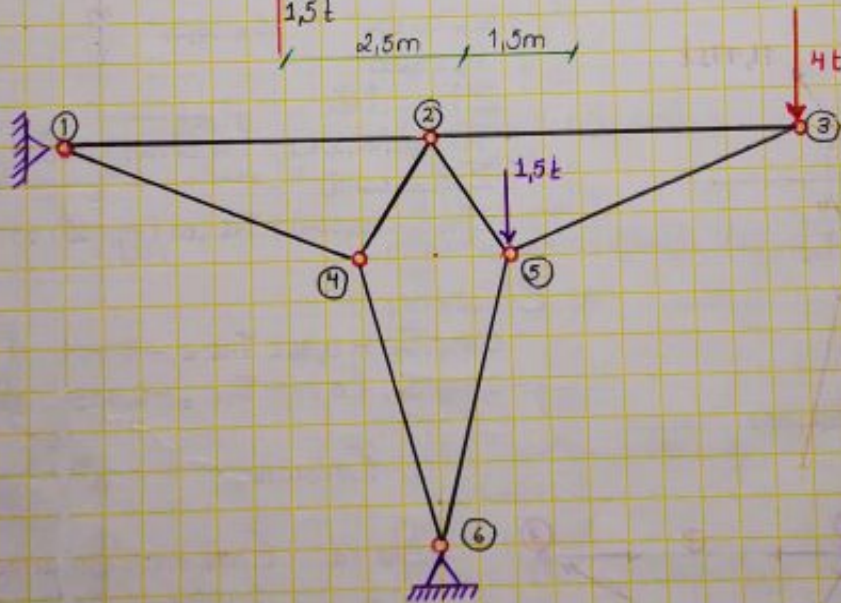


$$\begin{aligned} 2n - k &= 0 \\ 2(6) - 4 &= 8 \\ 8 - 7 &= 1 \end{aligned}$$

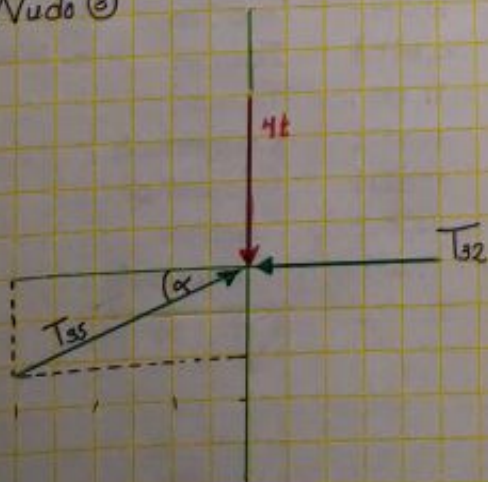
1.- Idealización de cargas Barra 3-5



$$\begin{aligned} \sum M_6 &= 0 \quad \curvearrowright \oplus \\ 4(2.5) - V_3(4) &= 0 \\ V_3 &= 2.5t \\ \sum F_v &= 0 \quad \uparrow \oplus \\ V_5 - 4 + 2.5 &= 0 \\ V_5 &= 1.5t \end{aligned}$$



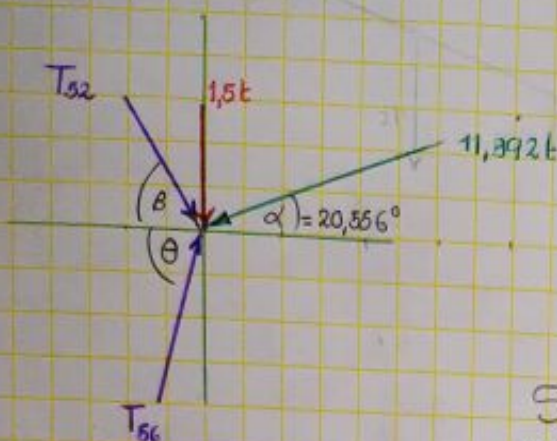
2.- Tensiones
* Nudo 3



$$\alpha = \arctg\left(\frac{1.5}{4}\right) = 20.556^\circ$$

$$\begin{aligned} \sum F_v &= 0 \quad \uparrow \oplus \\ -4 + T_{35} \sin(20.556) &= 0 \\ T_{35} &= 11.392t // \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ -T_{32} + 11.392 \cdot \cos(20.556) &= 0 \\ T_{32} &= 10.667t // \end{aligned}$$

Nudo ⑤



$$\beta = \arctg\left(\frac{1.5}{1}\right) = 56.31^\circ$$

$$\theta = \arctg\left(\frac{3.5}{1}\right) = 74.055^\circ$$

$$\sum F_v = 0 \uparrow \oplus$$

$$-T_{52} \cdot \text{Sen } \beta + T_{56} \cdot \text{Sen } \theta - 1.5 - 11.392 \cdot \text{Sen}(20.556) = 0$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$T_{52} \cdot \text{Cos } \beta + T_{56} \cdot \text{Cos } \theta - 11.392 \cdot \text{Cos}(20.556) = 0$$

Sistema de Ecuaciones

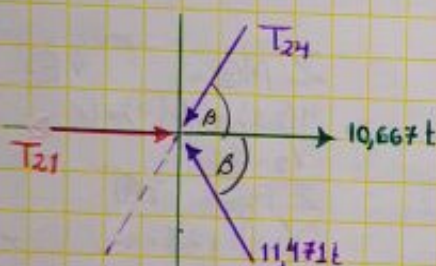
$$-0.832 T_{52} + 0.962 T_{56} = +5.5$$

$$0.555 T_{52} + 0.275 T_{56} = 10.667$$

$$\left. \begin{array}{l} T_{52} = 11.471 \text{ t} \\ T_{56} = 15.638 \text{ t} \end{array} \right\}$$

$$\left. \begin{array}{l} T_{52} = 11.471 \text{ t} \\ T_{56} = 15.638 \text{ t} \end{array} \right\}$$

Nudo ②



$$\sum F_v = 0 \uparrow \oplus$$

$$-T_{24} \cdot \text{Sen}(56.31) + 11.471 \cdot \text{Sen}(56.31) = 0$$

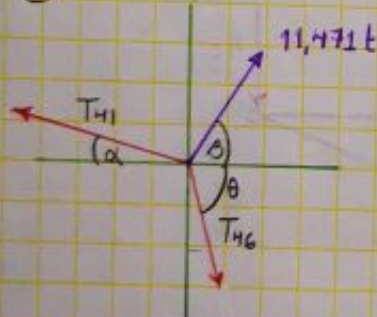
$$T_{24} = 11.471 \text{ t}$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$+T_{21} - 11.471 \cdot \text{Cos}(56.31) - 11.471 \cdot \text{Cos}(56.31) + 10.667 = 0$$

$$T_{21} = 2.058 \text{ t}$$

Nudo ④



$$\alpha = 20.556^\circ$$

$$\beta = 56.31^\circ$$

$$\theta = 74.055^\circ$$

$$\sum F_v = 0 \uparrow \oplus$$

$$T_{41} \cdot \text{Sen}(20.556) - T_{46} \cdot \text{Sen}(74.055) + 11.471 \cdot \text{Sen}(56.31) = 0$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$-T_{41} \cdot \text{Cos}(20.556) + T_{46} \cdot \text{Cos}(74.055) + 11.471 \cdot \text{Cos}(56.31) = 0$$

Ecuaciones

$$0.351 T_{41} - 0.962 T_{46} = -9.544$$

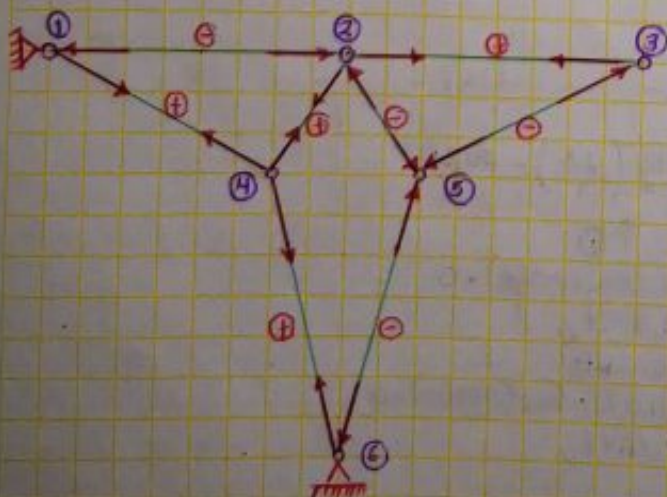
$$-0.936 T_{41} + 0.275 T_{46} = -6.363$$

$$\left. \begin{array}{l} T_{41} = 10.879 \text{ t} \\ T_{46} = 13.89 \text{ t} \end{array} \right\}$$

$$\left. \begin{array}{l} T_{41} = 10.879 \text{ t} \\ T_{46} = 13.89 \text{ t} \end{array} \right\}$$

8 Diagrama de tensiones

4. Tensiones



Barra

Tensión

Esfuerzo

1-2

2.058 t

Compresión

1-4

10.879 t

Tracción

2-3

10.667 t

Tracción

2-4

11.471 t

Tracción

2-5

11.471 t

Tracción

3-5

11.392 t

Compresión

4-6

13.89 t

Compresión

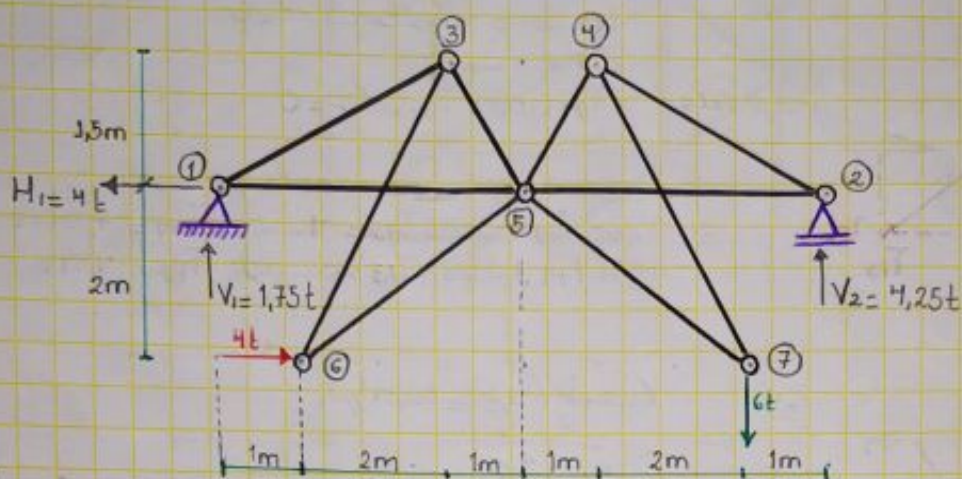
5-6

15.638 t

Tracción

Compresión

Determinar los esfuerzos internos

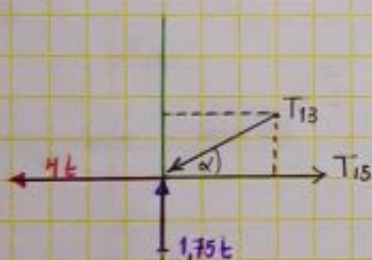


1.- Reacciones

$$\begin{aligned}\sum M_{(1)} &= 0 \quad \curvearrowright \oplus \\ -4(2) + 6(7) - V_2(8) &= 0 \quad \therefore V_2 = 4,25t \\ \sum F_v &= 0 \quad \uparrow \oplus \\ V_1 + 4,25 - 6 &= 0 \quad \therefore V_1 = 1,75t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ H_1 &= 4t\end{aligned}$$

2.- Tensiones por Nudos

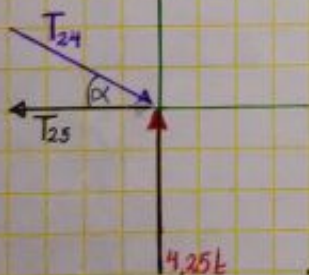
Nudo ①



$$\alpha = \arctg\left(\frac{1,5}{3}\right) = 26,565^\circ$$

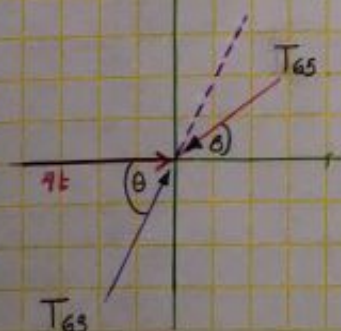
$$\begin{aligned}\sum F_v &= 0 \quad \uparrow \oplus \\ 1,75 - T_{13} \cdot \text{Sen}(26,565) &= 0 \\ T_{13} &= 3,913t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ -4 + T_{15} - 3,913 \cdot \text{Cos}(26,565) &= 0 \\ T_{15} &= 7,499t\end{aligned}$$

Nudo ②



$$\begin{aligned}\sum F_v &= 0 \quad \uparrow \oplus \\ 4,25 - T_{24} \cdot \text{Sen}(26,565) &= 0 \\ T_{24} &= 9,503t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ -T_{25} + (9,503) \text{Cos}(26,565) &= 0 \\ T_{25} &= 8,5t\end{aligned}$$

Nudo ⑥



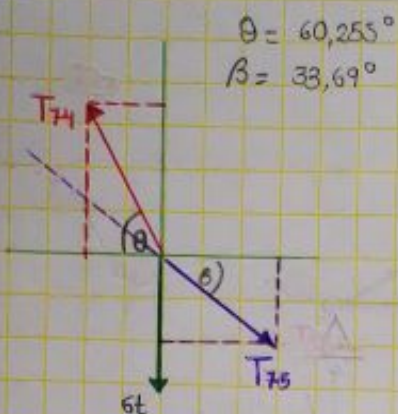
$$\beta = \arctg\left(\frac{2}{3}\right) = 33,69^\circ$$

$$\theta = \arctg\left(\frac{3,5}{2}\right) = 60,255^\circ$$

$$\begin{aligned}\sum F_v &= 0 \quad \uparrow \oplus \\ T_{63} \text{ Sen } \theta - T_{65} \text{ Sen } \beta &= 0 \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ 4 + T_{63} \text{ Cos } \theta - T_{65} \text{ Cos } \beta &= 0 \\ \begin{cases} 0,868 T_{63} - 0,555 T_{65} = 0 \\ 0,496 T_{63} - 0,832 T_{65} = -4 \end{cases} \end{aligned}$$

$$\begin{cases} T_{63} = 4,968t \\ T_{65} = 7,769t \end{cases}$$

Nudo ⑦

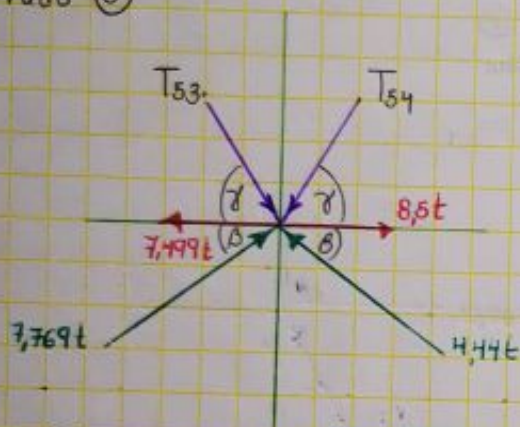


$$\begin{aligned}\sum F_v &= 0 \uparrow \oplus \\ T_{74} \text{ Sen } \theta - T_{75} \text{ Sen } \beta - 4 &= 0 \\ \sum F_h &= 0 \rightarrow \oplus \\ -T_{74} \text{ Cos } \theta + T_{75} \text{ Cos } \beta &= 0\end{aligned}$$

Sistema de Ecuaciones

$$\begin{cases} 0,868 T_{74} - 0,555 T_{75} = 4 \\ -0,496 T_{74} + 0,832 T_{75} = 0 \end{cases} \quad \begin{cases} T_{74} = 7,447t \\ T_{75} = 4,44t \end{cases}$$

Nudo ⑤



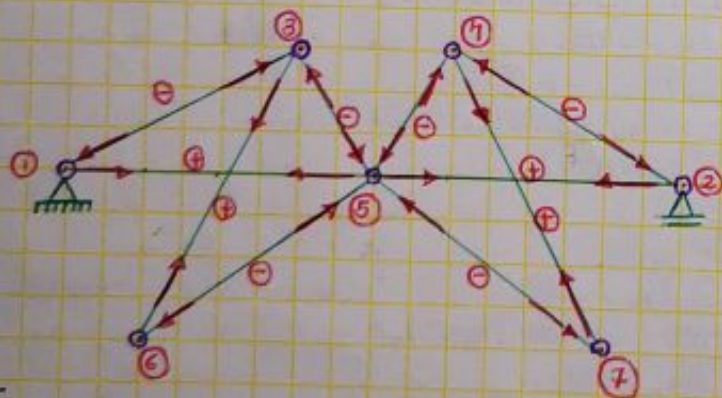
$$\gamma = \arctg\left(\frac{1,5}{1}\right) = 56,31^\circ //$$

$$\begin{aligned}\sum F_v &= 0 \uparrow \oplus \\ 7,769 \text{ Sen } \beta + 4,44 \text{ Sen } \beta - T_{53} \text{ Sen } \gamma - T_{54} \text{ Sen } \gamma &= 0 \\ \sum F_h &= 0 \rightarrow \oplus \\ 7,769 \text{ Cos } \beta - 4,44 \text{ Cos } \beta + 8,5 - 7,769 + T_{53} \text{ Cos } \gamma - T_{54} \text{ Cos } \gamma &= 0\end{aligned}$$

Sistema de Ecuaciones

$$\begin{cases} 0,832 T_{53} + 0,832 T_{54} = 6,772 \\ 0,555 T_{53} - 0,555 T_{54} = -3,771 \end{cases} \quad \begin{cases} T_{53} = 0,672t \\ T_{54} = 7,467t \end{cases}$$

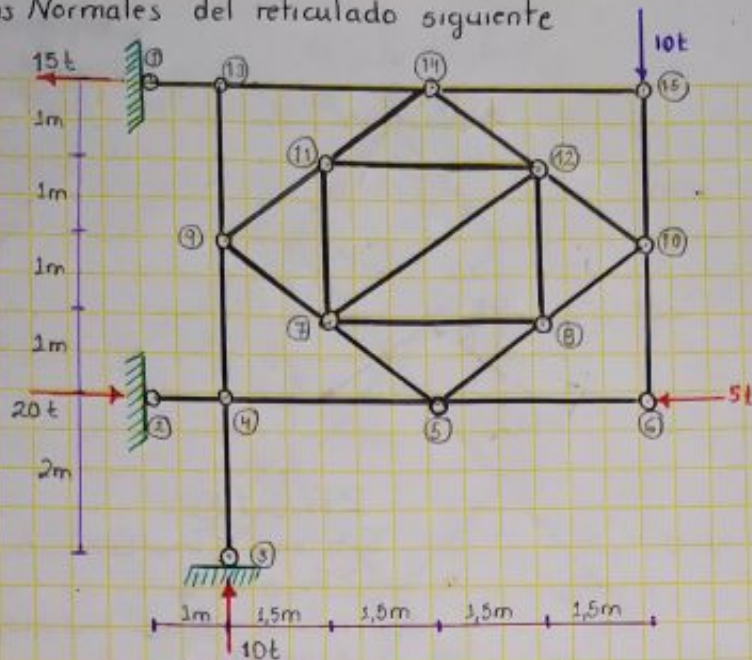
3 - Diagrama de tensiones



4 - Tensiones

Barra	Tensión	Esfuerzo
1-3	3,913t	Compresión
1-5	7,799t	Tracción
2-4	9,503t	Compresión
2-5	8,5t	Tracción
3-5	0,672t	Compresión
3-6	4,968t	Tracción
4-5	7,467t	Compresión
4-7	7,447t	Tracción
5-6	7,769t	Compresión
5-7	4,440t	Compresión

Calcular las Normales del reticulado siguiente

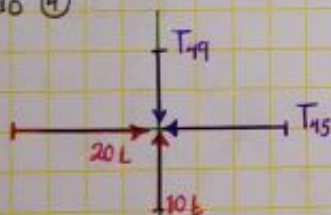


1. Reacciones

$$\begin{aligned} \sum M_H = 0 \quad \curvearrowright + \\ 10(6) - H_1(4) = 0 \quad \therefore H_1 = 15t // \\ \sum F_H = 0 \rightarrow + \\ -15 + H_2 - 5 = 0 \quad \therefore H_2 = 20t \\ \sum F_V = 0 \uparrow + \\ V_3 = 10t \end{aligned}$$

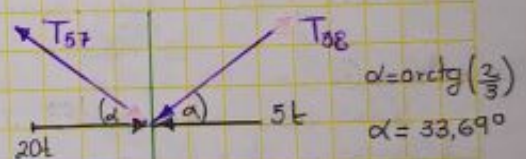
2. Tensiones

* Nudo 4



$$\begin{aligned} \sum F_V = 0 \uparrow + \\ T_{49} = 10t \\ \sum F_H = 0 \rightarrow + \\ T_{45} = 20t \end{aligned}$$

* Nudo 5

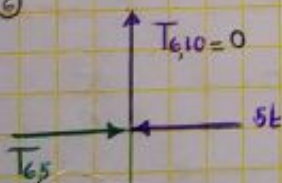


$$\begin{aligned} \sum F_V = 0 \uparrow + \\ + T_{57} \sin(33.69) - T_{58} \sin(33.69) = 0 \\ T_{57} - T_{58} = 0 \\ \sum F_H = 0 \rightarrow + \\ - T_{57} \cos(33.69) - T_{58} \cos(33.69) + 20 - 5 = 0 \\ T_{57} + T_{58} = 18.028 \end{aligned}$$

Sistema de Ecuaciones

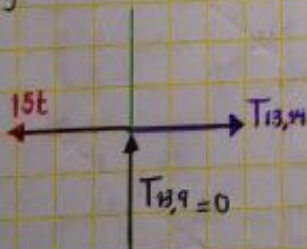
$$\begin{cases} T_{57} - T_{58} = 0 \\ T_{57} + T_{58} = 18.028 \end{cases} \Rightarrow \begin{aligned} T_{57} &= 9.014t // \\ T_{58} &= 9.014t // \end{aligned}$$

* Nudo 6



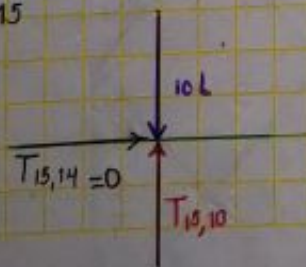
$$\begin{aligned} \sum F_V = 0 \uparrow + \\ T_{6,10} = 0 \\ \sum F_H = 0 \rightarrow + \\ T_{6,5} = 5t // \end{aligned}$$

* Nudo 13



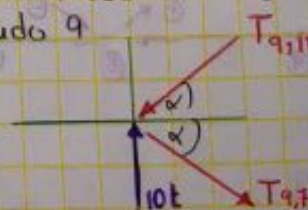
$$\begin{aligned} \sum F_V = 0 \uparrow + \\ T_{13,9} = 0 \\ \sum F_H = 0 \rightarrow + \\ T_{13,14} = 15t // \end{aligned}$$

* Nudo 15



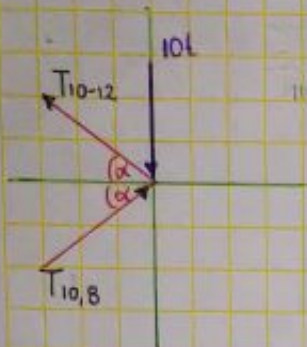
$$\begin{aligned} \sum F_V = 0 \uparrow + \\ T_{15,10} = 10t \\ \sum F_H = 0 \rightarrow + \\ T_{15,14} = 0t \end{aligned}$$

* Nudo 9



$$\begin{aligned} \sum F_V = 0 \uparrow + \\ - T_{9,11} \sin \alpha - T_{9,7} \sin \alpha + 10 = 0 \\ \sum F_H = 0 \rightarrow + \\ - T_{9,11} \cos \alpha + T_{9,7} \cos \alpha = 0 \\ \begin{cases} T_{9,7} + T_{9,11} = 18.028 \\ T_{9,7} - T_{9,11} = 0 \end{cases} \Rightarrow \begin{aligned} T_{9,7} &= 9.014t // \\ T_{9,11} &= 9.014t // \end{aligned} \end{aligned}$$

* Nudo 10 $\alpha = 33,69^\circ$



$$\sum F_v = 0 \uparrow \oplus$$

$$T_{10,8} \text{ Sen } \alpha + T_{10,12} \text{ Sen } \alpha - 10 = 0$$

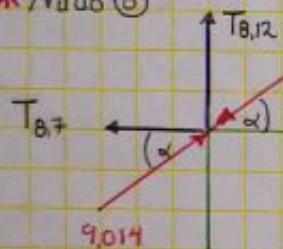
$$\sum F_h = 0 \rightarrow \oplus$$

$$T_{10,8} \text{ Cos } \alpha - T_{10,12} \text{ Cos } \alpha = 0$$

Sistema de Ecuaciones

$$\begin{cases} T_{10,8} + T_{10,12} = 18,028 \\ T_{10,8} - T_{10,12} = 0 \end{cases} \quad \begin{cases} T_{10,8} = 9,014 \text{ t} \\ T_{10,12} = 9,014 \text{ t} \end{cases}$$

* Nudo 8

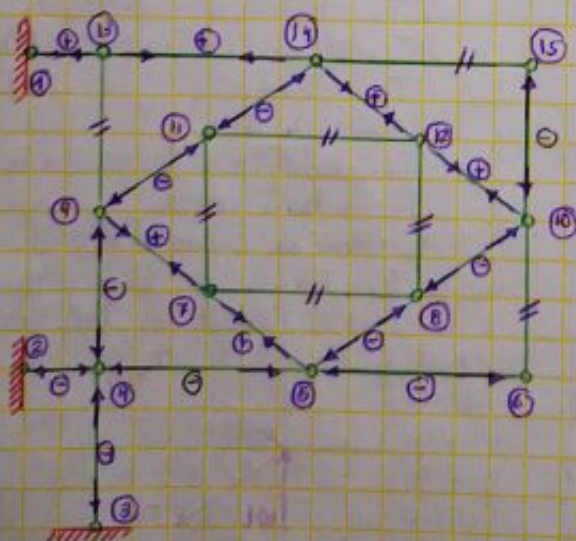


Como las fuerzas se equilibran las tensiones en las barras son nulas
 $T_{8,7} = 0$; $T_{8,12} = 0$

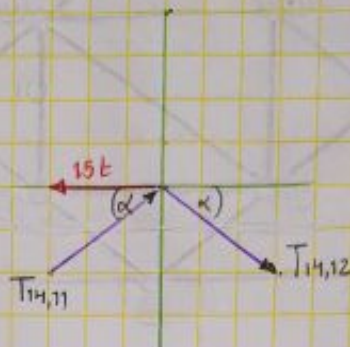
* Nudo 7

Idem a las anteriores $T_{7,12} = 0$

4. Diagrama de tensiones



* Nudo 14



$$\sum F_v = 0 \uparrow \oplus$$

$$T_{14,11} \text{ Sen } \alpha - T_{14,12} \text{ Sen } \alpha = 0$$

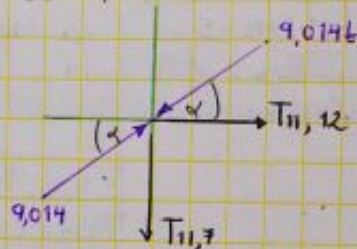
$$\sum F_h = 0 \rightarrow \oplus$$

$$T_{14,11} \text{ Cos } \alpha + T_{14,12} \text{ Cos } \alpha - 15 = 0$$

Ecuaciones

$$\begin{cases} T_{14,11} - T_{14,12} = 0 \\ T_{14,11} + T_{14,12} = 18,028 \end{cases} \quad \begin{cases} T_{14,11} = 9,014 \text{ t} \\ T_{14,12} = 9,014 \text{ t} \end{cases}$$

* Nudo 11



Idem al anterior

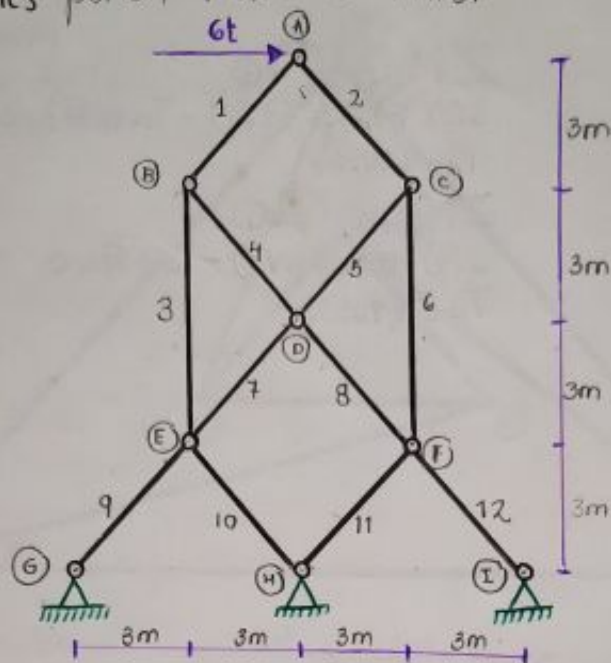
$$T_{11,12} = 0$$

$$T_{11,7} = 0$$

Borra Tension Esfuerzo

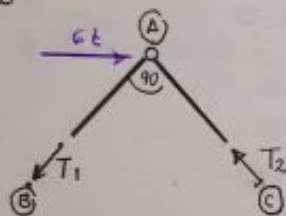
Borra	Tension	Esfuerzo
4-5	20	C
4-9	10	C
6-6	5	C
5-7	9,014	T
5-8	9,014	C
6-10	0	Nulo
7-8	0	Nulo
7-9	9,014	T
7-11	0	Nulo
8-10	9,014	C
8-12	0	Nulo
9-11	9,014	C
9-13	0	Nulo
10-12	9,014	T
10-15	10	C
11-12	0	Nulo
11-14	9,014	C
12-14	9,014	T
11-14	9,014	C
14-15	0	Nulo

Calcular las Normales por el Método de Ritter



$2n - R = 0$
 $2(9) - 6 = 12$
 $18 - 6 = 12$

Tensiones



$$\sum M_B = 0 \quad \curvearrowright +$$

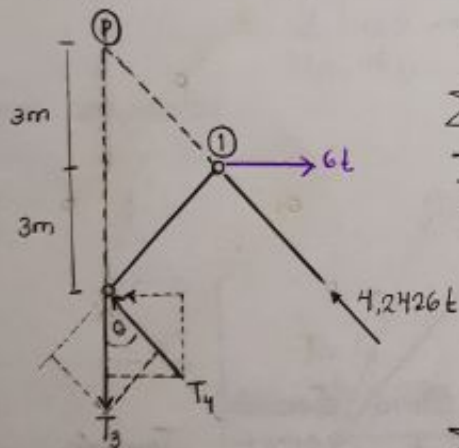
$$6(3) - T_2 \cdot 3\sqrt{2} = 0$$

$$T_2 = 4,2426t //$$

$$\sum M_C = 0 \quad \curvearrowright +$$

$$6(3) - T_1(3\sqrt{2}) = 0$$

$$T_1 = 4,2426t //$$



$$\theta = \arctg\left(\frac{3}{3}\right) = 45^\circ$$

$$\sum M_P = 0 \quad \curvearrowright +$$

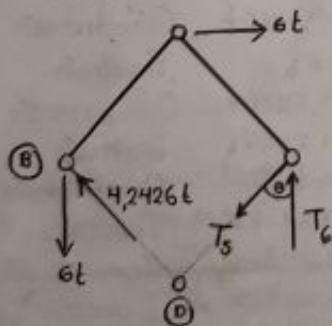
$$-6(3) + T_4 \sin 45^\circ (6) = 0$$

$$T_4 = 4,2426t //$$

$$\sum M_D = 0 \quad \curvearrowright +$$

$$4,2426(3\sqrt{2}) - T_3 \cos 45^\circ (3\sqrt{2}) = 0$$

$$T_3 = 6t //$$



$$\sum M_D = 0 \quad \curvearrowright +$$

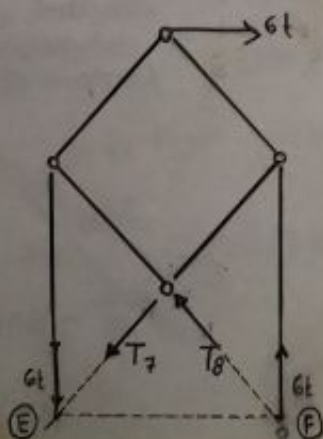
$$6(6) - 6(3) - T_6(3) = 0$$

$$T_6 = 6t //$$

$$\sum M_B = 0 \quad \curvearrowright +$$

$$6(3) - 6(6) + T_5 \cos 45^\circ \cdot 6 = 0$$

$$T_5 = 4,2426t$$



$$\sum M_F = 0 \quad \curvearrowright +$$

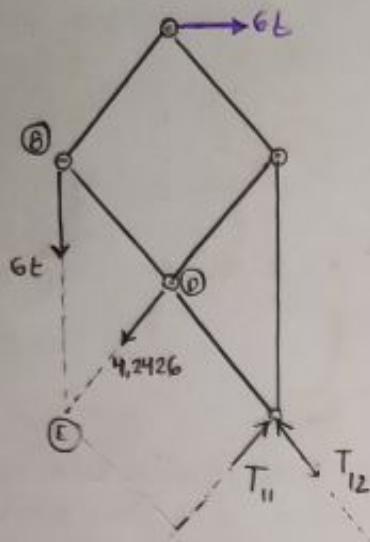
$$-6(6) + 6(9) - T_7(3\sqrt{2}) = 0$$

$$T_7 = 4,2426t$$

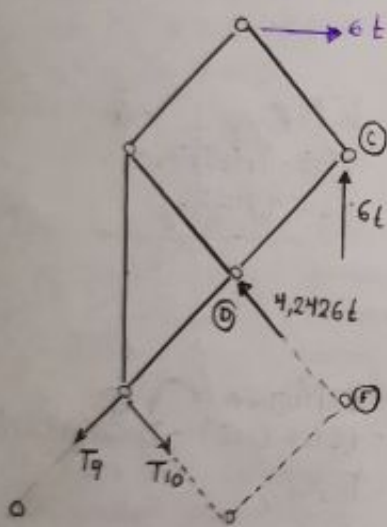
$$\sum M_E = 0 \quad \curvearrowright +$$

$$6(9) - 6(6) - T_8(3\sqrt{2}) = 0$$

$$T_8 = 4,2426t$$

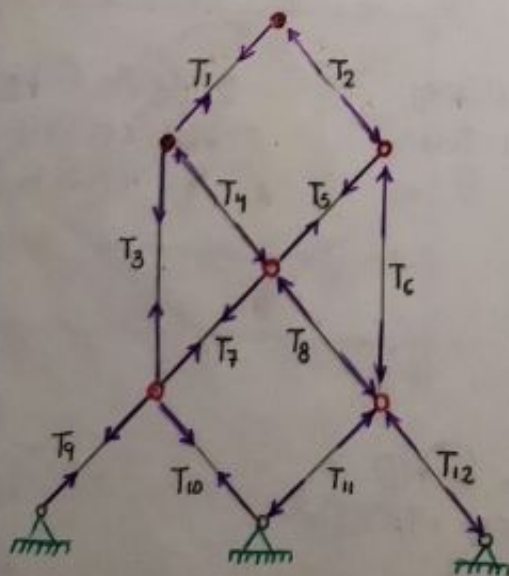


$$\begin{aligned}\sum M_B &= 0 \quad \curvearrowright \oplus \\ 6(3) + 4,2426(3\sqrt{2}) - T_{11}(6\sqrt{2}) &= 0 \\ T_{11} &= 4,2426t \\ \sum M_D &= 0 \quad \curvearrowright \oplus \\ 6(9) - 4,2426(3\sqrt{2}) + T_{12} \cdot 3\sqrt{2} &= 0 \\ T_{12} &= 8,4853t\end{aligned}$$



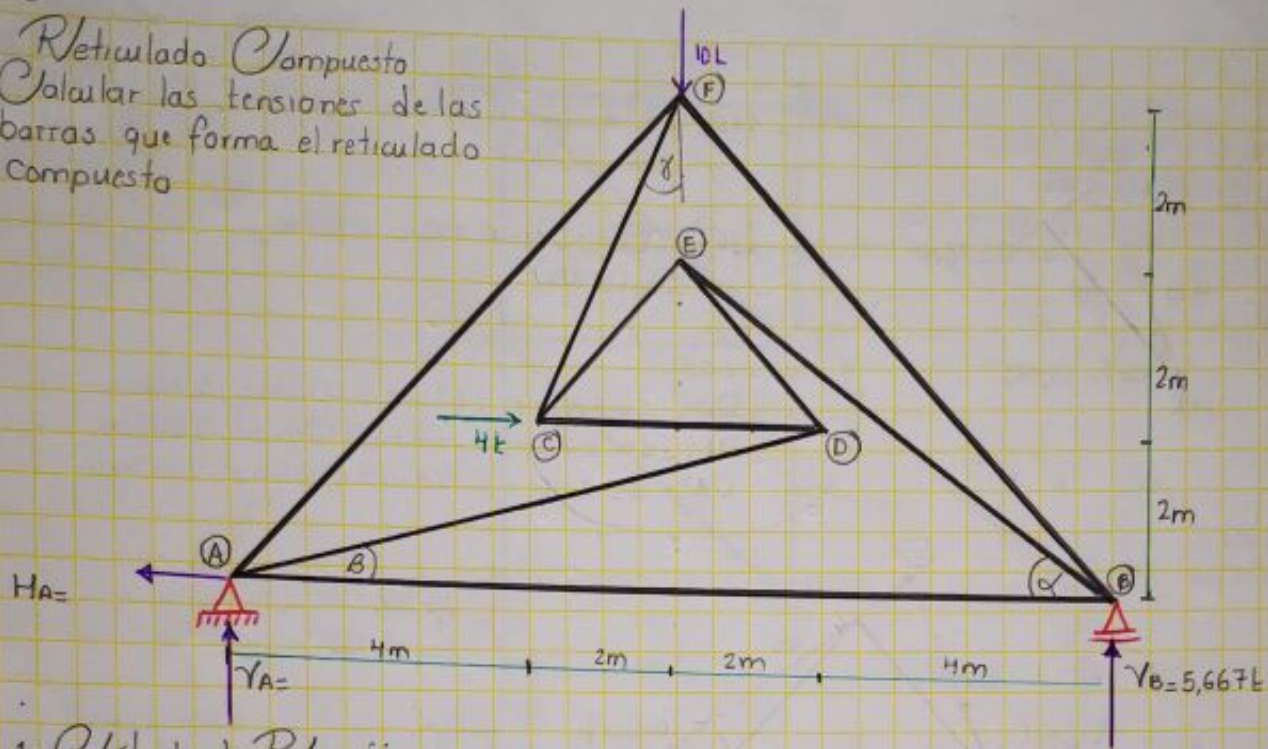
$$\begin{aligned}\sum M_C &= 0 \quad \curvearrowright \oplus \\ 6(3) + 4,2426(3\sqrt{2}) - T_{10}(6\sqrt{2}) &= 0 \\ T_{10} &= 4,2426t \\ \sum M_D &= 0 \quad \curvearrowright \oplus \\ 6(9) - 4,2426(3\sqrt{2}) - T_9(3\sqrt{2}) &= 0 \\ T_9 &= 8,4853t\end{aligned}$$

2- Diagrama de Tensiones



Barra	Tensión	
1	4,2426t	Tracción
2	4,2426t	Compresión
3	6t	Tracción
4	4,2426t	Compresión
5	4,2426t	Tracción
6	6t	Compresión
7	4,2426t	Tracción
8	4,2426t	Compresión
9	8,4853t	Tracción
10	4,2426t	Compresión
11	4,2426t	Tracción
12	8,4853t	Compresión

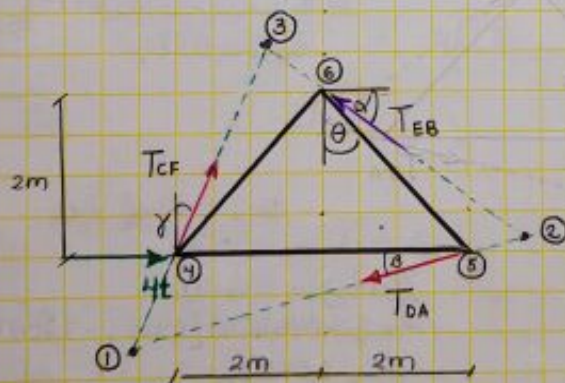
Reticulado Compuesto
Calcular las tensiones de las barras que forma el reticulado compuesto



1.- Cálculo de Reacciones

$$\begin{aligned}\sum M_A &= 0 \quad (\curvearrowright) \oplus \\ 4(2) + 10(6) - V_B(12) &= 0 \quad \therefore V_B = 5,667t \\ \sum F_v &= 0 \quad \uparrow \oplus \\ V_A - 10 + 5,667 &= 0 \quad \therefore V_A = 4,333t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ H_A &= 4t\end{aligned}$$

2.- Tensiones compuestas



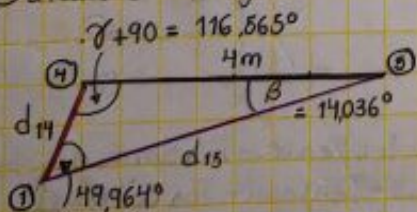
$$\alpha = \arctg\left(\frac{4}{6}\right) = 33,69^\circ$$

$$\beta = \arctg\left(\frac{2}{8}\right) = 14,036^\circ$$

$$\gamma = \arctg\left(\frac{2}{4}\right) = 26,565^\circ$$

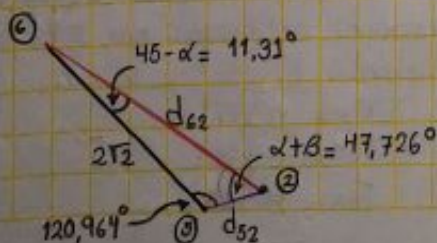
$$\theta = \arctg\left(\frac{2}{2}\right) = 45^\circ$$

a) Cálculo de Longitudes



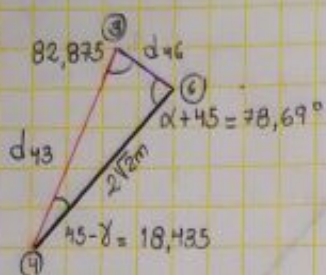
$$d_{14} = \frac{4m \cdot \text{Sen}(14,036)}{\text{Sen}(49,964)} = 1,267m$$

$$d_{15} = \frac{4 \cdot \text{Sen}(116,565)}{\text{Sen}(49,964)} = 4,673m$$



$$d_{52} = \frac{2\sqrt{2}}{\text{Sen}(47,726)} \cdot \text{Sen}(11,31) = 0,75m$$

$$d_{62} = \frac{2\sqrt{2}}{\text{Sen}(47,726)} \cdot \text{Sen}(120,964) = 3,278m$$



$$d_{43} = \frac{2\sqrt{2} \cdot \text{Sen}(78,69)}{\text{Sen}(82,875)} = 2,795 \text{ m} //$$

$$d_{36} = \frac{2\sqrt{2} \cdot \text{Sen}(18,435)}{\text{Sen}(82,875)} = 0,901 \text{ m} //$$

* Distancias

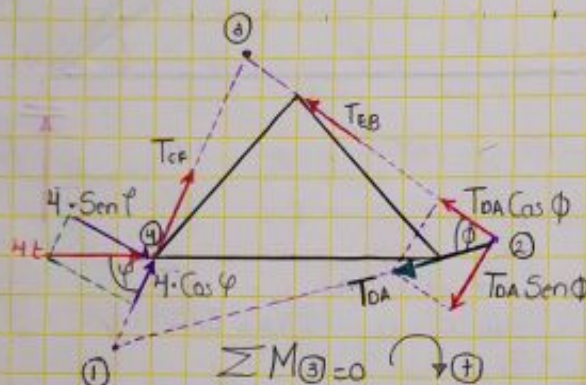
$$D_{12} = d_{13} + d_{23} = 4,673 \text{ m} + 0,75 \text{ m} = 5,423 \text{ m}$$

$$D_{13} = d_{14} + d_{43} = 1,267 + 2,795 \text{ m} = 4,062 \text{ m}$$

$$D_{23} = d_{26} + d_{63} = 3,278 \text{ m} + 0,901 \text{ m} = 4,179 \text{ m}$$

3. Tensiones

* Tensión DA



$$\phi = 47,726^\circ$$

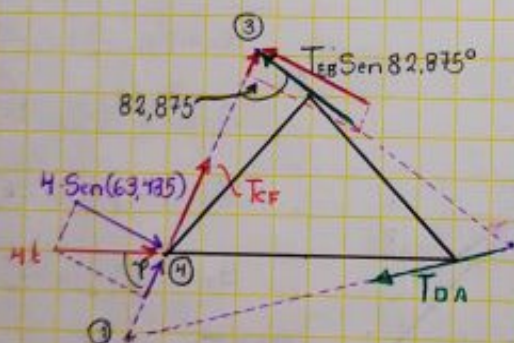
$$\rho = 63,435^\circ$$

$$\sum M_{(3)} = 0 \quad \curvearrowright (+)$$

$$T_{DA} \text{ Sen}(47,726)(4,179) - 4 \cdot \text{Sen}(63,435)(2,795) = 0$$

$$T_{DA} = 3,234 \text{ t} //$$

* Tensión EB



$$\sum M_{(1)} = 0 \quad \curvearrowright (+)$$

$$4 \cdot \text{Sen}(63,435)(1,267) - T_{EB} \text{ Sen} 82,875 (4,062) = 0$$

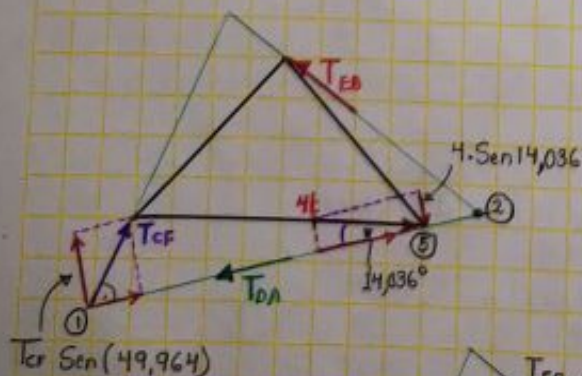
$$T_{EB} = 1,125 \text{ t} //$$

* Tensión CF

$$\sum M_{(2)} = 0 \quad \curvearrowright (+)$$

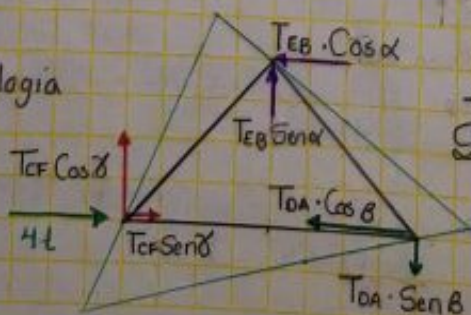
$$T_{CF} \text{ Sen}(49,964) [5,423] - 4 \text{ Sen}(14,036)(0,75) = 0$$

$$T_{CF} = 0,175 \text{ t} //$$



$$T_{CF} \text{ Sen}(49,964)$$

4.- 2da Metodología



$$\left. \begin{aligned} T_{CF} \cos \gamma + T_{EB} \text{ Sen} \alpha - T_{DA} \text{ Sen} \beta &= 0 \\ 4 + T_{CF} \text{ Sen} \gamma - T_{EB} \cos \alpha - T_{DA} \cos \beta &= 0 \\ -T_{EB} \text{ Sen} \alpha (2) - T_{EB} \cos \alpha (2) + T_{DA} \text{ Sen} \beta (4) &= 0 \end{aligned} \right\} 3 \text{ E}$$

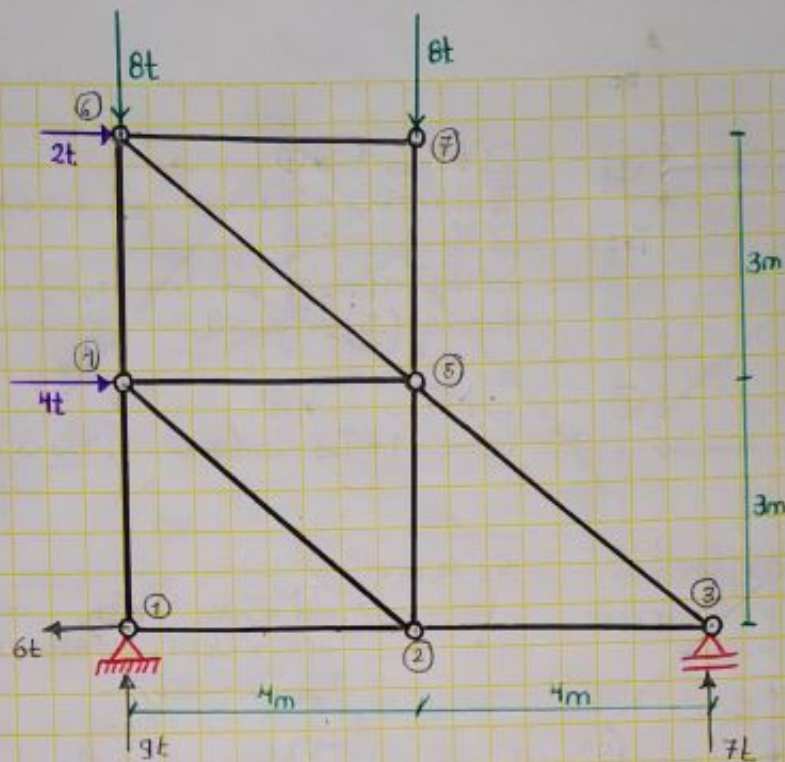
De las 3 ecuaciones obtenidas por $\sum F_v$, $\sum F_H$ y $\sum M$ obtenemos:

$$T_{CF} = 0,177 \text{ t}$$

$$T_{EB} = 1,131 \text{ t}$$

$$T_{DA} = 3,235 \text{ t}$$

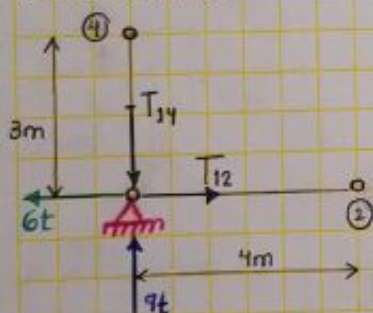
Calcular los esfuerzos



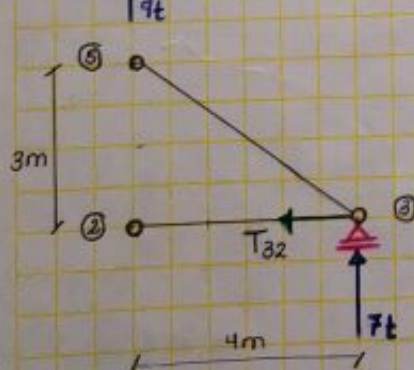
1- Reacciones

$$\begin{aligned}\sum M_1 &= 0 \quad \curvearrowright \oplus \\ 4(3) + 2(6) + 8(4) - V_3(8) &= 0 \quad \therefore V_3 = 7t \\ \sum F_v &= 0 \quad \uparrow \oplus \\ V_1 - 8 - 8 + 7 &= 0 \quad \therefore V_1 = 9t \\ \sum F_H &= 0 \quad \rightarrow \oplus \\ H_1 &= 6t\end{aligned}$$

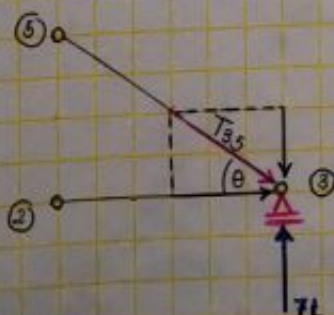
2- Tensiones



$$\begin{aligned}\sum M_4 &= 0 \quad \curvearrowright \oplus \\ 6 \cdot 3 - T_{12} \cdot 3 &= 0 \quad \therefore T_{12} = 6t // \\ \sum M_2 &= 0 \quad \curvearrowright \oplus \\ 9(4) - T_{14}(4) &= 0 \quad \therefore T_{14} = 9t //\end{aligned}$$

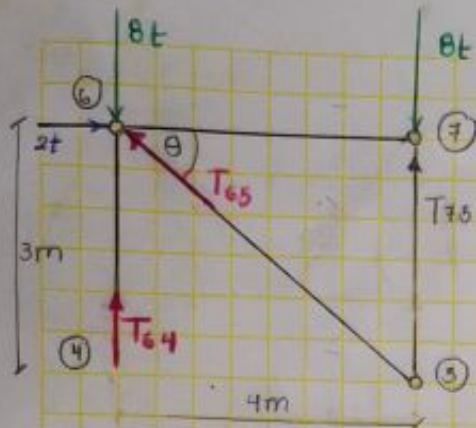


$$\begin{aligned}\sum M_2 &= 0 \quad \curvearrowright \oplus \\ T_{32}(3) - 7(4) &= 0 \\ T_{32} &= 9.333t //\end{aligned}$$



$$\theta = \arctg\left(\frac{3}{4}\right) = 36,8699^\circ$$

$$\begin{aligned}\sum M_2 &= 0 \quad \curvearrowright \oplus \\ T_{35} \cdot \text{Sen}(36,8699)(4) - 7(4) &= 0 \\ T_{35} &= 11,667t //\end{aligned}$$



$$\sum M_5 = 0 \quad (\curvearrowright \oplus)$$

$$T_{64}(4) - 8(4) + 2(3) = 0 \quad \therefore T_{64} = 6,5t //$$

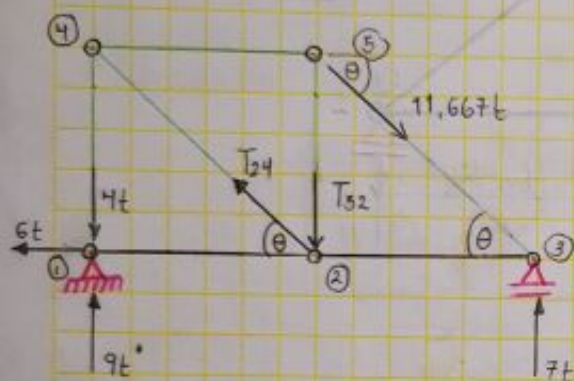
$$\sum M_6 = 0 \quad (\curvearrowright \oplus)$$

$$8(4) - T_{73}(4) = 0 \quad \therefore T_{73} = 8t //$$

$$\theta = \arctg\left(\frac{3}{4}\right) = 36,869^\circ$$

$$\sum M_7 = 0 \quad (\curvearrowright \oplus)$$

$$(6,5 - 8) \cdot 4 + T_{63} \cdot \text{Sen}(36,869) \cdot [4] = 0 \quad \therefore T_{63} = 2,5t //$$



$$\theta = 36,8699^\circ$$

$$\sum M_4 = 0 \quad (\curvearrowright \oplus)$$

$$6(3) + T_{25}(4) - 7(8) + 11,667 \cdot \text{Sen}(36,8699)(4) = 0$$

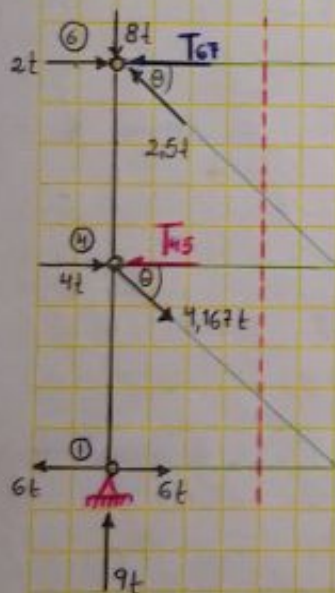
$$T_{25} = 2,4998t //$$

$$\sum M_1 = 0 \quad (\curvearrowright \oplus)$$

$$2,4998 \cdot (4) - 7(8) + 11,667 \cdot \text{Sen}(36,8699)(8) - T_{24} \cdot$$

$$\cdot \text{Sen}(36,8699)(4) = 0$$

$$T_{24} = 4,167t //$$



$$\sum M_3 = 0 \quad (\curvearrowright \oplus)$$

$$T_{45}(3) - 4(3) - 4,167(\text{Cos}(36,8699)) \cdot 3 = 0$$

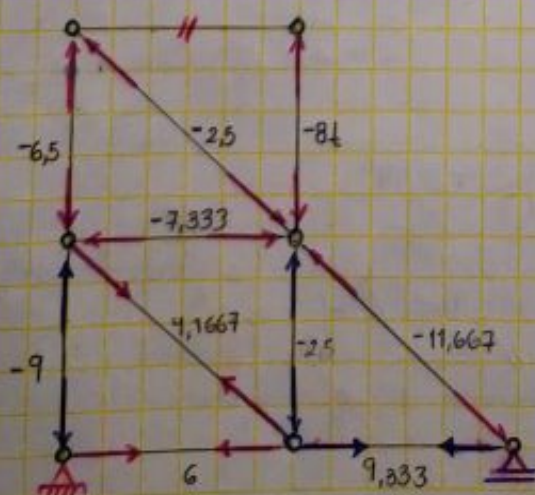
$$T_{45} = 7,333t //$$

$$\sum M_4 = 0 \quad (\curvearrowright \oplus)$$

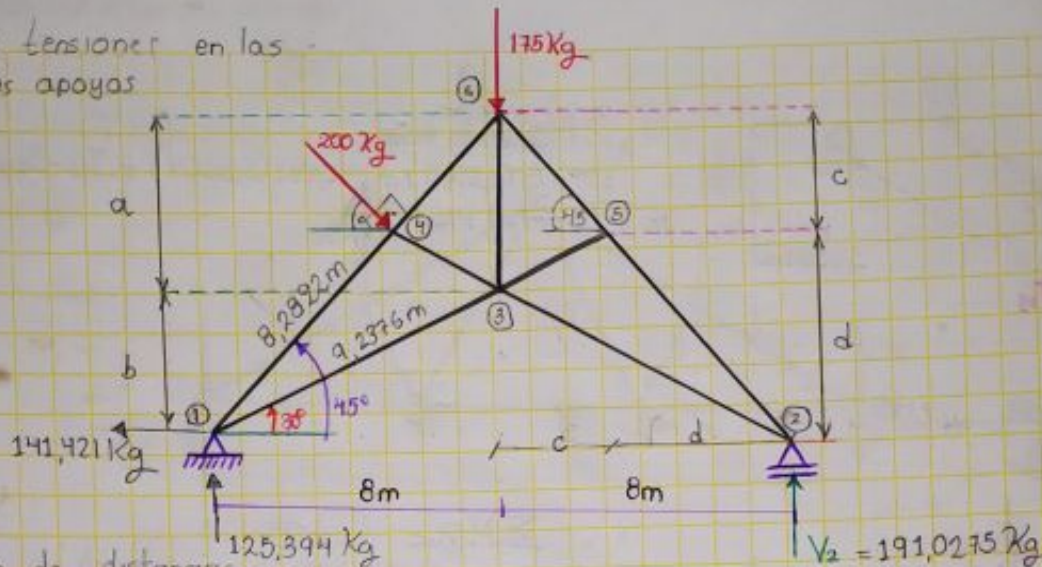
$$2(3) - 2,5 \cdot \text{Cos}(36,8699)(3) - T_{67}(3) = 0$$

$$T_{67} = 0 //$$

3 - Diagrama de Tensiones



hallar las tensiones en las barras de los apoyos



1.- Cálculo de distancias

$$\text{Tang } 30 = \frac{b}{8} \therefore b = 8 \text{ Tang } 30$$

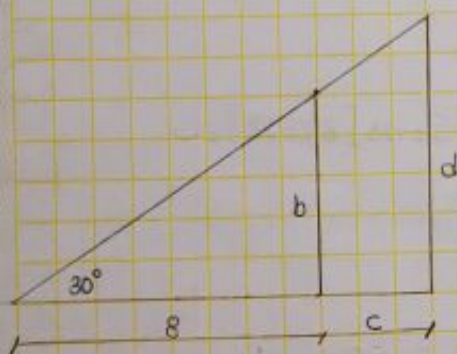
$$b = 4,6188 \text{ m} //$$

$$\text{Tang } 45 = \frac{a + 4,6168}{8} \therefore a = 8 \text{ Tang } 45 - 4,6168$$

$$a = 3,3812 \text{ m} //$$

$$c + d = a + b$$

$$\textcircled{1} \quad c + d = 8 \text{ m} \Rightarrow c = 8 - d$$



$$\frac{d}{b+c} = \frac{8 \text{ Tang } 30}{8}$$

$$8d = 8 \text{ Tang } 30 (b+c)$$

$$8d = 64 \text{ Tang } 30 + 8c \text{ Tang } 30$$

$$8d = 64 \text{ Tang } 30 + 8(8-d) \text{ Tang } 30$$

$$d = 5,8564$$

$$c = 8 - 5,8564$$

$$c = 2,1436 //$$

2. Reacciones

$$\sum M_{\textcircled{1}} = 0 \quad \curvearrowright \oplus$$

$$200(8,2822) + 175(8) - V_2(16) = 0$$

$$V_2 = 191,0275 \text{ t}$$

$$\sum F_v = 0 \quad \uparrow \oplus$$

$$V_1 - 200 \text{ Sen } 45 - 175 + 191,0275 = 0$$

$$V_1 = 125,3938 \text{ Kg}$$

$$\sum F_h = 0 \quad \rightarrow \oplus$$

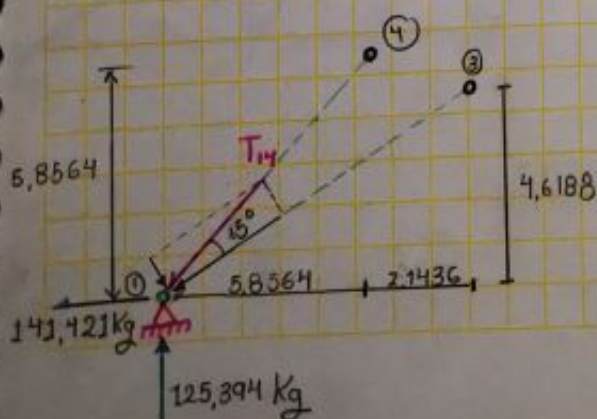
$$H_1 = 200 \text{ Cos } 45 = 141,421 \text{ Kg}$$

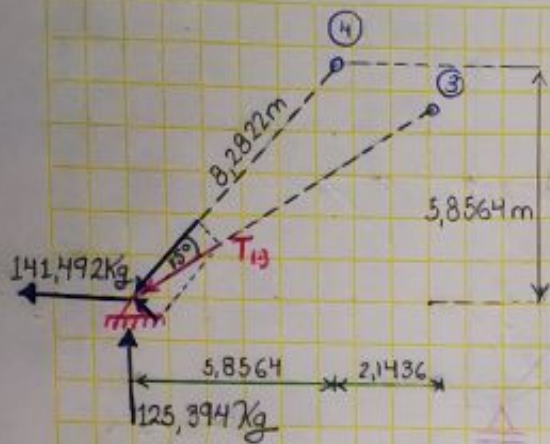
3. Tensiones

$$\sum M_{\textcircled{3}} = 0 \quad \curvearrowright \oplus$$

$$125,394(8) + 141,421(4,6188) - T_{14} \cdot \text{Sen } 15 \cdot (9,2376) = 0$$

$$T_{14} = 692,781 \text{ Kg}$$

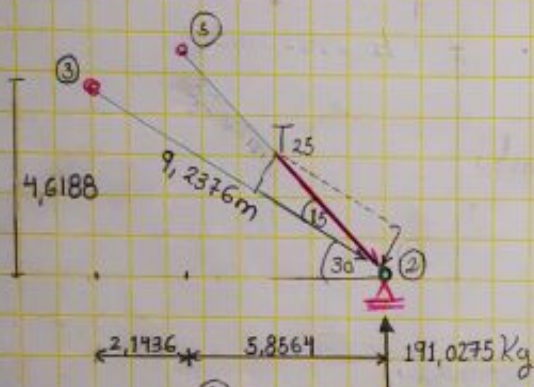




$$\sum M_{(4)} = 0 \quad (\curvearrowright)$$

$$125,394(5,8564) + 141,492(5,8564) + T_{13} \cdot \text{Sen } 15 (8,282) = 0$$

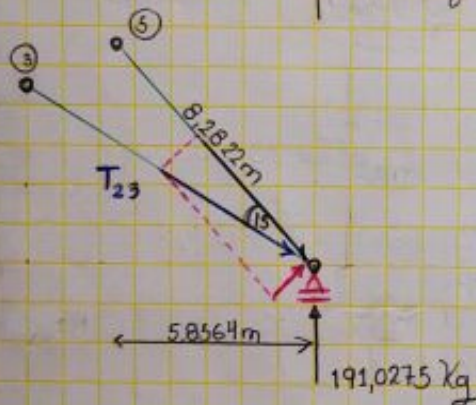
$$T_{13} = -729,146 \text{ Kg}$$



$$\sum M_{(3)} = 0 \quad (\curvearrowright)$$

$$-191,0275(8) + T_{25} \text{ Sen } 15 (9,2376) = 0$$

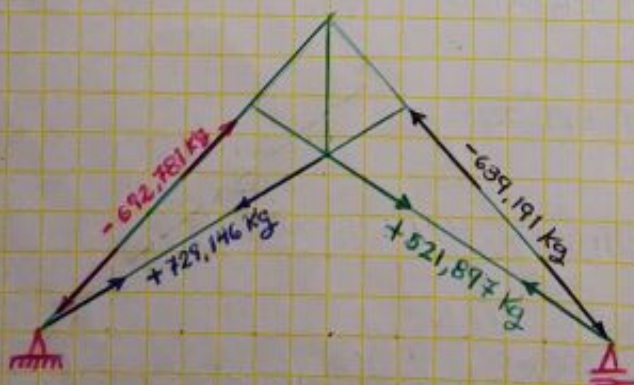
$$T_{25} = 639,191 \text{ Kg}$$



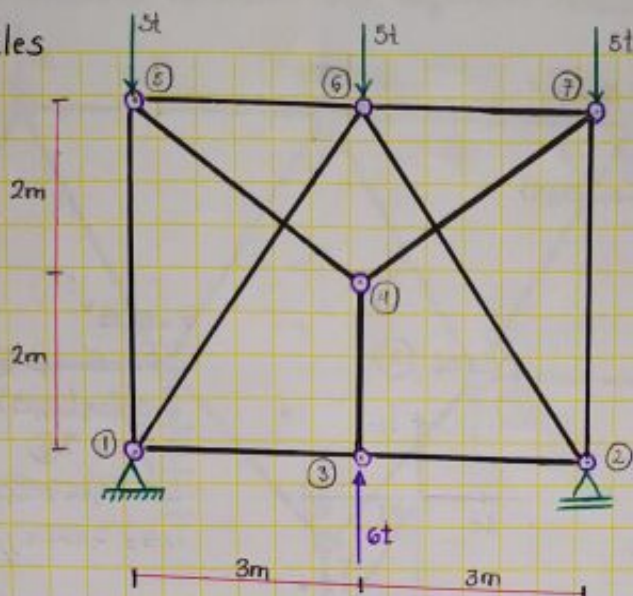
$$\sum M_{(5)} = 0 \quad (\curvearrowright)$$

$$-191,0275(5,8564) - T_{23} \text{ Sen } 15 [8,282] = 0$$

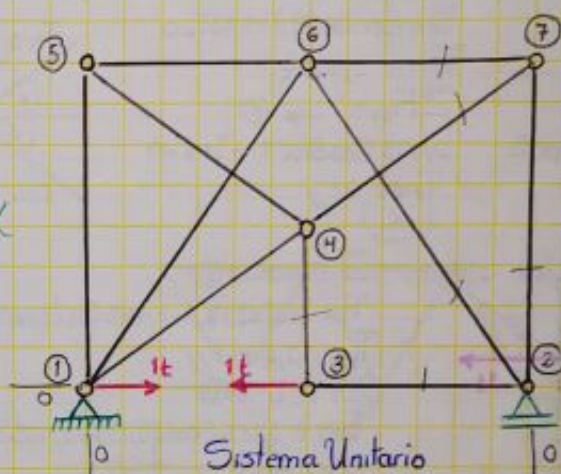
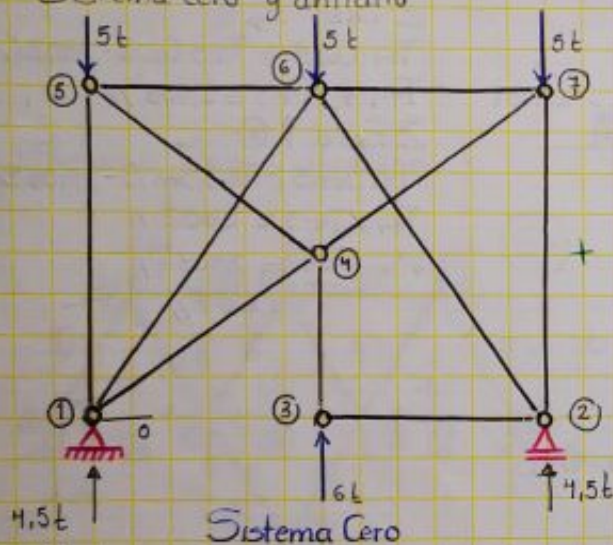
$$T_{23} = -521,897 \text{ Kg}$$



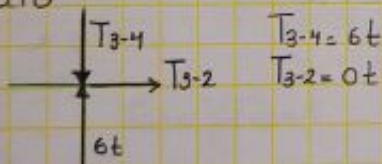
Calcular las Normales



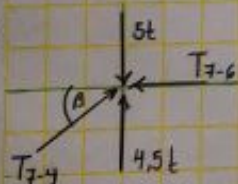
1- Sistema cero y unitario



2- Sistema Cero
Nudo ③



Nudo ⑦



$$\beta = \arctan(2/3) = 33,69^\circ$$

$$\sum F_v = 0 \uparrow \oplus$$

$$T_{7-4} \cdot \sin(33,69) + 4,5 - 5 = 0$$

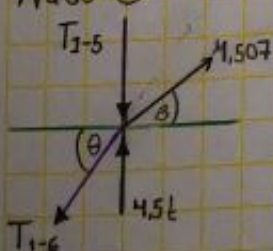
$$T_{7-4} = 0,901 \text{ t} //$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$+ T_{7-4} \cos(33,69) - T_{7-6} = 0$$

$$T_{7-6} = 0,7497 \text{ t} //$$

Nudo ①



$$\theta = \arctan(4/3) = 53,13^\circ$$

$$\sum F_H = 0 \rightarrow \oplus$$

$$- T_{1-6} \cos(53,13) + 4,507 \cdot \cos(33,69) = 0$$

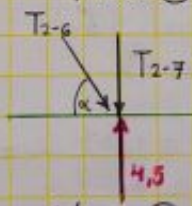
$$T_{1-6} = 6,25 \text{ t} //$$

$$\sum F_v = 0 \uparrow \oplus$$

$$- 6,25 \cdot \sin(53,13) + 4,5 + 4,507 \cdot \sin(33,69) - T_{1-5} = 0$$

$$T_{1-5} = 2 \text{ t} //$$

Nudo ②



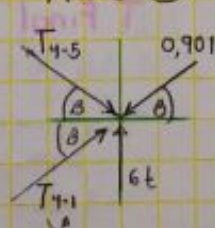
$$\sum F_H = 0 \rightarrow \oplus$$

$$T_{2-6} \cos \alpha = 0 \therefore T_{2-6} = 0$$

$$\sum F_v = 0 \uparrow \oplus$$

$$T_{2-7} = 4,5 \text{ t}$$

Nudo ④



$$\sum F_v = 0 \uparrow \oplus$$

$$T_{4-1} \sin \beta + T_{4-5} \sin \beta + 6 - 0,901 \cdot \sin \beta = 0$$

$$T_{4-1} - T_{4-5} = -9,916$$

$$\sum F_H = 0 \rightarrow \oplus$$

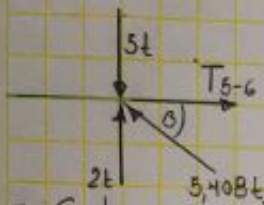
$$T_{4-1} \cos \beta + T_{4-5} \cos \beta - 0,901 \cdot \cos \beta = 0$$

$$T_{4-1} + T_{4-5} = 0,901$$

$$T_{4-1} = -4,507 //$$

$$T_{4-5} = 5,408 //$$

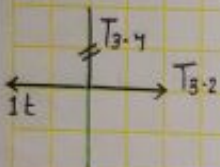
Nudo ⑤



$$\begin{aligned}\sum F_H = 0 &\rightarrow \oplus \\ T_{5-6} &= 5,408 \cos(33,69) \\ T_{5-6} &= 4,5t\end{aligned}$$

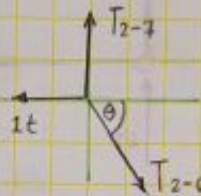
3- Sistema Unitario

Nudo ③



$$\begin{aligned}\sum F_V = 0 &\uparrow \oplus \\ T_{3-4} &= 0t \\ \sum F_H = 0 &\rightarrow \oplus \\ T_{3-2} &= 1t\end{aligned}$$

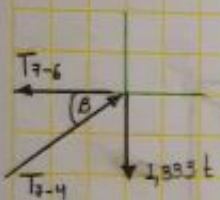
Nudo ②



$$\theta = 53,13^\circ$$

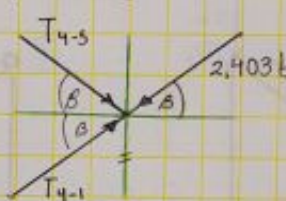
$$\begin{aligned}\sum F_H = 0 &\rightarrow \oplus \\ T_{2-6} \cos(53,13) - 1 &= 0 \therefore T_{2-6} = 1,667t \\ \sum F_V = 0 &\uparrow \oplus \\ T_{2-7} - 1,667 \sin(53,13) &= 0 \\ T_{2-7} &= 1,333t\end{aligned}$$

Nudo ⑦



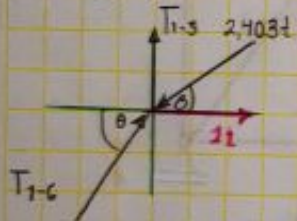
$$\begin{aligned}\beta &= 33,69^\circ \\ \sum F_V = 0 &\uparrow \oplus \\ T_{7-4} \sin(33,69) - 1,333 &= 0 \\ T_{7-4} &= 2,403t \\ \sum F_H = 0 &\rightarrow \oplus \\ 2,403 \cos(33,69) - T_{7-6} &= 0 \\ T_{7-6} &= 2t\end{aligned}$$

Nudo ④



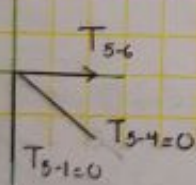
$$\begin{aligned}\sum F_H = 0 &\rightarrow \oplus \\ T_{4-1} \cos \beta + T_{4-5} \cos \beta - 2,403 \cos \beta &= 0 \\ T_{4-1} + T_{4-5} &= 2,403 \\ \sum F_V = 0 &\uparrow \oplus \\ T_{4-1} \sin \beta - T_{4-5} \sin \beta - 2,403 \sin \beta &= 0 \\ T_{4-1} - T_{4-5} &= 2,403 \\ \therefore T_{4-1} &= 2,403t \\ T_{4-5} &= 0t\end{aligned}$$

Nudo ①



$$\begin{aligned}\sum F_H = 0 &\rightarrow \oplus \\ T_{1-6} \cos(53,13) - 2,403 \cos(33,69) + 1 &= 0 \\ T_{1-6} &= 1,666t \\ \sum F_V = 0 &\uparrow \oplus \\ T_{1-3} + 1,666 \sin(53,13) - 2,403 \sin(33,69) &= 0 \\ T_{1-3} &= 0t\end{aligned}$$

Nudo ⑥



$$T_{5-6} = 0t$$

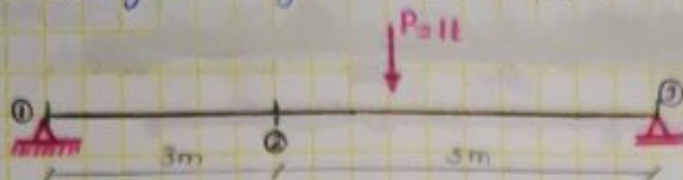
4- Cálculo de χ

5- Tensiones Finales

$$\chi = \frac{-T_{1-4}^0}{T_{1-4}^u} = \frac{-4,507}{-2,403} = 1,8756 //$$

Barra	T^0	χ	T^u	T. Final
1-3	0	1,8756	1	1,8756 t
1-5	-2	"	0	-2 t
1-6	-6,25	"	1,667	-8,1234 t
2-3	0	"	1	1,8756 t
2-6	0	"	-1,667	-3,126 t
2-7	-4,5	"	1,333	-2 t
3-4	-6	"	0	-6 t
4-5	-5,408	"	0	-5,408 t
4-7	-0,901	"	-2,403	-5,408 t
5-6	4,5	"	0	4,5 t
6-7	0,7497	"	2	4,5 t

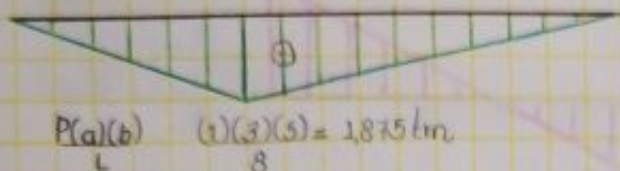
117.- Para la siguiente viga calcular las líneas de influencias sgtes:



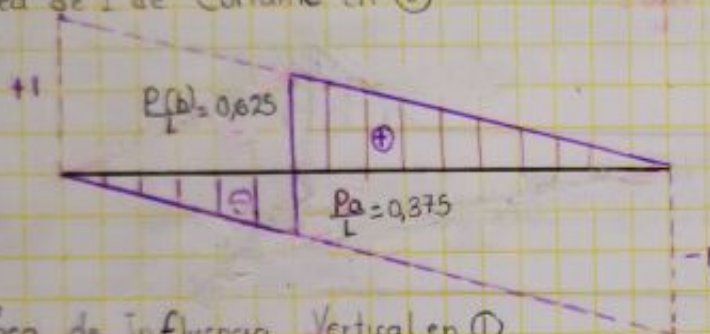
- a) M_{2-2}
- b) Q_{2-2}
- c) V_1
- d) V_3

Utilizar el método Gráfico - Numérico

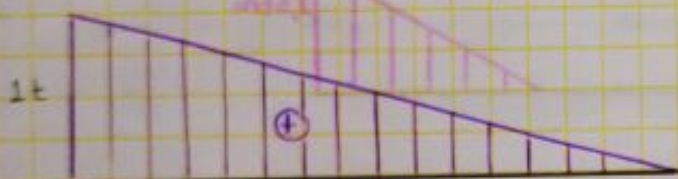
a) Línea de I. de Momentos en ②



b) Línea de I. de Cortante en ②

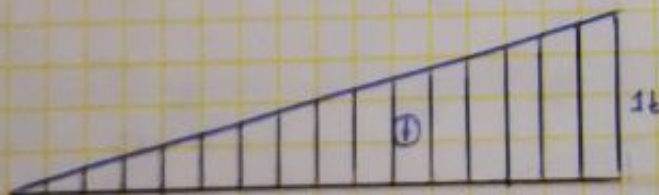


c) Línea de Influencia Vertical en ①



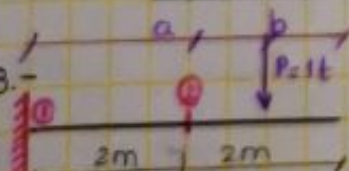
Si $P=1t$ esta en ① $\Rightarrow V_1 = 1t$
Si $P=1t$ " " ③ $\Rightarrow V_1 = 0t$

d) Línea de Influencia Vertical en ③



Si $P=1t$ esta en ① $\Rightarrow V_3 = 0$
Si $P=1t$ " " ③ $\Rightarrow V_3 = 1t$

118.-



Para la viga mostrada calcular las L.I. sgtes:

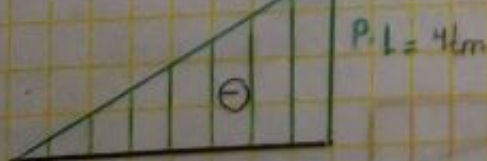
a) Línea de I. de V en ①



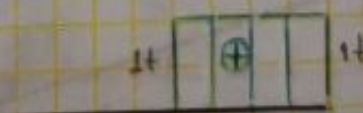
d) Línea de I. de M_{2-2}



b) Línea de I. de M en ①

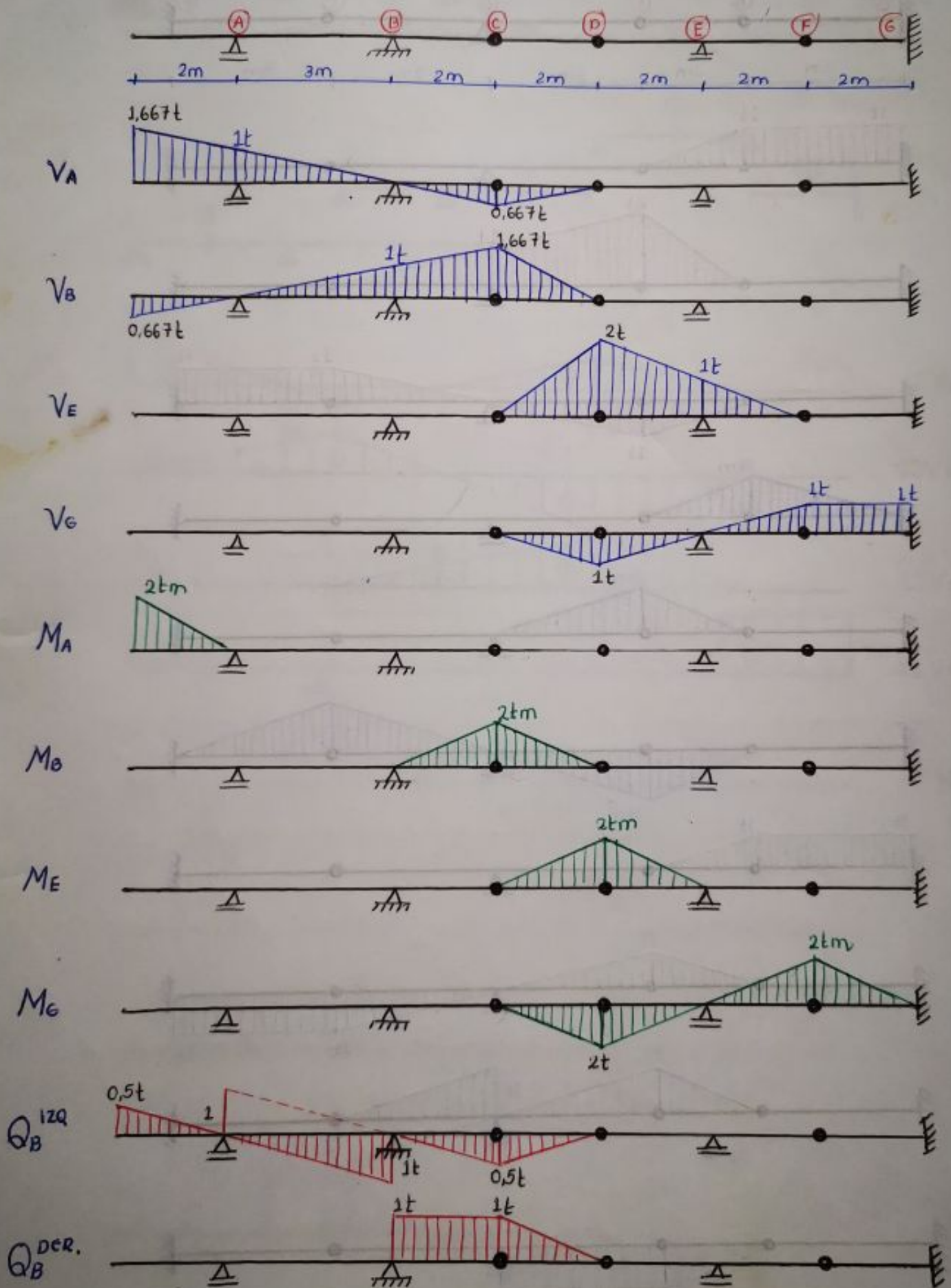


c) Línea de I. de Q_{2-2}

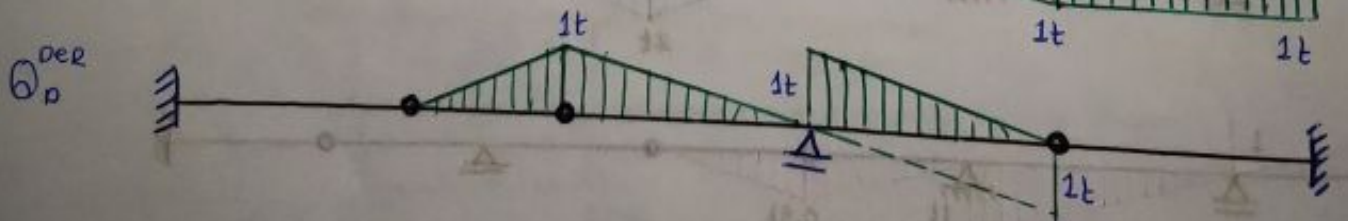
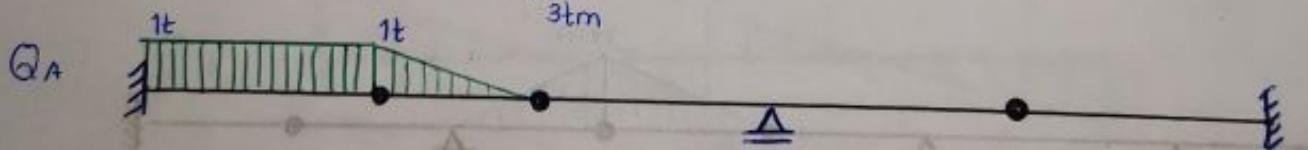
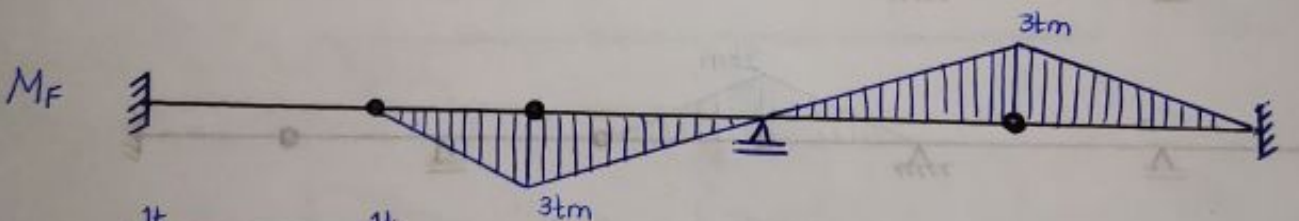
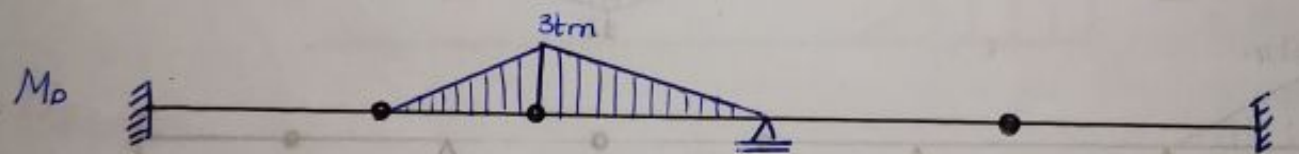
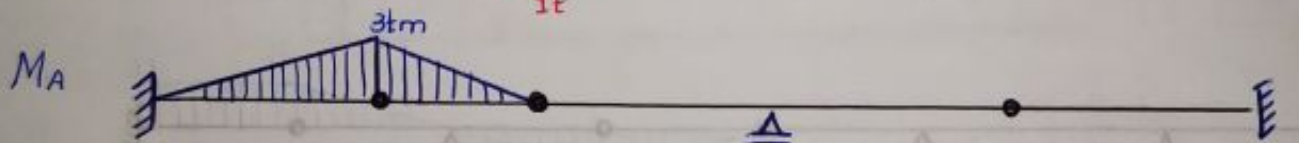
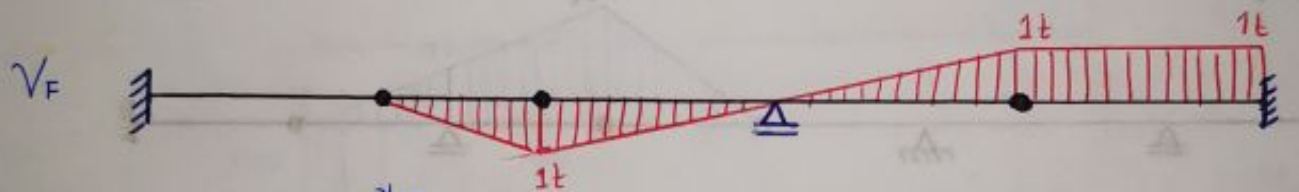
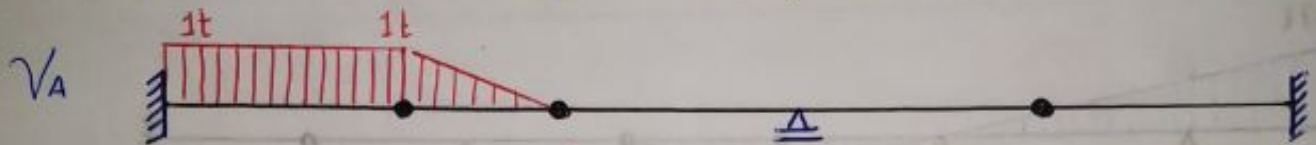
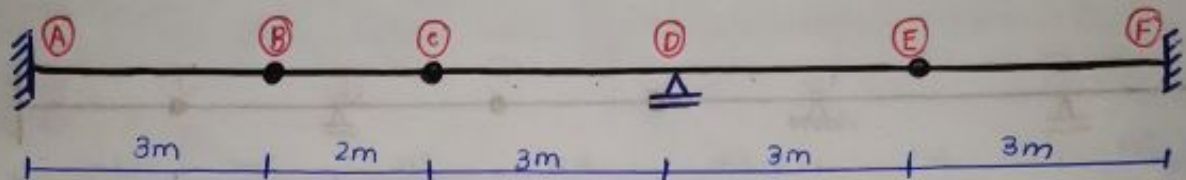


- a) V_1
- b) M_1
- c) M_{2-2}
- d) Q_{2-2}

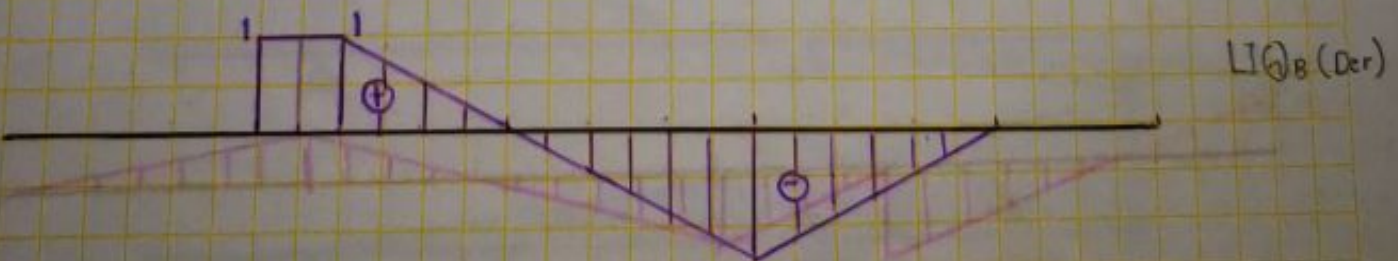
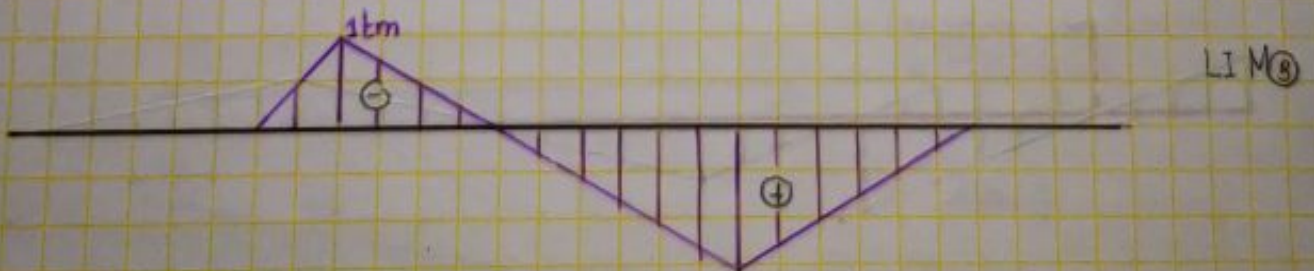
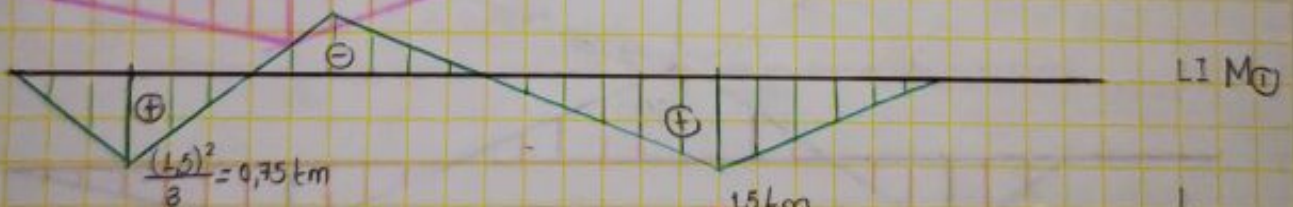
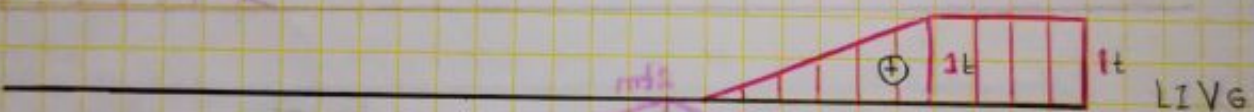
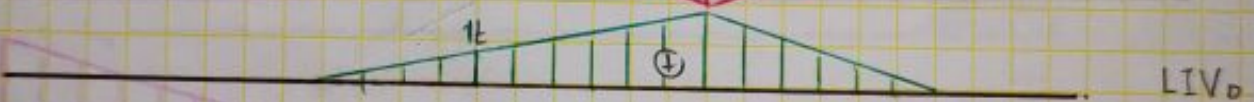
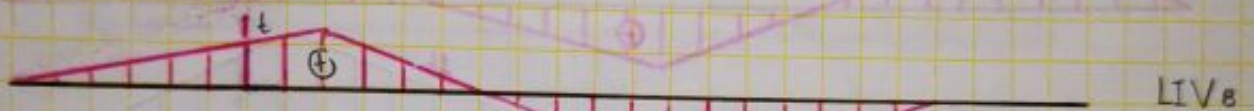
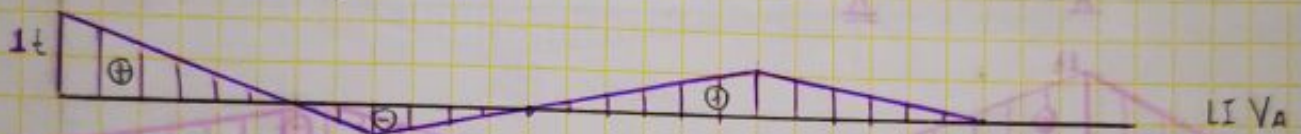
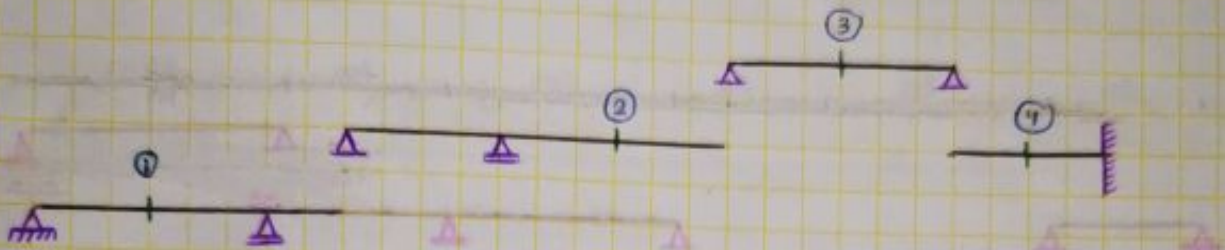
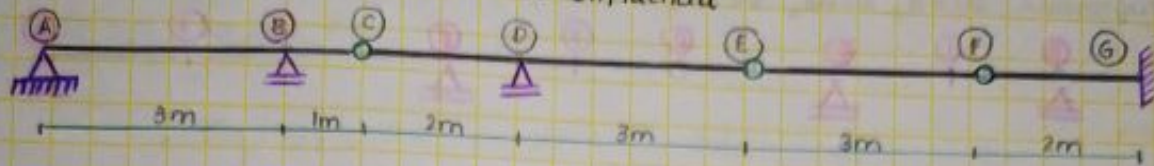
Hallar los siguientes diagramas de Línea de Influencia



Hallar las siguientes líneas de Influencia



120.- Diagramar las siguientes Líneas de Influencia



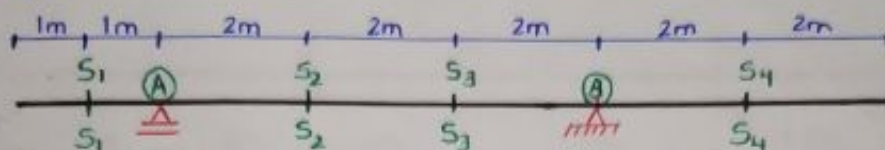
$$1t \cdot 3m = 3t \cdot m$$

Por el Método Gráfico-Numérico diagramar las Líneas de Influencias siguientes:

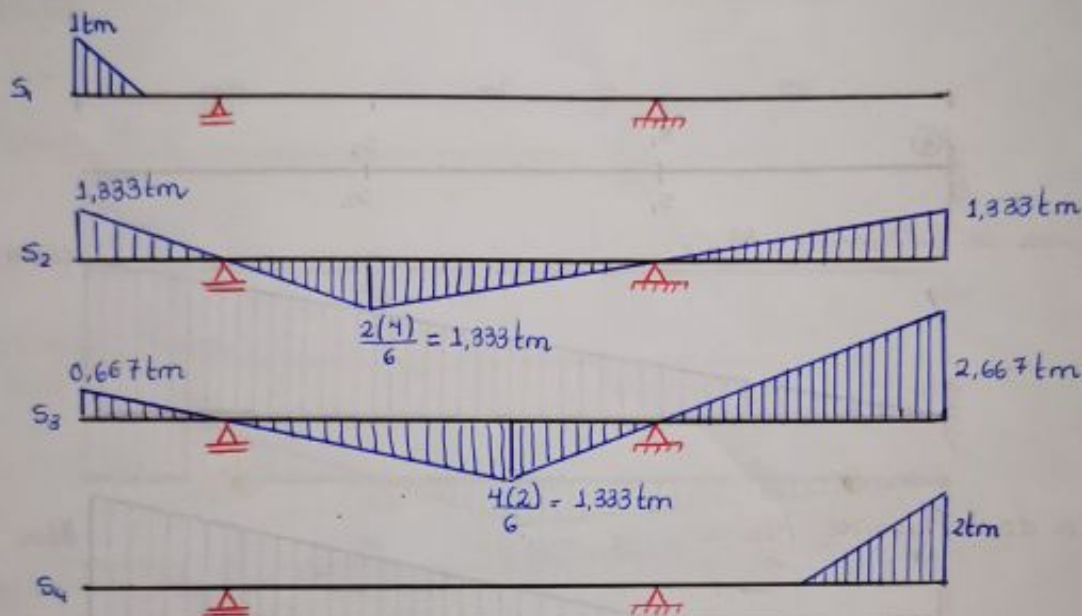
a) Momento - S_1, S_2, S_3, S_4

b) Cortante - S_1, S_2, S_3, S_4

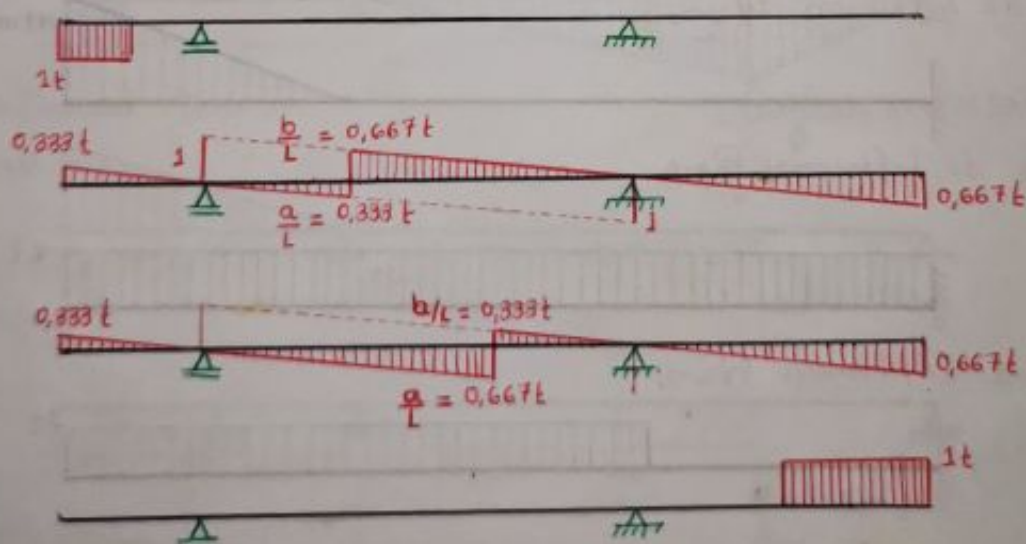
c) Reacción - A, B



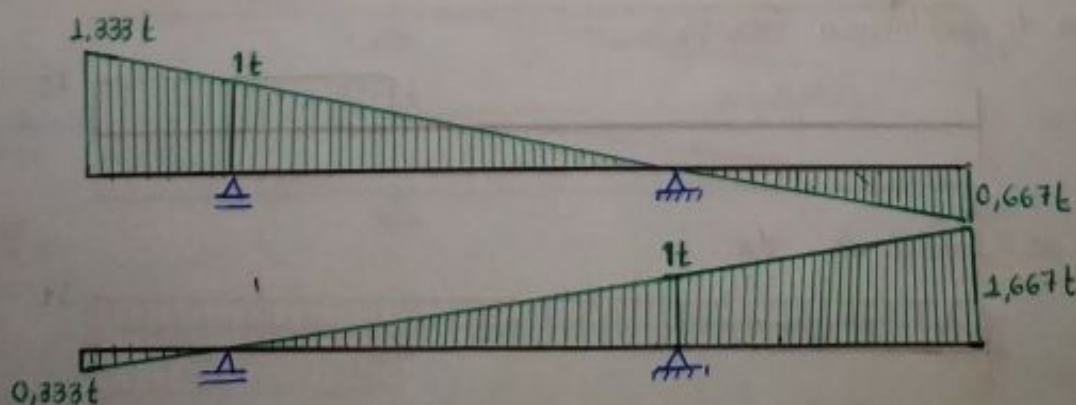
a) Momentos



b) Cortantes

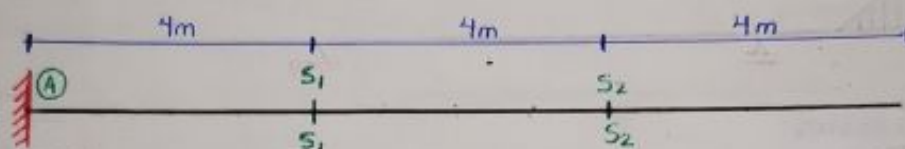


c) Reacción



Por el método gráfico numérico diagramar las siguientes líneas de influencia

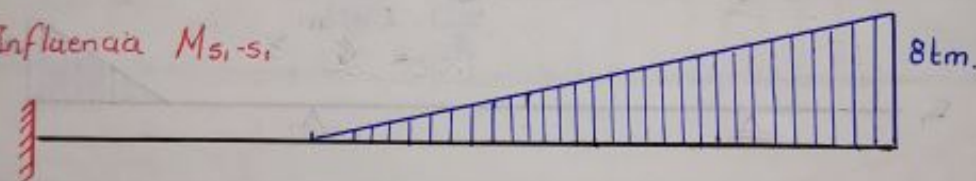
- M_{A-A}
- $M_{S_1-S_1}$
- $M_{S_2-S_2}$
- Q_{A-A}
- $Q_{S_1-S_1}$
- $Q_{S_2-S_2}$
- V_A



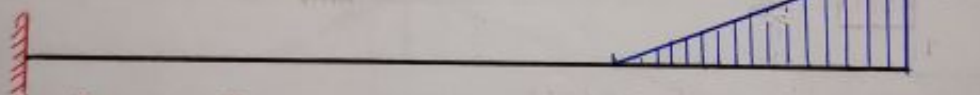
a) Línea de Influencia M_{A-A}



b) Línea de Influencia $M_{S_1-S_1}$



c) Línea de Influencia $M_{S_2-S_2}$



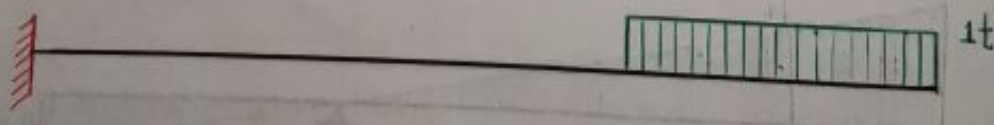
e) Línea de Influencia Q_{A-A}



f) Línea de Influencia $Q_{S_1-S_1}$



g) Línea de Influencia $Q_{S_2-S_2}$



g) Línea de Influencia V_A



Utilizando el método Gráfico-Numérico obtener las siguientes Líneas de Influencia:

a) $M_{S_1-S_1}$

b) $M_{S_2-S_2}$

c) $M_{S_3-S_3}$

d) M_{A-A}

e) $Q_{S_1-S_1}$

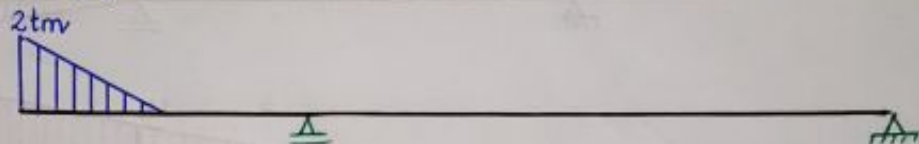
f) $Q_{S_3-S_3}$

g) $Q_{A-A}(\text{derecha})$

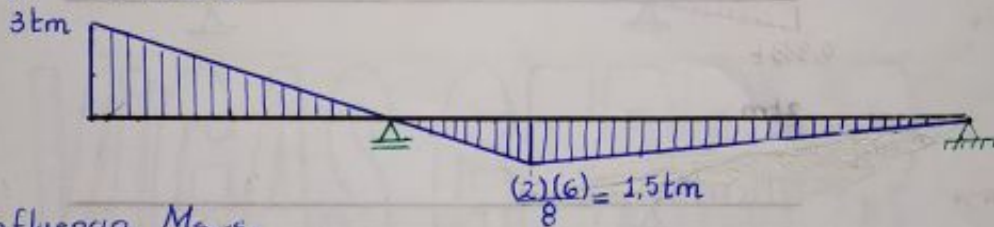
h) $Q_{A-A}(\text{izquierda})$



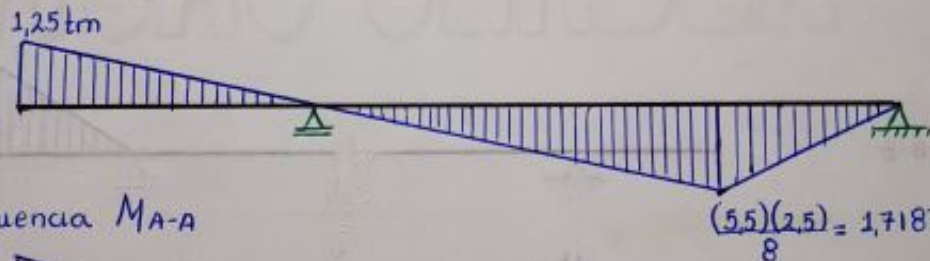
a) Línea de Influencia $M_{S_1-S_1}$



b) Línea de Influencia $M_{S_2-S_2}$



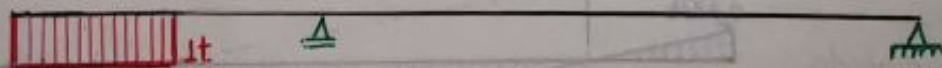
c) Línea de Influencia $M_{S_3-S_3}$



d) Línea de Influencia M_{A-A}



e) Línea de I. $Q_{S_1-S_1}$



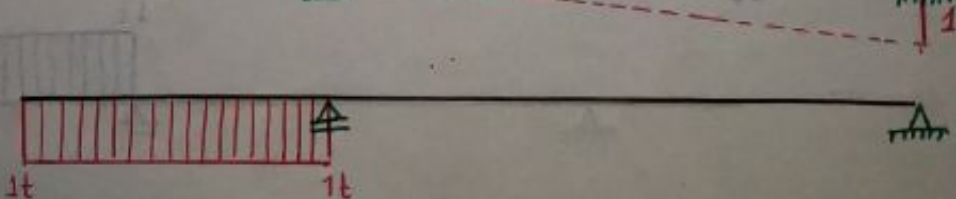
f) Línea de I. $Q_{S_3-S_3}$



g) Línea de I. Q_{A-A}^{derecha}

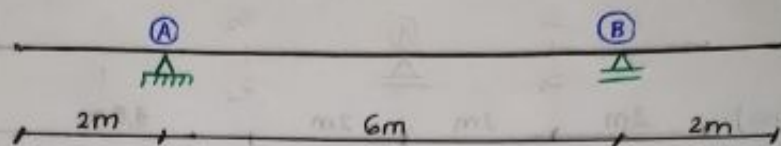


h) Línea de I. $Q_{A-A}^{\text{izq.}}$



Obtener los diagramas de linea de influencia siguientes!

- V_A
- V_B
- M_{A-A}
- M_{B-B}
- Q_{A-A} derecha
- Q_{B-B} izquierda
- Q_{A-A} izquierda
- Q_{B-B} derecha



a) L.I. V_A



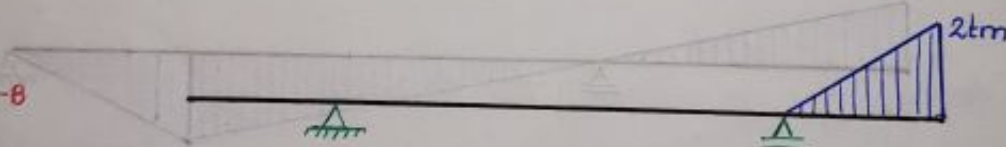
b) L.I. V_B



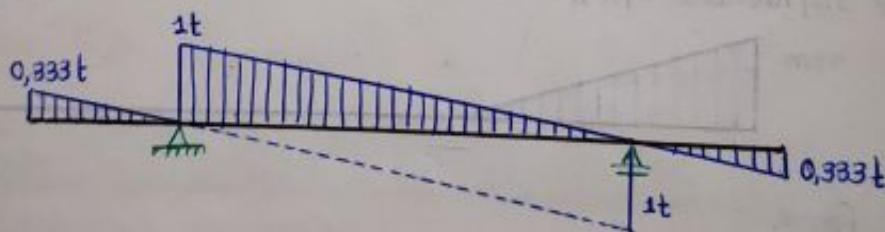
c) L.I. M_{A-A}



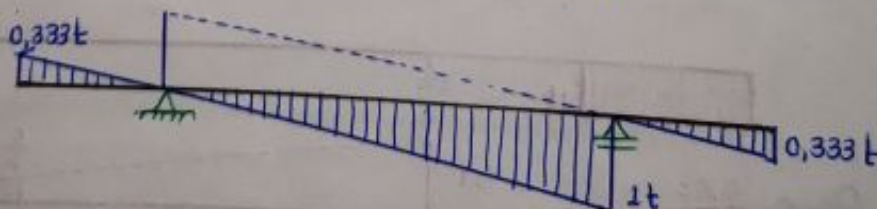
d) L.I. M_{B-B}



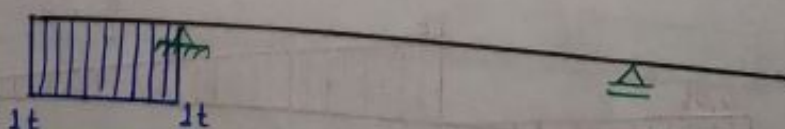
e) Q_{A-A} derecha



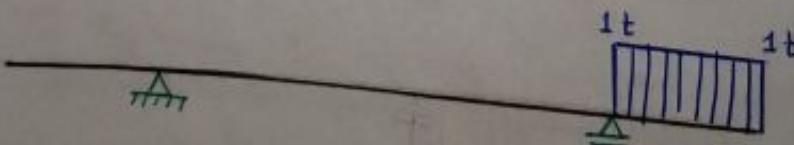
f) Q_{B-B} izquierda



g) Q_{A-A} izquierda

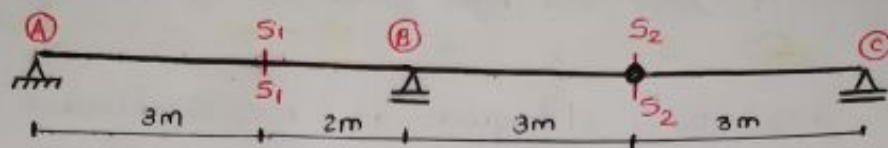


h) Q_{B-B} derecha

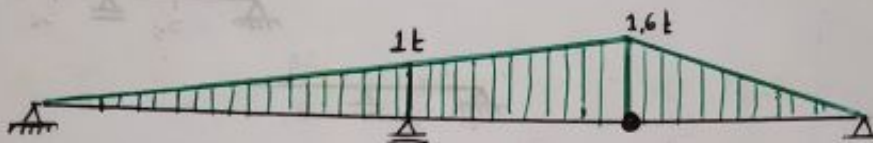


Diagramar las siguientes Líneas de Influencia

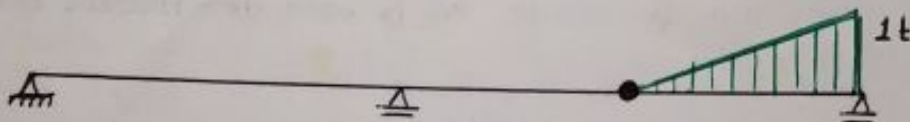
- a) V_A
- b) V_B
- c) V_C
- d) $M_{S_1-S_1}$
- e) $M_{S_2-S_2}$
- f) M_B
- g) Q_B^{IZQ}
- h) Q_B^{DER}



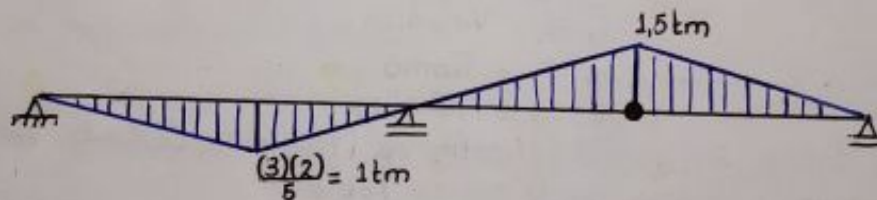
b) L.I. V_B



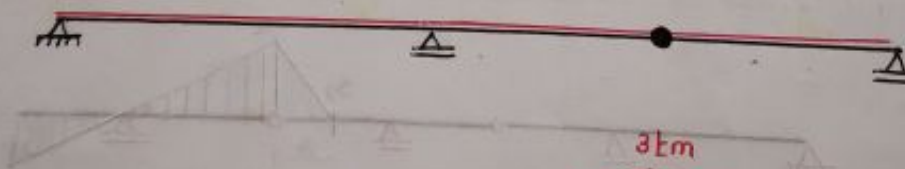
c) L.I. V_C



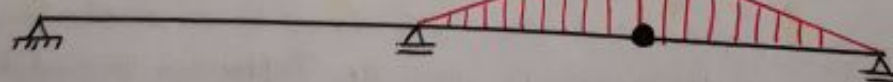
d) L.I. $M_{S_1-S_1}$



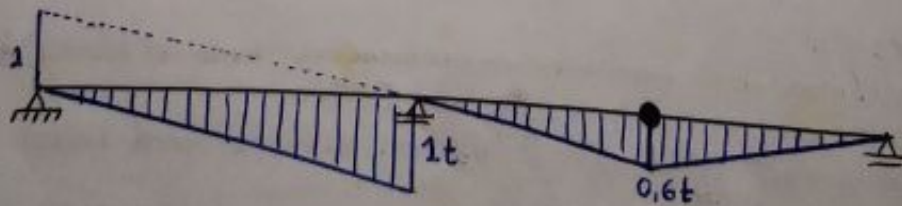
e) L.I. $M_{S_2-S_2}$



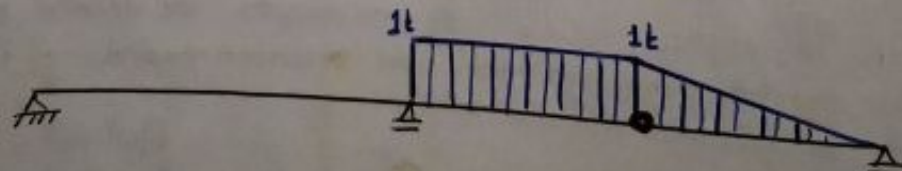
f) L.I. M_B



g) L.I. Q_B^{IZQ}

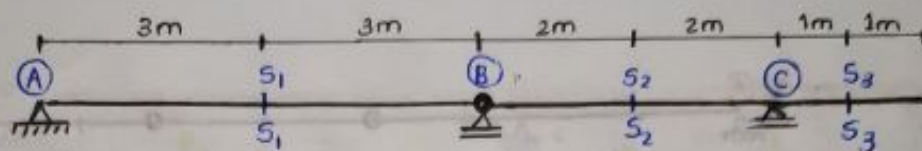


h) L.I. Q_B^{DER}



Hallar las siguientes Líneas de Influencia

- a) V_A
- b) V_B
- c) V_C
- d) $M_{S_1-S_1}$
- e) $M_{S_2-S_2}$
- f) $M_{S_3-S_3}$
- g) $Q_{S_2-S_2}$
- h) $Q_{S_1-S_1}$



a) L.I. V_A



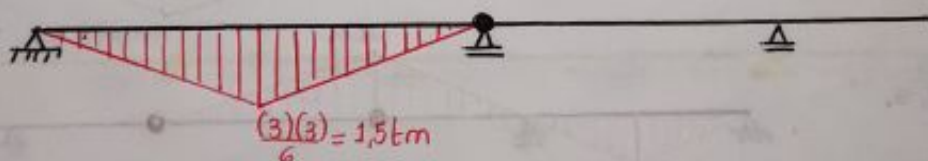
b) L.I. V_B



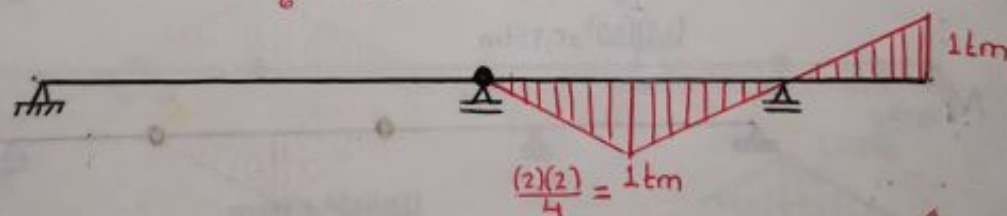
c) L.I. V_C



d) L.I. $M_{S_1-S_1}$



e) L.I. $M_{S_2-S_2}$



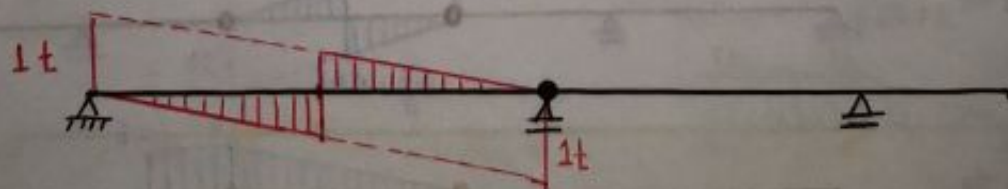
f) L.I. $M_{S_3-S_3}$



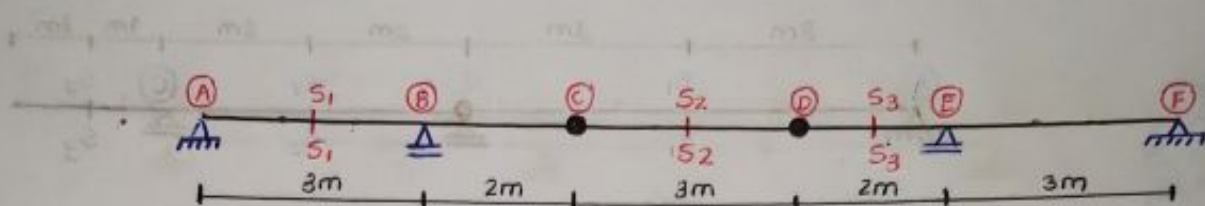
g) L.I. $Q_{S_2-S_2}$



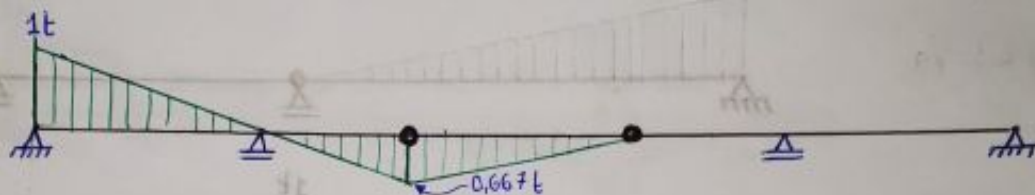
h) L.I. $Q_{S_1-S_1}$



Hallar las siguientes Líneas de Influencia



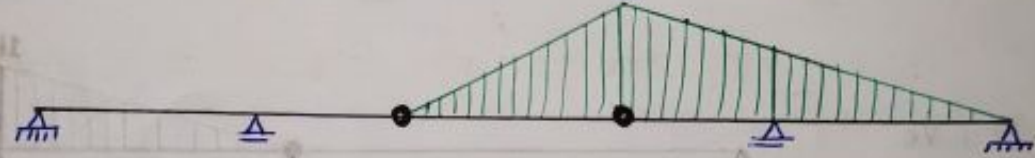
a) L.I. V_A



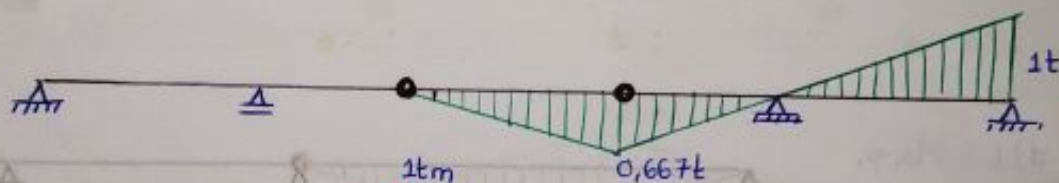
b) L.I. V_B



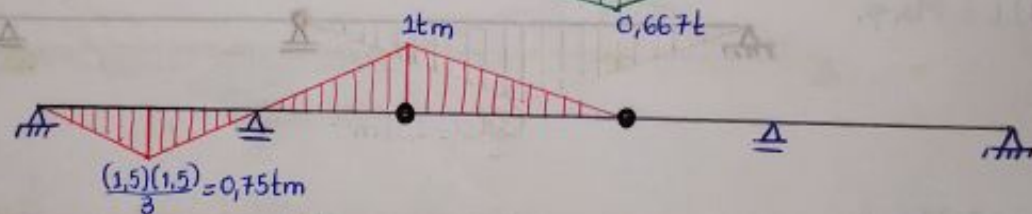
c) L.I. V_E



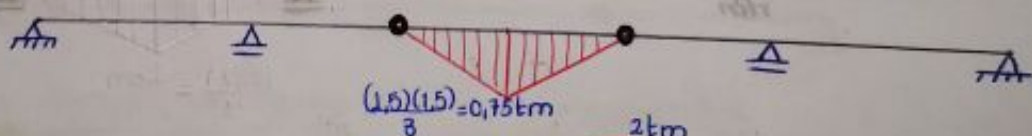
d) L.I. V_F



e) L.I. $M_{S_1-S_2}$



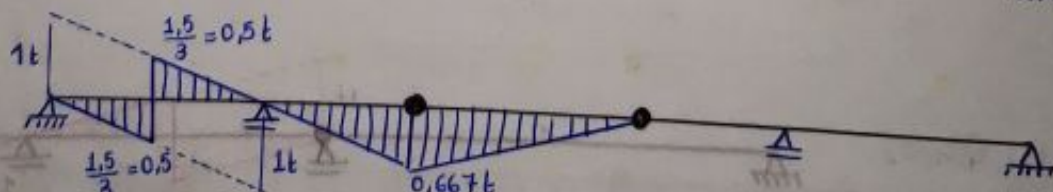
f) L.I. $M_{S_2-S_3}$



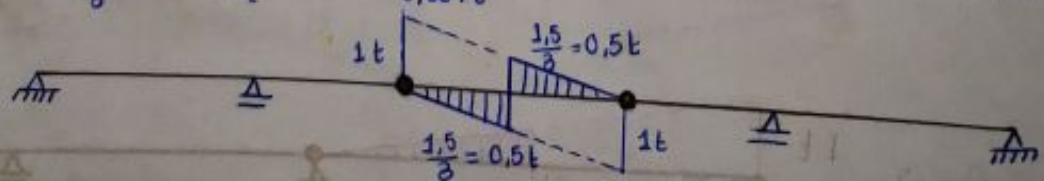
g) L.I. $M_{S_3-S_4}$



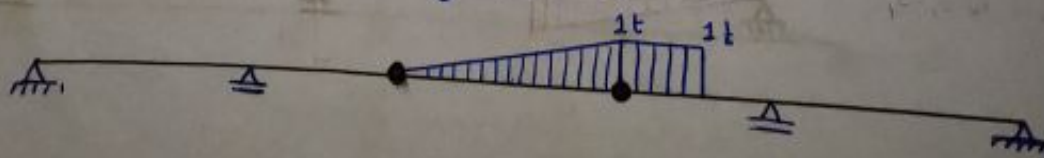
h) L.I. $Q_{S_1-S_2}$



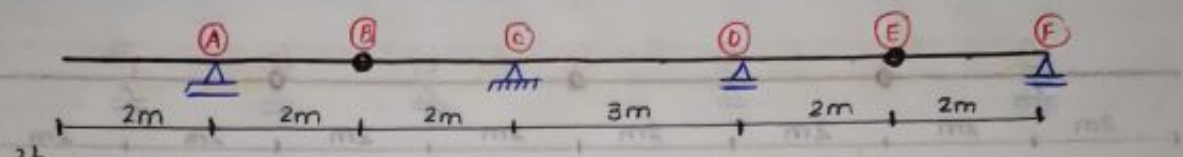
i) L.I. $Q_{S_2-S_3}$



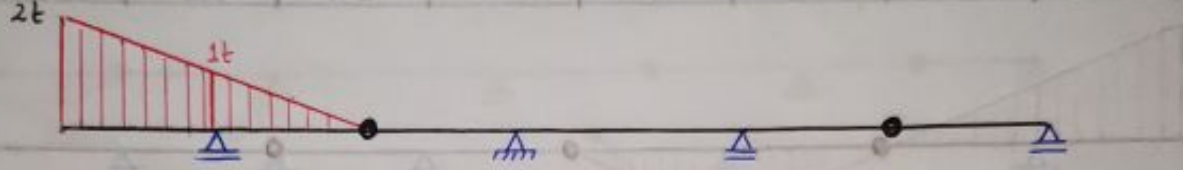
j) L.I. $Q_{S_3-S_4}$



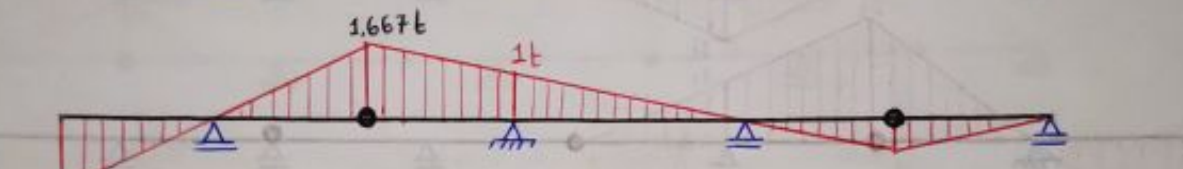
Hallar las siguientes líneas de Influencia



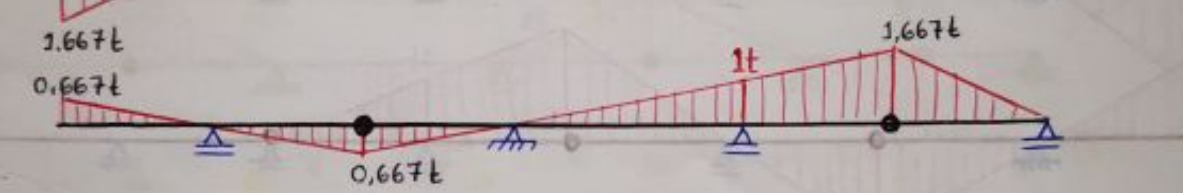
a) L.I. V_A



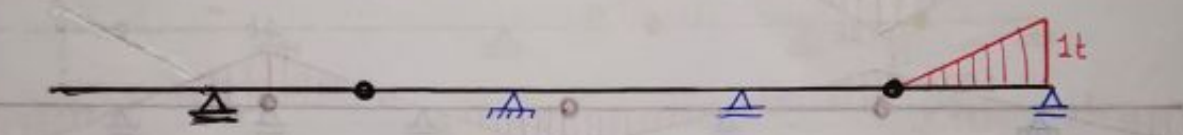
b) L.I. V_c



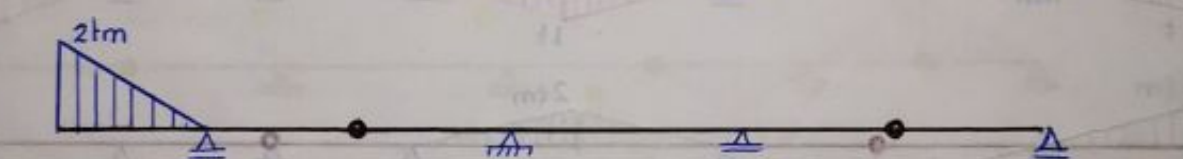
c) L.I. V_o



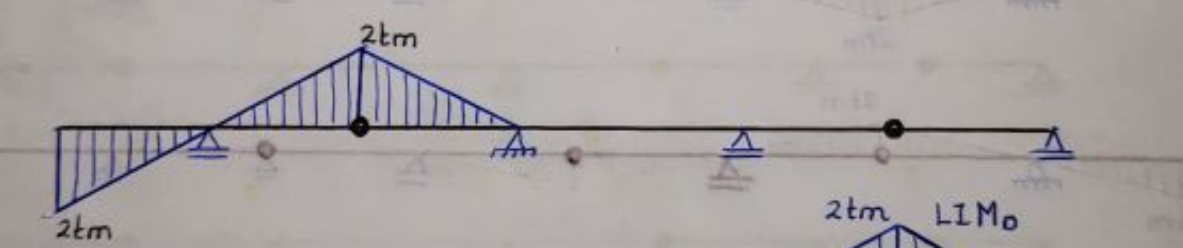
d) L.I. V_F



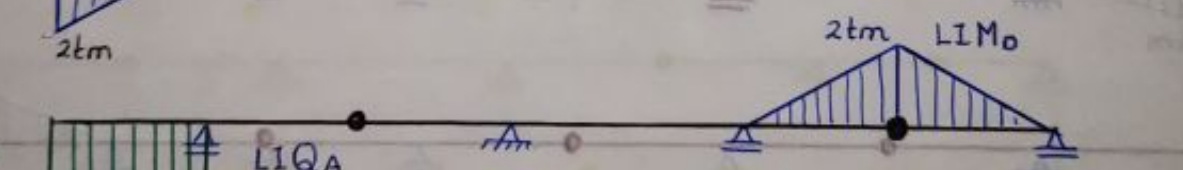
e) L.I. M_A



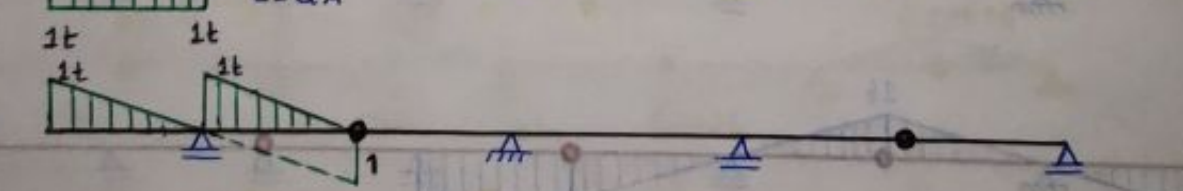
f) L.I. M_c



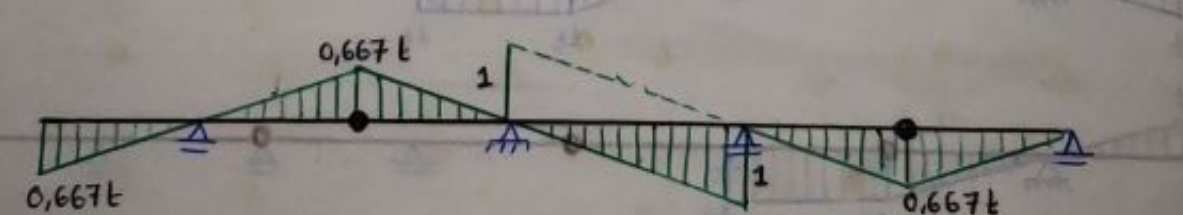
g) L.I. M_o



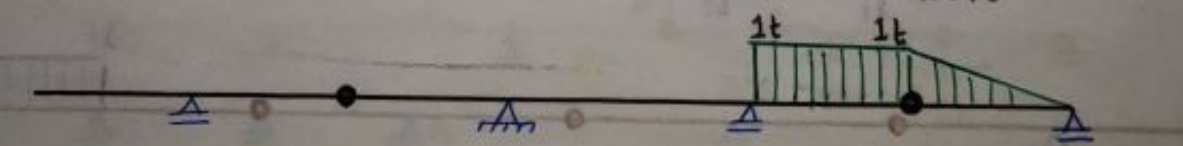
L.I. $Q_A(12Q)$



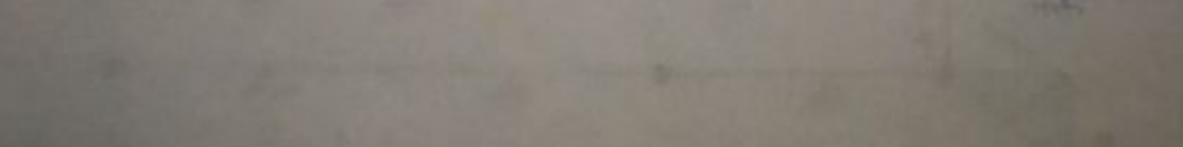
h) L.I. $Q_A(0ER)$



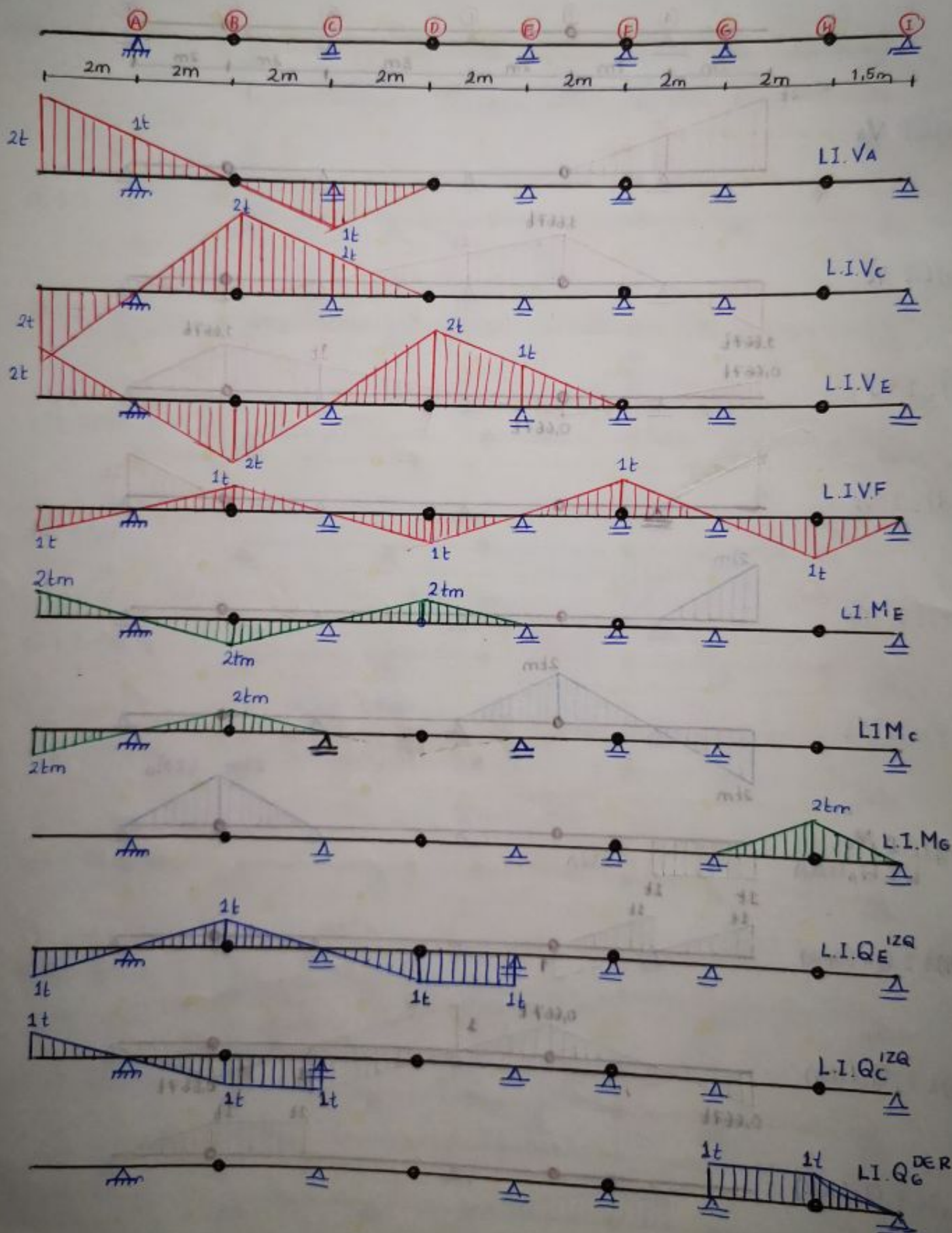
i) L.I. $Q_o(12Q)$



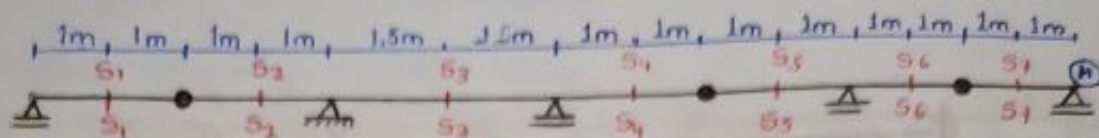
j) L.I. $Q_o(0ER)$



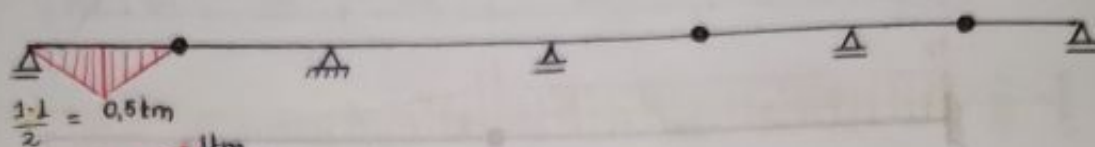
Hallar las siguientes Líneas de Influencia:



Calcular las siguientes Líneas de Influencia:



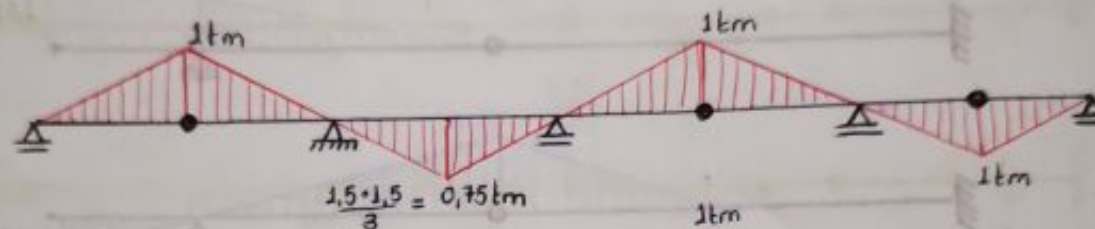
LI M_{S1}



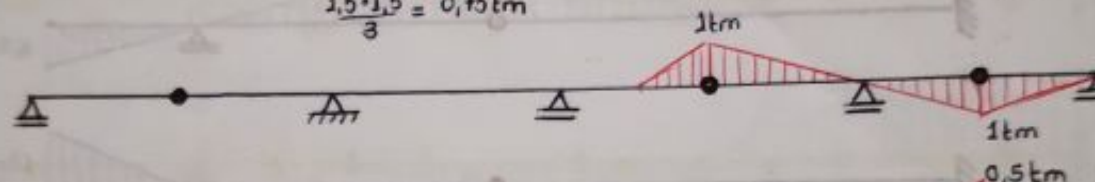
LI M_{S2}



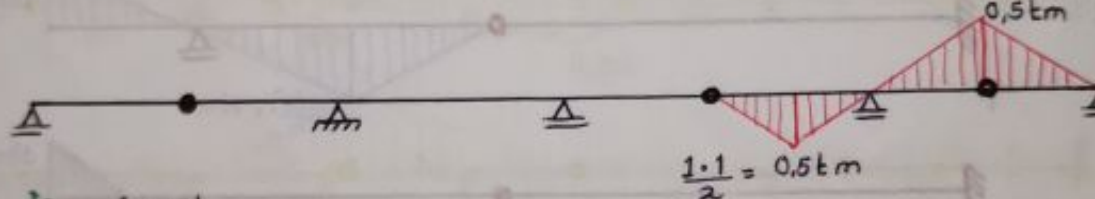
LI M_{S3}



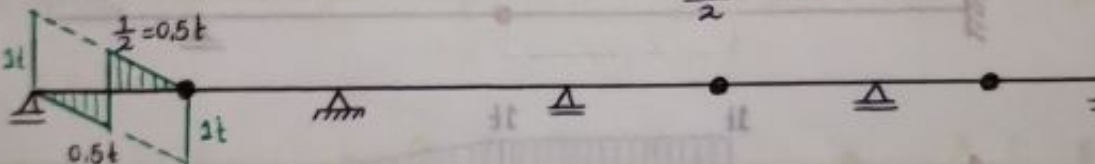
LI M_{S4}



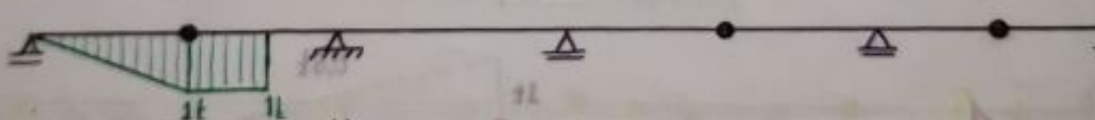
LI M_{S5}



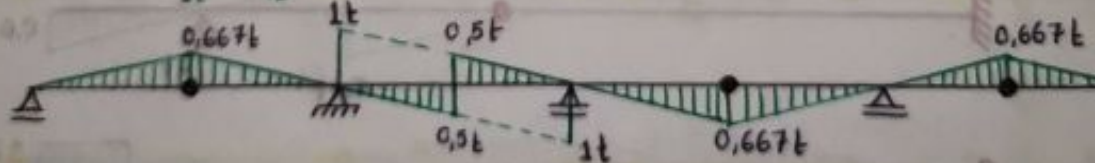
LI Q_{S1}



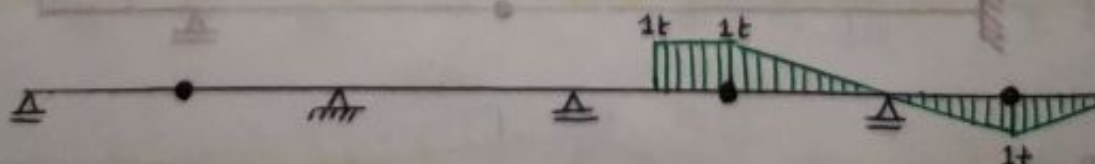
LI Q_{S2}



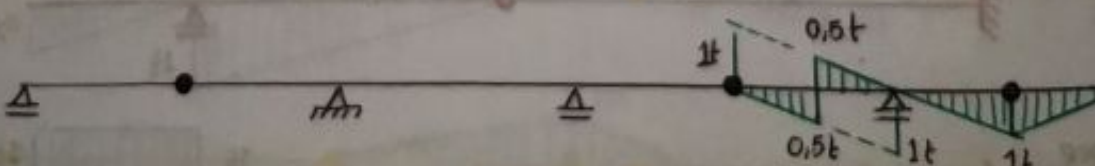
LI Q_{S3}



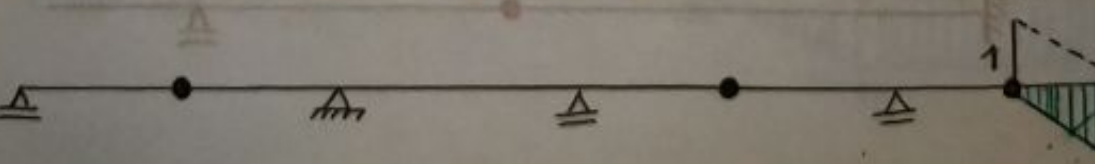
LI Q_{S4}



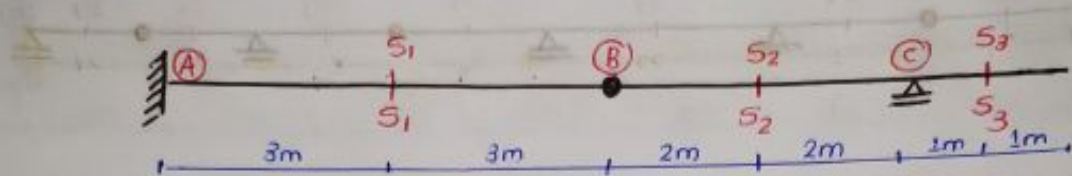
LI Q_{S5}



LI Q_H



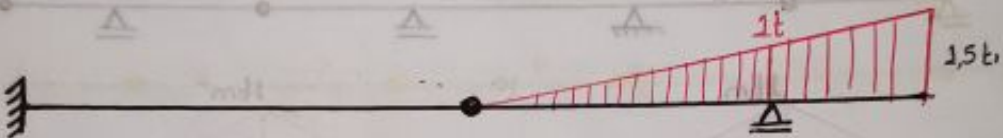
Hallar las siguientes Lineas de Influencia



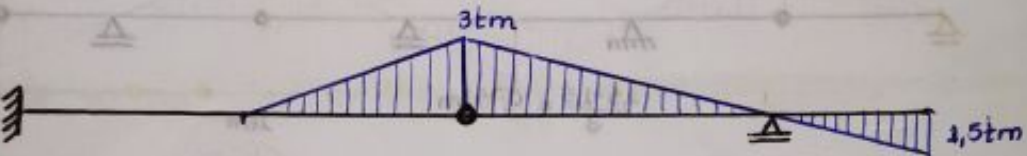
a) L.I. V_A



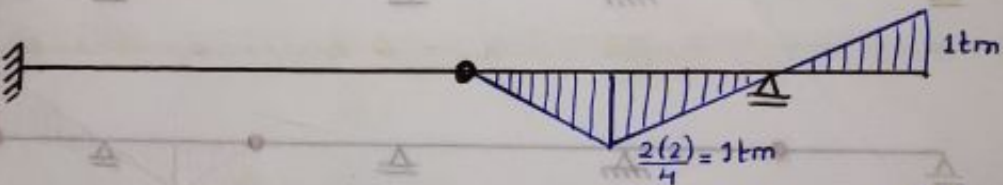
b) L.I. V_C



c) L.I. M_{S1}



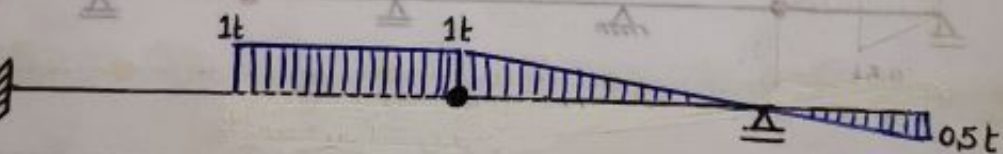
d) L.I. M_{S2}



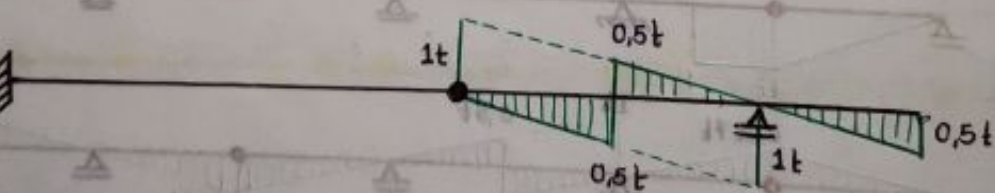
e) L.I. M_{S3}



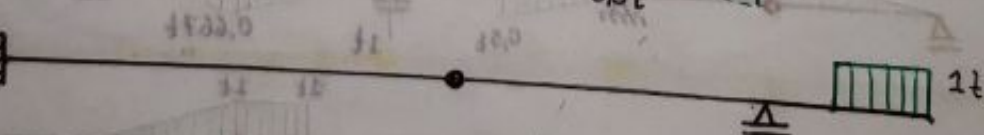
f) L.I. O



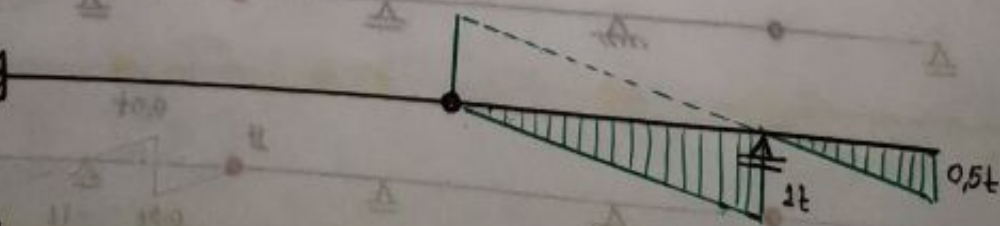
g) L.I. Q_{S2}



h) L.I. Q_{S3}



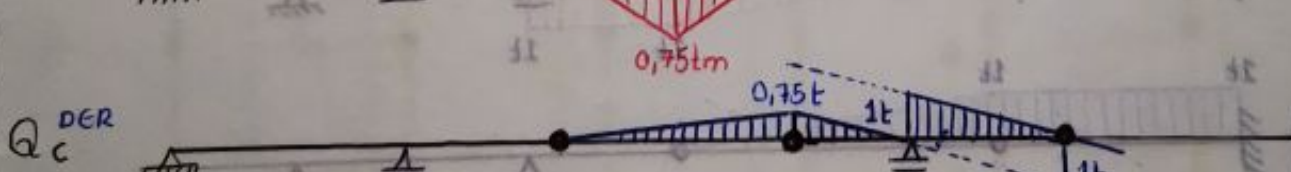
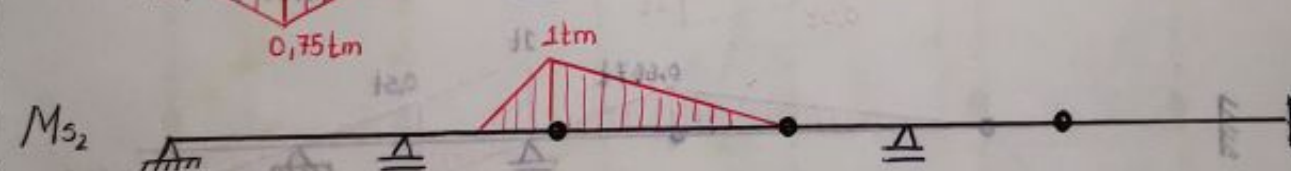
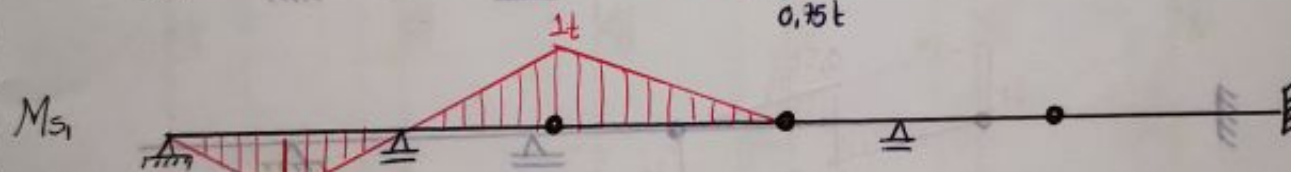
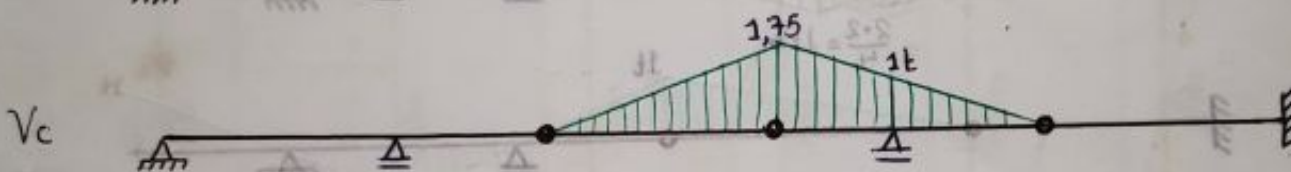
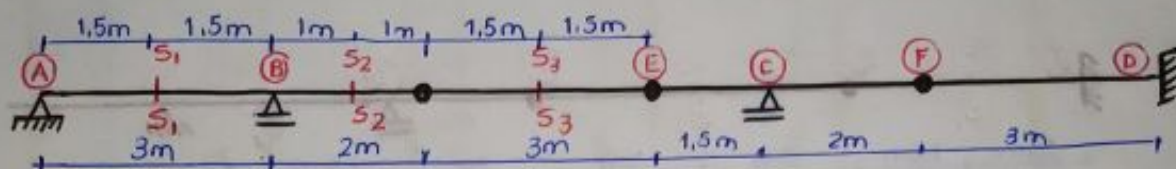
i) L.I. Q_c^{IzQ}



j) L.I. Q_c^{DER}



Hallar las siguientes Lineas de Influencia



Diagramar las siguientes Líneas de Influencia

